

47. In Angular , one can create a local HTML reference of an HTML tag using a variable , which starts with character \_\_\_\_ .

- A. @
- B. #
- C. \*
- D. &

48. If you provide a service in two Angular components, in the "providers" section of @Component decorator, how many instances of service shall get created?

- A. 4
- B. 2
- C. 3

49. In Angular routing , which of these tags is used to show the selected route component dynamically ?

- A. <router></router>
- B. <router-output></router-output>
- C. <router-outlet></router-outlet>
- D. <router-input></router-input>

ANS : C

50. Which method of Angular RouterModule should be called for providing all routes in AppModule?

- A. RouterModule.forChild
- B. RouterModule.forRoot
- C. RouterModule
- D. RouterModule.all

ANS : B

## JAVA SPRING :

Question: 1 Expression Language is the part of Core Container in Spring Framework.

Your Answer: None

Correct Answer: True

Description: None

Question: 2 AOP is the part of Core Container in Spring Framework.

Your Answer: None

Correct Answer: False

Description: None

Question: 3 In which spring version has Spring Expression Language been supported?

Your Answer: None

Correct Answer: Spring 3.0

Description: None

Question: 4 Which class represents the IoC container?

Your Answer: None

Correct Answer: ApplicationContext

Description: None

Question: 5 How to get the object of DAO in spring framework?

Your Answer: None

Correct Answer: Using Spring Dependency injection

Description: None

Question: 6 Which is correct for Spring Framework?

Your Answer: None

Correct Answer: Spring framework is a light-weight solution.

Description: None

Question: 7 In which spring version, java 5 features has been introduced?

Your Answer: None

Correct Answer: Spring 3.0

Description: None

Question: 8 Can we integrate spring with struts?

Your Answer: None

Correct Answer: Yes

Description: None

Question: 9 How to use idref in spring framework?

Your Answer: None

Correct Answer: With setter method and constructor argument both

Description: None

Question: 10 Which is the part of the Data Access layer in Spring Framework?

Your Answer: None

Correct Answer: JMS

Description: None

1. Does Spring provide programmatic transaction management? Select a unique answer.

a. Yes using the @Transactional bean post processor

**b. Yes with the TransactionTemplate class**

c. Yes with the TransactionService class

d. Yes using the @Transactional bean post processor

ans: b

Explanation

1. The TransactionTemplate class provides an execute(TransactionCallback) method

2. The TransactionService class does not exist

3. The @Transactional annotation is for declarative transaction management

2. How could you externalize constants from a Spring configuration file or a Spring annotation into a .properties file? Select one or more

**a. By using the <context:property-placeholder /> tag**

b. By using the <util:constant /> tag

c. By declaring the ConstantPlaceholderConfigurer bean post processor

d. By using the c: namespace

ans: a

3. To validate Java beans in a web application using annotations.

a) XML

b) Java Based

c) JAR-303 standard

**d) All of the mentioned**

ans: d

4. Annotation used to indicate a field has to have a minimum of 2 characters.

a) @NotNull

**b) @Size**

c) @MaxSize

d) @size

Answer: b

Explanation: @Size annotation used to indicate a field has to have a minimum of 2 characters.

5. Spring MVC supports generating Excel files using which of the following libraries.

a) Apache POI library

b) JExcelAPI library

**c) All of the mentioned**

d) None of the mentioned

ans: C

6. Interface for Dispatcher Servlet to auto detect view resolver beans.

- a) localeResolver
- b) tiles
- c) ViewResolver
- d) none of the mentioned.

ans: C

Explanation: Spring MVC comes with several ViewResolver implementations for you to resolve views using different strategies.

7. By default, XmlViewResolver loads view beans from /WEB-INF/views.xml which is final.

- a) True
- b) False

ans: b

8. Views that cannot be resolved by InternalResourceViewResolver.

- a) redirect
- b) redirect:prefix
- c) redirect:suffix
- d) all of the mentioned

ans: a

9. Annotation which allows a controller's field to be assigned using Spring Expression Language(SpEL)

- a) @Value
- b) @After
- c) @Default
- d) @None

ans: a

10. SpEL statements are recognizable.

- a) True
- b) False

View Answer

Answer: a

Explanation: They use a notation in the form "{ SpEL statement }".

11. To publish a REST service with Spring.

- a) publishing an application's data as a REST service
- b) accessing data from third-party REST services
- c) none of the mentioned
- d) all of the mentioned

ans: a b

12. Publishing an application's data as a REST service requires.

- a) @RequestMapping
- b) @PathVariable
- c) All of the mentioned
- d) None of the mentioned

ans: a b

13. Annotation added as an input parameter to the handler method.

- a) @PathVariable
- b) @Path
- c) @PathLocale
- d) None of the mentioned

ans: a

14. Which of the following statements is true regarding the @ResponseStatus annotation?

- a. @ResponseStatus is detected on nested exceptions
- b. TheExceptionHandlerResolver uses the @ResponseStatus annotation to map exception to HTTP status code
- c. A controller handler is annotated with the @ResponseStatus, the response status set by RedirectView takes precedence over the annotation value.
- d. The @ResponseStatus annotation can go on a @RequestMapping method or a @RestController class or a business exception class.

ans: a

15. Default localeResolver used by Spring.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) AcceptLocaleResolver

ans: c

16. Alternative way to resolve locales.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) SessionLocaleResolver

ans: d

17. General-purpose class that allows a response to be rendered using a marshaller.

- a) MarshallingView
- b) Marshaling
- c) View
- d) All of the mentioned

View Answer

Answer: a

Explanation: The member template view is defined as a MarshallingView type, which is a general-purpose class that allows a response to be rendered using a marshaller.

18. Annotation which allows the Jaxb2Marshaller marshaller to detect a class's (i.e., object's) fields.

- a) @XmlRootElement
- b) @XmlRoot
- c) @NotNull
- d) None of the mentioned

ans: a

19. How to auto-inject into a field a Spring bean by its name? Select one or more answer choices.

- a. By using both the @Autowired and the @Qualifier Spring annotations
- b. By using the @Autowired annotation and naming the field with the bean name
- c. With the name attribute of the @Autowired annotation
- d. By using the single @Qualifier annotation

ans : a

20. Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the following pointcut expression? execution(\* \*..AccountServiceImpl.update(..)) && execution(\* \*..ClientServiceImpl.update(..))

- a. No joint point is defined
- b. Matches public update methods of the 2 classes, whatever the arguments
- c. Matches any update methods of the 2 classes, whatever the arguments and method visibility
- d. Matches any update methods of the 2 classes, with one more arguments and whatever method visibility

ans: a

21. Select the right statement about referring a Spring configuration file inside the package com.example.myapp in the below example?

ApplicationContext context = new ClassPathXmlApplicationContext("classpath:/com.example.myapp.config.xml");

- a. all of the above
- b. The classpath: prefix could be omitted
- c. Package name using the dot character is not well formatted
- d. The slash character preceding com.example could be omitted

ans: a

22. Using the Spring AOP framework, what is the visibility of the method matched by the following join point?

```
@Pointcut("execution(* *(..))")
private void anyOperation() {};
```

- a. Public methods
- b. All methods, regardless of their visibility
- c. All methods, except private method
- d. Protected and public methods

ans: a

23. What is the name of the bean defined in the following configuration class? Select a single answer.

```

@Configuration
public class ApplicationConfig {
    @Autowired
    private DataSource dataSource;
    @Bean
    ClientRepository clientRepository() {
        ClientRepository accountRepository = new JpaClientRepository();
        accountRepository.setDataSource(dataSource);
        return accountRepository;
    }
}

```

- a. clientRepository
  - b. JpaClientRepository
  - c. jpaClientRepository
  - d. Two beans are defined: a data source and a repository
- ans: a

24. Which of the following is true regarding the below Spring controller?

```

@RestController
public class OwnerController {
    @RequestMapping(value = "/owner/{ownerId}", method = RequestMethod.POST)
    @ResponseBody
    public Owner findOwner(@PathVariable("ownerId") int ownerId) {
        return new Owner();
    }
}

```

- a. RequestMethod.GET method is more accurate than POST
  - b. @ResponseBody could be removed
  - c. @PathVariable should be replaced with the @PathParam annotation
  - d. Returning the 201 HTTP status code is better
- ans: a

25. What is advice? Select a unique answer.

- a. An action taken by an aspect at a particular join point
  - b. A point during the execution of a program
  - c. An aspect and a pointcut
  - d. A predicate that matches join points
- ans: a

26. What is the easiest method to write a unit test?

- a. @RequestMapping("/displayAccount")  
String displayAccount(@RequestParam("accountId") int id, Model model)
  - b. void displayAccount(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException
  - c. void displayAccount(HttpServletRequest req, HttpSession session) throws ServletException, IOException
  - d. @RequestMapping("/displayAccount")  
String displayAccount(@PathVariable("accountId") int id, Model model)
- ans: a

27. Select method's signatures that match with the following pointcut:

- ```
execution(* com.test.service..*.*(*))
```
- a. void com.test.service.MyServiceImpl#transfert(Money amount)
  - b. void com.test.service.account.MyServiceImpl#transfert(Money amount)
  - c. void com.test.service.MyServiceImpl#transfert(Account account, Money amount)
  - d. void com.test.service.account.MyServiceImpl#transfert(Account account, Money amount)
  - e. None of the above
- ans: a

28. Given the Spring configuration file, which are the correct statements?

- ```

<bean class="com.spring.service.BankServiceImpl"
    p:bankName="NationalBank">
</bean>

```
- a. The p namespace has to be declared
  - b. NationalBank is a scalar value
  - c. Bean id is bankServiceImpl
  - d. The BankServiceImpl references a NationalBank bean

ans: a

29. Given the following configuration class, what are the correct affirmations? Select one or more answers.

```
public class ApplicationConfig {  
    private DataSource dataSource;  
    @Autowired  
    public ApplicationConfig(DataSource dataSource) {  
        this.dataSource = dataSource;  
    }  
    @Bean(name="clientRepository")  
    ClientRepository jpaClientRepository() {  
        return new JpaClientRepository();  
    }  
}
```

- a. Configuration annotation is missing
- b. Default or no-arg constructor is missing
- c. @Bean name is ambiguous
- d. @Bean scope is prototype

ans: a

30. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object? Select a unique answer.

- a. ResultSetExtractor
- b. RowMapper
- c. RowCallbackHandler
- d. ResultSetMapper

ans: B

Q 1 - Which of the following is correct about dependency injection?

- A - It helps in decoupling application objects from each other.
- B - It helps in deciding the dependencies of objects.
- C - It stores objects states in a database.
- D - It stores object states in the file system.

Answer : A

Q 2 - Which are the correct implementation classes of ApplicationContext?

- A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext
- B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext
- C - AdvancedApplicationContext, FileApplicationContext
- D - FileSystemApplicationContext, ClassPathApplicationContext

Answer : A

Q 3 - What is true about <set> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

Answer : B

Q 4 - What is the constructor mode of autowiring?

- A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.
- B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.
- C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the bean name in the configuration file.
- D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

Answer : D

Q 5 - What is true about @Autowired annotation?

- A - The @Autowired annotation can be used to autowire bean on the setter method.
- B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.
- C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.
- D - All of the above.

Answer : B

Q 6 - What are the different points where weaving can be applied?

A - Compile time, load time  
B - Compile time, run time  
C - Run time  
D - Compile time, load Time, Run time  
Answer : D

Q 7 - What is Spring MVC framework?  
A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.  
B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.  
C - Spring MVC framework is used for Transaction management for Web Applications.  
D - Spring MVC framework is used for AOP for Web Applications.  
Answer : B

Q 8 - By default a bean is lazily loaded.  
A - True  
B - False  
Answer : B

Q 9 - What is the scope of bean in portlet context?  
A - session  
B - global-session  
C - prototype  
D - request  
Answer : B

Q 10 - Which of the following classes can be used to call Stored Procedures in spring?  
A - SPHelper  
B - JdbcTemplateCall  
C - JdbcTemplate  
D - SimpleJdbcCall  
Answer : D

Q 42 - What is a ContextRefreshedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the HTTP Response is returned.  
D - This event is published when the ApplicationContext is either initialized or refreshed.  
D

Q 43 - What is a ContextStartedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the ApplicationContext is started using the start method on the ConfigurableApplicationContext interface.  
D - This event is published when the HTTP Response is returned.  
C

Q 44 - What is ContextStoppedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the ApplicationContext is stopped using the stop method on the ConfigurableApplicationContext interface.  
C - This event is published when the HTTP Request is received.  
D - This event is published when the HTTP Response is returned.  
B

Q 45 - What is ContextClosedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the HTTP Response is returned.  
D - This event is published when the ApplicationContext is closed using the close method on the ConfigurableApplicationContext interface.  
D

Q 46 - What is RequestHandledEvent:event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
  - B - This event is published when the HTTP Request is received.
  - C - This event is published when the HTTP session is initialized or refreshed.
  - D - This event is published when the HTTP Request is serviced.
- C

Q 2 - What is Introduction?

- A - An introduction represents a point in your application where you can plug-in the AOP aspect.
- B - This is used to inject values in objects.
- C - This is not invoked during program execution by Spring AOP framework.
- D - An introduction allows you to add new methods or attributes to existing classes.

Answer: D

Q 3 - What is @Controller annotation?

- A - The @Controller annotation indicates that a particular class serves the role of a controller.
- B - The @Controller annotation indicates how to control the transaction management.
- C - The @Controller annotation indicates how to control the dependency injection.
- D - The @Controller annotation indicates how to control the aspect programming.

Answer: A

Q 4 - What is the prototype scope?

- A - This scopes a single bean definition to have any number of object instances.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

Answer: A

Q 5 - What is true about <props> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings

Answer: D

Q 6 - How to handle shut down of IoC containers?

- A - Using shutdownHook()
- B - Using shutdownHandler()
- C - Using registerHook()
- D - Using registerShutdownHook()

Answer: D

Q 7 - By default a bean is eagerly loaded.

- A - False
- B - True

Answer: B

Q 8 - What is true about BeanPostProcessor?

- A - It is a concrete class.
- B - It is an interface.
- C - It is an abstract class.
- D - None of the above.

Answer: B

Q 9 - How can you inject Java Collection in Spring?

- A - Using list, set, map or props tag.
- B - Using list, set, map or collection tag.
- C - Using list, set, props or collection tag.
- D - Using list, collection, map or props tag.

Answer: A

Q 10 - Which class is used to map a database row to a java object in spring?

- A - ResultSet
- B - RowMapper
- C - RowSetMapper
- D - ResultSetMapper

Answer: B



Q 11 - How can the bean life cycle be controlled?

- A - Using init() only
- B - Using InitializingBean class only
- C - Using DisposableBean class only
- D - Using All of above

Answer:D

Q 12 - What is the scope of bean in portlet context?

- A - session
- B - global-session
- C - prototype
- D - request

Answer:B

Q 13 - What is a ContextStoppedEvent event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the ApplicationContext is stopped using the stop() method on the ConfigurableApplicationContext interface.
- C - This event is published when the HTTP Request is received.
- D - This event is published when the HTTP Response is returned.

Answer:B

Q 14 - Which ORM Spring supports ?

- A - Hibernate
- B - iBatis
- C - JPA
- D - All of the above.
- E - None of the above.

Answer:D

Q 15 - What is true about <map> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This tag is not supported.

Answer:C

Q 16 - Core container has AOP as one of its modules.

- A - True
- B - False

Answer:B

Q 17-Can you inject null and empty string values in Spring?

- A - Yes
- B - No

Answer:A

Q 18 - What stands true for the spring framework?

- A - Spring framework is a lightweight framework.
- B - Spring framework is a heavy weight framework.

Answer:A

Q 19 - Which class acts as IoC Container?

- A - ServletContext
- B - DispatcherServlet
- C - ApplicationContext
- D - None of the above

Answer:C

Q 20 - What is the scope of stateless bean?

- A - global-session
- B - singleton
- C - prototype

D - request  
Answer:B

Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. true
2. false

ANS : A

Which of the following layers the @Controller annotation is used in?

1. Service layer.
2. Presentation layer.
3. Session layer.
4. Business layer.
5. None of these.

ANS : B

Which of the following statements is true?

1. ApplicationContext extends BeanFactory.
2. None of these.
3. ApplicationContext implements BeanFactory.
4. BeanFactory extends ApplicationContext.
5. BeanFactory implements ApplicationContext

ANS : A

Q 20 - What is singleton scope?

- A - This scopes the bean definition to a single instance per Spring IoC container.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 21 - What is the prototype scope?

- A - This scopes a single bean definition to have any number of object instances.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 22 - What is the request scope?

- A - This scopes a bean definition to an HTTP request.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP Session.
- D - This scopes the bean definition HTTP Application/ Global session.

A

Q 23 - What is session scope?

- A - This scopes a bean definition to an HTTP session.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Application/ Global session.

A

Q 24 - What is global-session scope?

- A - This scopes a bean definition to an HTTP Application/ Global session.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Session.

A

Q 30 - What is true about collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

ANS : C

Q 30 - What is true about <props> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

ANS : D

Q 48 - What is the Join point?

- A - This represents a point in your application which joins two objects.
- B - This represents a point in your object where you join values.
- C - This represents a point in your object where you join injected values.
- D - This represents a point in your application where you can plug-in the AOP aspect.

ANS :- D

Q 49 - What is Advice?

- A - This is the way to instruct an object to behave in a certain manner.
- B - This is used to inject values in objects.
- C - This is the actual action to be taken either before or after the method execution.
- D - This is not invoked during program execution by Spring AOP framework.

ANS :- C

Q 50 - What is Pointcut?

- A - This represents a point in your application where you can plug-in the AOP aspect.
- B - This is a set of one or more join points where an advice should be executed.
- C - This is used to inject values in objects.
- D - This is invoked during program execution by Spring AOP framework.

ANS :- B

Q 2 - What is the Target object?

- A - A represents an object in your application where you can plug-in AOP aspect.
- B - The object being advised by one or more aspects, this object will always be a proxy object, also referred to as the advised object.
- C - This is used to inject values in objects.
- D - This is not invoked during program execution by Spring AOP framework.

B

Q 3 - What is Weaving?

- A - Weaving is the process of injecting values in objects to create an advised object.
- B - Weaving is the process of linking aspects with other application types or objects to create an advised object.
- C - This is used to inject values in objects.
- D - Weaving is used to check object dependencies.

B

Q 10 - Which of the following aspects implementation spring supports?

- A - XML Schema based aspect implementation
- B - @AspectJ based aspect implementation
- C - Both of the above.
- D - None of the above.

C

Q 11 - What are the types of the transaction management Spring supports?

- A - Programmatic transaction management
- B - Declarative transaction management
- C - Both of the above.
- D - None of the above.

C

Q 12 - What is Spring MVC framework?

- A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.
- B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.
- C - Spring MVC framework is used for Transaction management for Web Applications.
- D - Spring MVC framework is used for AOP for Web Applications.

B

Q 43 - What is a DispatcherServlet?

- A - DispatcherServlet is used for transaction management.

B - DispatcherServlet is used for AOP.  
C - DispatcherServlet handles all the HTTP requests and responses.  
D - DispatcherServlet is used for Dependency injection.  
C

Q 44 - What is ACID in transactional management?

A - Accurate, Controlled, Isolation, Durability  
B - Atomicity, Consistency, Independent, Done  
C - Atomicity, Consistency, Isolation, Durability  
D - Accurate, Controlled, Independent, Done  
C

Q 45 - Where do you define DispatcherServlet?

A - In Beans configuration file.  
B - Web.xml file  
C - Meta-inf/dispatcher.xml  
D - Web-inf/ dispatcher.xml  
B

Q 46 - What is true about BeanPostProcessor?

A - It is a concrete class.  
B - It is an interface.  
C - It is an abstract class.  
D - None of the above.  
B

Q 47 - What BeanPostProcessor does?

A - It processes beans once a bean is initialized.  
B - It defines callback methods that you can implement to provide your own instantiation logic, dependency-resolution logic etc.  
C - It processes beans once a bean is loaded.  
D - It processes beans once a bean exits.  
B

Q 48 - Can a bean be configured to have an inner bean?

A - True  
B - False  
A

Q 49 - Can we inject value and ref both together in a bean?

A - True  
B - False  
A

Q 50 - Following class can be extended to create custom events in spring.

A - SpringEvent  
B - Event  
C - ApplicationEvent  
D - None of above  
C

Q 2 - Which of the following is the correct assertion about spring?

A - Spring enables developers to develop enterprise-class applications using POJOs.  
B - Spring is organized in a modular fashion.  
C - Testing an application written with spring is simple because environment-dependent code is moved into this framework.  
D - All of the above.  
D

Q 5 - What AOP stands for?

A - Aspect Oriented Programming  
B - Any Object Programming  
C - Asset Oriented Programming  
D - Asset Oriented Protocol  
A

Q 6 - What is true about cross-cutting concerns?

A - The functions that span multiple points of an application are called cross cutting concerns.  
B - Cross-cutting concerns are conceptually separate from the application's business logic.

C - Logging is one of the examples of cross cutting concerns.

D - All of the above.

A

Q 7 - Which are the modules of the core container?

A - Beans, Core, Context, SpEL

B - Core, Context, ORM, Web

C - Core, Context, Aspects, Test

D - Bean, Core, Context, Test

A

Q 8 - Which are the modules of the Data Access/ integration layer?

A - JDBC, ORM, OXM, JMS, Transactions

B - JDBC, ORM, OXM, JMS

C - JDBC, ORM, Web, Beans

D - JDBC, ORM, OXM, JMS

A

Q 9 - Which are the modules of the Web layer?

A - WebSocket, Servlet, Web, Portlet

B - WebSocket, Servlet, Web-MVC, Web

C - HTML, JSP, WEB, Portlet

D - HTML, Servlet, WEB, Portlet

A

Q 11 - Which of the statements is correct?

A - The JDBC module provides a JDBC-abstraction layer that removes the need to do tedious JDBC related coding.

B - The ORM module provides integration layers for popular object relational mapping APIs, including JPA, JDO, Hibernate, and iBatis.

C - The Java Messaging Service JMS module contains features for producing and consuming messages.

D - All of the above.

D

Q 13 - What types of Dependency injection does spring supports?

A - Constructor based, Setter based

B - Constructor based, Setter based, Getter Based

C - Setter based, Getter based, Properties based

D - Constructor based, Setter based, Properties based

A

Q 14 - Which are the IoC containers in Spring?

A - BeanFactory, ApplicationContext

B - BeanFactory, ApplicationContext, locContextFactory

C - BeanFactory, BeanContext, locContextFactory

D - BeanFactory, ApplicationContext, BeanContext

A

Q 17 - Which of the following stands true for spring beans?

A - Spring beans are managed by the Spring IoC container.

B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.

C - Spring beans are simple POJOs.

D - All of the above.

D

Q 18 - Which is the way to provide configuration metadata to spring?

A - XML Based configuration file.

B - Annotation based configuration.

C - Java based configuration.

D - All of the above.

D

Q 19 - What is bean scope?

A - Bean scope forces Spring to produce a new bean instance as per the scope defined.

B - Bean scope defines the accessibility of bean in a java class.

C - Bean scope defines the accessibility of bean in a java package.

D - Bean scope defines the accessibility of bean in a web application.

A

Q 33 - What is no mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Similar to byType, but type applies to constructor arguments.

A

Q 34 - What is byName mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Similar to byType, but type applies to constructor arguments.

B

Q 35 - What is byType mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in the configuration file.

D

Q 36 - What is the constructor mode of autowiring?

A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the bean names in the configuration file.

D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

D

Q 37 - What is autodetect mode of autowiring?

A - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in the configuration file.

C

Q 15 - Which is the correct implementation class of BeanFactory?

A - XmlBeanFactory

B - ClassPathBeanFactory

C - FileSystemBeanFactory

D - AdvancedBeanFactory

A

Q 40 - What does @Required annotation mean?

A - This annotation indicates that bean property must be populated by the user.

B - This annotation indicates that bean property is required while saving the bean data to the database.

C - This annotation simply indicates that the affected bean property must be populated at configuration time, through an explicit property value in a bean definition or through autowiring.

D - This annotation indicates that bean property is required while serializing the bean.

C

Q 41 - What is true about @Autowired annotation?

A - The @Autowired annotation can be used to autowire bean on the setter method.

B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.

C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.

D - All of the above.

B

Q 47 - What is the aspect?

- A - Aspect is a way to do the dependency injection.
  - B - A module which has a set of APIs providing cross-cutting requirements.
  - C - Aspect is used to log information about applications.
  - D - Aspect represents properties of spring based application.
- B

Q 48 - What is the Join point?

- A - This represents a point in your application which joins two objects.
  - B - This represents a point in your object where you join values.
  - C - This represents a point in your object where you join injected values.
  - D - This represents a point in your application where you can plug-in AOP aspect.
- D

Q 50 - What is Pointcut?

- A - This represents a point in your application where you can plug-in AOP aspect.
  - B - This is a set of one or more join points where an advice should be executed.
  - C - This is used to inject values in objects.
  - D - This is invoked during program execution by Spring AOP framework.
- B

Which of these operations instruct the IOC container to create a bean instance?

- The initialization method
- The destruction method
- The lazy-initialization mode
- The auto-wiring mode
- None of the above

ANS : E

Which interface in Spring MVC ensures that implementation is detected?

- WebApplicationInitializer
- WebInitializer
- XmlWebApplicationContext
- AbstractAnnotationConfigDispatcherServletInitializer
- None of the above

ANS : D

The getModeMap() method of ModelAndView returns a(n): \_\_\_\_\_ instance.

- Empty
- Http
- ModelMap
- Map

ANS : C

## SPRING AOP :-

Question: 1 What is an after returning advice? Select a unique answer.

Your Answer: None

Correct Answer: Advice to be executed after a join point completes without throwing an exception

Description: None

Question: 2 Select method's signatures that match with the following pointcut:

execution(\* com.test.service..\*.\*(\*))

Your Answer: None

Correct Answer: void com.test.service.MyServiceImpl#transfert(Money amount)

Description: None

Question: 3 Using the Spring AOP framework, what is the visibility of the method matches by the following join point?

```
@Pointcut("execution(* *(..))")
private void anyOperation() {};
```

Your Answer: None

Correct Answer: Public methods

Description: Due to the proxy-based nature of Spring's AOP framework, protected methods are by definition not intercepted, neither for JDK proxy nor for CGLIB proxies. As a consequence, any given pointcut will be matched against public methods only! To intercept private and protected methods, AspectJ weaving should be used instead of the Spring's proxy-based AOP framework.

Question: 4 Which one is the correct statement about AOP proxy?

Your Answer: None

Correct Answer: AOP proxies are created by Spring in order to implement the aspect contracts

Description: 1. An object created by the AOP framework in order to implement the aspect contracts

2. If the target object does not implement any interfaces then a CGLIB proxy will be created.

You could also use CGLIB proxy instead of JDK dynamic proxy

3. If the target object does not implement any interfaces then a CGLIB proxy will be created.

4. When CGLIB proxy is used, final methods cannot be advised, as they cannot be overridden.

Question: 5 What is a pointcut?

Your Answer: None

Correct Answer: An expression to identify join points

Description: None

Q 6 - How before advice works?

A - Run advice before a class loads.

B - Run advice before a method execution.

C - Run advice before http response is to be returned.

D - Run advice before http request is to be processed.

B

Q 7 - How after-returning advice works?

A - Run advice after a class loads only if class loads successfully.

B - Run advice after a method execution only if the method completes successfully.

C - Run advice after http response is returned only if http response is success.

D - Run advice after http request is processed with no exception.

B

Q 8 - How after-throwing advice works?

A - Run advice after a method execution only if the method exits by throwing an exception.

B - Run advice after a class loads only if the class throws an exception during load time.

C - Run advice after http response is returned with error status.

D - Run advice after http request is processed and an exception occurred.

A

Q 9 - How around advice works?

A - Run advice before and after the advised method is invoked.

B - Run advice before and after a class is loaded.

C - Run advice before and after http response is returned.

D - Run advice before and after http request is processed.

A

Q 12 - What is Spring MVC framework?

A - Spring MVC framework is Model-View-Controller architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

B

Question: 7 What are the unique correct answers about Spring AOP support?

Your Answer: None

Correct Answer: A point cut could select methods that have a custom annotation

Description: The @annotation designator enables to select methods that are annotated by a given annotation



Question: 8 What is advice? Select a unique answer.

Your Answer: None

Correct Answer: An action taken by an aspect at a particular join point

Description: None

Question: 9 Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the pointcut expressions?

execution(\* \*..AccountServiceImpl.update(..))

&& execution(\* \*..ClientServiceImpl.update(..))

Your Answer: None

Correct Answer: No joint point is defined

Description: Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the pointcut expressions?

execution(\* \*..AccountServiceImpl.update(..))

&& execution(\* \*..ClientServiceImpl.update(..))

Pointcut expression could not satisfy both first and second execution points. Do not confuse the && operator and || operator.

Question: 10 Using the Spring AOP framework, what are the joinpoint methods of the following pointcut expressions?

execution(public \* \*(..))

Your Answer: None

Correct Answer: The execution of all public method

Description: The execution of all public method

Question: 10 How many types of Dynamic proxy are available in Spring?

Your Answer: None

Correct Answer: two

Description: 1. JDK Dynamic Proxy

2. CGLIB Dynamic Proxy

Question: 11 What is the Type of Proxy in Spring Framework?

Your Answer: None

Correct Answer: two

Description: 1. static

2. Dynamic

Question: 6 How many types of the injection method in Spring?

Your Answer: None

Correct Answer: two

Which of the following is not a valid difference between @Inject and @Autowired annotations?

1. There is no difference and both can be used interchangeably.

2. @Autowired is a legacy annotation whereas @Inject is its latest version.

3. @Inject annotation is part of Java EE 7's Context and Dependency Injection framework while @Autowired is the Spring Frameworks own implementation.

4. None of these.

ANS :- B

Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. true

2. false

ANS : A

Question: 10 What is the role of ApplicationContextAware in spring?

Your Answer: None

Correct Answer: Make a bean aware to the container.

Description: None

Question: 9 Can the class which spring container makes subclass be final?

Your Answer: None

Correct Answer: No

Description: None

Question: 10 When a bean has scope limited to HTTP session, that is called

Your Answer: None

Correct Answer: Session scope

Description: None

Q 14 - What are the ways to access Hibernate by using Spring?

- A - Inversion of Control with a Hibernate Template and Callback.
  - B - Extending HibernateDAOSupport and Applying an AOP Interceptor node.
  - C - Both of above.
  - D - None of above.
- C

Q 15 - Which ORM Spring supports ?

- A - Hibernate
  - B - iBatis
  - C - JPA
  - D - All of above.
  - E - None of above.
- D

Q 16 - Which of the following database is not supported using jdbcTemplate?

- A - MySql
  - B - PostgresSql
  - C - NoSql
  - D - Oracle
- C

Q 24 - Expression Language/ SpEL was introduced in which version of spring framework.

- A - 1.0
  - B - 2.0
  - C - 3.0
  - D - 4.0
- C

Q 25 - Can we integrate Struts with Spring.

- A - Yes
  - B - No
- A

Q 26 - By default a bean is lazily loaded.

- A - True
  - B - False
- B

Q 27 - By default a bean is eagerly loaded.

- A - False
  - B - True
- B

Q 31 - If a bean can be created any number of times, scope is

- A - session
  - B - global-session
  - C - prototype
  - D - request
- C

Q 32 - What is the scope of stateless beans?

- A - global-session
  - B - singleton
  - C - prototype
  - D - request
- B

Q 33 - What is the scope of stateful beans?

- A - session
  - B - global-session
  - C - prototype
  - D - request
- C

Q 34 - If a bean is scoped to HTTP session, scope is

- A - global-session
  - B - session
  - C - prototype
  - D - request
- B

Q 41 - Which of the following classes can be used to execute Sql queries in spring?

- B - JDBCHelper
  - A - JdbcTemplate
  - C - DBHelper
  - D - DBTemplate
- A

Q 42 - Which of the following classes can be used to call Stored Procedures in spring?

- A - SPHelper
  - B - JdbcTemplateCall
  - C - JdbcTemplate
  - D - SimpleJdbcCall
- D

Q 50 - Following class can be extended to create custom events in spring.

- A - SpringEvent
  - B - Event
  - C - ApplicationEvent
  - D - None of above
- C

32. Annotation to add security to your spring boot app?

- a. @EnableWebSecurity
- b. @EnableSecurity
- c. @EnableSpringSecurity

ANS : A

2.Element used to specify access attributes for bean's methods.

- A. security:protect
- B. security:intercept
- C. security:intercept-security
- D. none of the mentioned

ANS : A

31. How to reload my changes on Spring Boot without having to restart the server?

- a. org.springframework.boot spring-boot-devtools true
- b. org.springframework.boot spring-boot-test true
- c. org.springframework.boot spring-boot-web true
- d. org.springframework.boot spring-boot-jasper true

ANS : A.

35. LoggerInterceptor which implements the HandlerInterceptor.The LoggerInterceptor will override the which following methods

- a. preHandlerBean(), postHandlerBean(), afterCompletion()
- b. PreProcessHandler(), PostProcessHandler(), afterCompletion()
- c. PreHandler(), PostHandler(), afterCompletion()
- d. none

ANS : C

34. How does the Spring Batch work?

- a. step, ItemReader, ItemProcessor, ItemStreamWriter
- b. step, ItemProcessor, ItemStreamWriter

c. step, ItemReader, ItemStreamWriter  
ANS : A.

Q - Which of the following stands true for spring beans?  
A - Spring beans are managed by the Spring IoC container.  
B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.  
C - Spring beans are simple POJOs.  
D - All of the above.  
Answer : D

## JSON :

31) Which of the following codes will not throw an error?

- A. JSON.parse("");
- B. JSON.parse(null);
- C. JSON.parse();
- D. JSON.parse({});

Ans: B

32) What does JSONP stand for?

- A. JSON Procedures
- B. JSON Parsing
- C. JSON with padding
- D. JSON Programming

Ans: C

33) What is the value of json in the following code?

```
var obj = { fruit: 'apple', toJSON: function () { return 'orange'; } }; var json = JSON.stringify(obj);
```

- A. {"x": "orange"}
- B. {"fruit": "apple"}
- C. {"x": "apple"}
- D. {"fruit": "orange"}

Ans: A

34) What is the MIME type of JSON?

- A. application/x-json
- B. text/json
- C. application/json
- D. application/javascript

Ans: C

35) What function will convert a JavaScript object to a JSON string?

- A. JSON.text()
- B. JSON.serialize()
- C. JSON.toString()
- D. JSON.stringify()

Ans: D

36) In the below notation, Student is of type { "Student": [ "Amy", "Bob", "John" ] }

- A. Not a valid JSON string
- B. Array
- C. Class
- D. Object

Ans: B

37) JSON-RPC is a remote procedure call protocol

38) To use observables, Angular uses a third-party library called . . . . .

D) Object.prototype

Ans: D

75.) The four kinds of class members are\_\_\_\_\_

- a) Instance methods, Instance fields, Static method, Dynamic method
- b) Instance fields, Instance methods, Class fields, class methods
- c) Instance fields, non-Instance fields, Dynamic methods, Global methods
- d) Global methods, Local methods, Dynamic methods, Static methods

Ans: B

84.) Which field is required in <input>

- A) required
- B) minlength
- c) Both

Ans: A

86.) How can we add/ delete in an array and return the removed item.

- A) slice
- B) shift
- c) reverse
- D) splice

Ans: D.splice

87.) How to get information about browser

- A) Browser
- B) Version
- C) navigator

Ans: C

## MONGODB :-

1) In MongoDB, what is the equivalent of SQL term row?

- A. Primary Key
- B. Index
- C. Field
- D. Document

ANS : D

2) What is the equivalent of the following SQL command in MongoDB?

```
UPDATE user  
SET email = 'NA'  
WHERE age < 18
```

- A. db.user.updateMany({ gt: { \$age: 18 } }, { \$set: { email: "NA" } })
- B. db.user.updateMany({ age: { \$lt: 18 } }, { \$set: { email: "NA" } })
- C. db.user.updateMany({ gt: { \$age: 18 } }, { \$set: { email: "NA" } })
- D. db.user.updateMany({ age: { \$gt: 18 } }, { \$set: { email: "NA" } })

ANS : B

3) How would you limit the results of a find to 10 items?

```
adb.collection.find({}, { limit: 10 })  
db.collection.find().limit(10)  
db.collection.limit(query, 10)
```

This is not possible in the node.js driver

ANS : B

4) Which of the following code will give an error on the MongoDB command shell?

- A. use databasename
- B. show dbs

C.USE databasename  
D.Db  
ANS : C

5) What theorem or principal MongoDB follows?  
A.APAC  
B.Always-Sync  
C.ASCII  
D.CAP  
ANS : D

6) What is NoSQL?  
A.No Object SQL  
B.No SQL Server  
C.Not Only SQL  
D.No Relational Database  
ANS : C

7) When there is no match of the document in the collection what do upsert: true do?  
A.Delete  
B.Nothing  
C.Insert  
D.Update  
ANS : C

8) Upon running the following command, what all the fields will be returned in the output?  
`db.collection.find( { qty: { $gt: 6 } }, { name: 1, _id: 0 } )`  
A.\_id  
B.name  
C.\_id, name  
D.name1, \_id0  
ANS : B

9) What method is used to remove a single item from a MongoDB collection?  
`db.collection.deleteOne()`  
`db.collection.delete()`  
`db.collection.removeOne()`  
`db.collection.remove()`  
ANS : A

10) In the following command, what does the second parameter represent?  
`db.collection.find( { qty: { $gt: 6 } }, { name: 1, wins: 1 } )`  
Read Concern  
Projection  
Write Concern  
Query  
ANS : B

## **NODE / EXPRESS :**

1) What is the command used to create a new item when using mongoose?  
A.Book.Save()  
B.Book.New()  
C.Book.Create()  
ANS : A

2)What is the HTTP verb used to update a resource in REST?  
A.DELETE

B.GET  
C.POST  
D.PUT  
ANS : D

3) Google's V8 Engine, also used in Chrome, provides what capability to node.js?  
A.JavaScript support  
B. 3rd party module repository  
C.Support for HTML rendering and DOM manipulation  
D.Asynchronous File and Network access  
ANS : A

4) What is the package you use to connect to MongoDB?  
A.Mongod  
B.MongoConnect  
C.Schema  
D.Mongoose  
ANS : D

5) Assigning a value to "module.exports" in a node.js JavaScript file  
A.allows that value to be imported into other JavaScript files in your project.  
B.publishes the exported value to npm, node's package repository.  
C.persists that value to disk, allowing you to retrieve it later.  
ANS : A

6) Running "npm init" will...  
A.initialize your project, creating a project.json file.  
B.initialize the npm cli on your system.  
C.download and add the "init" module to your node.js project.  
AND : A

7) What is the module for Node that allows you to run a web server?  
A.Sublime  
B.REST  
C.FS  
D.HTTP  
ANS : D

8) What is the mongoose command used to delete an item?  
A.delete()  
B.trash()  
C.remove()  
ANS : C

9) What verb do you use to do a partial update on an item?  
A.Post  
B.Patch  
C.Put  
ANS : B

10) What is the object that you use to pull query string params?  
A.res.query  
B.res.params  
C.req.query  
D.req.params  
ANS :- C

**JAVA:**

1) Given the variable `List<String> strings`, which of the following is valid?

- A. `strings.put("a");`
- B. `List.add(strings, "a");`
- C. `strings[0] = "a";`
- D. `strings.add("a");`

ANS : D

2) When is a method required to include a return statement?

- A. When the method has a non-void return type
- B. When the method contains more than one statement
- C. Never
- D. Always

ANS : A

3) What is the type of the class literal `Foo.class`?

- A. `Class<Foo>`
- B. `Class`
- C. `Class<?>`
- D. `Class<Class>`

ANS : A

4) How would you declare and instantiate a variable, given the generic class `Bag<T>`?

- A. `Bag bag = new Bag();`
- B. `Bag<T> bag = new Bag<T>();`
- C. `Bag bag<T> = new Bag<T>();`
- D. `Bag<T> bag = new Bag();`

ANS : B

5) What is the keyword used to transfer control to another part of the program?

- A. `catch`
- B. `throw`
- C. `try`
- D. `finally`

ANS : B

6) What indicates the end of a Java statement?

- A. Both a semicolon and a newline
- B. A newline
- C. A double-slash (`//`)
- D. A semicolon (`;`)

ANS : D

7) Which of the following will initialize the array `myVals` to contain two values: 1 and 2

- A. `int[] myVals = new int[2] {1, 2};`
- B. `int[] myVals = {1, 2};`
- C. `int[2] myVals = {1, 2};`
- D. `int[] myVals = [1, 2];`

ANS : B

8) Given the following code: `float f = 7.0f/2.0f; long l = 7L/2L;` Which statement is true?

- A. The values of `f` and `l` are equivalent
- B. The attempt to assign `l` the result of `7L/2L` will create an error
- C. The value of `f` is less than the value of `l`
- D. The value of `f` is greater than the value of `l`

ANS : D



9) Which of the following can be a valid exception constructor?

- A. Exception(Object cause)
- B. Exception(String message)
- C. Exception(int cause)
- D. Exception(String message, Object cause)

ANS : B

10) Which of these classes is a checked exception?

- A. ClassNotFoundException
- B. NumberFormatException
- C. ThreadDeath
- D. NoClassDefFoundError

ANS : A

11) Given the following code: String s1 = "hi"; String s2 = "h"; s2 += "i"; boolean b1 = s1 == s2; boolean b2 = s1.equals(s2); Which variable(s) is/are assured to be set to true?

- A. Both b1 and b2
- B. b1
- C. Neither b1 or b2
- D. b2

12) Given a type parameter T, what would assign to T[] ts;?

- A. ts = (T[]) new Object[SIZE];
- B. Its impossible.
- C. ts = new T[SIZE];
- D. ts = new Object[SIZE];

ANS : A

13) How would you declare a generic factory method that creates a new generic list with a given size?

- A. public static List<T> <T> newList(int size)
- B. <T> public static List<T> newList(int size)
- C. public <T> static List<T> newList(int size)
- D. public static <T> List<T> newList(int size)

ANS : D

14) Which instances of this class are treated as unchecked exceptions?

- A. Error
- B. Throwable
- C. Exception

ANS : A

15) In which of the following situations will the compiler generate an error?

- A. Catching an checked exception that is not thrown in a try block
- B. Mixing subclasses and superclasses in a throws clause
- C. Don't having at least one catch block after a try block
- D. Downcasting an instance of Exception to an incorrect type (e.g. RuntimeException re = (RuntimeException)e;

ANS : A

Q.1) Select the correct statement for the given code based on Polymorphism in Java.

```
class Bike{
    void run(){System.out.println("running");}
}
class Splendor extends Bike{
    void run(){System.out.println("running safely with 60km");}

    public static void main(String args[]){
```

```
Bike b = new Splendor();
b.run();
}
}
```

- A.Compile time polymorphism
- B.Encapsulation Runtime
- C.Java Runtime Polymorphism
- D.None of the above

**ANS :- A**

Q.2) In the Running phase of a program, JVM executes the \_\_\_\_\_ generated by the compiler.

- classcode
- bytecode
- plugin code
- Bitcode

**ANS :- B**

What is the premise of equality for IdentityHashMap?

- Name equality
- Hashcode equality
- Length equality
- Reference equality

**ANS :- D**

Which of the following options is incorrect in terms of the given statement?

If a class inherits from some other class, it should \_\_\_\_\_.

- Over-ride or add the minimum to accomplish the derived class' purpose
- Override all the methods of its parent class
- Make sure the result "contains" its base class
- Make use of the parent class's capabilities.

**ANS : B**

Find the output of the following program. Choose the correct option.

```
public class Solution{
    public static void main(String[] args){
        byte x = 127;
        x++;
        x++;
        System.out.print(x);
    }
}
```

- A. -128
- B. 128
- C. 129
- D. 2

**ANS :- A**

(Reason , the range is from [-128,127] )

How many classes of the java.lang package are known as wrapper classes in Java?

- 6
- 5
- 8
- 3

**ANS : C**

Rohini uses static keywords in Java.

Find the option in which she cannot apply a static keyword.

variables  
methods  
blocks  
nested classes  
class  
ANS : E

What is the main use of the keyword "this" in the given options?

To invoke current class method (implicitly)  
To return the current class instance from the method  
To refer to the current class instance variable  
All of the above.  
ANS : D

Predict the output of the code given below and select the correct option.

```
int i =20;  
int j =30;  
int k = j += i/5;  
System.out.println(i+":"+j+":"+k);  
A.0.19067129629629628  
B.0.857175925925926  
C.1.4406712962962962  
D.0.8573379629629629  
ANS : D
```

What is the output of the following code snippet?

```
long choice=2;  
switch(choice) {  
case 1: System.out.print("Hello");  
break;  
case 2: System.out.print(" Java");  
default:  
System.out.print("Try Again");  
}  
A.JavaTry Again  
B.Try Again  
C.Compilation Error  
D.Java  
ANS : A
```

What will be the output of the following code?

```
import java.util.*;  
class Student  
{  
    int id;  
    String name;  
    private Student(int i,String n)  
    {  
        id = i;  
        name = n;  
    }  
  
    void display(){System.out.println(id+" "+name);}  
}  
public class Example{  
  
    public static void main(String args[]){
```

```

Student s1 = new Student(111,"Karan");
Student s2 = new Student(222,"Aryan");

s1.display();
s2.display();
}
}

```

- A. 111 Karan  
222 Aryan
  - B. 111 "Karan"  
222 "Aryan"
  - C. 111  
Karan  
222  
Aryan
  - D. Error: Student4 (int,String) has private access in Student4
- ANS :- D

Which option is correct about the below code?

```

import java.util.*;
class ThreadSample extends Thread{
public void run(){
for(int time=1;time<5;time++){
try{Thread.sleep(1000);}catch(InterruptedException e)
{System.out.println(e);}
System.out.println(time);
}
}
}

public class ThreadExample{

public static void main(String args[]){
ThreadSample t1=new ThreadSample();
ThreadSample t2=new ThreadSample();

t1.run();
t2.run();
System.out.println("hello");
}
}

```

- A. 123412341234.....infinite
  - B. Hello will be printed after the run method of both the thread objects are executed.
  - C. It will give an error from try catch block
  - D. None of the above
- ANS : B

Which are the two classes that have high-level streams containing the methods for serializing and deserializing an object?

- A. ObjectInputStream and ObjectOutputStream
  - B. ObjectInputStream and OutputStream
  - C. InputStream and OutputStream
  - D. ObjectOutputStream and OutputStream
- ANS :- B

```

class output {
public static void main(String args[])

```

```

{
StringBuffer s1 = new StringBuffer("
Hello World");
s1.insert(6 , "Good ");
System.out.println(s1);
}
}

```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld
- d) Hello Good World

Answer: D

Explanation: The insert() method inserts one string into another. It is overloaded to accept values of all simple types, plus String and Objects. Sting is inserted into the invoking object at specified position. "Good " is inserted in "Hello World" T index 6 giving "Hello Good World".

10. Which of these keywords cannot be used for a class which has been declared final?

- a) abstract
- b) extends
- c) abstract and extends
- d) None of the mentioned

View Answer

Answer: a

23. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself. What design issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

```

import java.util.*;
class vector {
public static void main(String args[])
{
Vector obj = new Vector(4,2);
obj.addElement(new Integer(3));
obj.addElement(new Integer(2));
obj.addElement(new Integer(5));
obj.removeAll(obj);
System.out.println(obj.isEmpty());
}
}

```

- a) 0
- b) 1
- c) true
- d) false

Answer: C

```

import java.util.*;
class Array {
public static void main(String args[])

```

```

{
    int array[] = new int [5];
    for (int i = 5; i > 0; i--)
        array[5 - i] = i;
    Arrays.sort(array);
    for (int i = 0; i < 5; ++i)
        System.out.print(array[i]);
}

```

Ans) 12345

What is the output of this program?

```

import java.util.*;
class Maps {
    public static void main(String args[])
    {
        TreeMap obj = new TreeMap();
        obj.put("A", new Integer(1));
        obj.put("B", new Integer(2));
        obj.put("C", new Integer(3));
        System.out.println(obj.entrySet());
    }
}

```

- a) [A, B, C]
  - b) [1, 2, 3]
  - c) {A=1, B=2, C=3}
  - d) [A=1, B=2, C=3]
- ANS :- D

42. What is the output of this program?

```

import java.util.*;
class Bitset {

    public static void main(String args[])
    {
        BitSet obj = new BitSet(5);
        for (int i = 0; i < 5; ++i)
            obj.set(i);
        obj.clear(2);
        System.out.print(obj);
    }
}

```

- a) {0, 1, 3, 4}
  - b) {0, 1, 2, 4}
  - c) {0, 1, 2, 3, 4}
  - d) {0, 0, 0, 3, 4}
- View Answer  
Answer: a

43. What is the output of this program?

```

import java.util.*;
class Array {
    public static void main(String args[])
    {
        int array[] = new int [5];
    }
}

```

```
for (int i = 5; i > 0; i--)
```

```
array[5-i] = i;  
Arrays.fill(array, 1, 4, 8);  
for (int i = 0; i < 5 ; i++)  
System.out.print(array[i]);  
}  
}
```

advertisements

- a) 12885
- b) 12845
- c) 58881
- d) 54881

Answer: C , fill(start\_idx,end\_idx,value).

46. What is the output of this program?

```
import java.util.*;  
class Collection_Algos {  
    public static void main(String args[])  
    {  
        LinkedList list = new LinkedList();
```

```
        list.add(new Integer(2));  
        list.add(new Integer(8));  
        list.add(new Integer(5));  
        list.add(new Integer(1));  
        Iterator i = list.iterator();  
        Collections.reverse(list);  
        Collections.shuffle(list);  
        while(i.hasNext())  
            System.out.print(i.next() + " ");  
    }  
}
```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8
- d) Any random order

Answer: d

Explanation: shuffle – randomizes all the elements in a list.

47.What is the output of this program?

```
import java.util.*;  
class Collection_Algos {  
    public static void main(String args[])  
    {  
        LinkedList list = new LinkedList();  
  
        list.add(new Integer(2));  
        list.add(new Integer(8));  
        list.add(new Integer(5));  
        list.add(new Integer(1));  
        Iterator i = list.iterator();  
        Collections.reverse(list);  
        Collections.sort(list);  
        while(i.hasNext())  
            System.out.print(i.next() + " ");  
    }  
}
```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8
- d) 2 1 8 5

ANS : C , Sorting the list in the ascending order.

```
52. class output {
    public static void main(String args[])
    {
        StringBuffer c = new StringBuffer("Hello");
        StringBuffer c1 = new StringBuffer("World");
        c.append(c1);
        System.out.println(c);
    }
}
```

- a) Hello
- b) World
- c) Helloworld
- d) Hello World

ANS : D

```
53. class output {
    public static void main(String args[])
    {

        StringBuffer s1 = new StringBuffer("Hello");
        s1.setCharAt(1,'x');
        System.out.println(s1);
    }
}
```

- a) xello
- b) xxxxx
- c) Hxlllo
- d) Hexlo

ANS : C

```
59. class output {
    public static void main(String args[])
    {
        String s1 = "one";
        String s2 = s1 + " two";
        System.out.println(s2);
    }
}
```

- a) one
- b) two
- c) one two
- d) compilation error

View Answer

Answer: c

```
61. class exception_handling {
    public static void main(String args[])
    {
```

```
        try {
            System.out.print("Hello" + " "
                + 1 / 0);
```



```

    }
    catch(ArithmeticException e) {
        System.out.print("World");
    }
}
}

```

- a) Hello
- b) World
- c) HelloWorld
- d) Hello World

Answer: b

63. Which of the following describes the correct sequence of the steps involved in making a connection with a database.

1. Loading the driver
2. Process the results.
3. Making the connection with the database.
4. Executing the SQL statements.

(A) 1,2,3,4

(B) 1,3,4,2

(C) 2,1,3,4

(D) 4,1,2,3

Answer : 1,3,4,2

```

64. import java.io.*;
class fileinputoutput {
    public static void main(String args[]) {
        InputStream obj = new FileInputStream("inputoutput.java");
        System.out.print(obj.available());
    }
}

```

Note: inputoutput.java is stored in the disk.

- a) true
- b) false
- c) prints number of bytes in file
- d) prints number of characters in the file

View Answer

Answer: C

66. Which of these methods are used to read in from a file?

- a) get()
- b) read()
- c) scan()
- d) readfileinput()

Answer: B

67. Which of the following allows non repeatable read in JDBC Connection?

- a. TRANSACTION\_READ\_UNCOMMITTED
- b. TRANSACTION\_READ\_COMMITTED
- c. TRANSACTION\_SERIALIZABLE
- d. TRANSACTION\_REPEATABLE\_READ

Ans. D

## DOCKER & DEVOPS :-

1) After you initialize a new Git repository and create a file named git-quiz.html, which of the following commands will NOT work if issued?

git add git-quiz.html

git status  
git add .  
git commit -m "git quiz web file added"  
ANS : D

2) Which of the following are the benefits of using Gradle over Maven? Select all that apply.

- A. Gradle supports more languages than Maven.
- B. Gradle incorporates Maven features as well as additional functionality.
- C. Gradle builds are successful more often than Maven builds.
- D. A Gradle build is 1000 times faster than a Maven build.

ANS : A , B

3) State whether the statement is true or false.

Adopting Agile is necessary prior to embracing DevOps practices.

True

False

ANS : B

4) What is continuous testing?

Testing continuously for 24 hours

Manually testing each build, end-to-end

Automated testing

Automated testing on each continuous deployment

ANS : D

5) Adopting DevOps requires planning that spans which of the following? Select all that apply.

People

Efforts

Process

Tools/Technology

ANS : A,C,D

6) Which DevOps principle focuses on product and service thinking?

Customer-centric action

Continuous improvement

Create with the end in mind

Automate everything you can

ANS : C

7) Shift-left concept refers to moving everything to the earlier phase in the \_\_\_\_\_ cycle.

Requirement

Development

Design

Testing

ANS : B

8) In order to minimize risks to the business within DevOps, what is the main goal of version control?

Ensure the ability to alert when configuration changes from the desired state.

Ensure the ability to re-create the previous state of the test environment.

Ensure the ability to re-create the production environment and build processes.

Ensure the ability to share the source code between different developers teams.

ANS :- B

9) What is a continuous build? Select all that apply.

A build that runs non-stop, over and over again.

A build that has not failed in 10 continuous attempts.

A build that is executed every hour/day by a continuous integration server.

A build that is executed on every commit by a continuous integration server.

ANS : C , D

10) DevOps is primarily about:

Tools and their usage

Processes and adherence

Governance and audit processes

Tools, processes, and culture

ANS : D

11) Which statement best describes the relationship between DevOps and Continuous Delivery?

DevOps and Continuous Delivery are the same thing.

DevOps and Continuous Delivery share a background in Agile methods and LEAN thinking.

DevOps and Continuous Delivery share common processes.

DevOps and Continuous Delivery are not related and are mutually exclusive.

ANS : B

12) How does a DevOps organization act in principle when it comes to financing its work?

It finances special projects to serve its clients.

It finances products and services to serve its clients.

It finances teams in matrix organizations and these teams are responsible for handling their own budgets.

It finances development and operations teams separately, so they take care of their own business.

ANS : B

13) In Kubernetes, a node is:

A worker machine

A tool for starting a kubernetes cluster on a local machine

A machine that coordinates the scheduling and management of application containers on the cluster

A virtual machine

ANS : A

14) GIT can be integrated with Jenkins.

True

False

ANS : A

15) POM in Maven stands for \_\_\_\_\_.

Project Object Model

Process Object Model

Project Object Method

All of the above

ANS : A

16) What is a widely used reusable asset to reinforce information security of deliverables from your DevOps team?

Data storage systems

Handling the logging of sensitive client information

Data transfer between clients and software

All of the above

ANS : D

17) DevOps is NOT a set of \_\_\_\_\_ steps.

Development

Fixed

Testing

Design

ANS : B

18) What is the sequence in which Maven looks for the resources?

Local -> Remote - Maven Central

Remote → Maven Central - → Local

Remote → Local - → Maven Central

Maven Central - → Local - → Remote

ANS : A

19) Why is it recommended that production and development environments should be similar?

A. It is easier to manage for the ops team.

B. It improves the accuracy of testing, eases the task of deployment of software, and reduces chances of major problems when deploying software into production.

C. People can fix defects directly in the production environment.

D. Development environments are cheaper, hence it makes sense to make the production environment similar (and cheaper) to the development environments.

ANS : B

20) DevOps can be implemented in any \_\_\_\_\_.

Process

Environment

Defects

None of the above

ANS : B

21) What is continuous deployment? Select all that apply.

Deployment on every code push/commit

Deployment every hour/day

Local deployment on the developer's laptop

Deployment on IBM Cloud

ANS : A.

22) In DevOps, one of the significant ROI areas is:

Zero defects

Faster time to value

Less feedback

Less customer experience

ANS : B

23) How do you remove local (untracked) files and directories from the current Git working tree?

i. git clean -f

ii. git delete -f

iii. git remove -f -d

iv. git add --all

v. git reset --hard HEAD

A : Only i

B : Only ii and iii

C : Only i and iv

D : All of the above.

ANS : A

## PREV BATCH FSD QUESTIONS :-

1) Which event occurs when the user clicks on the HTML element ?

A. onmouseover

B. onchange

C. onmouseclick

D. onclick

ANS : D

2) The correct HTML tag to left align the content inside a table cell is :-

A. <td align = "left">

B.<td align = "right">  
C.<td leftAlign>  
D.<table align="left">  
ANS : A.

3) What is the output of the Java Code Snippet ?  
StringJoiner astring = new StringJoiner("-", " ( ", " ) ");  
astring.add("A");  
astring.add("B");  
astring.add("C");

- A. (A-B-C)
- B. (A)-(B)-(C)
- C. (-A-)(-B-)(-C-)

ANS : A

4) What are the Different Modes of Autowiring ?  
A. byName,byType,constructor,autodetect.  
B. byName,byType,constructor,autocorrect.  
C.byName,byContent,constructor,autodetect.  
D.byName,byContent,setter,autodetect.

ANS :- A

5) Which Annotation is used to refer to Pointcuts.  
A. @PointCut.  
B. @PointCutExecution.  
C. @PointCutBefore.

ANS :- A

6) Which of the below API Calls provides you the security information ?  
1) HttpServletRequest.isUserInRole()  
2) HttpServletRequest.getUserPrincipal()

- A. Only 1
- B. Only 2
- C. Both i and ii

ANS:- C

7) What does it mean by publishing a REST service in Spring ?  
i. Publishing an application's data as Json over a REST service.  
ii.Publishing an application's data as XML over a REST service.  
lii. Sharing Database within two microservices using Spring Data Access Object (DAO).  
A. Only 1  
B. Only 1 and 2  
C. Only 2  
D. All of the above.

ANS : B -> Chatgpt

8) Which element is used to restrict the access to a particular URL within Spring Security ?  
i. restrict-url  
ii.intercept-url  
iii.intercept-restrict.

ANS : B

9) What is used to dispatch requests to the WEB Application ?  
i. DispatchherServlet.  
li. ViewHandler.

iii. ViewResolver.

ANS:A

10) Which annotation is used to create Primary KEY : Foreign KEY relation between two tables ?

- i. ForeignKey.
- ii. JoinedKey.
- iii. JoinColumn.

ANS: C

11) You can only use Hibernate in ORM in Spring Boot Applications.

- i. TRUE
- II.FALSE

ANS : B

12) In your Data Access Object (DAO) method , you neither need to surround your code with a try/catch block nor declare throwing an exception in the method signature.

- i. TRUE
- ii. FALSE.

ANS : B

13) Which annotation is used as a substitute for the initialization method ?

- 1. @PostConstruct.
- 2. @PreDestroy.
- 3. @PreConstruct.

ANS : 1

14) A logger location where logging information is sent is called

- 1. appender.
- 2. logger
- 3. component.

ANS : appender A

15) By default, a bean is lazily loaded in Spring.

- 1. True
- 2. False

ANS : B

16) Sharding a MongoDB database across many server instances can be achieved with \_\_\_\_

- A.LAN
- B.SAN.
- C.MAN

ANS : LAN A

17) In MongoDB , which is the operation that uses an index often , having better performance than those who do NOT use an index ?

- A. SELECT.
- B. UPDATE
- C. DELETE.
- D. SORT.

ANS : D.

18) Which one is NOT a spring data MongoDB annotation ?

- A. @Document
- B. @Field
- C. @Entity
- D. @Query.

ANS : C @Entity

19) What type of management does "Capped Collections" provide for inserted documents in MongoDB ?

- A. FIFO.
- B. LIFO
- C. LRU
- D. MRU

ANS :- A FIFO updated

20) Identify the incorrect statement on MongoDB ?

- A. Secondary Indices are not available in MongoDB.
- B. MongoDB supports search by field, range queries and regular expression matches.
- C. MongoDB can store the business subject in the minimal number of documents.

ANS :- A. -> it can create

21) In MongoDB, which of the following methods return one document ?

- A. findOne()
- B. findOne1()
- C. selectOne()

ANS : A

22) Redux is an exclusive feature of the React store ?

- A. TRUE
- B. FALSE

ANS : B

23) Identify the correct ES6 statement.

- i. `var React = require('react');`
- ii. `import React from react;`
- iii. `module.exports = Component.`
- iv. `export default Component.`

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only ii and iv

ANS : D

24) You can also use an ES6 class to define a component in React.

- A. True.
- B. False.

ANS: A

25) Which of the below methods are NOT a part of React Component's lifecycle ?

- A. Initial Rendering.
- B. Render / Re-Render.
- C. unMounting.
- D. Remote URL calls.

ANS : D

26) Identify the correct statement in MongoDB.

- A. Queries specify criteria, or conditions , which identify the documents that MongoDB returns to the clients.
- B. Write Operations , or queries , retrieve data stored in the database.
- C. The selection limits the amount of data that MongoDB returns to the client over the network.

ANS : A -> mostly

27) Identify the incorrect statement on MongoDB.

- A. Secondary indices are not available in MongoDB.
- B. MongoDB supports search by field, range queries, and regular expression searches.
- C. MongoDB can store the business subject in the minimal number of documents.

ANS : A

28) Which of the following is not a callback method of React ?

- A. render()
- B. componentDidMount()
- C. componentFailedToMount()
- D. componentWillunMount()

ANS : C -> failedtomount

29) How can we render JSX in the browser ?

- i. Using Babel
- ii. Using Typescript.
- iii. Using Flux.
- iv. Using Redux.

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only iii and iv.

ANS : A

30) What is the command line tool introduced for the JS engine in Java 8 ?

- A. jjs
- B. jss
- C. jfs
- D. jbd

ANS : A

31) Which of the following is the immutable Java List static factory method ?

- A. List.of
- B. List.immutable.
- C. List.create

ANS : - A

32) Which of the following statements are true about Java ?

- i. A finally block is executed, only after the catch block is executed.
- ii. A finally block is executed whether an exception is thrown or not.
- iii. A finally block is NOT Mandatory.
- iv. A finally block is executed, only if an exception occurs.

- A. i
- B. iv
- C. i and iv
- D. ii and iii

ANs :- D

33) Which Java Class has natural sorting implemented ?

- A. LinkedHashSet.
- B. TreeSet
- C. HashSet

ANS :- B -> TreeSet

34) Is the following a valid Java code snippet ?

```
Optional< String > aValue = Optional.of("Not Empty");  
if(aValue.isPresent()) {  
    System.out.println("Got Not Empty");  
}
```



- A. True
- B. False

ANS :- True its an valid code

35) Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

- A. True
- B. False

ANS :- A -> True

36) In Java Optional.ofNullable() method of the Optional Class, returns a Non-empty Optional if the given object has a value, otherwise it returns an empty Optional.

- A. true
- B.false

ANS :- A->True

37) Which of the following types of Java methods cannot be type parameterized ?

- A. FINAL methods.
- B. Overriding methods.
- C. Overloading methods.

Ans:- C-> chatgpt updated by sir -> B Overriding

38) Which of the following statements is true about microservices architecture ?

- i. Microservices applications are required to be deployed on the same physical host.
- ii. Microservices architecture supports high availability of individual microservices.
- iii. Microservices cannot be used if a company is embracing DevOps.
- iv. Microservices are designed using a bounded context that can communicate with other bounded contexts.

- A. Only i
- B. Only ii
- C. Only ii and iv
- D. All of the above.

Ans -> D

39) Load Balancing improves the distribution of workloads across multiple computing resources , such as computers or a computer cluster.

- A. True.
- B. False.

ANS:- A

40) What are the core operations of DevOps with applications development and with infrastructure.

- A. Provisioning, Configuration and Orchestration.
- B. Code Building, Coverage and Unit Testing.
- C. Packaging and Deployment.
- D. All of the above.

ANS : D

41) Which CSS property is used to change the font of an element ?

- A. font
- B. font-family
- C. font-style
- D.font-weight

ANS :- B

42) The Bootstrap grid system is based on how many columns ?

- A. 12

- B. 3
  - C. 6
  - D. 9
- ANS:- A

43) After you initialize a new GIT repository and create a file named git-quiz.html , which of the following commands will NOT work if needed?

- A. git add git-quiz.html
- B. git status
- C. git add
- D. git commit -m "gut quiz web file added"

ANS: C git add (without argument it won't add anything)

44) Choose the correct HTML element to define important text.

- A. <strong>
- B. <important>
- C. <b>
- D. <pre>

ANS :- A

45) Which is the correct CSS Syntax ?

- A. body color-black
- B. {body color-black}
- C. body {color:black;}
- D. {body;color:black}

ANS : C

46) Which Bootstrap class adds zebra-stripes to a table ?

- A. table-bordered.
- B. table-zebra.
- C. table-striped
- D. even and odd

ANS :- C

47) The \_\_ Javascript method is like the mathematical "for all" quantifier , which resumes true if and only if the function returns true for all elements in the array.

- A. every()
- B. some()
- C. all()
- D. forall()

ANS : B Updated ans is A

48) How can you detect the client's browser name in Javascript ?

- A. class.navName
- B. navigator.appName
- C. browser.name
- D. window.browser.

ANS : B

49) Which of the following methods CANNOT be used to add array elements in Javascript ?

- A. push()
- B. splice()
- C. shift()
- D. All can be used.

ANS : D

50) In Javascript \_\_ \_\_ is a comma separated list of colon separated name value pairs enclosed within curly braces.

- A. An object prototype.

- B. A map object.
  - C. An class
  - D. An object literal.
- Ans. D

51) In Javascript , what is the observation made of the statement if (!a[i]) continue ?

- i. Skips the undefined elements.
- ii. Skips the null elements.
- iii. Skips the non existent elements.

ANS :-

- A. i and ii only.
- B. ii and iii only.
- C. i and iii only.
- D. i,ii and iii

ANS :- D

52) Which of these methods is the ultimate element selection method and a very powerful technique by which client side Javascript programs can select the document elements that they are going to manipulate ?

- A. querySelectAll()
- B. querySelector()
- C. querySelect()
- D. querySelectorAll()

ANS :- D

53) To use the Angular HttpClient component, you must import the \_\_\_\_ module.

- A. HttpClientModule.
- B. HttpModule.
- C. Http

Ans :- A

54) In Angular , one can create local HTML reference or HTML tag using variable which starts with character \_\_\_\_

- A. @
- B. #
- C. "
- D. &

ANS :- B

55) Which of these Angular services can be injected in your component to the dynamic mode navigation ?

- A. Routing.
- B. RouterService.
- C. RoutingService
- D. Router

ANS :- D

56) In Angular , which of the following wild card route paths will you use to define a 404 route ?

- A. \*\*
- B. /
- C. ##
- D. default

ANS :- A

# 1. Core Java

## From First Folder

### 1. What is the basis for equality state in Java Identity HashMap?

- a. Hash Code and equals
- b. Name
- c. **Reference**

▼ Explanation:

- The question is asking about the basis for equality in Java Identity Hash Map.
- In Java, the Identity Hash Map class uses reference equality (==) rather than object equality (equals()) to determine if two keys are the same.
- This means that two keys are considered equal if they refer to the same memory location, rather than having equal content.
- This is useful when you want to compare objects based on their memory addresses rather than their content.

### 2. In Java, which of the following can be marked static?

- a. Methods, Variables, Initialization Blocks, Outer Classes, and Nested Classes
- b. **Methods, Variables, Initialization Blocks, and Nested Classes**
- c. Methods, Variables, Initialization Blocks, and Outer Classes
- d. Methods, Variables, and Initialization Block.

▼ Explanation:

- The question is asking which elements in Java can be marked as static.
- The answer is "Methods, Variables, Initialization Blocks, and Nested Classes" (option 2).
- This is because static can be applied to methods, variables, and initialization blocks to make them associated with the class rather than instances of the class.
- Nested classes can also be marked as static to indicate they are nested within the class but do not require an instance of the enclosing class to be instantiated.

▼ Extra:

- In Java, it is not possible to mark a top-level class as static.
- Static classes are only allowed for nested classes. This means that a class can only be declared static if it is declared inside another class.
- There are a few reasons why top-level classes cannot be static.
- First, it would make it difficult to create instances of the class.
- Second, it would make it difficult to manage the memory used by the class.
- Third, it would make it difficult to use the class in other programs.

▼ Static Inner Class Example:

```
public class OuterClass {

    public static class InnerClass {
        private static final String[] MONTHS = {"January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"};

        public static String getMonth(int month) {
            return MONTHS[month - 1];
        }
    }
}

public class AnotherClass {

    public static void main(String[] args) {
        int monthNumber = 3;
        OuterClass.InnerClass obj = new OuterClass.InnerClass();
        String monthName = obj.getMonth(monthNumber);
        System.out.println("The month for the number " + monthNumber + " is " + monthName);
    }
}

//The month for the number 3 is March
```

```
public class OuterClass {

    public static class InnerClass {
        private static int count = 0;
    }
}
```

```

        public static int getCount() {
            return count;
        }

        public static void incrementCount() {
            count++;
        }
    }

    public static void main(String[] args) {
        // Increment the count
        OuterClass.InnerClass.incrementCount();

        // Get the count
        int count = OuterClass.InnerClass.getCount();

        // Print the count to the console
        System.out.println("The count is: " + count);
    }
}
//The count is: 1

```

### Why use Static Classes?

- To improve performance:
  - Static inner classes are loaded into memory when the outer class is loaded, which can improve the performance of your program.
- To save memory:
  - Static inner classes do not require an instance of the outer class to be created, which can save memory.
- To improve encapsulation:
  - Static inner classes can only access the static members of the outer class, which can improve the encapsulation of your code.
- To make the code more readable and maintainable:
  - Static inner classes can be used to group related code together, which can make the code more readable and maintainable.

### ▼ Inner Class Example:

- Inner classes are classes that are declared inside another class. They are also known as nested classes. Inner classes have access to the private members of the outer class, and they can also be used to encapsulate related code.
- Here are some reasons why you might want to use an inner class:
- To access the private members of the outer class, To encapsulate related code, To create a local class, and To create an anonymous class.
- Here are some specific examples of when you might want to use an inner class:
  - To create a helper class: You can use an inner class to create a helper class that is specific to the outer class. For example, you could create an inner class to provide helper methods for working with the data in the outer class.
  - To create a callback class: You can use an inner class to create a callback class that is used to notify the outer class when an event occurs. For example, you could create an inner class to listen for changes to the data in the outer class.
  - To create a local class: You can use an inner class to create a local class that is only used within the scope of the outer class. This can be useful for reducing the amount of code that you need to write.
  - To create an anonymous class: You can use an inner class to create an anonymous class. Anonymous classes are classes that do not have a name. They are often used to implement callback interfaces.

```
// MainClass.java
public class MainClass {

    public static void main(String[] args) {
        OuterClass outerObj = new OuterClass();

        // Creating an instance of the inner class us
        OuterClass.InnerClass innerObj = outerObj.new

        // Calling the display method of the inner cl
        innerObj.display();
    }
}
```

```

    }
}

public class MainClass {

    public static void main(String[] args) {
        OuterClass outerObj = new OuterClass();

        // Creating an instance of the inner class us
        OuterClass.InnerClass innerObj = outerObj.new

        // Calling the display method of the inner cl
        innerObj.display();
    }
}

```

### 3. What is the output of the following Java code snippet?

```

public class home implements Runnable {
    public void run() {
        System.out.printf("%d",3);
    }
    public static void main(String[]args)throws InterruptedException
    {
        Thread thread = new Thread(new home());
        thread.start();
        System.out.printf("%d",1);
        thread.join();
        System.out.printf("%d",2);
    }
}

```

- a. 132
- b. 123
- c. 213
- d. 312

#### ▼ Explanation:

1. Inside the `main` method, a new thread is created and started. This thread executes the `run()` method of the `Test` class concurrently.



2. In the main thread, `System.out.printf("%d", 1);` prints "1" to the console.
3. `thread.join();` is called, which waits for the new thread to finish executing.
4. Inside the `run()` method, `System.out.printf("%d", 3);` prints "3" to the console.
5. After the new thread finishes, `System.out.printf("%d", 2);` prints "2" to the console.

So, the correct output sequence is "132".

#### 4. Which of the following statements is false about Java main method?

- a. **We can override main method.**
- b. Main method should be declared public and static.
- c. Main method should have only 1 argument of type String array.
- d. We can overload main method.

##### ▼ Explanation:

The main method in Java cannot be overridden because it's a static method, and static methods cannot be overridden in Java. Instead, they are hidden when a subclass defines a method with the same signature.

#### 5. HTTP2 Client API is introduced in Java9.

- a. **True**
- b. False

##### ▼ Explanation:

HTTP/2 Client API was indeed introduced in Java 9. This API provides support for HTTP/2, which is a major revision of the HTTP network protocol used by the World Wide Web. It offers features like multiplexing, header compression, and server push, among others, aiming to improve performance and efficiency compared to HTTP/1.1.

#### 6. Pick the right statement about creating a custom Java annotation.

- a. Annotations are created by using `@interface`, followed by annotation name.
- b. An annotation can have elements as well. They look like methods. We should not provide implementation for these elements.

c. All annotations extends `java.lang.annotation.Annotation` interface. Annotations can not include any extends clause.

d. **All of the above**

▼ Explanation:

1. **Annotations are created by using @interface, followed by annotation name:**

In Java, custom annotations are created using the

`@interface` keyword followed by the annotation name. For example:

```
@interface MyAnnotation {  
    // Elements and optional default values can be declared here  
}
```

2. **An annotation can have elements as well. They look like methods. We should not provide implementation for these elements:**

Annotations can indeed have elements, which are similar in appearance to methods. These elements define the data that can be associated with the annotation. However, unlike methods, these elements do not have implementations provided within the annotation declaration. For example:

```
@interface MyAnnotation {  
    String value(); // An element declaration, looks like a method signature  
}
```

3. **All annotations extend `java.lang.annotation.Annotation` interface.**

**Annotations cannot include any extends clause:**

All annotations implicitly extend the

`java.lang.annotation.Annotation` interface. This is an inherent feature of Java's annotation system. However, annotations cannot explicitly extend other interfaces or classes. For example:

```
@interface MyAnnotation {  
    // Annotation elements  
}  
// MyAnnotation implicitly extends java.lang.annotation.Annotation
```

Therefore, all of the statements are correct, as they accurately describe the process and characteristics of creating custom Java annotations.

## 7. What is the max capacity of a Java Blocking Queue?

- a. Integer.INFINITY
- b. **Integer.MAX\_VALUE**
- c. BigDecimal.MAX\_VALUE

### ▼ Explanation:

The maximum capacity of a Java Blocking Queue depends on the specific implementation being used. However, for most standard implementations such as `ArrayBlockingQueue` or `LinkedBlockingQueue`, the maximum capacity is `Integer.MAX_VALUE`, which is  $2^{31} - 1$ , because these implementations internally use arrays or linked structures which are limited by the maximum array size or the maximum integer value.

So, the correct answer is:

### 2. Integer.MAX\_VALUE

### ▼ Extra:

A Java BlockingQueue is a type of queue that supports operations that wait for the queue to become non-empty when retrieving an element and wait for space to become available in the queue when adding an element. It is part of the `java.util.concurrent` package and is designed for concurrent programming scenarios where multiple threads need to communicate and synchronize data exchange.

## 8. Which of the following is the immutable Java List static factory method?

- a. **List.of**
- b. List.immutable
- c. List.create

### ▼ Explanation:

`List.of` is a standard method introduced in Java 9 specifically for creating immutable lists. It's important to note that while there are other ways to create immutable lists in Java, such as using `Collections.unmodifiableList`, `List.of` provides a more concise and readable way to achieve immutability without the need for explicit instantiation or copying.

The other options are not standard methods or such standard methods do not exist.

9. What is the command line tool introduced for JS engine in Java8?

- a. **jjs**
- b. jss
- c. jfs
- d. jbd

▼ Explanation:

**jjs** is the command-line tool provided in Java 8 for running JavaScript code via the Nashorn JavaScript engine. Nashorn was introduced in Java 8 as a modern JavaScript engine, allowing Java developers to execute JavaScript code seamlessly within their Java applications.

10. Which of these Java code snippets is a correct way of defining generic method?

- ☒ a. `public<T1,T2,...,Tn>methodName(){`
- b. `T1,T2,...,Tn>methodName(T1,T2,...,Tn){`
- c. All of the above

11. In Java9, private methods are allowed in Interfaces.

- a. **True**
- b. False

▼ Explanation:

- In Java 9, private methods are indeed allowed in interfaces. This feature was introduced as part of the Java Enhancement Proposal (JEP) 213: "MilliTick #2: Small Language Enhancements". Private methods in interfaces allow interface developers to create helper methods or default method implementations that are not part of the public API but can be reused within the interface itself.
- Before Java 9, interface methods could only be constant variables or abstract methods. However, Java 9 introduced the ability to add private methods to interfaces, which can be implemented static or non-static.

12. Java system properties can be set or modified at runtime.

- a. **True**
- b. False

▼ Explanation:

The statement is true. In Java, system properties can indeed be set or modified at runtime. These properties can be accessed and modified using the `System.setProperty(String key, String value)` method. For example:

```
System.setProperty("my.property", "my_value");
```

This code sets the value of the system property `"my.property"` to `"my_value"` at runtime.

**13. Identify the Java class used to create a new Process.**

- a. **ProcessBuilder**
- b. ProcessFactory
- c. Process
- d. Thread

▼ Explanation:

The Java class used to create a new process is `java.lang.ProcessBuilder`.

The `ProcessBuilder` class provides a way to construct processes and their associated operating system commands. It allows you to specify the command and its arguments, as well as other process attributes such as the working directory, environment variables, and redirection of standard input, output, and error streams.

Here's a basic example of how you can use `ProcessBuilder` to create a new process:

```
import java.lang.ProcessBuilder;

public class CreateProcess {
    public static void main(String[] args) {

        try{
            // Create a new ProcessBuilder object.
            ProcessBuilder builder = new ProcessBuilder();

            // Set the command to be executed.
            builder.command("ls", "-l");

            // Start the process.
```

```

        Process process = builder.start();

        // Wait for the process to finish.
        process.waitFor();

        // Get the exit value of the process.
        int exitValue = process.exitValue();

        // Print the exit value.
        System.out.println("Exit value: " + exitValue);
    }
    catch (IOException | InterruptedException e) {
        e.printStackTrace();
    }
}
//OP: Exit value: 0

```

In this example, we create a new `ProcessBuilder` instance with the command `"ls"`, which lists the files in the current directory. We then start the process using `start()` method, and finally, we wait for the process to complete using `waitFor()` method.

#### 14. Pick the correct Java method reference usages.

- a. `object::instanceMethod`
- b. `Class::staticMethod`
- c. `Class::new`
- d. **All of the above**

##### ▼ Explanation:

The correct Java method reference usages are:

1. **`object::instanceMethod`**: This refers to an instance method of an object.
2. **`Class::staticMethod`**: This refers to a static method of a class.
3. **`Class::new`**: This refers to a constructor of a class.

So, the correct answer is:

#### 4. All of the above

#### 15. Which is the correct Java code snippet to be used to create a Functional Interface?

- a. interface MyFunctionalInterface implements Functional{
- b. **"@FunctionalInterface interface MyFunctionalInterface{"**
- c. "@Functional interface MyFunctionalInterface{"

▼ Extra:

Function Interface in Java with Examples - GeeksforGeeks

A Computer Science portal for geeks. It contains well written, well thought and well explained computer science and programming articles, quizzes and practice/competitive programming/company

<https://www.geeksforgeeks.org/function-interface-in-java-with-examples/>



**16. Which one will allow traversing a Java collection in each direction?**

- a. **ListIterator**
- b. Iterator
- c. MapIterator
- d. SetIterator

▼ Explanation:

The question is asking about Java interfaces that allow traversal of collections in both forward and backward directions.

The correct answer is 1. ListIterator.

Explanation:

1. ListIterator: This interface specifically supports bidirectional traversal of lists in Java collections, allowing traversal both forwards and backwards.
2. Iterator: This interface only supports forward traversal of collections.
3. MapIterator: This interface does not exist in Java; it seems like a distractor.
4. SetIterator: This interface also does not exist in Java; it seems like another distractor.

So, ListIterator is the correct choice because it's designed specifically for traversing lists bidirectionally.

▼ Extra:

```
import java.util.ArrayList;
import java.util.List;
import java.util.ListIterator;
```

```

public class ListIteratorExample {
    public static void main(String[] args) {
        // Creating a list of integers
        List<Integer> numbers = new ArrayList<>();
        numbers.add(1);
        numbers.add(2);
        numbers.add(3);
        numbers.add(4);
        numbers.add(5);

        // Using ListIterator to traverse the list forward
        System.out.println("Forward Traversal:");
        ListIterator<Integer> forwardIterator = numbers.listIterator()
        while (forwardIterator.hasNext()) {
            System.out.println(forwardIterator.next());
        }

        // Using ListIterator to traverse the list backward
        System.out.println("\nBackward Traversal:");
        ListIterator<Integer> backwardIterator = numbers.listIterator()
        while (backwardIterator.hasPrevious()) {
            System.out.println(backwardIterator.previous());
        }
    }
}

```

Output:

Forward Traversal:

1  
2  
3  
4  
5

Backward Traversal:

5  
4  
3



```
2
1
```

- `numbers.size()` returns the number of elements in the list.
- When you pass `numbers.size()` as the parameter to `listIterator(int index)`, it creates a `ListIterator` positioned at the specified index, which in this case is the end of the list (since indices are zero-based, `numbers.size()` returns the index just beyond the last element).
- Therefore, the `backwardIterator` is positioned at the end of the list, and when you call `previous()` method on it, it iterates backward through the list.

So, by initializing the `ListIterator` with `numbers.size()`, you're starting the iteration from the end of the list, which effectively allows you to traverse the list backward.

**17. In Java, which method is used to inform the observer of the change in observed object?**

- a. Check
- ☒ b. Update
- c. Inform
- d. notify

**18. In Java, it is NOT possible to override default serialization.**

- a. True
- b. **False**

▼ Explanation:

The question is asking whether it's possible to override default serialization in Java.

The correct answer is 2. False.

Explanation:

1. True: This would imply that it's not possible to override default serialization in Java, which is incorrect. It is indeed possible to customize serialization behavior in Java.
2. False: This is the correct answer. In Java, you can override default serialization behavior by implementing special methods (`writeObject` and

`readObject` ) in your class, providing custom serialization and deserialization logic. This allows you to control how instances of your class are serialized and deserialized, overriding the default behavior.

So, it is possible to override default serialization in Java, making option 2 the correct choice.

#### ▼ Extra:

Serialization in Java refers to the process of converting an object into a stream of bytes, which can be easily stored in a file, sent over the network, or persisted in a database. Deserialization, on the other hand, is the process of reconstructing an object from its serialized form.

Here's a brief explanation of key concepts related to serialization in Java:

##### 1. Default Serialization:

- In Java, every serializable class automatically supports default serialization.
- Default serialization is done by Java runtime, which means you don't have to write any additional code to serialize or deserialize objects of serializable classes.
- To make a class serializable, you simply implement the `java.io.Serializable` interface. This interface acts as a marker interface, indicating that the objects of the class can be serialized.

##### 2. Custom Serialization and Deserialization:

- Sometimes, you might need more control over the serialization process, such as excluding certain fields from serialization, handling transient fields, or encrypting sensitive data before serialization.
- Java allows you to customize the serialization and deserialization process by providing special methods in your class:
  - `private void writeObject(ObjectOutputStream out) throws IOException` : This method allows you to define custom serialization logic. You can write specific fields to the `ObjectOutputStream` or perform any custom operations before serialization.
  - `private void readObject(ObjectInputStream in) throws IOException, ClassNotFoundException` : This method allows you to define custom deserialization logic. You can read fields from the `ObjectInputStream` and reconstruct the object accordingly.

- By implementing these methods, you can override the default serialization behavior and provide custom serialization and deserialization logic for your class.

### 3. Object Serialization Stream Protocol:

- Java serialization uses a protocol to serialize and deserialize objects. This protocol defines how objects are written to and read from streams.
- The serialization stream protocol ensures that objects are written and read in a consistent format, regardless of the JVM or platform used.

Overall, serialization in Java provides a convenient way to store and transfer object data. Default serialization offers automatic serialization support for serializable classes, while custom serialization allows you to tailor the serialization process to your specific requirements.

## 19. Pick the class that is NOT part of Java collection framework.

- a. Map
- b. List
- c. Queue
- d. **Struct**

### ▼ Explanation:

The question is asking which class is NOT part of the Java Collection Framework.

The correct answer is 4. Struct.

Explanation:

1. Map: This is a key-value pair data structure in Java, represented by interfaces such as `java.util.Map`. Maps are part of the Java Collection Framework.
2. List: Lists represent an ordered collection of elements, and they are part of the Java Collection Framework, defined by interfaces like `java.util.List`.
3. Queue: Queues represent a data structure where elements are inserted at the rear and removed from the front. They are part of the Java Collection Framework, defined by interfaces like `java.util.Queue`.
4. Struct: Unlike the other options, `Struct` is not a standard class in the Java Collection Framework. While there might be custom implementations or

libraries that provide a `Struct` class, it's not part of the standard Java Collection Framework.

So, the correct answer is option 4. Struct.

20. Identify the parts of lambda expression in Java.

- a. Name
  - b. Parameterlist
  - c. Body
  - d. ReturnType
- 1.i
  - 2.iv
  - 3.i and iv
  - 4.
- ii and iii**

▼ Explanation:

The question is asking about the components of a lambda expression in Java.

The correct answer is 4. ii and iii.

Explanation:

- 1. Name: Lambda expressions do not have a name; they are anonymous functions.
- 2. Parameter list: This is a required part of a lambda expression, where you specify the parameters enclosed within parentheses. It corresponds to option ii.
- 3. Body: This is another required part of a lambda expression, where you define the implementation of the function. It corresponds to option iii.
- 4. Return type: Lambda expressions in Java can infer their return type from the context or be explicitly declared if needed. However, the return type itself is not explicitly specified within the lambda expression syntax.

So, the correct parts of a lambda expression in Java are option ii (Parameter list) and option iii (Body).

▼ Extra:

Lambda expressions in Java are a feature introduced in Java 8. They provide a concise way to represent anonymous functions: small blocks of code that can be passed around as arguments to methods or stored in variables.

Here's a brief explanation of lambda expressions and their syntax:

### 1. Syntax:

- Lambda expressions consist of three main parts: parameters, an arrow `>`, and a body.
- Syntax: `(parameters) -> body`
- Parameters: Enclosed within parentheses, specifying the input parameters (if any) to the function.
- Arrow (`>`): Separates the parameters from the body.
- Body: Contains the code to be executed. It can be a single expression or a block of statements enclosed within curly braces `{ }`.

### 2. Example:

```
// Lambda expression to square a number
(int x) -> x * x
```

- Here, `(int x)` represents the parameter list, `>` is the arrow, and `x * x` is the body.

### 3. Usage:

- Lambda expressions are often used in combination with functional interfaces. A functional interface is an interface with only one abstract method.
- Lambda expressions provide a way to implement the abstract method of a functional interface concisely.
- They are commonly used in Java's collection framework with methods like `forEach`, `map`, `filter`, etc.

### 4. Benefits:

- Conciseness: Lambda expressions enable writing shorter and more readable code compared to anonymous classes.
- Readability: They make code more readable by allowing you to express behavior directly where it's needed.
- Flexibility: Lambda expressions promote functional programming practices and enable a more functional style of coding in Java.

Lambda expressions are a powerful addition to Java, especially for writing code that is more expressive and concise, particularly when working with

collections and streams.

21. **Java Functional Interface can be defined as\_\_\_\_\_.**

- a. Simple Abstract Markup
- b. Simple Active Markup
- c. Single Ambivalue Method
- d. **Single Abstract Method**

22. **Which is the correct API to manage reactive streams in Java?**

- a. `java.util.concurrent.Flow`
- b. `java.util.concurrent.Flow.Publisher`
- c. `java.util.concurrent.Flow.Processor`
- d. **All of the above**

▼ Explanation:

1. `java.util.concurrent.Flow.Publisher` : This interface represents a provider of a potentially unbounded number of sequenced elements, publishing them according to the demand received from its `Subscriber(s)` . It is one of the building blocks of reactive streams.
2. `java.util.concurrent.Flow.Subscriber` : This interface represents a consumer of elements received from a `Publisher` . It receives them via the `onNext` method and can also handle errors and completion signals.
3. `java.util.concurrent.Flow.Processor` : This interface represents a processing stage that sits between the `Publisher` and the `Subscriber` . It receives elements from the `Publisher` , processes them in some way, and then forwards them to the `Subscriber` . It can also act as a `Publisher` and/or a `Subscriber` .
4. The `java.util.concurrent.Flow` package provides the necessary interfaces and classes to work with reactive streams, including publishers, subscribers, and processors. Therefore, all the options listed are correct. Thank you for pointing out the discrepancy.

▼ Extra:

### Java 9 Reactive Streams | Baeldung

Quick and practical introduction to Reactive Streams in Java 9.

<https://www.baeldung.com/java-9-reactive-streams>



## 23. In Java, which of the following data structure classes CANNOT be type parameterized?

- a. List
- b. Set
- c. **Array**
- d. Map

### ▼ Explanation:

The correct answer is:

1. Array

Explanation:

1. List: In Java, `List` can be type parameterized. For example, you can have `List<Integer>` or `List<String>`, where the type parameter specifies the type of elements in the list.
2. Set: Similarly, in Java, `Set` can be type parameterized. You can have `Set<Integer>` or `Set<String>`, specifying the type of elements the set can contain.
3. Array: Unlike `List` and `Set`, arrays in Java cannot be type parameterized. When you create an array, you specify the type of its elements using square brackets (`[]`). For example, `int[]`, `String[]`, etc. Arrays are covariant and reified, meaning they retain information about their component type at runtime.
4. Map: Java allows type parameterization for `Map`. You can have `Map<Integer, String>`, `Map<String, Integer>`, etc., where the type parameters specify the types of keys and values in the map.

So, the correct answer is option 3. Array. Arrays in Java cannot be type parameterized.

## 24. Which of these OOPS concept is enabled by method overriding in Java?

- a. **Polymorphism**
- b. Abstraction

- c. Inheritance
- d. Encapsulation

25. What is the output of the following Java code snippet?

```
classFirst {
    publicFirst() {
        System.out.println("F");
    }
}
classSecondextendsFirst {
    publicSecond() {
        System.out.println("S");
    }
}
classThirdextendsSecond {
    publicThird() {
        System.out.println("T");
    }
}
publicclassTestClass {
    publicstaticvoidmain(String[]args) {
        Thirde=newThird();
    }
}
```

- a. TSF
- b. T
- c. F
- d. **FST**

26. How does one assign system property values to variable using Java class fields?

- a. "@Autowired("\${aProperty}") private String aSysProp;"
- b. "@Environment("\${aProperty}") private String aSysProp;"
- c. "@Property("\${aProperty}") private String aSysProp;"
- d. **"@Value("\${aProperty}") private String aSysProp;"**

▼ Explanation:



```
@Value("${aProperty}") private String aSysProp;
```

Explanation:

In Spring Framework, you can assign system property values to Java class fields using the `@Value` annotation with the syntax `@Value("${propertyName}")`. Here's how it works:

1. `@Value`: This annotation is used to inject values into Spring beans.
2. `"${aProperty}"`: This expression denotes a placeholder for a property value. The property value is resolved from the Spring environment, which includes system properties, configuration files, and other sources.
3. `private String aSysProp;`: This declares a private field `aSysProp` in the Java class to hold the injected property value.

Therefore, the correct option is 4. `@Value("${aProperty}") private String aSysProp;`.

▼ Extra:

Let's go through the other options:

1. `@Autowired("${aProperty}") private String aSysProp;`
  - This line is incorrect because `@Autowired` is used for dependency injection, typically for injecting beans into other beans, not for injecting property values.
  - The correct annotation for injecting property values is `@Value`.
2. `@Environment("${aProperty}") private String aSysProp;`
  - There is no annotation `@Environment` in Spring Framework.
  - The correct annotation for injecting property values is `@Value`.
3. `@Property("${aProperty}") private String aSysProp;`
  - There is no annotation `@Property` in Spring Framework.
  - The correct annotation for injecting property values is `@Value`.

So, options 1, 2, and 3 are incorrect because they use incorrect or nonexistent annotations for injecting property values in Spring. The correct annotation for this purpose is `@Value`.

## 27. Which of the following types of Java methods CANNOT be Type parameterized?

- a. Final methods
- ✓ b. Overriding methods

c. Overloaded methods

**28. Pick the correct Java Stream API's?**

- a. Stream.allMatch
- b. Stream.noneMatch
- c. Stream.anyMatch
- d. **All the above**

▼ Explanation:

The correct answer is:

- 4. All the above

Explanation:

1. `Stream.allMatch` : This method returns true if all elements of the stream match the given predicate, otherwise, it returns false.
2. `Stream.noneMatch` : This method returns true if no elements of the stream match the given predicate, otherwise, it returns false.
3. `Stream.anyMatch` : This method returns true if any elements of the stream match the given predicate, otherwise, it returns false.

All three methods are part of the Java Stream API, and they are used to perform matching operations on the elements of a stream based on a given predicate. Therefore, the correct answer is option 4. All the above.

Predicates are like filter → Predicate is a functional interface defined in the java.util.function package.

```
Predicate<Integer> isEven = n -> n % 2 == 0;
```

**29. Which class is introduced in Java 8 to get the current milliseconds?**

- a. **Clock**
- b. Current Time
- c. Date Time
- d. Time

▼ Explanation:

The class introduced in Java 8 to get the current milliseconds is:

- 1. Clock

Explanation:

Java 8 introduced the `Clock` class in the `java.time` package as part of the new Date and Time API. The `Clock` class provides access to the current date and time based on the system clock. You can use the `Clock` class to obtain the current time in milliseconds using its `millis()` method.

Here's an example of how you can use the `Clock` class to get the current time in milliseconds:

```
import java.time.Clock;

public class Main {
    public static void main(String[] args) {
        // Get the system clock
        Clock clock = Clock.systemDefaultZone();

        // Get the current time in milliseconds
        long currentTimeMillis = clock.millis();

        // Print the current time in milliseconds
        System.out.println("Current time in milliseconds:
" + currentTimeMillis);
    }
}
```

This will output the current time in milliseconds since the epoch (January 1, 1970, 00:00:00 GMT).

So, the correct answer is option 1. Clock.

▼ Extra:

1. **Current Time:** This option is not a specific class introduced in Java 8. It's a generic term used to refer to the current time, but there's no dedicated class named "Current Time" in Java 8.
2. **Date Time:** This option refers to the `DateTime` class, which is not introduced in Java 8. Prior to Java 8, the `Date` and `Time` classes were used separately, and Java 8 introduced the `LocalDateTime` class to represent a date-time without a time zone.
3. **Time:** Similarly, there's no specific class named "Time" introduced in Java 8 to get the current milliseconds. Prior to Java 8, the `Time` class was

part of the `java.sql` package for handling time values in databases. However, it's not used for getting the current milliseconds in Java 8.

30. **In Java, Which of these class types supports sequential and parallel processing of data?**

- a. Queue
- b. Lambda
- c. ExecutorService
- d. **Streams**

▼ Explanation:

Streams in Java are specifically designed to support both sequential and parallel processing of data. They allow for functional-style operations on collections, arrays, or I/O resources, and can exploit multicore architectures for parallel execution when appropriate. Therefore, Streams are the correct choice for supporting both sequential and parallel processing of data in Java.

▼ Extra:

1. **Queue**: Queues in Java are data structures that follow the FIFO (First In, First Out) principle, used for storing elements in a linear sequence. They are not specifically designed for processing data sequentially or in parallel.
2. **Lambda**: Lambdas in Java are anonymous functions that can be passed as arguments to methods or stored in variables. While lambdas can be used in conjunction with streams for processing data, they don't inherently support sequential or parallel processing on their own.
3. **ExecutorService**: The `ExecutorService` interface in Java provides a higher-level abstraction for managing threads and executing tasks asynchronously. It allows for concurrent execution of tasks, which can support parallel processing of data.
4. **Streams**: Streams in Java, introduced in Java 8, are a sequence of elements supporting sequential and parallel aggregate operations. They provide a high-level abstraction for processing data in a functional style, allowing for easy parallelization of operations

31. **Which Java Set implementation is sorted and synchronized?**

- a. TreeSet

- b. LinkedHashSet
- c. HashSet
- d. **ConcurrentSkipListSet**

▼ Explanation:

The ConcurrentSkipListSet implementation in Java is both sorted and synchronized. It maintains its elements in sorted order using a skip list data structure and provides thread-safe access, making it suitable for concurrent usage without the need for external synchronization.

▼ Extra:

1. **TreeSet:** TreeSet is an implementation of the Set interface in Java that uses a Red-Black tree to store elements in sorted order. However, it is not inherently synchronized.
2. **LinkedHashSet:** LinkedHashSet is an implementation of the Set interface that maintains the insertion order of elements. It is not sorted, but it can be synchronized externally if needed.
3. **HashSet:** HashSet is an implementation of the Set interface that does not guarantee any specific order of elements. It is not sorted or synchronized by default.
4. **ConcurrentSkipListSet:** ConcurrentSkipListSet is an implementation of the Set interface that uses a skip list data structure to store elements in sorted order. It is also designed to be thread-safe and can support concurrent access from multiple threads without external synchronization.

32. Which of the following statements is NOT correct for JVM?

- a. JVM acts as a translator that translates different Machine code (on the basis of Host Machine) for a common Byte Code.
- b. JVM is a Virtual Machine that acts as a intermediary between Java Application and Host Operating System.
- c. JVM reads Byte Code and generates Machine Code.
-  d. JVM reads Source Code and generates Byte Code.

33. Is the following a valid Java code Snippet?

```
Optional < String >aValue = Optional.of ("Not Empty");
```

```
If (aValue.isPresent()) {  
  
System.out.println( "Got Not Empty");  
  
}
```

a. True

b. **False**

▼ Explanation:

No, the provided Java code snippet contains syntax errors and would not compile. Here's the corrected version:

```
import java.util.Optional;  
  
public class Main {  
    public static void main(String[] args) {  
        Optional<String> aValue = Optional.of("Not Empty");  
  
        if (aValue.isPresent()) {  
            System.out.println("Got Not Empty");  
        }  
    }  
}
```

Here's the explanation:

1. The method `Optional.of("Not Empty")` is used to create an `Optional` object containing the string "Not Empty", which is valid syntax.
2. The method `isPresent()` is a method in the `Optional` class that takes a `Consumer` as an argument and executes it if the `Optional` contains a non-null value. However, in the given code snippet, it's used incorrectly within the `if` condition, which will result in a compilation error.
3. Instead, the correct method to check whether an `Optional` contains a value is `isPresent()`. So, the correct usage would be `if (aValue.isPresent())`.
4. Lastly, `System.out.println` is a typo. It should be `System.out.println` to print "Got Not Empty" to the console.

34. **Java Stream does NOT store the elements. It simply performs the aggregate operations to get the desired stream of data.**

a. **True**

b. False

▼ Explanation:

The statement is **true**.

Java Streams do not store elements themselves. Instead, they operate on a source of elements (such as a collection, array, or I/O channel) and perform aggregate operations (such as filter, map, reduce, etc.) to process the elements and produce a desired stream of data. Streams are designed to process elements lazily and can be considered as a pipeline of operations. They do not have their own storage for elements, and they do not modify the underlying data source. Therefore, the statement is true.

35. **Which Java class is used to apply reduction or aggregate functions on streams?**

a. Filters

b. Reduction

c. **Collectors**

▼ Explanation:

The correct answer is:

1. **Collectors**

**Explanation:**

In Java, the `Collectors` class is used to apply reduction or aggregate functions on streams. It provides various methods like `toList()`, `toSet()`, `joining()`, `summarizingInt()`, etc., which are used with the `collect()` method of streams to perform reduction operations and collect the results into a container like a `List`, `Set`, or `Map`. These reduction operations are crucial for processing stream elements and producing a result, and the `Collectors` class provides a convenient way to perform them.

36. **Which of the following statements are true about Java?**

i. **A finally block is executed, only after the catch block is executed.**

ii. **A finally block is executed whether an exception is thrown or not.**

iii. **A finally block is NOT mandatory.**

iv. **A finally block is executed, only if an exception occurs.**

- a. i
- b. iv
- c. i and iv
- d. **ii and iii**

37. Which of these is a standard Java Interface?

- a. **Serializable**
- b. Long
- c. System
- d. Collections

▼ Explanation:

Serializable is a standard Java interface provided by the Java API, used for serializing objects. It's commonly used in scenarios where objects need to be converted into a stream of bytes for storage or transmission.

▼ Extra:

1. **Long**: This is a wrapper class for the primitive type `long`. It's not an interface.
2. **System**: This is a final class in Java that provides access to system properties and environment variables. It's not an interface.
3. **Collections**: This is a utility class in Java that consists of static methods for operating on collections, such as sorting, searching, and synchronizing. It's not an interface.

38. What is the replacement for PermGen space in java8?

- a. FixedGen
- b. **Metaspace**
- c. MetaGenspace

▼ Explanation:

In Java 8, the Permanent Generation (PermGen) was replaced by Metaspace. Metaspace is a memory space where class metadata is stored, including information about classes, methods, fields, annotations, etc. Unlike PermGen, Metaspace is not a fixed size and can dynamically expand or contract depending on the application's needs. It helps mitigate issues related to PermGen space such as `OutOfMemoryError` due to class metadata exceeding the allocated space.



▼ Extra:

**FixedGen:** This option is not the replacement for PermGen space in Java 8. There's no concept of "FixedGen" in Java memory management.

**MetaGenspace:** Similarly, "MetaGenspace" is not a valid term or concept in Java memory management. The correct replacement for PermGen space in Java 8 is Metaspace, not MetaGenspace.

39. What is the output of the following Java code snippet?

```
StringJoiner astring = new StringJoiner("-", "{", "}");
```

```
astring.add("A");
```

```
astring.add("B");
```

```
astring.add("C");
```

- a. {A-B-C}
- b. {A}-{B}-{C}
- c. {-A-}{-B-}{-C-}

▼ Explanation:

The `StringJoiner` class in Java is used to construct a sequence of characters separated by a delimiter, with an optional prefix and suffix. Let's analyze the given code:

```
import java.util.StringJoiner;

public class Main {
    public static void main(String[] args) {
        StringJoiner astring = new StringJoiner("-", "{", "}");
        astring.add("A");
        astring.add("B");
        astring.add("C");

        System.out.println(astring);
    }
}
```

Here, we are creating a `StringJoiner` object `astring` with a delimiter "-" and prefix "{" and suffix "}". Then, we add three strings "A", "B", and "C" to the `StringJoiner` object.

- The `StringJoiner` object `asString` is initialized with a delimiter "-" and prefix "{" and suffix "}".
- The `add()` method adds the strings "A", "B", and "C" to the `StringJoiner` object, separated by the delimiter "-".
- The resulting string is "{A-B-C}", where the elements "A", "B", and "C" are joined together with the delimiter "-".

40. What is the return type of Java Lambda expressions?

- a. Void
- b. Lambda
- ☒ c. Function
- d. Regex

41. Name the two types of streams in Java.

- a. Sync and async
- b. **Sequential and parallel**
- c. Random and ordered

▼ Explanation:

- In Java, there are two types of streams: sequential streams and parallel streams.
- Sequential streams process elements in a single thread, one element at a time, maintaining the order of elements as they appear in the source.
- Parallel streams, on the other hand, enable parallel processing of elements using multiple threads, potentially improving performance for operations that can be parallelized.

Therefore, the correct types of streams in Java are sequential and parallel.

42. What is the difference between `peek()` and `dequeue()`?

- a. **`Peek()` returns the next item in line but `dequeue()` removes and returns the next item in line**
- b. `peek()` and `dequeue()` return the next item in line
- c. `peek()` and `dequeue` remove and return the next item in line

43. Which one of the following statements on Java String is correct?

- a. `replace()` method replaces only first occurrences of a character in invoking string with the given character.

- b. `replace()` method replaces last occurrence of a character in invoking string with the given character.
- c. **`replace()` method replaces all occurrences of one character in invoking string with the given character.**
- d. `replace()` method replaces all the characters in invoking string with the given character.

## From Second Folder

### 1. What is the result?

```
class Animal {
    public String noise() {
        return "peep";
    }
}

class Dog extends Animal {
    public String noise() {
        return "bark";
    }
}

class Cat extends Animal {
    public String noise() {
        return "meow";
    }
}
...
30. Animal animal = new Dog();
31. Cat cat = (Cat)animal;
32. System.out.println(cat.noise());
//Exception in thread "main" java.lang.ClassCastException: cla
```

- a. peep
- b. bark
- c. meow
- d. Compilation fails.

e. **An exception is thrown at runtime.**

## 2. What is the output?

```
class A {
    void foo() throws Exception {
        throw new Exception();
    }
}

class SubB2 extends A {
    void foo() {
        System.out.println("B ");
    }
}

class HelloWorld {
    public static void main(String[] args) {
        A a = new SubB2();
        a.foo(); // in the question this is the line 15
    }
}
```

- A. B
- B. B, followed by an Exception.
- C. Compilation fails due to an error on line 9.

**D. Compilation fails due to an error on line 15.**

E. An Exception is thrown with no other output

### ▼ Explanation:

The foo() of parent class throws the exception. When overriding the exception in the child class the same exception gets carried, so it needs to be handled where we are calling the foo(). (The foo() of child class is called.

### ▼ Extra:

Rule of exception while overriding:

- If parent class foo() does not throws any exception then the child class overridden foo() should either not throw the exception or if it throws then it should throw the unchecked exception.

- If parent class foo() throws an exception then either the child class overridden foo() should throw the same exception or it should not throw any exception or it should throw the specific exception of that type.

## 2. What is the output?

```
class multithreaded_programing{
    public static void main(String args[]){
        Thread t = Thread.currentThread();
        t.setName("new thread");
        System.out.println(t);
    }
}
```

**OUTPUT:- Thread[new thread,5,main]**

### ▼ Explanation:

- **Thread** : This indicates that the object being printed is of type **Thread**.
- **[new thread,5,main]** : This part represents the state and properties of the thread.
  - **new thread** : This is the name of the thread. It was set using `t.setName("new thread")`.
  - **5** : This represents the thread priority. The default priority of a thread is 5.
  - **main** : This indicates that the thread is running in the main thread group.

## 3. Observe the code:

```
public class MyRunnable implements Runnable{
    public void run(){
        //some code
    }
}
```

Which of these will create and start this thread?

**ANSWER:- new Thread(new MyRunnable()).start();**

### ▼ Explanation:

Because the class implements Runnable, an instance of it has to be passed to the Thread constructor and then the instance of the Thread has to be

started.

#### 4. Observe the code below

```
class Mammal { }  
class Raccoon extends Mammal {  
    Mammal m = new Mammal();  
}  
class BabyRaccoon extends Mammal { }
```

**Which four statements are true? (Choose four)**

- ☒ A. Raccoon is-a Mammal.
- ☒ B. Raccoon has-a Mammal.
- ☒ C. Baby Raccoon is-a Mammal.
- ☐ D. Baby Raccoon is-a Raccoon.
- ☐ E. Baby Raccoon has-a Mammal.
- ☒ F. Baby Raccoon is-a Baby Raccoon.

#### 5. Observe the following code

```
import java.io.*;  
public class fileinputoutput {  
    public static void main(String[] args) {  
        String obj = "abc";  
        byte b[] = obj.getBytes();  
        ByteArrayInputStream obj1 = new ByteArrayInputStream(b  
  
        for (int i = 0; i < 2; ++ i) {  
            int c;  
            while ((c = obj1.read()) != -1) {  
                if (i == 0) {  
                    System.out.print(Character.toUpperCase((ch  
                }  
            }  
        }  
    }  
}  
ANS: - ABC
```

▼ Explanation:

Let's analyze the code snippet step by step:

1. `String obj = "abc";` : This line initializes a `String` object named `obj` with the value "abc".
2. `byte b[] = obj.getBytes();` : This line converts the string `obj` into a byte array `b` using the `getBytes()` method. Each character in the string is converted to its corresponding ASCII value.
3. `ByteArrayInputStream obj1 = new ByteArrayInputStream(b);` : This line creates a new `ByteArrayInputStream` named `obj1` using the byte array `b` as its input source.
4. The code then enters a loop `for (int i = 0; i < 2; ++i)` which runs twice:
  - a. In the first iteration (`i == 0`):
    - It reads bytes from the `ByteArrayInputStream obj1` using `obj1.read()` and converts each byte to its corresponding character.
    - The characters are converted to uppercase using `Character.toUpperCase()` and then printed.
  - b. In the second iteration (`i == 1`):
    - Nothing is done because the loop body is empty for `i != 0`.

However, there is an issue in this code. The `obj1.read()` method advances the stream pointer every time it's called. So, after the first iteration of the outer loop, the stream pointer will have reached the end of the stream, and subsequent reads will return -1 indicating the end of the stream. Therefore, the loop will not execute any further.

The output of this code will be: `ABC`, printed once.

#### 6. Predict the output of the code:

```
class output {  
    public static void main(String args[]) {  
        StringBuffer s1 = new StringBuffer("Hello World");  
        s1.insert(6 , "Good ");  
        System.out.println(s1);  
    }  
}
```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld

**d) Hello Good World**

▼ Explanation:

String buffer are mutable.

7. **Which of these keywords cannot be used for a class which has been declared final?**

a) **abstract**

b) extends

c) abstract and extends

d) None of the mentioned

▼ Explanation:

A abstract class is incomplete by itself and relies upon its subclasses to provide complete implementation. If we declare a class final then no class can inherit that class, an abstract class needs its subclasses hence both final and abstract cannot be used for a same class.

8. **sos('1'+new integer(2)+3);**

ANS-123

9. **Observe the code and predict the output**

```
int a[]=new int[]{1,2,6,4,5};
int b[]=new int[5];
b[2]=a[2];
a[2]=10;
System.out.println(b[2]);
```

ANS: - 6

10. **What is the output of this program?**

```
import java.util.*;
class vector {
    public static void main(String args[]) {
        Vector obj = new Vector(4,2); // for initial size and
        obj.addElement(new Integer(3));
        obj.addElement(new Integer(2));
        obj.addElement(new Integer(5));
        obj.removeAll(obj); // removeAll() accepts a collectio
        System.out.println(obj.isEmpty());
```



```
}  
}
```

a) 0

b) 1

**c) true**

d) false

#### 11. What is the output?

```
import java.util.*;  
  
class Maps {  
    public static void main(String args[]) {  
        TreeMap obj = new TreeMap();  
        obj.put("A", new Integer(1));  
        obj.put("B", new Integer(2));  
        obj.put("C", new Integer(3));  
        System.out.println(obj.entrySet());  
    }  
}
```

OUTPUT:- [A=1, B=2, C=3]

a) [A, B, C]

b) [1, 2, 3]

c) {A=1, B=2, C=3}

d)

**[A=1, B=2, C=3]**

▼ Explanation:

obj.entrySet() method is used to obtain a set that contains the entries in the map. This method provides set view of the invoking map.

#### 12. What is the output?

```
import java.util.*;  
  
class Bitset {  
    public static void main(String args[]) {  
        BitSet obj = new BitSet(5);  
        for (int i = 0; i < 5; ++i)
```

```

        obj.set(i);
    obj.clear(2);
    System.out.print(obj);
}
}

```

a) {0, 1, 3, 4}

b) {0, 1, 2, 4}

c) {0, 1, 2, 3, 4}

d) {0, 0, 0, 3, 4}

▼ Explanation:

1. `BitSet obj = new BitSet(5);` : This line creates a new instance of the `BitSet` class named `obj`. The constructor `BitSet(int nbits)` is called with the argument `5`, which specifies the initial size of the bit set to be 5 bits. A `BitSet` is a collection of bits, where each bit has a value of either 0 or 1.
2. `for (int i = 0; i < 5; ++i) obj.set(i);` : This loop iterates from `0` to `4` (inclusive) and sets each bit at index `i` in the `obj` bit set using the `set(int bitIndex)` method. This effectively sets all bits in the `obj` bit set.
3. `obj.clear(2);` : This line clears the bit at index `2` in the `obj` bit set using the `clear(int bitIndex)` method. After this operation, the bit at index `2` will be set to `0`.
4. `System.out.print(obj);` : This line prints the string representation of the `obj` bit set using the `toString()` method. The string representation shows the bits in the set, where a set bit is represented as `1` and an unset bit is represented as `0`.

The output of the code will be a representation of the `obj` bit set after setting all bits initially and then clearing the bit at index `2`. Depending on the implementation of `BitSet.toString()`, the output could vary, but it will generally show the bits in the set, such as `[11100]` for a `BitSet` with 5 bits after the operations described above.

▼ Extra:

`BitSet` it's a class in Java's standard library ( `java.util.BitSet` ) that implements a set of bits or flags. It is often used to perform operations on bit patterns or to represent sets of integers.

13. What is the output?

```

import java.util.*;

class Array {
    public static void main(String args[]) {
        int array[] = new int [5];
        for (int i = 5; i > 0; i--)
            array[5-i] = i;

        Arrays.fill(array, 1, 4, 8); // reference, start, end,

        for (int i = 0; i < 5 ; i++)
            System.out.print(array[i]);
    }
}

```

a) 12885

b) 12845

**c) 58881**

d) 54881

14. What is the output of the following program?

```

import java.util.*;

class Collection_Algos {
    public static void main(String args[]) {
        LinkedList list = new LinkedList();
        list.add(new Integer(2));
        list.add(new Integer(8));
        list.add(new Integer(5));
        list.add(new Integer(1));
        Iterator i = list.iterator();
        Collections.reverse(list);
        Collections.shuffle(list);

        while(i.hasNext())
            System.out.print(i.next() + " ");
    }
}

```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8

**d) Any random order**

▼ Explanation:

reverse — reverse the list.

shuffle – randomizes all the elements in a list.

15. What is the output?

```
import java.util.*;
class Collection_Algos {
    public static void main(String args[]) {
        LinkedList list = new LinkedList();
        list.add(new Integer(2));
        list.add(new Integer(8));
        list.add(new Integer(5));
        list.add(new Integer(1));

        Iterator i = list.iterator();

        Collections.reverse(list);
        Collections.sort(list);

        while(i.hasNext())
            System.out.print(i.next() + " ");
    }
}
```

- a) 2 8 5 1
- b) 1 5 8 2

**c) 1 2 5 8**

- d) 2 1 8 5

16. What is the output?

```
import java.util.*;
public class genericstack <E> {
    Stack <E> stk = new Stack <E>();
```

```

        public void push(E obj) {
            stk.push(obj);
        }

        public E pop() {
            E obj = stk.pop();
            return obj;
        }
    }

    class Output {
        public static void main(String args[]) {
            genericstack <String> gs = new genericstack<String>();
            gs.push("Hello");
            System.out.print(gs.pop() + " ");
            genericstack <Integer> gs = new genericstack<Integer>();
            gs.push(36);
            System.out.println(gs.pop());
        }
    }
}

```

- a) Error // error because gs string and gs integer will give error, and even generic class not be public
- b) Hello
- c) 36
- d) Hello 36 // if the everything syntactically is correct then this is the output.

#### 17. What is the output?

```

class output {
    public static void main(String args[]) {
        StringBuffer c = new StringBuffer("Hello");
        StringBuffer c1 = new StringBuffer(" World");
        c.append(c1);
        System.out.println(c);
    }
}

```

OUTPUT:- Hello World

- a) xello
- b) xxxxx
- c) Hxlllo
- d) Hexlo

18. Questions on this:

```
import java.util.*;

class HelloWorld {
    public static void main(String[] args) {
        Calendar calendar = new GregorianCalendar();
        int year = calendar.get(Calendar.YEAR);
        int month = calendar.get(Calendar.MONTH);
        int dayOfMonth = calendar.get(Calendar.DAY_OF_MONTH);

        System.out.println(year+" "+month+" "+dayOfMonth);

        int dayOfWeek = calendar.get(Calendar.DAY_OF_WEEK);
        int weekOfYear = calendar.get(Calendar.WEEK_OF_YEAR);
        int weekOfMonth= calendar.get(Calendar.WEEK_OF_MONTH);
        int hour = calendar.get(Calendar.HOUR); // 12 hour clo

        System.out.println(dayOfWeek+" "+weekOfYear+" "+weekOf

        int hourOfDay = calendar.get(Calendar.HOUR_OF_DAY); //
        int minute = calendar.get(Calendar.MINUTE);
        int second = calendar.get(Calendar.SECOND);
        int millisecond= calendar.get(Calendar.MILLISECOND);
    }
}
```

19. What is the output?

```
class exception_handling {
    public static void main(String args[]) {
        try {
            System.out.print("Hello" + " " + 1 / 0);
        }
        catch(ArithmeticException e) {
```

```

        System.out.print("World");
    }
}

```

a) Hello

**b) World**

c) HelloWorld

d) Hello World

20. What is the output?

```

class output {

    public static void main(String args[]) {
        char c[]={'a', '1', 'b' , ' ' , 'A' , '0'};
        for (int i = 0; i < 5; ++i) {
            if(Character.isDigit(c[i]))
                System.out.println(c[i]+" is a digit");

            if(Character.isWhitespace(c[i]))
                System.out.println(c[i]+" is a Whitespace char");

            if(Character.isUpperCase(c[i]))
                System.out.println(c[i]+" is an Upper case Lett");

            if(Character.isLowerCase(c[i]))
                System.out.println(c[i]+" is a lower case Lett");

            i=i+3;
        }
    }
}

```

a) a is a lower case Letter

is White space character

b) b is a lower case Letter

is White space character

**c) a is a lower case Letter**

**A is a upper case Letter**

d) a is a lower case Letter

0 is a digit

**21. What is the output?**

```
import java.io.*;
class filesinputoutput {
    public static void main(String args[]) {
        InputStream obj = new FileInputStream("inputoutput.jav
        System.out.print(obj.available());
    }
}
```

Note: inputoutput.java is stored in the disk.

a) true

b) false

☒ c) prints number of bytes in file

d) prints number of characters in the file

▼ Explanation:

obj.available() returns the number of bytes.

**22. What is the output?**

```
import java.io.*;

public class filesinputoutput {
    public static void main(String[] args) {
        String obj = "abc";
        byte b[] = obj.getBytes();
        ByteArrayInputStream obj1 = new ByteArrayInputStream(b

        for (int i = 0; i < 2; ++ i) {
            int c;
            while((c = obj1.read()) != -1) {
                if(i == 0) {
                    System.out.print(Character.toUpperCase((ch
                    obj2.write(1);
                }
            }
        }
    }
}
```



```

        System.out.print(obj2);
    }
}

```

23. Which of these methods are used to read in from file?

a) get()

**b) read()**

c) scan()

d) readFileInput()

24. What is the output of the code?

```

public class Test8 {
    public static void main (String args[]) {
        Number numberRef = new Integer(0);
        Double doubleRef = (Double)numberRef;
    }
}

```

ANS:- Throw exception

Exception in thread "main" java.lang.ClassCastException: class java.lang.Integer cannot be cast to class java.lang.Double (java.lang.Integer and java.lang.Double are in module java.base of loader 'bootstrap')

25. What is the output?

```

import java.io.*;

class HelloWorld {
    public static void main(String[] args) {
        StringBuffer s1 = new StringBuffer("Hello");
        StringBuffer s2 = s1.reverse();
        System.out.println(s2);
    }
}

```

OUTPUT:- olleH

26. Which Pattern

```
import java.util.*;

class HelloWorld {
    public static void main(String args[]) {
        List<String> List = new ArrayList<String>();
        //add string
        List.add("cricket");
        List.add("football");
        List.add("hockey");
        Iterator it = List.iterator();

        while(it.hasNext()) {
            String s = (String) it.next();
        }
    }
}
```

Ans:- Iterator

## 27. **java.util.enumeration is an example of which pattern?**

Ans:- In short:- iterator

`java.util.Enumeration` is an example of the Iterator pattern.

The Iterator pattern provides a way to access elements of an aggregate object (such as a collection) sequentially without exposing its underlying representation. In Java, `Enumeration` serves this purpose by allowing traversal of elements in legacy collections like `Vector` and `Hashtable`. It provides methods like `hasMoreElements()` to check if there are more elements, and `nextElement()` to retrieve the next element.

By using the Iterator pattern, code that relies on collections can be decoupled from the specific implementation of those collections, promoting flexibility and code reuse.

## 28. **Observe the following code:**

```
Class x implements Runnable {
    Public static void main(String args[]) {
        /*missing code? */
    }
}
```

```
    public void run() {}  
}
```

Ans:- x run = new x();

Thread t = new Thread(run); t.start();

**29. Predict the output of the code:**

```
import java.util.*;  
  
class HelloWorld {  
    public static void main(String args[]) {  
        try {  
            int a = 5;  
            int b = 0;  
            int c = a/b;  
            System.out.println("World");  
        }  
        catch(Exception e) {  
            System.out.println("hello");  
        }  
    }  
}  
OUTPUT:- hello
```

**30. What is the output?**

```
Int a = 10;  
Int b = 0;  
Int c = a/b;  
SOP("c");  
OUTPUT:- runtime exception
```

**31. Which will contain the body of the thread?**

Answer : run()

**32. Which two of the following methods are defined in class thread?**

Start() 2. Wait() 3. Notify() 4. Run()

Ans:- start()

run()

▼ Explanation:

- `start()` : This method is used to begin the execution of a thread. When `start()` is called on a `Thread` object, it creates a new thread of execution and invokes the `run()` method on that thread.
- `run()` : This method contains the code that constitutes the thread's execution. When a thread is started using `start()`, the JVM calls the `run()` method of the thread, and the code inside `run()` is executed on the new thread.

`wait()` and `notify()` methods are defined in the `Object` class, not directly in the `Thread` class. They are used for inter-thread communication and synchronization in Java's concurrency mechanism.

33. Assume the following method is properly synchronized and called from a thread A on an object B: `wait(2000);`

After calling this method, when will the thread A become a candidate to get another turn at the CPU?

- After thread A is notified, or after two seconds.
- After the lock on B is released, or after two seconds.
- Two seconds after thread A is notified.
- Two seconds after lock B is released.

▼ Explanation:

- When `wait(2000)` is called, it causes the current thread (thread A) to wait for at most 2000 milliseconds (2 seconds) for another thread to notify it using `notify()` or `notifyAll()` on object B or until the specified time period elapses.
- If thread A is notified before the 2-second timeout expires, it will resume execution immediately after being notified.
- If thread A is not notified within the 2-second timeout, it will resume execution after the specified time period elapses, even if it wasn't notified.
- Therefore, the thread A becomes a candidate to get another turn at the CPU after it is notified or after two seconds, whichever happens first.

34. Predict the output:

```
Class{  
PSVM
```

```
String str = new String( "....");
}
Do{
str = "Hello Stop World ";
SOP(str);
}
While(str!=Strong);
{
.....
}
Answer : Hello Stop World
```

36. The code snippet below is an example of which of the following?

```
Long myLong = 211;
```

**A Auto boxing**

B Auto unboxing

C Auto casting

D Auto instancing

▼ Explanation

Autoboxing is the automatic conversion of a primitive type to its corresponding wrapper class object. In the given code snippet, the primitive value `21` is automatically boxed into a `Long` object when assigned to the variable `myLong`. This is an example of autoboxing because a primitive `long` value is being automatically converted into its corresponding wrapper class `Long`.

36. What is the result?

```
interface TestA {
    String toString();
}
public class Test {
    public static void main(String[] args) {
        System.out.println(new TestA() {
            public String toString() {
                return "test";
            }
        });
    }
}
```

```
    });  
  }  
}
```

What is the result?

1.

**test**

2. null

3. An exception is thrown at runtime. D. Compilation fails because of an error in line 1.

4. Compilation fails because of an error in line 4.

5. Compilation fails because of an error in line 5.

▼ Explanation:

This code snippet demonstrates the creation of an anonymous class that implements the `TestA` interface and overrides its `toString()` method. Here's an explanation of the code:

#### 1. Interface Declaration:

```
interface TestA {  
    String toString();  
}
```

An interface named `TestA` is declared. It contains a single method `toString()`.

#### 2. Anonymous Class Definition:

```
new TestA() {  
    public String toString() {  
        return "test";  
    }  
}
```

An anonymous class is defined that implements the `TestA` interface. It overrides the `toString()` method, providing its implementation that returns the string `"test"`.

#### 3. `main()` Method:

```
public static void main(String[] args) {  
    System.out.println(new TestA() {
```

```

        public String toString() {
            return "test";
        }
    });
}

```

The `main()` method creates an instance of the anonymous class that implements the `TestA` interface. It then calls the `toString()` method of this instance and prints the returned value using `System.out.println()`.

The output of running this code will be:

```
test
```

This is because the overridden `toString()` method in the anonymous class returns the string `"test"`, which is then printed to the console.

37. Output?

```

public class Test {
    public static void main(String[] args) {
        GenericTest<String> arr[] = new GenericTest[5]; //line 1
        arr[0] = new GenericTest("Java"); //line 2
        arr[1] = new GenericTest(1); //line 3
        arr[2] = (GenericTest<String>)new GenericTest(1); //line 4
        arr[3] = (GenericTest<String>)new GenericTest<Integer>(1); //line 5

        for(GenericTest o:arr) {
            System.out.println(o);
        }
    }
}

```

38. Output?

```

public class Test {
    public static void main(String[] args) {
        String word = "IDEAL";
        System.out.println(word.substring(0,word.length()-1)+word.charAt(word.length()-1));
    }
}

```

OUTPUT:- IDEAL

**39. How many thread one from exception main class?**

Answer : is only 1 thread(main thread)

**40. Start same thread two times**

```
t1.start()
```

```
t1.start()
```

Answer: compilation fails, compile time error. In Java, once a thread is started, it cannot be started again. If you attempt to start the same thread more than once using the

`start()` method, it will result in an `IllegalThreadStateException`.

**41. What is the output?**

```
class Output{
    public static void main(String args[]) {
        Object obj = new Object();
        System.out.print(obj.getClass());
    }
}
```

Answer:- class java.lang.Object

**42. What is the output?**

```
class A{
    int i;
    int j;

    A(){
        i = 1;
        j = 2;
    }
}

class output{
    public static void main(String args[]) {
        A obj1 = new A();
        System.out.println(obj1.toString());
    }
}
```



Answer:- String associated with obj1 and output will be A@5e265ba4

▼ Explanation:

In Java, the `toString()` method is a method defined in the `Object` class, which is the root class of all Java classes.

When you call `toString()` on an object, it returns a string representation of that object. By default, the `toString()` method in the `Object` class returns a string that consists of the class name followed by the "@" symbol and the hexadecimal representation of the object's hash code.

43. Output?

```
class output{
    public static void main(String args[]) {
        public void divide(int a, int ) {
            try{
                int c = a/b;
            }
            catch(Exception e) {
                System.out.println(Exception);
            }
            finally{
                System.out.println("finally")
            }
        }
    }
}
```

Answer:- Whether the exception occurs or not the finally is executed and finally is printed.

44. Insert the appropriate code

```
public class sprite {
    public int fubar(Foo foo){
        return foo.bar();
    }
    public void testFoo(){
        fubar(
            //insert code here
        );
    }
}
```

```
}  
}
```

Answer:

```
new Foo(){public int bar(){return 1;}}
```

55. What is the output of following code:

```
class Main {  
    String getValue(String word) {  
        System.out.println(word);  
        if (word.length() == 1) {  
            return "";  
        }  
        else {  
            return getValue( word.substring(0, word.length() -  
        }  
    }  
  
    public static void main(String[] args) {  
        Main obj = new Main();  
        System.out.println(obj.getValue("DEMOS"));  
    }  
}  
  
getValue("DEMOS")  
getValue("DEMO") + S  
getValue("DEM") + O  
getValue("DE") + M  
getValue("D") + E  
return "" + E  
return E + M  
return EM + O  
return EMO + S  
return EMOS
```

OUTPUT:- EMOS

56. What sort of compilation error you will get ? How to fix the compilation error

```
abstract class xyz {
    abstract abc (int a, int b) { }
}
```

Answer: Give return type void. and remove empty parenthesis.

57. Will this code compile?

```
interface ATOB {
    String s = "yo";
    public void meth();
}
interface BTOC { }
interface C extends ATOB, BTOC {
    public void meth();
    public void meth(int x);
}
```

Answer: Yes this code will compile without any error.

58. How Can I inherit both the interface and abstract class ?

```
interface Data { public void load(); }
abstract class Info { public abstract void load(); }
```

Answer:

```
class xyz extends Info implements interface{ void load() {/
```

59. Will this code compile?

```
class ATOB {
    static public String str = "abc";
}

class BTOC extends ATOB {
    static public String str = "bcr";
}

class Main {
```

```

    public static void main(String[] args) {
        BTOC obj = new BTOC();
        System.out.println(obj.str);
    }
}

```

Ans: Yes. will compile fine, but str is being overwritten in B class.

30. Which of these handles the exception when no catch is used?

- a. **Default handler**
- b. Finally
- c. Throw handler
- d. Java run time system

▼ Explanation:

In Java, when an exception is thrown and no `catch` block matches the exception type, the exception is propagated up the call stack until it reaches a matching `catch` block or until it reaches the top-level of the program. If the exception is not caught anywhere along the call stack, it ultimately leads to termination of the program and Java's default exception handler displays the exception details including the stack trace.

31. What is the output of the code

```

class Main {

    static boolean foo(char c) {
        System.out.print(c);
        return true;
    }

    public static void main(String[] args) {
        int i = 0;
        for(foo('A'); foo('B') && (i < 2); foo('C')) {
            i++;
            foo('D');
        }
    }
}

```

```
}  
}
```

OUTPUT: ABDCBDCB

32. What is the OUTPUT of the code

```
import java.util.*;  
  
class Main {  
  
    public static void main(String[] args) {  
        //Creating and adding elements  
        TreeSet<Integer> al=new TreeSet<Integer>();  
        al.add("7"); //line 5  
        al.add(9);  
  
        //Traversing elements  
        Iterator<Integer> itr = al.iterator();  
        while(itr.hasNext()){  
            System.out.println(itr.next());  
        }  
    }  
}
```

Ans: error: incompatible types: String cannot be converted to Integer al.add("7");  
//line 5

33. Which implementation of set would you choose if you want the iterator of set would give you object in the order it were inserted?

a.

**Linked Hash Set**

b. Tree Set

c. Hash Set

▼ Explanation:

If you want the iterator of a set to give you objects in the order they were inserted, the appropriate choice would be:

a. **LinkedHashSet**

**LinkedHashSet** maintains the insertion order of elements. When you iterate over a **LinkedHashSet**, you will get the elements in the order they were

inserted. This is because `LinkedHashSet` maintains a doubly-linked list internally to preserve the insertion order of elements.

`TreeSet` orders elements according to their natural ordering or a comparator, and `HashSet` does not guarantee any specific order during iteration.

Therefore, for your requirement, `LinkedHashSet` is the most suitable choice.

34. What is the output

```
import java.util.*;
class TestHashMaps{
    public static void main(String args[]) {
        HashMap<Integer,String> hm= new HashMap<Integer,String>();
        hm.put(100, "John");
        hm.put(101, "Paul");
        hm.put(102, "George");
        hm.put(103, "Ringo");
        for (Map.Entry m : hm.entrySet()) {
            System.out.println(m.getKey() + " " + m.getValue());
        }
    }
}
```

OUTPUT:

```
100 John
101 Paul
102 George
103 Ringo
```

35. What is the output?

```
import java.util.*;
public class SortOf {
    public static void main(String[] args) {
        ArrayList<Integer> a = new ArrayList<Integer>();
        a.add(1);
        a.add(5);
        a.add(3);
        Collections.sort(a);
        a.add(2);
        Collections.reverse(a);
        System.out.println(a);
    }
}
```

```
}  
}
```

What is the result?

1. [1, 2, 3, 5]
2. [2, 1, 3, 5]
3. **[2, 5, 3, 1]**
4. [5, 3, 2, 1]
5. [1, 3, 5, 2]
6. Compilation fails.
7. An exception is thrown at runtime.

▼ Explanation:

1. An `ArrayList` named `a` is created to store `Integer` objects.
2. Integers 1, 5, and 3 are added to the `ArrayList` in that order.
3. The `Collections.sort(a)` method is called to sort the elements of the `ArrayList` in ascending order. After this step, the elements in `a` become [1, 3, 5].
4. Integer 2 is added to the `ArrayList` after sorting.
5. The `Collections.reverse(a)` method is called to reverse the order of elements in the `ArrayList`. After this step, the elements in `a` become [5, 3, 2, 1].
6. Finally, the `System.out.println(a)` statement prints the elements of the `ArrayList`, resulting in the output [2, 5, 3, 1].

36. . Which Man class properly represents the relationship "Man has a best friend who is a Dog"?

- a. `class Man extends Dog { }`
- b. `class Man implements Dog { }`
- c. `class Man { private BestFriend dog; }`
- d. **`class Man { private Dog bestFriend; }`**
- e. `class Man { private Dog<bestFriend>; }`
- f. `class Man { private BestFriend<dog>; }`

▼ Explanation:

The correct representation of the relationship "Man has a best friend who is a Dog" would be:

1. `class Man { private Dog bestFriend; }`

In this representation:

- `Man` is a class that has a private member variable named `bestFriend`.
- The type of `bestFriend` is `Dog`, indicating that the best friend of a man is a `Dog`.

Option 3 could also potentially represent the relationship, but it's more common to use the type of the relationship (in this case, `Dog`) as the type of the member variable (`bestFriend`).

37. What is the output?

```
class Alpha {
    public void foo() {
        System.out.print("Afoo ");
    }
}

class Beta extends Alpha {
    public void foo() {
        System.out.print("Bfoo ");
    }

    public static void main(String[] args) {
        Alpha a = new Beta();
        Beta b = (Beta)a;
        a.foo();
        b.foo();
    }
}
```

- a. Afoo Afoo
- b. Afoo Bfoo
- c. Bfoo Afoo
- d. **Bfoo Bfoo**
- e. Compilation fails.



f. An exception is thrown at runtime.

38. Which code fragment, inserted at line 23, allows the code to compile?

```
import java.util.Date;
import java.text.DateFormat;
DateFormat df;
Date date = new Date();
// insert code here
String s = df.format(date);
```

1. `df = new DateFormat();`
2. `df = Date.getFormat();`
3. `df = date.getFormat();`
4. `df = DateFormat.getFormat();`
5. **`df = DateFormat.getInstance();`**

▼ Extra:

To ensure the code compiles and functions correctly, you need to initialize the `DateFormat` object `df`. Here are two possible ways to initialize `df`:

1. Using `DateFormat.getDateInstance()`:

```
df = DateFormat.getDateInstance();
```

2. Using `DateFormat.getInstance()`:

```
df = DateFormat.getInstance();
```

Either of these options will initialize the `DateFormat` object `df`, allowing you to use it to format the `Date` object `date`.

39. Output?

```
public class Base {
    public static final String F00 = "foo";
    public static void main(String[] args) {
        Base b = new Base();
        Sub s = new Sub();
        System.out.print(Base.F00);
        System.out.print(Sub.F00);
    }
}
```

```

        System.out.print(b.F00);
        System.out.print(s.F00);
        System.out.print(((Base)s).F00);
    }
}
class Sub extends Base {
    public static final String F00="bar";
}

```

- a. foofoofoofoofoo
- b. foobarfoobarbar
- c. foobarfoofoofoo
- d. **foobarfoobarfoo**
- e. barbarbarbarbar
- f. foofoofoobarbar
- g. foofoofoobarfoo

▼ Explanation:

Let's delve deeper into how explicit type casting works in this scenario.

In Java, when you perform explicit type casting, you're essentially telling the compiler to treat the reference as if it were of the specified type. However, the actual object remains the same.

In the provided code:

```

javaCopy code
System.out.print(((Base)s).F00);

```

Here, `((Base)s)` explicitly casts `s`, which is of type `Sub`, to type `Base`. After the explicit casting, the reference `((Base)s)` is treated as if it were of type `Base`, but the actual object it points to remains a `Sub` object.

However, since `F00` is a `static` field, it's accessed based on the reference type, not the object type. Therefore, despite the explicit casting, `((Base)s).F00` accesses the `F00` field from the `Base` class, not the `Sub` class. This is because static members are resolved at compile-time based on the reference type.

As a result, `((Base)s).F00` prints `"foo"`, even though `s` points to a `Sub` object.

In summary, explicit type casting in Java affects only the reference type, not the actual object, and static members are resolved based on the reference type.

70. **A company has a business application that provides its users with many different reports: receivables reports, payables reports, revenue projects, and so on. The company has just purchased some new, state-of-the-art, wireless printers, and a programmer has been assigned the task of enhancing all of the reports to use not only the company's old printers, but the new wireless printers as well. When the programmer starts looking into the application, the programmer discovers that because of the design of the application, it is necessary to make changes to each report to support the new printers. Which two design concepts most likely explain this situation?**

(Choose two.)

- a. Inheritance
- b. **Low cohesion**
- c. **Tight coupling**
- d. High cohesion
- e. Loose coupling
- f. Object immutability

▼ Explanation:

Thank you for the clarification. Let's reassess the options:

1. Low cohesion: Cohesion refers to the degree to which elements within a module or component belong together. Low cohesion means that the elements within a module are not closely related to each other. In this scenario, low cohesion might indicate that the printing functionality is scattered across multiple parts of the application, making it difficult to change the printing behavior without impacting various reports.
2. Tight coupling: Tight coupling means that modules or components are highly dependent on each other. In this scenario, if the reports are tightly coupled with the printer implementation, any changes to the printer require corresponding changes to all reports. This implies that the reports are tightly coupled with the printer implementation, leading to the necessity of making changes to each report to support the new printers.

Given this clarification, the two design concepts that most likely explain the situation are indeed low cohesion (option 2) and tight coupling (option 3).

These concepts suggest that the application lacks a cohesive design where

printing functionality is centralized and independent of the reports, and it is tightly coupled, requiring changes to each report when the printer implementation changes. Thank you for pointing out the correct options.

**71. A team of programmers is reviewing a proposed API for a new utility class. After some discussion, they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?**

- a. **Looser coupling**
- b. Tighter coupling
- c. Lower cohesion
- d. Higher cohesion
- e. Weaker encapsulation
- f. Stronger encapsulation

▼ Explanation

Apologies for the oversight. Let's reconsider the options with a focus on loose coupling:

1. **Looser coupling:** Loose coupling refers to reducing the dependencies between components or modules in a system. When components are loosely coupled, changes to one component are less likely to require changes to other components.

Reducing the number of methods in the API without losing functionality can indeed promote looser coupling because it reduces the surface area of interaction between different parts of the system. With fewer methods, there are fewer opportunities for dependencies between components to form, leading to looser coupling.

So, the correct answer is indeed:

1. Looser coupling

**72. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself. What design issue has the team discovered?**

- a. Tight coupling
- b. Low cohesion

- c. High cohesion
- d. Loose coupling
- e. **Weak encapsulation**
- f. Strong encapsulation

▼ Explanation:

The design issue discovered by the team is:

1. Weak encapsulation

Weak encapsulation occurs when internal details of a class are not properly encapsulated, allowing other classes to access methods or fields that should be kept private or protected. In this scenario, the team realizes that the current design of the utility class allows other classes to access methods that should only be accessible within the utility class itself, indicating weak encapsulation.

Strong encapsulation, on the other hand, refers to the practice of properly encapsulating the internal details of a class, hiding them from external access and ensuring that they are only accessible through well-defined interfaces. In this scenario, the team has identified a lack of strong encapsulation, leading to the design issue.

73. **Output?**

```
public class TestString1 {  
    public static void main(String[] args) {  
        String str = "420";  
        str += 42;  
        System.out.print(str);  
    }  
}
```

- a. 42
- b. 420
- c. 462
- d. **42042**
- e. Compilation fails.
- f. An exception is thrown at runtime.

74. Which three are valid on line ?

```
public interface Status {  
    /* insert code here */ int MY_VALUE = 10;  
}
```

(Choose three.)

- a. **final**
- b. **static**
- c. native
- d. **public**
- e. private
- f. abstract
- g. protected

▼ Explanation:

Among the options provided, the following three are valid modifiers for the line:

1. **static** : Declaring a variable as **static** means it belongs to the class rather than instances of the class. It can be accessed using the class name directly.
2. **final** : Declaring a variable as **final** means its value cannot be changed after initialization. It behaves as a constant.
3. **public** : Declaring a variable as **public** allows it to be accessed from any other class. It's part of the class's public interface.

So, the valid options are:

1. static
2. final
3. public

75. A programmer has an algorithm that requires a `java.util.List` that provides an efficient implementation of `add(0, object)`, but does NOT need to support quick random access. What supports these requirements?
- a. `java.util.Queue`
  - b. `java.util.ArrayList`
  - c. `java.util.LinearList`

#### d. `java.util.LinkedList`

##### ▼ Explanation:

The requirement for efficient implementation of `add(0, object)` suggests the need for a data structure where adding elements at the beginning is efficient. However, the requirement does not mandate quick random access, implying that iterating through the elements sequentially is sufficient.

Among the options provided:

1. `java.util.Queue` : Queues typically prioritize adding elements at the end rather than at the beginning. Although some implementations may support adding elements at the beginning, it's not a primary characteristic of queues.
2. `java.util.ArrayList` : While `ArrayList` supports adding elements at any position, inserting an element at the beginning is not efficient because it requires shifting all existing elements to the right.
3. `java.util.LinearList` : There is no such class in the Java standard library named `LinearList`.
4. `java.util.LinkedList` : `LinkedList` is a doubly linked list implementation in Java. It provides efficient insertion at the beginning ( `addFirst()` method) and does not require shifting elements. However, it does not support quick random access like an `ArrayList` does.

Therefore, the correct choice is:

1. `java.util.LinkedList`

76. Which three are true?(Choose Three) A. Cat is-a Animal

```
interface Jumper {
    public void jump();
}
class Animal {}
class Dog extends Animal {
    Tail tail;
}
class Beagle extends Dog implements Jumper{
    public void jump() {}
}
class Cat implements Jumper{
```

```
public void jump() {}  
}
```

- a. Cat is-a Animal
- b. **Cat is-a Jumper**
- c. **Dog is-a Animal**
- d. Dog is-a Jumper
- e. Cat has-a Animal
- f. **Beagle has-a Tail**
- g. Beagle has-a Jumper

▼ Explanation

Let's evaluate each statement:

1. **Cat is-a Jumper:** Correct. Both `Cat` and `Beagle` classes implement the `Jumper` interface, so `Cat` is-a `Jumper`.
2. **Dog is-a Animal:** Correct. The `Dog` class extends the `Animal` class, so `Dog` is-a `Animal`.
3. **Dog is-a Jumper:** Incorrect. While `Beagle` (a subclass of `Dog`) implements `Jumper`, `Dog` itself does not implement the `Jumper` interface.
4. **Cat has-a Animal:** Incorrect. There's no direct relationship between `Cat` and `Animal` in terms of composition. `Cat` is not defined to contain or have an `Animal`.
5. **Beagle has-a Tail:** Correct. The `Dog` class contains a `Tail` instance variable, so `Beagle` (a subclass of `Dog`) inherits this relationship and has-a `Tail`.
6. **Beagle has-a Jumper:** Incorrect. `Beagle` implements the `Jumper` interface, but it's not composed of or contains any `Jumper` instance.

So, the correct options are:

1. Cat is-a Jumper
2. Dog is-a Animal
3. Beagle has-a Tail

**77. Which class correctly uses the Data interface and Info class?**



```
interface Data { public void load(); }  
abstract class Info { public abstract void load(); }
```

a. **Correct option**

```
public class Employee extends Info implements Data {  
    public void load() { /*do something*/ }  
}
```

b.

```
public class Employee implements Info extends Data {  
    public void load() { /*do something*/ }  
}
```

c.

```
public class Employee extends Info implements Data {  
    public void load(){ /*do something*/}  
    public void Info.load(){ /*do something*/}  
}
```

d.

```
public class Employee implements Info extends Data {  
    public void Data.load(){ /*do something*/}  
    public void load(){ /*do something*/ }  
}
```

e.

```
public class Employee implements Info extends Data {  
    public void load(){ /*do something*/}  
    public void Info.load(){ /*do something*/}  
}
```

f.

```
public class Employee extends Info implements Data{  
    public void Data.load() { /*do something*/}
```

```
public void Info.load() { /*do something*/ }  
}
```

▼ Explanation

The correct class that uses the `Data` interface and `Info` class is:

```
public class Employee extends Info implements Data {  
    public void load() { /* do something */ }  
}
```

Explanation:

In this class:

- `Employee` extends `Info`, which is an abstract class.
- `Employee` implements `Data`, which is an interface.
- The `load()` method in `Employee` satisfies both the `Info` class's abstract method requirement and the `Data` interface's method requirement. This means that the `load()` method defined in `Employee` is implementing both the abstract method from `Info` and the method from the `Data` interface.

78. Output?

```
class TestA {  
    public void start() {  
        System.out.println("TestA");  
    }  
}  
  
class TestB extends TestA {  
    public void start() {  
        System.out.println("TestB");  
    }  
    public static void main(String[] args) {  
        ((TestA)new TestB()).start();  
    }  
}
```

- TestA
- TestB**
- Compilation fails.

d. An exception is thrown at runtime.

▼ Explanation:

Sure, let's break down the code and focus on the line `((TestA)new TestB()).start();`.

**1. Class Definitions:**

- `TestA`: This class defines a method `start()` that prints `"TestA"`.
- `TestB`: This class extends `TestA` and overrides the `start()` method to print `"TestB"`.

**2. Main Method:**

- In the `main()` method of the `TestB` class, an instance of `TestB` is created and then casted to `TestA`: `((TestA)new TestB())`.
- This cast is explicitly converting the object of type `TestB` to type `TestA`. Since `TestB` is a subclass of `TestA`, this casting is allowed.

**3. Method Invocation:**

- After casting, the `start()` method of the resulting object is invoked: `((TestA)new TestB()).start();`.
- Despite the casting, the `start()` method that gets invoked is determined at runtime based on the actual type of the object. Since the actual object is of type `TestB`, the overridden `start()` method from `TestB` is called.

**4. Output:**

- When the `start()` method is called, it prints `"TestB"`, as the object is of type `TestB`.

In summary, the line `((TestA)new TestB()).start();` is casting an object of type `TestB` to `TestA` and then invoking the `start()` method. However, because the object is still of type `TestB`, the overridden `start()` method from `TestB` is called, resulting in the output `"TestB"`.

79. Output?

```
import java.io.*;
class files {
    public static void main(String args[]) {
        File obj = new File("/java/system");
        System.out.print(obj.getName());
    }
}
```

```
}  
}
```

- a. java
- b. **system**
- c. java/system
- d. /java/system

▼ Explanation

In the provided code, the `files` class has a `main` method that attempts to create a `File` object named `obj` using the path `"/java/system"`. Then, it prints the name of the file represented by this `File` object.

However, there's a misunderstanding in the code. The `File` constructor doesn't represent a file but rather a directory path or a file path. If the path does not exist or is not a directory or file, the `getName()` method will return an empty string.

In this case, `"/java/system"` is provided as a directory path. If this directory exists in the file system, the code will print the name of the directory represented by the `File` object `obj`. Otherwise, it will print an empty string if the directory doesn't exist or if there's any other issue accessing it.

If you want to represent a file, you should provide the complete path to a file, including the filename and its extension. For example, `"/java/system/file.txt"`. Then, the `getName()` method will return `"file.txt"`.

Keep in mind that the behavior of this code depends on the existence of the specified directory or file in the file system. If the directory doesn't exist, it will print an empty string.

### 30. Output?

```
import java.io.*;  
class files {  
    public static void main(String args[]) {  
        File obj = new File("/java/system");  
        System.out.print(obj.getAbsolutePath());  
    }  
}
```

Note: file is made in c drive.

- a. java

- b. system
- c. java/system
- d. **/java/system**

▼ Explanation:

In the provided code, the `files` class has a `main` method that attempts to create a `File` object named `obj` using the path `"/java/system"`. Then, it prints the absolute path of the `File` object.

The `getAbsolutePath()` method of the `File` class returns the absolute path of the file or directory represented by the `File` object. The absolute path is the complete path from the root directory to the specified file or directory.

In this case, `"/java/system"` is provided as a directory path. If this directory exists in the file system, the code will print its absolute path. If the directory doesn't exist or if there's any other issue accessing it, the code will print an empty string or throw an exception.

For example, if `"/java/system"` exists and is located in the root directory `"/"`, the output will be something like `"/java/system"`. However, if the directory doesn't exist, the output will be an empty string or an exception will be thrown.

### 31. Output?

```
import java.lang.*;

class exception_handling {
    public static void main(String args[]) {
        try {
            int a = args.length;
            int b = 10 / a;
            System.out.print(a);
            try {
                if (a == 1)
                    a = a / a - a;
                if (a == 2) {
                    int[] c = {1}; // [] were not there in the
                    c[8] = 9;
                }
            }
        }
        catch(ArrayIndexOutOfBoundsException e) {
            System.out.println("TypeA");
        }
    }
}
```

```

        }
    }
    catch (ArithmeticException e) {
        System.out.println("TypeB");
    }
}
}

```

- a. TypeA
- b. **TypeB**
- c. 0TypeA

▼ Explanation:

In the provided code, the `exception_handling` class contains a `main` method that demonstrates nested try-catch blocks to handle exceptions.

Here's a breakdown of the code:

1. The outer try block:

- It attempts to perform division by zero (`10 / a`) where `a` is the length of the command-line arguments (`args` array).
- If the length of `args` is 0, it will throw an `ArithmeticException`.
- If the length of `args` is non-zero, it will execute the inner try block.

2. The inner try block:

- It contains conditional statements based on the value of `a` (the length of `args`).
- If `a` is 1, it attempts to perform a division operation that will cause an `ArithmeticException`.
- If `a` is 2, it attempts to access an element of an array `c` at index 8, which is out of bounds and will throw an `ArrayIndexOutOfBoundsException`.

3. Catch blocks:

- If an `ArrayIndexOutOfBoundsException` occurs within the inner try block, it will be caught and "TypeA" will be printed.
- If an `ArithmeticException` occurs within the outer try block, it will be caught and "TypeB" will be printed.

Let's discuss the potential outcomes:

- If no command-line arguments are provided ( `args.length == 0` ), an `ArithmeticException` will be thrown in the outer try block and "TypeB" will be printed.
- If one command-line argument is provided ( `args.length == 1` ), an `ArithmeticException` will be thrown in the inner try block and "TypeB" will be printed.
- If two or more command-line arguments are provided ( `args.length >= 2` ), an `ArrayIndexOutOfBoundsException` will be thrown in the inner try block and "TypeA" will be printed.

Overall, the output will vary based on the number of command-line arguments provided and the exceptions thrown.

## Important link from the pdf:

<http://javabysri.blogspot.in/2012/06/jdbc-objective-questions.html>

# 3. Spring

## First

### Spring AOP

**1. What is advice?**

- a. A concern that cuts across multiple classes
- b. The process of linking aspects with other application types
- c. Action taken by an aspect at a particular join point
- d. A method to implement transaction management

▼ Answer

- c. Action taken by an aspect at a particular join point

▼ Extra

AOP:- Aspect Oriented Programming

**2. Which annotation represents an advice to be executed after the method execution only if method exits by throwing an exception?**

- a. @Before
- b. @After
- c. @Around
- d. @AfterThrowing

▼ Answer

- d. @AfterThowring

**3. Which of the following terminologies are typically used with regards to AOP?**

- a. **Concerns**
- b. **Advice**
- c. **Float**



d. **Pointcut**

- i. A, B, and D
- ii. A, C, and D
- iii. A, B, and C
- iv. B, C, and D

▼ Answer

- i. A, B and D

## Spring Auto-Wiring

1. **What attribute is added in the bean.xml to reduce or eliminate the need of <property> and <constructor-arg>?**

- a. Autodiscovery
- b. Autowire
- c. Scope
- d. Auto-detect

▼ Answer

- b. Autowrie

## Spring Core Concepts and Life Cycle

1. **Which of the following examples summarizes Program to an interface rather than an implementation?**

- a. Map m = new HashMap();
- b. ArrayList list = new ArrayList();
- c. ArrayList<Integer> = new ArrayList<Integer>();
- d. Strings = "HelloWorld";

▼ Answer

- c. ArrayList<Integer> = new ArrayList();

## Spring Dependency Check, Inner beans and Scopes

**1. If any of the beans implements the BeanPostProcessor interface, Spring will call which of these methods?**

- a. processBeforeInitialization()
- b. BeanPostProcessor()
- c. postProcessInterface()
- d. postProcessBeforeInitialization()

▼ Answer

- d. postProcessBeforeInitializatoin()

**2. Which Collections elements does Spring support?**

- a. List, hashmap, arraylist, andset
- b. List, props, hashmaps, and arraylist
- c. List, set, map, and props
- d. Hashmaps, arraylist, list, and treemap

▼ Answer

- c. List, set, map, and props

**3. What will be the result if you use Web-aware Spring Application Context scopes in a regular Spring IoC container?**

- a. It will return nothing.
- b. It will return a NullPointerException
- c. It will return a IllegalStateException
- d. It will work correctly.

▼ Answer

- a. It will return nothing.

**4. Which Bean scopes are valid only in the context of a web-aware Spring Application Context?**

- a. Singleton and Prototype
- b. Session, Global Session, and Request
- c. Singleton, Prototype, Request, and Session
- d. Session, Singleton, and Global Session

▼ Answer

b. Session, Global Session, and Request

5. **To use `java.util.List`, what is the Spring Collections element that you need?**

- a. `<list>`
- b. `<List>`
- c. `<lists>`
- d. `util:list`

▼ Answer

a. `<list>`

6. **Which of the below are benefits of using DI?**

- a. Tight coupling
- b. Loose coupling
- c. Easy testing
- d. Removing run time error

▼ Answer

b. Loose coupling and c. Easy testing

7. **What is the difference between `ClassPathXMLApplicationContext` and `FilePathXMLApplicationContext`?**

- a. First one reads from application class path and second one reads from absolute file path.
- b. First one reads from application class path and second one reads from relative file path.
- c. First one reads from relative application class path and second one reads from absolute file path.
- d. None of the above.

▼ Answer

c. First one reads from relative application class path and second one reads from absolute file path.

8. The Dependency Look up is a methodology where we get the resource after demand. There can be various ways to get the resource. Types of Dependency Look up include:

- a. Factory Method
- b. JNDI Method
- c. Key Word Method
- d. All of the above

▼ Answer

- d. All of the above

9. Which of the following declarations of inner Beans are correct

```
A.<beanid="outerBean"class="...">
    <propertyname="target">
        <beanid="innerBean"class="..."/>
    </property>
</bean>
```

```
B.<beanid="outerBean"class="...">
    <beanid="innerBean"class="...">
        <propertyname="target">
            </property>
        </bean>
    </bean>
```

```
C.<beanid="outerBean"class="...">
    <beanid="innerBean"class="...">
        <propertyname="target">
            </property>
        </bean>
    </bean>
```

```
D.<beanid="outerBean"class="...">
    <propertyname="target">
        <beanclass="...">
            <propertyname="name"value="HelloWorld"/>
        </bean>
```

```
        </property>
    </bean>
```

- a. A and B
- b. B and C
- c. B and D
- d. A and D

10. **When do you use Dependency Injection?**

- A. DI is very useful for decoupling your system.**
- B. DI is most useful when you are aiming for code reuse.**
- C. DI helps in designing tightly coupled code.**
- D. DI does NOT help in unit testing the code.**

- a. A and C
- b. B and C
- c. A and B
- d. A and D

▼ Answer

- c. A and B

## Spring Java Configuration

1. **For Java-based configuration, which is the class that sets the Spring Context?**

- a. Annotationconfig
- b. ComponentScan
- c. AnnotationConfigApplicationContext
- d. AnnotationConfigApplication

▼ Answer

- c. AnnotationConfigApplicationContext

2. **Which of the following is an example of java-based configuration?**

- a. context:annotation-config/

- b. @Configuration public class HelloWorldConfig{..}
- c. @Autowired
- d. ApplicationContext context=new  
    ClassPathXmlApplicationContext("Beans.xml");

▼ Answer

- b. @Configuration public class HelloWorldConfig{..}

## Spring ORM and MVC

### 1. Which is the main component in Spring Web MVC?

- a. Faces Servlet
- b. Action Servlet
- c. View Renderer Servlet
- d. Dispatcher Servlet

▼ Answer

- d. Dispatcher Servlet

### 2. Which element has to be used in order to configure Spring MVC?

- a. <mvc:annotation-driven/>
- b. <aop:aspect/>
- c. <tx:advice/>
- d. <oxm:jaxb2-marshaller/>

▼ Answer

- a. <mvc:annotation-driven/>

## Spring Stereotype Annotations

### 1. What annotation wiring element is added to scan packages to find and register beans with in the Application Context?

- a. <context:annotation-config/>
- b. <context:annotation-scan/>
- c. <context:component-config/>

d. <context:component-scan>

▼ Answer

d. <context:component-scan>

**2. In the following example, what type of autowire will be used?**

```
@Autowired(required=false)
public Employee AutowiredByConstructorService(@Qualifier("employee")
    this.employee=emp;
}
```

- a. ByName
- b. ById
- c. @Autowired
- d. Constructor

▼ Answer

d. Constructor

**3. What are different valid types of Spring bean autowiring?**

- a. byName, byId, and constructor
- b. byName, byType, constructor, @Autowired, and @Qualifier annotations
- c. by Scope, by Constructor, and @Autowired
- d. ById, byNameScope, constructor, and @Autowire

▼ Answer

b. byName, byType, constructor, @Autowire, @Qualifier annotations

**4. Spring handles object creation in order to be non-invasive. Can you identify which definition creates a spring-managed object?**

- a. String myObj;
- b. @Autowired private String myObj;
- c. private String myObj = new String("spring");
- d. @Spring String myObj=null;

▼ Answer

c. `private String myObj = new String("spring");`

**5. How many types of autowiring modes are supported in Spring?**

- a. Three
- b. Five
- c. Six
- d. Four

▼ Answer

b. Five

## Other

**1. The use of Autodiscovery and Autowire will reduce the amount of \_\_\_\_\_.**

- a. Java code generated
- b. Methods generated
- c. XML Spring configurations
- d. <bean> elements

▼ Answer

d. <bean> elements

**2. Dependency Injection (DI) or Inversion of Control (IOC) is based on which Hollywood principle ?**

- a. Do not call us. We will call you.
- b. Understand the problem first.
- c. The customer is always right.
- d. What you see is what you get.

▼ Answer

a. Do not call us. We will call you.

**3. From where does the Spring container gets its instructions?**

- a. Properties File



- b. Spring JARS
- c. XML File
- d. Annotations

▼ Answer

c and d

**4. The use of the prototype scope is recommended for \_\_\_\_\_ type of beans.**

- a. Stateless
- b. Stateful
- c. Restless
- d. Restful

▼ Answer

b. Stateful

**5. Which is the alternative to create, load, and manage beans in a Spring Application?**

- a. Application Factory
- b. Context Factory
- c. Bean Factory
- d. Spring Bean Factory

▼ Answer

c. Bean Factory

**6. Java- based configuration allows bean definition and Spring configuration to move out of \_\_\_\_\_.**

- a. XML files
- b. Java code
- c. Property files
- d. All of the above

▼ Answer

a. XML files

**7. Which one of these statements about XML declaration of transaction manager bean is NOT true?**

- a. The tx namespace provides JTA transaction manager declaration shortcut syntax.
- b. The Id of the bean has to be transactionManager.
- c. Depending on the application persistence technology, the HibernateTransactionManager or the DataSourceTransactionManager could be used as bean class.
- d. A default transaction timeout could be given.

▼ Answer

- b. The Id of the bean has to be transactionManager.

**8. Which one of the following is one of the challenge of Traditional Transaction Management?**

- a. Transaction Management at Database Connection Level
- b. Programmatic Management of Transactions
- c. Error-Prone Connection Management
- d. All of the above

▼ Answer

- d. All of the above

**9. What does the JDBC Template users to connect to the database?**

- a. A simple java bean with connection properties
- b. A data source bean configuration
- c. An XML with DB connection parameters
- d. A JDBC connection object

▼ Answer

- b. A data source bean configuration

**10. What is the use of Autodiscovery?**

- a. Eliminate the use of the <Property> element
- b. Reduce the use of the <Constructor-arg> element

- c. Eliminate the use of the <Props> element
- d. Reduce the use of the <bean> element

▼ Answer

- d. Reduce the use of the <bean> element

**11. The ----- packages are the basis of spring Framework's IOC container**

- a. org.springframework.beans and org.springframework.Application
- b. org.springframework.properties and org.springframework.context
- c. org.springframework.beans and org.springframework.properties
- d. org.springframework.beans and org.springframework.context

▼ Answer

- d. org.springframework.beans and org.springframework.context

**12. Which of the following attributes are valid in a Spring bean definition?**

- A. ID**
- B. Map**
- C. Set**
- D. Class**

▼ Answer

- A.ID and D.Class

**13. What Spring Collection can be used to inject a collection of name-value pairs, where name and value can be of any type?**

- a. <set>
- b. <List>
- c. <map>
- d. <props>

▼ Answer

- c. <map>

**14. Which attributes are applicable for an inner bean?**

- a. Context, Parent

- b. Id, Class
- c. Class, Inner
- d. Interface, Id

▼ Answer

c. Class, Inner

**15. To use bean definition template, you should add what attribute in the bean?**

- a. abstract="true"
- b. interface="true"
- c. class ="com.example.helloworld"
- d. public ="true"

▼ Answer

a. abstract= "true"

**16. A collection value of the type java.util.Properties is defined by what element?**

- a. <props>
- b. <prop>
- c. <properties>
- d. <property>

▼ Answer

a. <props>

**17. Which of the following is an annotation for Spring Java-based configuration?**

- a. @Component
- b. @SpringConfiguration
- c. @SpringXml
- d. @Configuration

▼ Answer

@Configuration

**18. What does the JDBC Template uses to connect to the database?**

- a. A simple java bean with connection properties
- b. A data source bean configuration
- c. An XML with DB connection parameters
- d. A JDBC connection object

▼ Answer

- b. A data source bean configuration

**19. Which of the following tasks are performed by a JDBCTemplate?**

- a. **Execute SQL query.**
- b. **Execute Update statements.**
- c. **Execute a Java program.**
- d. **Execute stored procedure calls.**

▼ Answer

- a, b and d

**20. Does Spring provide programmatic transaction management?**

- a. Yes, with the TransactionTemplate class
- b. Yes, with the TransactionService class
- c. Yes, using the @Transactional bean post processor
- d. No, Spring does not provide transactional management

▼ Answer

- a. Yes, with the Transaction Template class

**21. Which of the following are MVC – Specific annotations:**

- a. **@Controller**
- b. **@RequestMapping**
- c. **@PathVariable**
- d. **@Autowired**

▼ Answer

a, b and c only

22. To annotate all your Data Access Object (DAO) classes, use the -----

- a. @Service
- b. @Classes
- c. @Repository
- d. @Component

▼ Answer

c. @Repository

## Second

### Imp4

1. Which of the following statement is false ?

- a. The Corepackage is the most fundamental part of the framework and provides the loc an dDependency Injection feature.
- b. TheDAO package provides a JDBC-abstraction layer that removes the need to do tedious JDBC coding and parsing of database-vendor specific error code
- c. The ORM package provides integration layers for popular object-relational mapping APIs, including JPA,JDO, Hibernate, iBatis
- d. AOP takes out the direct dependency of crosscutting tasks from classes which, we can achieve through normal object oriented programming model. or example, method-interceptors and pointcuts to cleanly decouple code implementing functionality that
- e. Spring's Web package provides basic web-oriented integration features, such as multipart file-upload functionality
- f. Spring's MVC package provides a Model-View-Controller implementation for web-application MVC package provides a Model-

View-Controller implementation for web-application

g. None of these

▼ Answer

g. None of these

**2. Which of the following is false?**

- a. A BeanFactory pretty much just instantiates and configures beans.
- b. An ApplicationContext also instantiates bean, and it provides the supporting infrastructure to enable lots of enterprise- specific features such as transactions and AOP, message resource handling (for use in internationalization), event propagation.
- c. Application-layer specific contexts such as the WebApplicationContext use in web application
- d. none of the above

▼ Answer

d. none of the above

**3. Which of the following is actual representation of the Spring loc container?**

- a. Application Context
- b. Xml Bean Factory
- c. Bean Factory
- d. Web Application Context

▼ Answer

c. Bean Factory

**4. A bean is simply an object that is instantiated, assembled and otherwise managed by a Spring loc container.**

- a. true
- b. false

▼ Answer

a. true

**5. How container load configuration metadata from a variety of external resources such as the local file system from the Java CLASSPATH?**

- a. Application Context constructor
- b. Init()
- c. Context Contructor
- d. XmlClassPath Contructor

▼ Answer

Application Context constructor

**6. Which of following is correct way to load bean definitions from another file?**

- a. <bean id="bean1" class="..." import="something.xml"/>
- b. <resources name="something.xml" import="true"/>
- c. <import resource="something.xml"/>
- d. all the above

▼ Answer

- a. <import resource="something.xml"/>

**7. What is the correct way to write constructor injection?**

```
package com.bullraider;

public class TestBean {

    private int year;

    private String happy;

    public TestBean(String happy,int year) {

        this.year=year;

        this.happy = happy;
    }
}
```



```
}  
}
```

a.

```
<bean name="testClass" class="com.bullraider. TestBean">  
  
<constructor-arg type="java.lang.String" value="Happy new  
  
<constructor-arg type="int" value="2015"/>  
  
</bean>
```

b.

```
<bean name="testClass" class="com.bullraider.TestBean">  
  
<constructor-arg index="0" value="Happy new year"/>  
  
<constructor-arg index="1" value="2015"/>
```

c.

```
<bean name="testClass" class="com.bullraider. TestBean">  
  
<constructor-arg index=" 1 " value="Happy new year"/>  
  
<constructor-arg index=" 0 " value =^ prime prime 2015 ^ p  
  
</bean>
```

d.

```
<bean name="testClass" class="com.bullraider TestBean">  
  
<constructor-arg type="int" value="2015"/>  
  
<constructor-arg type="java.lang.String" value="Happy new  
  
</bean>
```

e. All of the above

▼ Answer

e. All of the above

**8. Which of the following are correct statements?**

- a. Form Controllers handle form interaction
- b. Multipart resolver to handle file uploads
- c. Dynamic binding of data to domain model
- d. Interaction with velocity.jsp, thymeleaf, pdf, Excel
- e. Sending email remote accessing via Soap, Burlap

▼ Answer

All of the above

**9. What do you mean by 'bean'?**

- a. component
- b. object
- c. class
- d. container
- e. xml file

▼ Answer

a. component and b. object

**10. What are responsibilities included for bean factory do ?**

- a. Instantiating Instantiating or sourcing application objects
- b. configuring such objects
- c. Assembling the dependencies between these objects
- d. Manage lifecycle of spring

▼ Answer

a, b and c

**11. What are the responsibilities of XmlBeanFactory included ?**

- a. This implementation allows you to express the objects that compose your application
- b. Resolve inter-dependencies between such objects, in terms of XML
- c. The XmlBeanFactory takes this XML configuration metadata and uses it to create a fully configured system or application
- d. None of this above

▼ Answer

a, b, c(if it is multi option or else a)

12. Which of the following are the correct way to nstantiating a Spring loc container?

```
ApplicationContext context = new ClassPathXmlApplicationCo
```

```
XmlBeanFactory beanFactory=new XmlBeanFactory(new ClassPat
```

```
BeanFactory fac=new ClassPathXmlApplicationContext("beans-
```

```
Resource res=new Classpathresource("beans-definitions.xml"
```

▼ Answer

all of the above

13. **What are the correct ways to instantiate bean?**

- a. Using default (empty) constructor
- b. Instantiation using a static factory method
- c. Instantiation using an instance factor method
- d. Using the container
- e. All these above are true

▼ Answer

e. All these above are true

14. **What is Bean Factory?**

- a. A BeanFactory is essentially maintains a registry of different beans and their dependencies.
- b. A BeanFactory enables you to read bean definitions and access them using the bean factory.
- c. Using just the BeanFactory we can create and read in some bean definitions in the XML format as following
  - i. Resource definition1 = new FileSystemResource("beans-def.xml");
  - ii. BeanFactory factory = new XmlBeanFactory(definition1)
- d. We can get ApplicationContext and WebApplicationContext object using BeanFactory

▼ Answer

a, b and c

#### 15. Which statements are true about DI(Dependency Injection)?

**DI says your components and services should tightly coupled in code.**

- a. DI says that no need create your objects but describe how they should be created.
- b. We should not directly connect your components and services together in code but describe which services are needed by which components in a configuration file
- c. Inversion of control relies on dependency injection
- d. We should not directly connect your components and services together in code but describe which services are needed by Which components in a configuration file

▼ Answer

a, c, d

#### 16. Which statements are true about IOC(Inversion of Control) ?

- a. Decoupling of the execution of a certain task from implementation.
- b. Each module can focus on what it is designed for.
- c. Replacing modules should not affect to other modules.
- d. Modules should observe about what other systems do

e. All the above

▼ Answer

a, b, c

**17. What do you mean by inner bean?**

- a. A <bean/> element inside the <bean/> elements is used to define a so-called Inner bean.
- b. A <bean/> element inside the <property/> elements is used to define a so-called inner bean.
- c. A <bean/> element inside the <constructor-arg/> elements is used to define a so-called inner bean.
- d. A <bean/> element inside the property or <constructor-arg/> elements is used to define a so-called inner bean.

▼ Answer

d. A <bean/> element inside the property or <constructor-arg/> elements is used to define a so-called inner bean.

**18. Which of the following are basic techniques to implement inversion of control?**

- a. using a factory pattern
- b. using a service locator pattern
- c. a constructor injection
- d. a setter injection
- e. an interface injection
- f. all the above

▼ Answer

f. all of the above

**19. Which of the following statements are false?**

**Constructor-based DI is effected by invoking Setter Injection each representing a dependency**

**Constructor-based DI is accomplished when the container invokes a class constructor with a number of arguments**

**@Autowire annotation annotation on constructor for constructor based autowiring.**

- a. Constructor-based DI is effected by invoking a constructor with a number of arguments, each representing a dependency
- b. We can aslo use @Autowire annotation on constructor for constructor based autowiring.
- c. Constructor-based DI is accomplished when the container invokes a class constructor with a number of arguments
- d. Constructor-based DI is effected by invoking Setter Injection each representing a dependency

▼ Answer

- d. Constructor-based DI is effected by invoking Setter Injection each representing a dependency

**20. What are the correct ways to define constructor injection?**

- a.

```
<bean d ="salute" class=^ prime prime com . com.bullraider
    <constructor-arg>
        <value>Salute</value>
    </constructor-arg>
</bean>
```
- b.

```
<bean id="salute" class="com.bullraider.app.beans.Salutati
    <constructor-arg le ="Salute">
    </constructor-arg>
</bean>
```
- c.

```
<bean id="salute" class="com.bullraider.app.beans.Salutati
    <constructor-arg value="Slute"/>
</bean>
```
- d. All of the above

▼ Answer

d. all of the above

## SS

1. **Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.**

a. true

b. false

▼ Answer

a. true

2. **Which of the among is not a special bean in WebApplicationContext?**

a. Locale resolver

b. Theme resolver

c. multipart file

d. resolver

e. Handler exception resolver

f. Handler mappings

g. None of the above

▼ Answer

d. resolver

3. **How the Spring DispatcherServlet supports for returning the last-modification-date, as specified by the Servlet API?**

a. By implementing the LastModified interface

b. By implementing the LastSessionModified interface

c. By implementing the LastDateModified interface

d. By implementing the LastModifiedDate interface

▼ Answer

4. **WebApplicationContext has more scopes like request, session, globalSession to the standard Application context (which only provides Singleton and Prototype scopes)**

a. true

b. false

▼ Answer

a. true

5. **All Spring's various controller inherit from AbstractController?**

a. true

b. false

▼ Answer

a. true

6. **You can write your own controller by using**

**handleRequestInternal(HttpServletRequest, HttpServletResponse) method from abstract controller and a ModelAndView object.**

a. true

b. false

▼ Answer

a. true

7. **Which of the following controller inspects the URL and retrieves the filename of the file request and uses that as a viewname ? For example, the filename of <http://www.springframework.org/welcome.html> request is welcome.**

a. UrlFilenameViewController

b. MultiActionMultiActionController

c. CommandController

d. AbstractController

▼ Answer

a. UrlFilenameViewController



**8. Which statements are true about DispatcherServlet?**

- a. DispatcherServlet is an expression of the "Front Controller" design pattern.
- b. The DispatcherServlet is the actual Servlet.
- c. DispatcherServlet is declared in the web.xml of your web application.
- d. The Spring DispatcherServlet has a couple of special beans it uses in order to be able to process requests and render the appropriate views. These beans can be configured in WebApplicationContext
- e. All of the above

▼ Answer

- e. All of the above

**9. Which statements are true about WebApplicationContext?**

- a. The WebApplicationContext is an extension of the plain ApplicationContext
- b. The WebApplicationContext is bound in the ServletContext, and by using static methods on the RequestContextUtils class.
- c. WebApplicationContext differs from a normal ApplicationContext in that it is capable of resolving themes
- d. We can get WebApplicationContext reference by using ApplicationContext.
- e. All the above

▼ Answer

- a, c, d (doubt)

**10. Which of the following controller combines multiple request-handling methods into one controller?**

- a. UrlFilenameViewController
- b. MultiActionController
- c. CommandController
- d. MultiRequestController

▼ Answer

11. **'https://localhost:8080/testing.view' will result in the method `testing(HttpServletRequest, HttpServletResponse)` being invoked.**

Which is the correct view resolver for this?

- a. ParameterMethodName Resolver
- b. Properties MethodName Resolver
- c. InternalViewResolver
- d. InternalPathMethodName Resolve

▼ Answer

- d. InternalPathMethodName Resolve

12. **'/index/welcome.html=dolt' and a request to /index/welcome.html comes in, the `dolt(HttpServletRequest, HttpServletResponse)` method will be invoked.**

**Which is the correct view resolver for this?**

- a. InternalPathMethodName Resolver
- b. ParameterMethodName Resolver
- c. Properties MethodName Resolver

▼ Answer

13. **Which of the following controller provide a way to interact with data objects and dynamically bind parameters from the `HttpServletRequest` to the data object specified?**

- a. MultiActionController
- b. AbstractController
- c. FormActionControllor
- d. CommandController

▼ Answer

- d. CommandController

14. **Which of the following is not a view resolver?**

- a. BeanNameViewResolver
- b. VelocityViewResolver
- c. JasperReportsViewResolver
- d. UriBasedViewResolver
- e. Resour Resource BundleViewResolver
- f. None of above

▼ Answer

- c. JasperReportsViewResolver

jsp, servlet, exceptions, syntaxes

```
statement = connection.createStatement(  
    java.sql.ResultSet.TYPE_FORWARD_ONLY,  
    java.sql.ResultSet.CONCUR_READ_ONLY);  
statement.setFetchSize(Integer.MIN_VALUE);
```

Q: Which isolation level prevents dirty read in JDBC, connection class.

(A) TRANSACTION\_READ\_COMMITTED

(B) TRANSACTION\_UNREAD\_COMMITTED

| Isolation Level | Table-Level Locking | Row-Level Locking |
|-----------------|---------------------|-------------------|
|-----------------|---------------------|-------------------|

|                              |                                                              |                                                              |
|------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| TRANSACTION_READ_UNCOMMITTED | Dirty reads, nonrepeatable reads, and phantom reads possible | Dirty reads, nonrepeatable reads, and phantom reads possible |
|------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|

|                            |                                                |                                                |
|----------------------------|------------------------------------------------|------------------------------------------------|
| TRANSACTION_READ_COMMITTED | Nonrepeatable reads and phantom reads possible | Nonrepeatable reads and phantom reads possible |
|----------------------------|------------------------------------------------|------------------------------------------------|

|                             |                                                           |                        |
|-----------------------------|-----------------------------------------------------------|------------------------|
| TRANSACTION_REPEATABLE_READ | Phantom reads not possible because entire table is locked | Phantom reads possible |
|-----------------------------|-----------------------------------------------------------|------------------------|

|                          |      |      |
|--------------------------|------|------|
| TRANSACTION_SERIALIZABLE | None | None |
|--------------------------|------|------|

TYPE\_SCROLL\_INSENSITIVE:

The result can be scrolled;  
its cursor can move both forward and backward relative to the current position,  
and it can move to an absolute position. The result set is insensitive to changes made  
to the underlying data source while it is open. It contains the rows that satisfy the query  
at either the time the query is executed or as the rows are retrieved.

TYPE\_SCROLL\_SENSITIVE:

The result can be scrolled; its cursor can move both forward and backward relative  
to the current position, and it can move to an absolute position. The result set reflects  
changes made to the underlying data source while the result set remains open.

Q: What is the effect of executing the following JSP statement, assuming a class with name Employee exists in classes package.

```
<%@ page import = "classes.Employee" %> <jsp:useBean id="employee"  
class="classes.Employee" scope="session"/> <jsp:setProperty  
name="employee" property="*" />
```

The code does not compile as there is no property attribute of setProperty tag.

The code does not compile as property attribute cannot take \* as a value.

The code sets value of all properties of employee bean to "\*".

Ans) The code sets the values of all properties of employee bean to matrchng parameters in request object.

bookmark is in get.

Q:

```
1. class Animal {  
    public String noise() {  
        return "peep";  
    }  
}  
class Dog extends Animal {  
    public String noise() {  
        return "bark";  
    }  
}  
class Cat extends Animal {  
    public String noise() {  
        return "meow";  
    }  
}  
...  
30. Animal animal = new Dog();  
31. Cat cat = (Cat)animal;  
32. System.out.println(cat.noise());
```

What is the result?

- A. peep
- B. bark
- C. meow
- D. Compilation fails.
- E. An exception is thrown at runtime.

Answer: E

Q:

```
1. class A {  
2. void foo() throws Exception {  
3. throw new Exception();  
4. }  
5. }  
6.  
7. class SubB2 extends A {  
8. void foo() {  
9. System.out.println("B ");  
10. }  
11. }  
12. class Tester {  
13. public static void main(String[] args) {  
14. A a = new SubB2();  
15. a.foo();  
16. }  
17. }
```

What is the result?

- A. B
- B. B, followed by an Exception.
- C. Compilation fails due to an error on line 9.
- D. Compilation fails due to an error on line 15.
- E. An Exception is thrown with no other output

Answer: D  
Section: All

Explanation/Reference:  
Unhandled exception type Exception

2. `legal s3=s2+s1`
3. when we are sending data in URL in get method how to separate parameters? `ANs) &`  
when we are sending data in URL in get method how to separate parameter and separate path? `ANs) ?`
4. 

```
class multithreaded_programing{
    public static void main(String args[]){
        Thread t = Thread.currentThread();
        t.setName("new thread");
        System.out.println(t);
    }
}
```

`ans- thread[new thread,5,main]`

5. current thread
- ```
public class MyRunnable implements runnable{
    public void run(){
        //some code
    }
}
```
- which of these will create and start this thread?

Ans) `new Thread(new MyRunnable()).start();`

Description: Because the class implements Runnable, an instance of it has to be passed to the Thread constructor and then the instance of the Thread has to be started.

6. 

```
class Mammal { }

class Raccoon extends Mammal {

    Mammal m = new Mammal();
}
```
- ```
class BabyRaccoon extends Mammal { }
```

Which four statements are true? (Choose four.)

- `A. Raccoon is-a Mammal.`
- `B. Raccoon has-a Mammal.`
- `C. BabyRaccoon is-a Mammal.`
- D. BabyRaccoon is-a Raccoon.
- E. BabyRaccoon has-a Mammal.

**F. BabyRaccoon is-a BabyRaccoon.**

Answer: A,B,C,F

Explanation :Check is-A and Has-A relationship between both classes.

7. What is the output of this program?

```
import java.io.*;

public class filesinputoutput {

    public static void main(String[] args) {

        String obj  = "abc";

        byte b[] = obj.getBytes();

        ByteArrayInputStream obj1 = new ByteArrayInputStream(b);

        for (int i = 0; i < 2; ++ i) {

            int c;

            while ((c = obj1.read()) != -1) {

                if (i == 0) {

                    System.out.print(Character.toUpperCase((char)c)); }}}}

ans) ABC
```

8. class output {

```
    public static void main(String args[])

    {

        StringBuffer s1 = new StringBuffer("Hello World");

        s1.insert(6 , "Good ");

        System.out.println(s1);

    }

}
```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld
- d) Hello Good World

**Answer: d**

Explanation: The insert() method inserts one string into another. It is overloaded to accept values of all simple types, plus String and Objects. Sting is inserted into invoking object at

specified position. "Good " is inserted in "Hello World" T index 6 giving "Hello Good World".

output:

```
$ javac output.java
```

```
$ java output
```

```
Hello Good World
```

9. `string index(i)`

10. Which of these keywords cannot be used for a class which has been declared final?

a) `abstract`

b) `extends`

c) `abstract` and `extends`

d) None of the mentioned

View Answer

Answer: a

Explanation: A abstract class is incomplete by itself and relies upon its subclasses to provide complete

implementation. If we declare a class final then no class can inherit that class, an abstract class needs its subclasses

hence both final and abstract cannot be used for a same class.

11. `import all- import .*`

12. abstract class abc{

`abstract ... (int a, int b, int c)`

Q= method declared right?

13. `sos('1'+new integer(2)+3);`

ANS-123

14. `int a[]=new int[]{1,2,6,4,5};`

`int b[]=new int[5];`

`b[2]=a[2];`

`a[2]=10;`

`System.out.println(b[2]);`

ans: 6

15. GENERICS STRING TAG QUESTION

16. What gets printed when the following JSTL code fragment is executed?

Select the one correct answer.

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
```

```
<c:set var="item" value="2"/>
```

```
<c:if test="${var==1}" var="result" scope="session">
```

```
<c:out value="${result}"/>
```

```
</c:if>
```

A. The JSTL code does not compile as attribute for if tag are not correct.

B. true

C. false

D. Nothing gets printed.

17. ideal

removin loop

-operations

a)idea



b) deal

18. The sendError method defined in the HttpServlet class is equivalent to invoking the setStatus method with the following parameter. Select the one correct answer.

4.

- A. SC\_OK
  - B. SC\_MOVED\_TEMPORARILY
  - C. SC\_NOT\_FOUND Ans) C
  - D. SC\_INTERNAL\_SERVER\_ERROR
  - E. ESC\_BAD\_REQUEST
5. The sendRedirect method defined

The sendRedirect method defined in the HttpServlet class is equivalent to invoking the setStatus method with the following parameter and a Location header in the URL. Select the one correct answer.

- A. SC\_OK
- B. SC\_MOVED\_TEMPORARILY Ans) B
- C. SC\_NOT\_FOUND
- D. SC\_INTERNAL\_SERVER\_ERROR
- E. ESC\_BAD\_REQUEST

19. doGet is in which? a) request.httpresponse ? b) response.http?

20. jsp-attributes: differentiate a), b) & c)?

21. independently vary=bridge

22. factory set

23. A team of programmers is involved in reviewing a proposed design for a new utility class.

After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself.

What design issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

24. instanceof

25. man dog best friend ans: class man { private dog best friend; }

26. push pop stack

27. blue mmix deploy cloud foundry ans) cf push

28. 1..\* At least one instance

29. window if class=underline

30. abstract class- italics

31. slant line for delay

32. hollow diamond on collection side for aggregation, filled for composition and hollow triangle for generalization.

33. jstl

34. arrow towards mammal

- 35. iterator question
- 36. hasNext(), traverse
- 37. junit testing in which order
- 38. scroll insensitive
- 39. What is the output of this program?

```
import java.util.*;

class vector {

    public static void main(String args[]) {

        Vector obj = new Vector(4,2);

        obj.addElement(new Integer(3));

        obj.addElement(new Integer(2));

        obj.addElement(new Integer(5));

        obj.removeAll(obj);

        System.out.println(obj.isEmpty());

    }

}
```

- a) 0
- b) 1
- c) true
- d) false

View Answer

Answer: c

Explanation: firstly elements 3, 2, 5 are entered in the vector obj, but when obj.removeAll(obj); is executed all the elements are deleted and vector is empty, hence obj.isEmpty() returns true.

Output:

```
$ javac vector.java
$ java vector
true
```

- 40. 

```
import java.util.*;

class Array {

    public static void main(String args[]) {

        int array[] = new int [5];

        for (int i = 5; i > 0; i--)

            array[5 - i] = i;

        Arrays.sort(array);

        for (int i = 0; i < 5; ++i)

            System.out.print(array[i]);;
```

```
}
```

```
} Ans) 12345
```

41. What is the output of this program?

```
import java.util.*;

class Maps {

    public static void main(String args[]) {

        TreeMap obj = new TreeMap();

        obj.put("A", new Integer(1));

        obj.put("B", new Integer(2));

        obj.put("C", new Integer(3));

        System.out.println(obj.entrySet());

    }

}
```

- a) [A, B, C]
- b) [1, 2, 3]
- c) {A=1, B=2, C=3}
- d) [A=1, B=2, C=3]**

Answer: d

Explanation: obj.entrySet() method is used to obtain a set that contains the entries in the map. This method provides set view of the invoking map.

Output:

```
$ javac Maps.java
$ java Maps
[A=1, B=2, C=3]
```

42. What is the output of this program?

```
import java.util.*;

class Bitset {

    public static void main(String args[]) {

        BitSet obj = new BitSet(5);

        for (int i = 0; i < 5; ++i)

            obj.set(i);

        obj.clear(2);

        System.out.print(obj);

    }

}
```

a) {0, 1, 3, 4}

b) {0, 1, 2, 4}

c) {0, 1, 2, 3, 4}

d) {0, 0, 0, 3, 4}

View Answer

Answer: a

Explanation: None.

Output:

```
$ javac Bitset.java
```

```
$ java Bitset
```

```
{0, 1, 3, 4}
```

43. What is the output of this program?

```
import java.util.*;

class Array {

    public static void main(String args[]) {

        int array[] = new int [5];

        for (int i = 5; i > 0; i--)

            array[5-i] = i;

        Arrays.fill(array, 1, 4, 8);

        for (int i = 0; i < 5 ; i++)

            System.out.print(array[i]);

    }

}
```

advertisements

a) 12885

b) 12845

c) 58881

d) 54881

View Answer

Answer: c

Explanation: array was containing 5,4,3,2,1 but when method Arrays.fill(array, 1, 4, 8) is called it fills the index location starting with 1 to 4 by value 8 hence array becomes 5,8,8,8,1.

Output:

```
$ javac Array.java
```

```
$ java Array
```

```
58881
```

44. prefix uri

45. last; or;last

46. What is the output of this program?

```
import java.util.*;

class Collection_Algos {
```

```

        public static void main(String args[]) {

            LinkedList list = new LinkedList();

            list.add(new Integer(2));

            list.add(new Integer(8));

            list.add(new Integer(5));

            list.add(new Integer(1));

            Iterator i = list.iterator();

            Collections.reverse(list);

Collections.shuffle(list);

            while(i.hasNext())

                System.out.print(i.next() + " ");

        }

    }

```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8

d) Any random order

Answer: d

Explanation: shuffle - randomizes all the elements in a list.

Output:

```
$ javac Collection_Algos.java
```

```
$ java Collection_Algos
```

```
1 5 2 8
```

(output will be different on your system)

47.What is the output of this program?

```

import java.util.*;

class Collection_Algos {

    public static void main(String args[]) {

        LinkedList list = new LinkedList();

        list.add(new Integer(2));

        list.add(new Integer(8));

        list.add(new Integer(5));

        list.add(new Integer(1));

        Iterator i = list.iterator();

        Collections.reverse(list);

```

```

        Collections.sort(list);

        while(i.hasNext())

            System.out.print(i.next() + " ");

    }

}

```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8
- d) 2 1 8 5

Answer: c

Explanation: Collections.sort(list) sorts the given list, the list was 2->8->5->1 after sorting it became 1->2->5->8.

Output:

when reverse 1 5 8 2

but sorting 1 2 5 8 ans is C

48. URL from a remote location protocol http is used. a)http b)https

c)both d)none [Ans. c]

49. <%@ taglib uri="http://java.sun.com/jsp/jstl/core"

prefix="c" %>

<c:set var="item" value="2"/>

<c:forEach var="item" begin="0" end="0" step="2">

<c:out value="{item}" default="abc"/>

</c:forEach>

The JSTL code does not compile as an attribute for forEach tag is not correct. Ans A

A. 0

B. 2

C. ABC

D. Nothing gets printed as c.out statement does not get executed.

50.<%@ taglib uri="http://java.sun.com/jsp/jstl/core"

prefix="c" %>

<c:set var="j" value="4,3,2,1"/>

<c:forEach items="{j}" var="item" varStatus="status">

<c:if test="{status.first}">

<c:out value="{status.index}" default="abc"/>

</c:if>

</c:forEach>

A. 1

B. 2

C. 3

D. 4

E. abc

F. The program does not compile

51.What is the output of this program?

```
import java.util.*;
```

```
public class genericstack <E> {
```

```

        Stack <E> stk = new Stack <E>();

    public void push(E obj) {

        stk.push(obj);

    }

    public E pop() {

        E obj = stk.pop();

        return obj;

    }

}

class Output {

    public static void main(String args[]) {

        genericstack <String> gs = new genericstack<String>();

        gs.push("Hello");

        System.out.print(gs.pop() + " ");

        genericstack <Integer> gs = new genericstack<Integer>();

        gs.push(36);

        System.out.println(gs.pop());

    }

}

```

- a) Error
- b) Hello
- c) 36
- d) Hello 36

Answer: d

Explanation: None.

Output:

\$ javac Output.java

\$ java Output

Hello 36

```

52. class output {

    public static void main(String args[])

    {

        StringBuffer c = new StringBuffer("Hello");

        StringBuffer c1 = new StringBuffer(" World");

```

```

        c.append(c1);

        System.out.println(c);
    }
}

```

- a) Hello
- b) World
- c) Helloworld
- d) Hello World    Ans-d.

```

53.class output {
    public static void main(String args[])
    {
        StringBuffer s1 = new StringBuffer("Hello");

        s1.setCharAt(1, 'x');

        System.out.println(s1);
    }
}

```

- a) xello
- b) xxxxx
- c) Hxlllo
- d) Hexlo

View Answer

Answer: c

Explanation: None.

Output:

```
$ javac output.java
```

```
$ java output
```

```
Hxlllo
```

54. switch case

55. array of string

56. "welcome.trim()"= when compiled ans) equals

57. builder pattern

58. Question on this:

```
Calendar calendar = new GregorianCalendar();
```

```

int year      = calendar.get(Calendar.YEAR);
int month     = calendar.get(Calendar.MONTH);
int dayOfMonth = calendar.get(Calendar.DAY_OF_MONTH); // Jan = 0, not 1
int dayOfWeek  = calendar.get(Calendar.DAY_OF_WEEK);
int weekOfYear = calendar.get(Calendar.WEEK_OF_YEAR);
int weekOfMonth = calendar.get(Calendar.WEEK_OF_MONTH);

```

```

int hour      = calendar.get(Calendar.HOUR);           // 12 hour clock
int hourOfDay = calendar.get(Calendar.HOUR_OF_DAY);    // 24 hour clock
int minute    = calendar.get(Calendar.MINUTE);

```



```
int second      = calendar.get(Calendar.SECOND);  
int millisecond= calendar.get(Calendar.MILLISECOND);
```

```
59. class output {  
  
    public static void main(String args[])  
    {  
  
        String s1 = "one";  
  
        String s2 = s1 + " two";  
  
        System.out.println(s2);  
  
    }  
  
}
```

- a) one
  - b) two
  - c) one two
  - d) compilation error
- View Answer  
Answer: c

60. Total number of users using listners - [ans. httpsessionlistner](#)

```
61. class exception_handling {  
  
    public static void main(String args[]) {  
  
        try {  
  
            System.out.print("Hello" + " " + 1 / 0);  
  
        }  
  
        catch(ArithmeticException e) {  
  
            System.out.print("World");  
  
        }  
  
    }  
  
}
```

- a) Hello
- b) World
- c) HelloWorld
- d) Hello World

Answer: b

```
62. class output {  
  
    public static void main(String args[])
```

```

    {
        char c[]={ 'a', '1', 'b' , ' ' , 'A' , '0' };
        for (int i = 0; i < 5; ++i)
        {
            if(Character.isDigit(c[i]))
                System.out.println(c[i]+" is a digit");
            if(Character.isWhitespace(c[i]))
                System.out.println(c[i]+" is a Whitespace
character");
            if(Character.isUpperCase(c[i]))
                System.out.println(c[i]+" is an Upper case
Letter");
            if(Character.isLowerCase(c[i]))
                System.out.println(c[i]+" is a lower case
Letter");

            i=i+3;
        }
    }
}

```

a) a is a lower case Letter  
is White space character  
b) b is a lower case Letter  
is White space character  
c) a is a lower case Letter  
A is a upper case Letter  
d) a is a lower case Letter  
0 is a digit  
View Answer  
Answer:c

62. Two questions on these.  
Interfaces for Event Listeners-

```

Session Serialization--HttpSession
Request Lifecycle Event--ServletRequestListener
Session Attribute Binding--HttpSessionAttributeListener
Session LifecycleEvent--HttpSessionListener
Web Application Lifecycle Event--ServletContextListener

```

63. Which of the following describes the correct sequence of the steps involved in making a connection with a database.

1. Loading the driver
2. Process the results.

3. Making the connection with the database.  
4. Executing the SQL statements.

(A) 1,2,3,4  
(B) 1,3,4,2  
(C) 2,1,3,4  
(D) 4,1,2,3

Answer : 1,3,4,2

```
64. import java.io.*;
    class filesinputoutput {
        public static void main(String args[])
            InputStream obj = new
FileInputStream("inputoutput.java");
            System.out.print(obj.available());
        }
    }
```

Note: inputoutput.java is stored in the disk.

- a) true  
b) false  
c) prints number of bytes in file  
d) prints number of characters in the file

View Answer

Answer: c

Explanation: obj.available() returns the number of bytes.

Output:

```
$ javac filesinputoutput.java
$ java filesinputoutput
1422
```

(Output will be different in your case)

```
65. import java.io.*;
    public class filesinputoutput {
        public static void main(String[] args) {
            String obj = "abc";
            byte b[] = obj.getBytes();
            ByteArrayInputStream obj1 = new ByteArrayInputStream(b);
            for (int i = 0; i < 2; ++ i) {
                int c;
                while((c = obj1.read()) != -1) {
                    if(i == 0) {
                        System.out.print(Character.toUpperCase((char)c));
                        obj2.write(1);
                    }
                }
                System.out.print(obj2);
            }
        }
    }
```

- a) AaBaCa  
b) ABCaaa  
c) AaaBaaCaa  
d) AaBaaCaaa

Answer: d

Explanation: None.

Output:

```
$ javac filesinputoutput.java
$ java filesinputoutput
AaBaaCaaa
```

66. Which of these methods are used to read in from file?

- a) get()
- b) read()
- c) scan()
- d) readFileInput()

View Answer

Answer: b

67. toString question

68. SC\_OK

69. Which of the following allows non repeatable read in JDBC Connection?

- a. TRANSACTION\_READ\_UNCOMMITTED
- b. TRANSACTION\_READ\_COMMITTED
- c. TRANSACTION\_SERIALIZABLE
- d. TRANSACTION\_REPEATABLE\_READ Ans. D

70. dirty unread- ans)read.

71.package pkg;

```
class output {  
    public static void main(String args[])  
    {  
        StringBuffer s1 = new StringBuffer("Hello");  
        s1.setCharAt(1, x);  
        System.out.println(s1);  
    }  
}
```

- a) xello
- b) xxxxx
- c) Hxlllo
- d) Hexlo

Answer: c

```
72.public class Test8 {  
    public static void main (String args[]) {  
        Number numberRef = new Integer(0);  
        Double doubleRef = (Double)numberRef;  
    }  
}
```

It shows me exception at runtime:

Exception in thread "main" java.lang.ClassCastException:  
java.lang.Integer cannot be cast to java.lang.Double

73. command design pattern ans)d (history...)

74. Which of these is not a valid value for dispatcher element of filtermapping.

Select the one correct answer.

A. FORWARD

**B. RESPONSE** Ans) B

C. INCLUDE

D. ERROR

75. Which of the following is legal JSP syntax to print the value of i.  
Select the one correct answer

A. `<%int i = 1;%>`  
`<%= i; %>`

B. `<%int i = 1;`  
`i; %>`

C. `<%int i = 1%>`  
`<%= i %>`

**D. `<%int i = 1;%>`**  
**`<%= i %>`**

E. `<%int i = 1%>`  
`<%= i; %>`

Ans) D.

76. `<% int sum = 0;`  
`for(j = 0; j < 10; j++) { %>`  
`// XXX --- Add j to sum`  
`<% } %>`  
`// YYY --- Display ths sum. What must be added at the location YYY to`  
`print the sum of ten numbers. Select the one correct statement`

A. `<% sum %>`

B. `<% sum; %>`

**C. `<%= sum %>`**

D. `<%= sum; %>` ANS) C.

77. `<%@ taglib uri="http://java.sun.com/jsp/jstl/core"`  
`prefix="c" %>`  
`<c:set var="j" value="4,3,2,1"/>`  
`<c:forEach items="{j}" var="item" begin="1" end="2">`  
`<c:out value="{item}" default="abc"/>`  
`</c:forEach>`

A. 1

**B. 2**

**C. 3**

D. 4

E. abc

F. The program does not compile.

Which numbers gets pri

Ans) 2,3

78. Which number gets printed when the following JSTL code  
fragment is executed? Select the one correct answers.

`<%@ taglib uri="http://java.sun.com/jsp/jstl/core"`  
`prefix="c" %>`  
`<c:set var="j" value="4,3,2,1"/>`

```
<c:forEach items="${j}" var="item" varStatus="status">
<c:if test="${status.first}">
<c:out value="${status.index}" default="abc"/>
</c:if>
</c:forEach>
```

Ans) A

A. 1

B. 2

C. 3

D. 4

E. abc

F. The program does not compile.

79. Which of the following is not a valid HTTP/1.1 method. Select the one correct answer.

15.

A. CONNECT method

B. COMPARE method

C. OPTIONS method

D. TRACE method

Name the http method used

answer: B

80. Name the location of compiled class files within a war file? Select the one correct answer.

19.

A. /META-INF/classes

B. /classes

C. /WEB-INF/classes

D. /root/classes

Answers- web inf

81. What get printed when the following JSTL code fragment is executed?

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<c:forEach var="item" begin="0" end="12" step="3">
${item}
</c:forEach>
```

options

A) 0, 3, 6, 9, 12

B) 3, 6, 9, 12

C) 0, 6, 9, 12

D) 0, 3, 6, 9

<http://www.sanfoundry.com/java-questions-answers-freshers-experienced/>

The four techniques used in session tracking are:

Java Language Keywords

Here is a list of keywords in the Java programming language. You cannot use any of the following as identifiers in your programs. The keywords `const` and `goto` are reserved, even though they are not currently used. `true`, `false`, and `null` might seem like keywords, but they are actually literals; you cannot use them as identifiers in your programs.

|                        |                           |                         |                         |                           |
|------------------------|---------------------------|-------------------------|-------------------------|---------------------------|
| <code>abstract</code>  | <code>continue</code>     | <code>for</code>        | <code>new</code>        | <code>switch</code>       |
| <code>assert***</code> | <code>default</code>      | <code>goto*</code>      | <code>package</code>    | <code>synchronized</code> |
| <code>boolean</code>   | <code>do</code>           | <code>if</code>         | <code>private</code>    | <code>this</code>         |
| <code>break</code>     | <code>double</code>       | <code>implements</code> | <code>protected</code>  | <code>throw</code>        |
| <code>byte</code>      | <code>else</code>         | <code>import</code>     | <code>public</code>     | <code>throws</code>       |
| <code>case</code>      | <code>enum****</code>     | <code>instanceof</code> | <code>return</code>     | <code>transient</code>    |
| <code>catch</code>     | <code>extends</code>      | <code>int</code>        | <code>short</code>      | <code>try</code>          |
| <code>char</code>      | <code>final</code>        | <code>interface</code>  | <code>static</code>     | <code>void</code>         |
| <code>class</code>     | <code>finally</code>      | <code>long</code>       | <code>strictfp**</code> | <code>volatile</code>     |
| <code>const*</code>    | <code>float</code>        | <code>native</code>     | <code>super</code>      | <code>while</code>        |
| <code>*</code>         | <code>not used</code>     |                         |                         |                           |
| <code>**</code>        | <code>added in 1.2</code> |                         |                         |                           |
| <code>***</code>       | <code>added in 1.4</code> |                         |                         |                           |
| <code>****</code>      | <code>added in 5.0</code> |                         |                         |                           |

1. How to import all package?

Answer : `import java.*`

2. Which of the following is correct about junit?

Answer: - `It is an open source framework.`  
- `It provides Annotation to identify the test methods.`  
- `It provides Assertions for testing Expected results`

3. Pattern which follows sequential order?

Answer : `Iterator`

4. class Animal

```
{
    public String noise()
    {
        return "peep";
    }
}
class Dog extends Animal
{
    public String noise()
    {
        return "bark";
    }
}
class cat extends Animal
{
    public String noise()
    {
        return "meow";
    }
}
}.....
Animal animal = new Dog();
Cat cat= (Cat)animal;
SOP(cat.noise());
```

Answer : `Exception is thrown at the Runtime`

5. class output {

```
public static void main(String args[]){
    String buffer s1 = new StringBuffer("Hello world")
    s1.insert(6,"Good");
    SOP(s1);
}
```



```
}  
}
```

Answer: **Hello Goodworld**

6. SOP ('1'+new integer (2) +3);

Answer: **123**

7. A team of programmer is involved in reviewing a proposed design for a new utility class after some discussion.....

Answer: **weak encapsulation**

8. Man dog best friend

Answer : **class man private dog best friend;**

9. Bluemix deploy cloud foundry

Answer: **cf push**

10. Class vector {

```
PSVM(String args[])  
Vector obj = new vector(4,2);  
Obj.add element(new Interger(3));  
Obj.add element(new Interger(2));  
Obj.add element(new Interger(5));  
Obj.removeAll(obj);  
SOP(obj.isEmpty());  
}
```

Answer : **True**

11. StringBuffer s1 = new StringBuffer("Hello");

```
StringBuffer s2 = reverse(s1);
```

```
SOP(s2);
```

Answer : **olleH**

12. How to access the resultsetobject?

Answer : **by using special getter method.....**

13. X= something

```
While(X<10&&X>24)
```

Answer : **condition false**

14. RunWith(Suit.class)

```
@SuiteClasses({  
    annotationTest.class,  
    AssertTest.class  
    CalculateTest.class,  
    EmployeeTest.class})  
Public class SuiteTest{
```

```
//the class remains empty
//used only as a holder for the above annotation
}
```

What is the sequence of junit test ?

Answer : `AnnotationTest, AsserTest, CalculateTest,EmployeeTest`

15. `Public static void main(String args[])`

```
{
List<string> List = new ArrayList<string>();
//add string
List.add("cricket");
List.add("football");
List.add("hockey");
Iterator it = List.iterator();
While(it.hasNext())
{
String s = it.next();
}
}
```

Which pattern

Answer : `Iterator`

16. Class x implements Runnable {

```
Public static void main(String args[])
{
/*missing code? */
}
Public void run() {
}}
```

Which of the following line of code to start thread?

Answer : `x run = new x();`

`Thread t = new Thread(run); t.start();`

17. `PSVM()`

```
{
try{
int a = 5;
int b = 0;
int c = a/b;
SOP("World");
}
Catch(exception e)
{
SOP("hello");
}}
```

Answer : `hello`

18. Int a = 10;

Int b = 0;

Int c = a/b;

SOP("c");

Output :

Answer: runtime exception not arithmetic exception

19. Which will contain the body of the thread?

Answer : run()

20. Which two of the following methods are defined in class thread?

1. Start() 2. Wait() 3. Notify() 4. Run()

Answer : start() and run()

21. To send binary o/p?

Answer : getOutputStream

22. Unit testing?

Answer : Single Entity

23. Assume the following method is properly synchronized and called from a thread A on an object B:  
wait(2000);

After calling this method, when will the thread A become a candidate to get another turn at the CPU?

- A. After thread A is notified, or after two seconds.
- B. After the lock on B is released, or after two seconds.
- C. Two seconds after thread A is notified.
- D. Two seconds after lock B is released.

24. Invoked only once?

Answer : @BeforeClass

25. Invoked before each test?

Answer : @Before

26. Invoked after each test?

Answer : @After

27. Only once invoked finishing all tests?

Answer : @AfterClass

28. What will be proper order of session?

Answer : loading, instantiation, init, service, destroy

29. Proper order of access modifier

Answer : private default protected public

30. If the options will be like

New class foo{print(.....)},

Answer: new foo(){print(.....)}

31. Insert(6, "Good")

Answer : Hello Good Word

32. Class{  
 PSVM  
 String str = new String( ".....");  
 }  
 Do{  
 str = "Hello Stop World ";  
 SOP(str);  
 }  
 While(str!=Strong);  
 {  
 .....  
 }  
 Answer : Hello Stop World

33. The code snippet below is an example of which of the following?

Long myLong = 21l;  
 A Autoboxing  
 B Autounboxing  
 C Autocasting  
 D Autoinstanting

Answer: Autoboxing

34. clear(2) in the string of index 2

Answer : 0134

35.

```
1. interface TestA { String toString(); }
2. public class Test {
3. public static void main(String[] args) {
4. System.out.println(new TestA() {
5. public String toString() { return "test"; }
6. });
7. }
8. }
```

What is the result?

A. test  
 B. null  
 C. An exception is thrown at runtime.  
 D. Compilation fails because of an error in line 1.  
 E. Compilation fails because of an error in line 4.  
 F. Compilation fails because of an error in line 5.

Answer: A

36. 2+3\*5

Answer 17 integer

17 byte

25 byte

25 integer

37. Identify the correct signature of the main method?

Answer : public static void main(String args[])

38. public static void main(String[] args)

```
{
    GenericTest<String> arr[] = new GenericTest[5];           //line 1
    arr[0] = new GenericTest("Java");                       //line 2
    arr[1] = new GenericTest(1);                             //line 3
    arr[2] = (GenericTest<String>)new GenericTest(1);        //line 4
    arr[3] = (GenericTest<String>)new GenericTest<Integer>(1); //line 5
    for(GenericTest o:arr)
    {
        System.out.println(o);
    }
}
```

Output: Error in Line 1

39. Answer: AaBaaaCaaaa

40. Multiple exception catch block..

Answer : **superclass exception can not be caught first**

Can we caught either super class or subclass

41. l.add(2)

l.add(5)

l.add(8)

l.add(1)

l.reverse();

Answer: 1852

42. getvalue(IDEAL)

SOP(word.substring(0,word.length()-1)+word.charAt(word.length()-1))

Answer: DEAL

43. in javabeen reference id define?

a. Classname

b. Package

**c. Instance**

d. None

44. Int x=6;

Int y = 12;

If(x>y)

{

X++;

```
y--;  
}  
SOP(....);
```

Answer: 2

45. Datatype of the no 9.6352

Answer: float

double

Float

Double

46. How many thread one from exception mainclass

Answer : is only 1 thread(main thread)

47. Start same thread two times

```
t1.start()
```

```
t1.start()
```

Answer: compilation fails, compile time error

48. Decouples in which design pattern?

Answer: Bridge

49. String s1 = abc;

```
String s2 = def;
```

```
String s3 = s1.concat(s2.toUpperCase)
```

Answer:: abcdefabcDEF

50. what is the output of this program

```
class output{  
    public static void main(String args[])  
    {  
        Object obj = new object();  
        System.out.print(obj.getClass());  
    }  
}
```

Answer: class java.lang.object

51. declare array of string

```
string[]s;
```

```
string s[]
```

```
string []s;b
```

Answer: ABD

52. class A{

```
int i;
```

```
int j;
```

```
A(){
```

```
    I =1;
```

```
    J=2;
```

```
}
```

```

}
Class output{
Public static void main(String args[])
{
A obj1 = new A();
SOP(obj1.toString());
}
}

```

Answer: String associated with obj1

And output will be A@1cde5f

53. public void divide(int a, int )

```

{
Try{
Int c = a/b;
}
Catch(Exception e)
{
SOP(Exception);
}
Finally{
SOP("finally")
}
}

```

Output :

Answer if a,b not initialized then error and if b = 0; then compile time error with finally will work.

54. Current thread

```

Public class MyRunnable implements runnable {
Public void run()
{
//some code
}
}

```

Which of these will create and start this thread?

Answer:

new Thread(new MyRunnable()).start();

55. To execute in the parameterized statement?

Answer: prepared statement

56. Given

```

10 interface Foo{int bar();}
11 public class sprite{
12 public int fubar(Foo foo){return foo.bar();}
13 public void testFoo(){
14 fubar(

```

```
15 //insert code here
```

```
16 );
```

```
17 }}
```

```
Answer: new Foo(){public int bar(){return 1;}}
```

57. A: unit testing is the testing of single entity

B: unit testing is the process of checking the functionality of the application whether it is working as per requirements

Options :

a. A is true

b. B is true

c. Both are true

d. None of the above

What will be the output of this -

58. `String river = new String("TRAINING");`

```
System.out.println(river.length());
```

answer-8

59. `String getValue(String word)`

```
{
    if (word.length() == 1)
        return "";
    else
        return getValue( word.substring(0, word.length() - 1) ) + word.charAt(word.length() - 1);
}
```

What is the value of the string returned by `getValue("DEMOS")`

Answer: EMOS

60. `public class Question {`

```
public static void main(String args[]) {
```

```
    String s1 = "uvw";
```

```
    String s2 = "xyz";
```

```
    String s3 = s1.concat(s2.toUpperCase( ));
```

```
    System.out.println(s1+s2+s3);
```

```
}}
```

answer: uvwxyzuvwXYZ

61. `int a = 9;`

```
int b = 14;
```

```
while(a<b) {
```

```
    System.out.println("In the loop");
```

```
    a+=2;
```

```
    b-=2;
```

```
}
```

answer- In the loop



In the loop

62. abstract class xyz

```
{  
    abstract abc (int a, int b) { }  
}
```

- What sort of compilation error you will get ?
- How to fix the compilation error

Answer-- give return type void. and remove empty parentheses.

63. interface ATOB {

```
    String s = "yo";  
    public void meth();  
}  
interface BTOC { }  
interface C extends ATOB, BTOC {  
    public void meth();  
    public void meth(int x);  
}
```

- Will this compile ?

answer--- interface can also be extended (when inheriting from two interfaces.)

64. int i = -1;

```
int b = 10;  
int val = b/ i;
```

answer- -10

```
interface Data { public void load(); }  
abstract class Info { public abstract void load(); }
```

65. How Can i inherit both the interface and abstract class ?

answer= class xyz extends Info implements interface{ void load}

66. public class GenericTest<G> {

G g;

GenericTest(G g){

```
    this.g =g;
```

}

public static void main(String[] args)

```
{  
    GenericTest<String> arr[] = new GenericTest[5];           //line 1  
    arr[0] = new GenericTest("Java");                       //line 2  
    arr[1] = new GenericTest(1);                             //line 3  
    arr[2] = (GenericTest<String>)new GenericTest(1);        //line 4  
    arr[3] = (GenericTest<String>)new GenericTest<Integer>(1); //line 5  
    for(GenericTest o:arr)  
    {  
        System.out.println(o);
```

```

    }
}

```

Answer-- can not cast <Integer> to <String> line5

and default value is null in an array

```

67. class A {
    static String str = "STRING_IN_A";
}
class B extends A {
    static String str = "STRING_IN_B";
}

```

//

in void main

B b1=new B();

sop b1.str

//

Answer--- will compile fine.bt str is being overwritten in B class.

```

68. int j=8;
while( (j < 10) && (j > 24)) {
    System.out.println(" Am here");
}

```

answer-while() condition is always false

```

69. class output {
    public static void main(String args[]){
        StringBuffer c = new StringBuffer("Hello");
        StringBuffer c1 = new StringBuffer("World");
        c.append(c1);
        System.out.print(c);
    }
}

```

answer-helloworld

70. are resultset updateable?

Answer: yes but only if you indicate a concurrency strategy when executing the statement and if the driver and database support this option.

71. How can you start a database transaction in the database

Answer: by setting the autocommit property of the connection to false & execute a statement in the database

72. The object that does not contain the stored procedure itself but contains only a call to the store procedure?

Answer: callableStatement

73. To execute a stored procedure "total stack" in a database server which of the following code snippet is used?

Answer: callablestatement clbstmnt = con.preparecall("{call totoal stack}");  
cs.executeQuery();

74. Relative path

Answer: c:/test/\*\*\*

75. Which type of driver converts jdbc calls into the Network protocol used by the datamanagement system?

Answer: Type 4

76. SOP(index(i)+index(0)+lastIndex(i)+lastIndex(0));

Answer: 6469

77. When a select statement returns only one record with a single column of integer datatype on invoking resultset getinteger(0) causes\_\_\_\_\_

Answer:

78. Which listener class will close() method?

Answer: windowlistner

79. Class exception\_Handling{

```
Public static void main(String args[]){  
    Try{  
        SOP("Hello"+" " +1/0);  
    }  
    Catch(ArithmeticException e)  
    {  
        SOP("World");  
    }  
}}
```

Answer: World

80. What is the output of this program?

```
Import java.util.*  
Public static void main(String args[]){  
    TreeMap obj = new Treemap();  
    Obj.put("A", new Integer(1));  
    Obj.put("b", new Integer(2));  
    Obj.put("c", new Integer(3));  
    SOP(obj.entrySet());  
}
```

Answer: [A=1,B=2,C=3]

```
Public class genericstack<E>{
```

```
Stack<E>{
Stack<E>stk = new stack<E>();
}
}
```

81. Which of the following describes the purpose of JUnit?

- A It is a framework to help with writing unit tests for your code.
- B It is a framework to help with writing code involving unit conversions.
- C It is a framework to help with packaging your code into units for distribution.
- D It is a framework to convert all days/dates in your code to June.

Answer: A

82. Which of these keywords must be used to monitor for exceptions?

- a) try
- b) finally
- c) throw
- d) catch

View Answer

Answer: a

83. Treemap question and options will be like

(7,9)

(9,7)

Answer : 7,9

84. Testing Amit Gupta and Ravi Kumar wala question?

- a) Amit Gupta False
  - b) Ravi Kumar False
  - c) Amit Gupta False(Expected name as Ravi Kumar)
  - d) Ravi Kumar False(Expected name as Amit Gupta)
- use

85. Fill in the blank

```
Exception _____{
printstack e;
}
Exception_____ {
printStack e;
}
```

Answer class not found

**Sql exception**

86. Get value from resultset

Answer: getBoolean(),getClob()  
getBlob()

87. class exception\_handling {

```

public static void main(String args[]) {
    try {
        int a, b;
        b = 0;
        a = 5 / b;
        System.out.print("A");
    }
    catch(ArithmeticException e) {
        System.out.print("B");
    }
}

```

- a) A
- b) B
- c) Compilation Error
- d) Runtime Error

Ans:B

88. class A implements foo {

```

16  foo bar()
17  {
18  Return 1;
    }
    }
    Class A implements foo
    {
    25 foobar()
    26 {
    27 return 2;
    }
    }

```

Answer: return 2;

If eliminate 16,17,18 return 2

If eliminate 25 26 27 return 1

89. Which of these handles the exception when no catch is used?

- a. Default handler
- b. Finally
- c. Throw handler
- d. Java run time system

90. class exception\_handling {

```

public static void main(String args[]) {
    try {
        int a, b;

```

```

b = 0;
a = 5 / b;
System.out.print("A");
}
catch(ArithmeticException e) {
System.out.print("B");
}
finally {
System.out.print("C");
}
}
}

```

- a) A
- b) B
- c) AC
- d) BC

Answer:BC

91. public class Delta

```

{ static boolean foo(char c)
{
    System.out.print(c);
    return true;
}
public static void main( String[] argv )
{
    int i = 0;
    for (foo('A'); foo('B') && (i < 2); foo('C'))
    {
        i++;
        foo('D');
    }
}
}

```

Answer: A.  
ABDCBDCB

92. public class TreeSet {

```

public static void main(String args[]){
    //Creating and adding elements
    TreeSet<Integer> al=new TreeSet<Integer>();
    al.add("7"); //line 5
    al.add(9);
}
}

```

```

//Traversing elements
Iterator<Integer> itr=al.iterator();
while(itr.hasNext()){
    System.out.println(itr.next());
}
}
}

```

Answer: compilation error at line 5

93. Which implementation of set would you choose if you want the iterator of set would give you object in the order it were inserted?

**LinkedHashSet**

TreeSet

HashSet

94. Push(7);

Push(2);

Pop();

Push(9);

Answer: print[7,9]

95. (first,two,three,four,,five;) delimits(,)

Answer: firsttwothreefour;five;

143. Which packages contain the JDBC classes?

(a) java.jdbc and javax.jdbc

(b) java.jdbc and java.jdbc.sql

**(c) java.sql and javax.sql**

(d) java.rdb and javax.rdb

144. Which type of driver provides JDBC access via one or more ODBC drivers?

**(a) Type 1 driver**

(b) Type 2 driver

(c) Type 3 driver

(d) Type 4 driver

145. Which type of driver converts JDBC calls into the network protocol used by the database management system directly?

(a) Type 1 driver

(b) Type 2 driver

(c) Type 3 driver

**(d) Type 4 driver**

146. How can you retrieve information from a ResultSet?

- (a) By invoking the method `get(..., String type)` on the ResultSet, where type is the database type
- (b) By invoking the method `get(..., Type type)` on the ResultSet, where Type is an object which represents a database type
- (c) By invoking the method `getValue(...)`, and cast the result to the desired Java type.
- (d) By invoking the special getter methods on the ResultSet: `getString(...)`, `getBoolean (...)`, `getClob(...)`,...

147. How can you execute DML statements (i.e. insert, delete, update) in the database?

- (a) By making use of the `InsertStatement`, `DeleteStatement` or `UpdateStatement` classes
- (b) By invoking the `execute(...)` or `executeUpdate(...)` method of a normal Statement object or a sub-interface object thereof
- (c) By invoking the `executeInsert(...)`, `executeDelete(...)` or `executeUpdate(...)` methods of the `DataModificationStatement` object
- (d) By making use of the `execute(...)` statement of the `DataModificationStatement` object

148. How do you know in your Java program that a SQL warning is generated as a result of executing a SQL statement in the database?

- (a) You must catch the checked `SQLException` which is thrown by the method which executes the statement
- (b) You must catch the unchecked `SQLWarningException` which is thrown by the method which executes the statement
- (c) You must invoke the `getWarnings()` method on the Statement object (or a sub interface thereof)
- (d) You must query the ResultSet object about possible warnings generated by the database

149. What is, in terms of JDBC, a DataSource?

- (a) A DataSource is the basic service for managing a set of JDBC drivers
- (b) A DataSource is the Java representation of a physical data source
- (c) A DataSource is a registry point for JNDI-services
- (d) A DataSource is a factory of connections to a physical data source

150. What is the meaning of `ResultSet.TYPE_SCROLL_INSENSITIVE`

- (a) This means that the ResultSet is insensitive to scrolling
- (b) This means that the Resultset is sensitive to scrolling, but insensitive to updates, i.e. not updateable
- (c) This means that the ResultSet is sensitive to scrolling, but insensitive to changes made by others
- (d) The meaning depends on the type of data source, and the type and version of the driver you use with this data source

151. Are ResultSets updateable?



(a) Yes, but only if you call the method `openCursor()` on the `ResultSet`, and if the driver and database support this option

(b) Yes, but only if you indicate a concurrency strategy when executing the statement, and if the driver and database support this option

(c) Yes, but only if the `ResultSet` is an object of class `UpdateableResultSet`, and if the driver and database support this option

(d) No, `ResultSet`s are never updateable. You must explicitly execute DML statements (i.e. insert, delete and update) to change the data in the underlying database

152. How can you start a database transaction in the database?

(a) By asking a `Transaction` object to your `Connection`, and calling the method `begin()` on it

(b) By asking a `Transaction` object to your `Connection`, and setting the `autoCommit` property of the `Transaction` to false

(c) By calling the method `beginTransaction()` on the `Connection` object

(d) By setting the `autoCommit` property of the `Connection` to false, and execute a statement in the database

153. What is the meaning of the transaction isolation level `TRANSACTION_REPEATABLE_READ`

(a) Dirty reads, non-repeatable reads and phantom reads can occur

(b) Dirty reads are prevented; non-repeatable reads and phantom reads can occur

(c) Dirty reads and non-repeatable reads are prevented; phantom reads can occur

(d) Dirty reads, non-repeatable reads and phantom reads are prevented

154. What statements are correct about positioned updates (i.e. cursor updates) in `ResultSet`s? (2 correct answers)

[a] Using the cursor technique is currently the only possible way to change the data in the current row of a `ResultSet`

[b] Insert statements are only supported when using scrollable cursors.

[c] Only scrollable updateable `ResultSet`s can use this approach to change the data in the current row of a `ResultSet`

[d] The name of the cursor is specified by the `setCursorName(String name)` method the `Statement` object.

155. How can you execute a stored procedure in the database?

(a) Call method `execute()` on a `CallableStatement` object

(b) Call method `executeProcedure()` on a `Statement` object

(c) Call method `execute()` on a `StoredProcedure` object

(d) Call method `run()` on a `ProcedureCommand` object

156. What happens if you call the method `close()` on a `ResultSet` object?

(a) the method `close()` does not exist for a `ResultSet`. Only `Connections` can be closed.

(b) the database and JDBC resources are released

(c) you will get a `SQLException`, because only `Statement` objects can close `ResultSet`s

(d) the ResultSet, together with the Statement which created it and the Connection from which the Statement was retrieved, will be closed and release all database and JDBC resources

157. What happens if you call deleteRow() on a ResultSet object?

(a) The row you are positioned on is deleted from the ResultSet, but not from the database.

(b) The row you are positioned on is deleted from the ResultSet and from the database

(c) The result depends on whether the property synchronizeWithDataSource is set to true or false

(d) You will get a compile error: the method does not exist because you can not delete rows from a ResultSet

158. What statements are correct about batched insert and updates? (2 answers)

[a] To create a batch of insert and update statements, you create an object of type Batch, and call the method

addStatement(String statement) for each statement you want to execute in the batch

[b] Batch insert and updates are only possible when making use of parameterized queries.

[c] To do a batched update/insert, you call addBatch(String statement) on a Statement object for each statement you want to execute in the batch

[d] To execute a batched update/insert, you call the executeBatch() method on a Statement object

159. What is correct about DDL statements (create, grant,...)?

(a) DDL statements are treated as normal SQL statements, and are executed by calling the execute() method on a Statement (or a sub interface thereof) object

(b) To execute DDL statements, you have to install additional support files

(c) DDL statements can not be executed by making use of JDBC, you should use the native database tools for this.

(d) Support for DDL statements will be a feature of a future release of JDBC

160. The JDBC-ODBC Bridge supports multiple concurrent open statements per connection?

a. True

b. False

162. Which of the following allows non repeatable read in JDBC Connection?

a. TRANSACTION\_READ\_UNCOMMITTED

b. TRANSACTION\_READ\_COMMITTED

c. TRANSACTION\_SERIALIZABLE

d. TRANSACTION\_REPEATABLE\_READ

163. Which of the following statements is false as far as different type of statements is concern in JDBC?

a. Regular Statement

b. Prepared Statement

c. Callable Statement

d. Interim Statement

164. Which of the following methods are needed for loading a database driver in JDBC?

- a. registerDriver() method
- b. Class.forName()
- c. Both A and B
- d. getConnection()

165. Which of the following is false as far as type 4 driver is concern?

- a. Type 4 driver is "native protocol, pure java" driver
- b. Type 4 drivers are 100% Java compatible
- c. Type 4 drivers uses Socket class to connect to the database
- d. Type 4 drivers can not be used with Netscape

166. To execute a stored procedure "totalStock" in a database server, which of the following code snippet is used?

- a. Statement stmt = connection.createStatement();stmt.execute("totalStock()");
- b. CallableStatement clbstmnt = con.prepareCall("{call totalStock}");cs.executeQuery();
- c. StoreProcedureStatement  
stmt=connection.createStoreProcedure("totalStock()");spstmt.executeQuery();
- d. PreparedStatement pstmt = connection.prepareStatement("totalStock()");pstmt.execute();

167. Which driver is efficient and always preferable for using JDBC applications?

- a. Type – 4
- b. Type – 1
- c. Type – 3
- d. Type – 2

168. JDBC facilitates to store the java objects by using which of the methods of PreparedStatement  
setObject () 2. setBlob() 3. setClob()

- a. 1, 2
- b. 1,2,3
- c. 1,3
- d. 2,3

169. Which statement is static and synchronized in JDBC API?

- a. executeQuery()
- b. executeUpdate()
- c. getConnection()
- d. prepareCall()

170. The JDBC-ODBC bridge is

- a. Three tiered
- b. Multithreaded
- c. Best for any platform
- d. All of the above

171. All raw data types (including binary documents or images) should be read and uploaded to the database as an array of

- a. byte
- b. int
- c. boolean
- d. char

172. The class `java.sql.Timestamp` has its super class as

- a. `java.sql.Time`
- b. `java.util.Date`
- c. `java.util.Time`
- d. None of the above

173. BLOB, CLOB, ARRAY and REF type columns can be updated in

- a. JDBC 1.0
- b. JDBC 4.0
- c. JDBC 2.0
- d. JDBC 3.0

174. Which of the following methods finds the maximum number of connections that a specific driver can obtain?

- a. `Database.getMaxConnections`
- b. `Connection.getMaxConnections`
- c. `DatabaseMetaData.getMaxConnections`
- d. `ResultSetMetaData.getMaxConnections`

175. Are prepared statements actually compiled?

- a. Yes, they compiled
- b. No, they are bound by the JDBC driver

176. When the message "No Suitable Driver" occurs?

- a. When the driver is not registered by `Class.forName()` method
- b. When the user name, password and the database does not match
- c. When the JDBC database URL passed is not constructed properly
- d. When the type 4 driver is used

177. Which driver is called as thin-driver in JDBC?

- a. Type-4 driver
- b. Type-1 driver
- c. Type-3 driver
- d. Type-2 driver

178. How many transaction isolation levels are defined in `java.sql.Connection` interface?

- a. 4
- b. 3
- c. 5
- d. 2

179. Which method is used to perform DML statements in JDBC?

- a. execute()
- b. executeQuery()
- c. executeUpdate()
- d. executeResult()

180. What is the disadvantage of Type-4 Native-Protocol Driver?

- a. At client side, a separate driver is needed for each database.
- b. Type-4 driver is entirely written in Java
- c. The driver converts JDBC calls into vendor-specific database protocol
- d. It does not support to read MySQL data.

```
181. import java.util.*;
class TestHashMaps{
public static void main(String args[]) {
HashMap<Integer,String> hm= new HashMap<Integer,String> ();
hm.put(100, "John");
hm.put(101, "Paul");
hm.put(102, "George");
hm.put(103, "Ringo");
for (Map.Entry m: hm.entrySet()) {
System.out.println(m.getKey() + " " + m.getValue());
}
}
}
```

Answer:

```
100 John
101 Paul
102 George
103 Ringo
```

```
182. import java.util.Map;
import java.util.TreeMap;
public class TestTreeMap {
public static void main(String args[]) {
TreeMap< Integer, String > hm= new TreeMap< Integer, String > ();
hm.put(100, "John");
hm.put(102, "Paul");
hm.put(101, "George");
```

```

hm.put(103, "Ringo");
for (Map.Entry m: hm.entrySet()) {
    System.out.println(m.getKey() + " " + m.getValue());
}
}
}

```

Answer

100 John

101 George

102 Paul

103 Ringo

183. Given:

```

5. import java.util.*;
6. public class SortOf {
7.     public static void main(String[] args) {
8.         ArrayList<Integer> a = new ArrayList<Integer>();
9.         a.add(1); a.add(5); a.add(3);
11.        Collections.sort(a);
12.        a.add(2);
13.        Collections.reverse(a);
14.        System.out.println(a);
15.    }
16. }

```

What is the result?

A. [1, 2, 3, 5]

B. [2, 1, 3, 5]

C. [2, 5, 3, 1]

D. [5, 3, 2, 1]

E. [1, 3, 5, 2]

F. Compilation fails.

G. An exception is thrown at runtime.

Answer: C

184. class BabyRaccoon extends Mammal { }

Which four statements are true? (Choose four.)

A. Raccoon is-a Mammal.

B. Raccoon has-a Mammal.

C. BabyRaccoon is-a Mammal.

D. BabyRaccoon is-a Raccoon.

E. BabyRaccoon has-a Mammal.

F. BabyRaccoon is-a BabyRaccoon.

Answer: A,B,C,F

185. Which Man class properly represents the relationship "Man has a best friend who is a Dog"?

- A. class Man extends Dog { }
- B. class Man implements Dog { }
- C. class Man { private BestFriend dog; }
- D. class Man { private Dog bestFriend; }
- E. class Man { private Dog<bestFriend>; }
- F. class Man { private BestFriend<dog>; }

Answer: D

186. Given:

```
11. class Alpha {  
12. public void foo() { System.out.print("Afoo "); }  
13. }  
14. public class Beta extends Alpha {  
15. public void foo() { System.out.print("Bfoo "); }  
16. public static void main(String[] args) {  
17. Alpha a = new Beta();  
18. Beta b = (Beta)a;  
19. a.foo();  
20. b.foo();  
21. }  
22. }
```

What is the result?

- A. Afoo Afoo
- B. Afoo Bfoo
- C. Bfoo Afoo
- D. Bfoo Bfoo
- E. Compilation fails.
- F. An exception is thrown at runtime.

Answer: D

187. Given:

```
5. import java.util.Date;  
6. import java.text.DateFormat;  
21. DateFormat df;  
22. Date date = new Date();  
23. // insert code here  
24. String s = df.format(date);
```

Which code fragment, inserted at line 23, allows the code to compile?

- A. df = new DateFormat();
- B. df = Date.getFormat();
- C. df = date.getFormat();

D. `df = DateFormat.getFormat();`  
E. `df = DateFormat.getInstance();`

Answer: E

188. Given:

```
1. public class Base {  
2.     public static final String FOO = "foo";  
3.     public static void main(String[] args) {  
4.         Base b = new Base();  
5.         Sub s = new Sub();  
6.         System.out.print(Base.FOO);  
7.         System.out.print(Sub.FOO);  
8.         System.out.print(b.FOO);  
9.         System.out.print(s.FOO);  
10.        System.out.print(((Base)s).FOO);  
11.    } }  
12. class Sub extends Base {public static final String FOO="bar";}
```

What is the result?

- A. foofoofoofoofoo
- B. foobarfoobarbar
- C. foobarfoofoofoo
- D. foobarfoobarfoo
- E. barbarbarbarbar
- F. foofoofoobarbar
- G. foofoofoobarfoo

Answer: D

189 A company has a business application that provides its users with many different reports: receivables reports, payables reports, revenue projects, and so on. The company has just purchased some new, state-of-the-art, wireless printers, and a programmer has been assigned the task of enhancing all of the reports to use not only the company's old printers, but the new wireless printers as well. When the programmer starts looking into the application, the programmer discovers that because of the design of the application, it is necessary to make changes to each report to support the new printers. Which two design concepts most likely explain this situation? (Choose two.)

- A. Inheritance
- B. Low cohesion
- C. Tight coupling
- D. High cohesion
- E. Loose coupling
- F. Object immutability

Answer: B,C



190. A team of programmers is reviewing a proposed API for a new utility class. After some discussion, they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?

- A. Looser coupling
- B. Tighter coupling
- C. Lower cohesion
- D. Higher cohesion
- E. Weaker encapsulation
- F. Stronger encapsulation

Answer: A

191. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself. What design issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

192. Given:

1. public class TestString1 {
2. public static void main(String[] args) {
3. String str = "420";
4. str += 42;
5. System.out.print(str);
6. }
7. }

What is the output?

- A. 42
- B. 420
- C. 462
- D. 42042
- E. Compilation fails.
- F. An exception is thrown at runtime.

Answer: D

193. Given

11. public interface Status {
  12. /\* insert code here \*/ int MY\_VALUE = 10;
  13. }
- Which three are valid on line

12?

(Choose three.)

- A. final
- B. static
- C. native
- D. public
- E. private
- F. abstract
- G. protected

Answer: A,B,D

194. Given:

```
10. interface Foo { int bar(); }
11. public class Sprite {
12.     public int fubar( Foo foo ) { return foo.bar(); }
13.     public void testFoo() {
14.         fubar(
15.             // insert code here
16.         );
17.     }
18. }
```

Which code, inserted at line 15, allows the class Sprite to compile?

- A. Foo { public int bar() { return 1; } }
- B. new Foo { public int bar() { return 1; } }
- C. new Foo() { public int bar() { return 1; } }
- D. new class Foo { public int bar() { return 1; } }

Answer: C

195. Given:

```
11. class Animal { public String noise() { return "peep"; } }
12. class Dog extends Animal {
13.     public String noise() { return "bark"; }
14. }
15. class Cat extends Animal {
16.     public String noise() { return "meow"; }
17. } ...
30. Animal animal = new Dog();
31. Cat cat = (Cat)animal;
32. System.out.println(cat.noise());
```

What is the result?

- A. peep
- B. bark
- C. meow
- D. Compilation fails.

E. An exception is thrown at runtime.

Answer: E

196. A programmer has an algorithm that requires a `java.util.List` that provides an efficient implementation of `add(0, object)`, but does NOT need to support quick random access. What supports these requirements?

- A. `java.util.Queue`
- B. `java.util.ArrayList`
- C. `java.util.LinearList`
- D. `java.util.LinkedList`

Answer: D

197. Which `Man` class properly represents the relationship "Man has a best friend who is a Dog"?

- A. `class Man extends Dog { }`
- B. `class Man implements Dog { }`
- C. `class Man { private BestFriend dog; }`
- D. `class Man { private Dog bestFriend; }`
- E. `class Man { private Dog<bestFriend>; }`
- F. `class Man { private BestFriend<dog>; }`

Answer: D

199. A team of programmers is reviewing a proposed API for a new utility class. After some discussion,

they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?

- A. Looser coupling
- B. Tighter coupling
- C. Lower cohesion
- D. Higher cohesion
- E. Weaker encapsulation
- F. Stronger encapsulation

Answer: A

200. Given:

- 10. `interface Jumper { public void jump(); } ...`
- 20. `class Animal { } ...`
- 30. `class Dog extends Animal {`
- 31. `Tail tail; 32. } ...`
- 40. `class Beagle extends Dog implements Jumper{`
- 41. `public void jump() { }`
- 42. `} ...`
- 50. `class Cat implements Jumper{`
- 51. `public void jump() { }`
- 52. `}`

Which three are true?(Choose Three)

- A. Cat is-a Animal
  - B. Cat is-a Jumper
  - C. Dog is-a Animal
  - D. Dog is-a Jumper
  - E. Cat has-a Animal
  - F. Beagle has-a Tail
  - G. Beagle has-a Jumper
- Answer: B,C,F

201. Given:

10. interface Data { public void load(); }

11. abstract class Info { public abstract void load(); }

Which class correctly uses the Data interface and Info class?

- A. public class Employee extends Info implements Data {  
public void load() { /\*do something\*/ }  
}
- B. public class Employee implements Info extends Data {  
public void load() { /\*do something\*/ }  
}
- C. public class Employee extends Info implements Data {  
public void load(){ /\*do something\*/ }  
public void Info.load(){ /\*do something\*/ }  
}
- D. public class Employee implements Info extends Data {  
public void Data.load(){ /\*do something\*/ }  
public void load(){ /\*do something\*/ }  
}
- E. public class Employee implements Info extends Data {  
public void load(){ /\*do something\*/ }  
public void Info.load(){ /\*do something\*/ }  
}
- F. public class Employee extends Info implements Data{  
public void Data.load() { /\*do something\*/ }  
public void Info.load() { /\*do something\*/ }  
}

Answer: A

202. Given:

- 1. class TestA {
- 2. public void start() { System.out.println("TestA"); }
- 3. }
- 4. public class TestB extends TestA {
- 5. public void start() { System.out.println("TestB"); }
- 6. public static void main(String[] args) {

```
7. ((TestA)new TestB()).start();
8. }
9. }
```

What is the result?

- A. TestA
- B. TestB
- C. Compilation fails.
- D. An exception is thrown at runtime.

Answer: B

```
203. import java.io.*;
class files {
public static void main(String args[]) {
File obj = new File("/java/system");
System.out.print(obj.getName());
}
}
```

- a) java
- b) system
- c) java/system
- d) /java/system

Answer: B

```
204. import java.io.*;
class files {
public static void main(String args[]) {
File obj = new File("/java/system");
System.out.print(obj.getAbsolutePath());
}
}
```

Note: file is made in c drive.

- a) java
- b) system
- c) java/system
- d) /java/system

View Answer

Answer: d

Explanation: None.

Output:

```
$ javac files.java
$ java files
\java\system
```

205. What is the output of this program?

```
import java.io.*;
class files {
public static void main(String args[]) {
File obj = new File("/java/system");
System.out.
}
}
```

Note: file is made in c drive.

- a) java
- b) system
- c) java/system
- d) /java/system

Answer:D

```
206. class exception_handling {
public static void main(String args[]) {
try {
int a = args.length;
int b = 10 / a;
System.out.print(a);
try {
if (a == 1)
a = a / a - a;
if (a == 2) {
int c = {1};
c[8] = 9;
}
}
catch (ArrayIndexOutOfBoundsException e) {
System.out.println("TypeA");
}
catch (ArithmeticException e) {
System.out.println("TypeB");
}}}}
}}}
```

- a) TypeA
- b) TypeB
- c) 0TypeA

Important link:

<http://javabysri.blogspot.in/2012/06/jdbc-objective-questions.html>

MOCK TEST 1

Q 1 - What is spring?

- A - Spring is an open source development framework for enterprise Java.
- B - Spring is a proprietary framework.
- C - Spring is a development framework for .Net applications.
- D - Spring is a development framework for PHP based applications.

A

Q 2 - Which of the following is correct assertion about spring?

- A - Spring enables developers to develop enterprise-class applications using POJOs.
- B - Spring is organized in a modular fashion.
- C - Testing an application written with spring is simple because environment-dependent code is moved into this framework.
- D - All of above.

D

Q 3 - What is Dependency Injection?

- A - It is a design pattern which implements Inversion of Control for software applications.
- B - It is one of the spring module.
- C - It is a technique to get dependencies of any project.
- D - It is used to promote tight coupling in code.

A

Q 4 - Which of the following is correct about dependency injection?

- A - It helps in decoupling application objects from each other.
- B - It helps in deciding the dependencies of objects.
- C - It stores objects states in database.
- D - It stores object states in file system.

A

Q 5 - What AOP stands for?

- A - Aspect Oriented Programming
- B - Any Object Programming
- C - Asset Oriented Programming
- D - Asset Oriented Protocol

A

Q 6 - What is true about cross-cutting concerns?

- A - The functions that span multiple points of an application are called cross cutting concerns.
- B - Cross-cutting concerns are conceptually separate from the application's business logic.
- C - Logging is one of the examples of cross cutting concerns.
- D - All of the above.

A

Q 7 - Which are the modules of core container?

- A - Beans, Core, Context, SpEL
- B - Core, Context, ORM, Web
- C - Core, Context, Aspects, Test
- D - Bean, Core, Context, Test

A

Q 8 - Which are the modules of Data Access/ integration layer?

- A - JDBC, ORM, OXM, JMS, Transactions
- B - JDBC, ORM, OXM, JMS
- C - JDBC, ORM, Web, Beans
- D - JDBC, ORM, OXM, JMS

A

Q 9 - Which are the modules of Web layer?

- A - WebSocket, Servlet, Web, Portlet
- B - WebSocket, Servlet, Web-MVC, Web
- C - HTML, JSP, WEB, Portlet
- D - HTML, Servlet, WEB, Portlet

A

Q 10 - Which of the statement is not correct?

- A - Core and beans modules provide the fundamental parts of the framework, including Dependency Injection feature.
- B - The SpEL module provides a powerful Expression Language for querying and manipulating an object graph at runtime.
- C - Aspects module provides integration with AspectJ.
- D - None of the above.

D

Q 11 - Which of the statement is correct?

- A - The JDBC module provides a JDBC-abstraction layer that removes the need to do tedious JDBC related coding.
- B - The ORM module provides integration layers for popular object-relational mapping APIs, including JPA, JDO, Hibernate, and iBatis.
- C - The Java Messaging Service JMS module contains features for producing and consuming messages.
- D - All of the above.

D

Q 12 - Which of the statement is correct?

- A - The AOP module provides aspect-oriented programming implementation allowing you to



define method-interceptors and pointcuts to cleanly decouple code that implements functionality that should be separated.

B - The Aspects module provides integration with AspectJ - Which is again a powerful and mature aspect oriented programming AOP framework.

C - The Instrumentation module provides class instrumentation support and class loader

implementations to be used in certain application servers.

D - All of the above.

D

Q 13 - What types of Dependency injection does spring supports?

A - Constructor based, Setter based

B - Constructor based, Setter based, Getter Based

C - Setter based, Getter based, Properties based

D - Constructor based, Setter based, Properties based

A

Q 14 - Which are the IoC containers in Spring?

A - BeanFactory, ApplicationContext

B - BeanFactory, ApplicationContext, IocContextFactory

C - BeanFactory, BeanContext, IocContextFactory

D - BeanFactory, ApplicationContext, BeanContext

A

Q 15 - Which is the correct implementation class of BeanFactory?

A - XmlBeanFactory

B - ClassPathBeanFactory

C - FileSystemBeanFactory

D - AdvancedBeanFactory

A

Q 16 - Which are the correct implementation classes of ApplicationContext?

A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext

B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext

C - AdvancedApplicationContext, FileApplicationContext

D - FileSystemApplicationContext, ClassPathApplicationContext

A

Q 17 - Which of the following stands true for spring beans?

A - Spring beans are managed by the Spring IoC container.

B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.

C - Spring beans are simple POJOs.

D - All of the above.

D

Q 18 - Which is the way to provide configuration metadata to spring?

- A - XML Based configuration file.
- B - Annotation based configuration.
- C - Java based configuration.
- D - All of the above.

D

Q 19 - What is bean scope?

- A - Bean scope forces Spring to produce a new bean instance as per the scope defined.
- B - Bean scope defines the accessibility of bean in a java class.
- C - Bean scope defines the accessibility of bean in a java package.
- D - Bean scope defines the accessibility of bean in a web application.

A

Q 20 - What is singleton scope?

- A - This scopes the bean definition to a single instance per Spring IoC container.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 21 - What is prototype scope?

- A - This scopes a single bean definition to have any number of object instances.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 22 - What is request scope?

- A - This scopes a bean definition to an HTTP request.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP Session.
- D - This scopes the bean definition HTTP Application/ Global session.

A

Q 23 - What is session scope?

- A - This scopes a bean definition to an HTTP session.
- B - This scopes the bean definition to Spring IoC container.

- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Application/ Global session.

A

Q 24 - What is global-session scope?

- A - This scopes a bean definition to an HTTP Application/ Global session.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Session.

A

Q 25 - What is default scope of bean in Spring framework?

- A - singleton
- B - prototype
- C - request
- D - session

A

Q 26 - How can you inject Java Collection in Spring?

- A - Using list, set, map or props tag.
- B - Using list, set, map or collection tag.
- C - Using list, set, props or collection tag.
- D - Using list, collection, map or props tag.

A

Q 27 - What is true about <list> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

A

Q 28 - What is true about <set> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

B

Q 29 - What is true about <map> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This tag is not supported.

C

Q 30 - What is true about <props> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

D

Q 31 - What is bean autowiring?

- A - Autowiring lets Spring resolve collaborators otherbeans for your bean by inspecting the contents of the BeanFactory without using <constructor-arg> and <property> elements.
- B - Autowiring injects values in spring beans.
- C - Autowiring injects one bean into another.
- D - Autowiring helps in wiring a list of values, allowing duplicates.

A

Q 32 - Which are the different modes of autowiring?

- A - no, byName, byType, constructor, autodetect
- B - no, byName, byType, constructor, autocorrect
- C - byName, byContent, constructor, autodetect
- D - byName, byContent, setter, autodetect

A

Q 33 - What is no mode of autowiring?

- A - Default setting which means no autowiring and you should use explicit bean reference for wiring.
- B - Autowiring by property name.
- C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.
- D - Similar to byType, but type applies to constructor arguments.

A

Q 34 - What is byName mode of autowiring?

- A - Default setting which means no autowiring and you should use explicit bean reference for

wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Similar to byType, but type applies to constructor arguments.

B

Q 35 - What is byType mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in configuration file.

D

Q 36 - What is constructor mode of autowiring?

A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in configuration file.

D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

D

Q 37 - What is autodetect mode of autowiring?

A - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with

exactly one of the beans name in configuration file.

C

Q 38 Can you inject null and empty string values in Spring?

A - Yes

B - No

A

Q 39 - How do you turn on annotation wiring?

A - Add <annotation-context:config /> to bean configuration.

B - Add <annotation-config /> to bean configuration.

C - Add <annotation-context-config /> to bean configuration.

D - Add <context:annotation-config/> to bean configuration.

D

Q 40 - What does @Required annotation mean?

A - This annotation indicates that bean property must be populated by the user.

B - This annotation indicates that bean property is required while saving the bean data to database.

C - This annotation simply indicates that the affected bean property must be populated at configuration time, through an explicit property value in a bean definition or through autowiring.

D - This annotation indicates that bean property is required while serializing the bean.

C

Q 41 - What is true about @Autowired annotation?

A - The @Autowired annotation can be used to autowire bean on the setter method.

B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.

C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.

D - All of above.

B

Q 42 - What is ContextRefreshedEvent event?

A - This event is published when the Servlet Context is either initialized or refreshed.

B - This event is published when the HTTP Request is received.

C - This event is published when the HTTP Response is returned.

D - This event is published when the ApplicationContext is either initialized or refreshed.

D

Q 43 - What is ContextStartedEvent event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the HTTP Request is received.
- C - This event is published when the ApplicationContext is started using the start method on the ConfigurableApplicationContext interface.
- D - This event is published when the HTTP Response is returned.

C

Q 44 - What is ContextStoppedEvent event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the ApplicationContext is stopped using the stop method on the ConfigurableApplicationContext interface.
- C - This event is published when the HTTP Request is received.
- D - This event is published when the HTTP Response is returned.

B

Q 45 - What is ContextClosedEvent event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the HTTP Request is received.
- C - This event is published when the HTTP Response is returned.
- D - This event is published when the ApplicationContext is closed using the close method on the ConfigurableApplicationContext interface.

D

Q 46 - What is RequestHandledEvent:event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the HTTP Request is received.
- C - This event is published when the HTTP session is initialized or refreshed.
- D - This event is published when the HTTP Request is serviced.

C

Q 47 - What is aspect?

- A - Aspect is a way to do the dependency injection.
- B - A module which has a set of APIs providing cross-cutting requirements.
- C - Aspect is used to log information of application.
- D - Aspect represents properties of spring based application.

B

Q 48 - What is Join point?

- A - This represents a point in your application which joins two objects.
- B - This represents a point in your object where you join values.
- C - This represents a point in your object where you join injected values.
- D - This represents a point in your application where you can plug-in AOP aspect.

D

Q 49 - What is Advice?

- A - This is the way to instruct object to behave in certain manner.
- B - This is used to inject values in objects.
- C - This is the actual action to be taken either before or after the method execution.
- D - This is not invoked during program execution by Spring AOP framework.

C

Q 50 - What is Pointcut?

- A - This represents a point in your application where you can plug-in AOP aspect.
- B - This is a set of one or more joinpoints where an advice should be executed.
- C - This is used to inject values in objects.
- D - This is invoked during program execution by Spring AOP framework.

B



## MOCK TEST 2

Q 1 - What is Introduction?

- A - An introduction represents a point in your application where you can plug-in AOP aspect.
- B - This is used to inject values in objects.
- C - This is not invoked during program execution by Spring AOP framework.
- D - An introduction allows you to add new methods or attributes to existing classes.

D

Q 2 - What is Target object?

- A - A represents a object in your application where you can plug-in AOP aspect.
- B - The object being advised by one or more aspects, this object will always be a proxy object, also referred to as the advised object.
- C - This is used to inject values in objects.
- D - This is not invoked during program execution by Spring AOP framework.

B

Q 3 - What is Weaving?

- A - Weaving is the process of injecting values in objects to create an advised object.
- B - Weaving is the process of linking aspects with other application types or objects to create an advised object.
- C - This is used to inject values in objects.
- D - Weaving is used to check object dependencies.

B

Q 4 - What are the different points where weaving can be applied?

- A - Compile time, load time
- B - Compile time, run time
- C - Run time
- D - Compile time, load Time, Run time

D

Q 5 - What are the types of advice?

- A - then, after, after-returning, after-throwing, around
- B - When, after, after-returning, around
- C - Where, after, after-returning, after-throwing, around
- D - Before, after, after-returning, after-throwing, around

D

Q 6 - How before advice works?

- A - Run advice before a class loads.
- B - Run advice before a method execution.
- C - Run advice before http response is to be returned.
- D - Run advice before http request is to be processed.

B

Q 7 - How after-returning advice works?

A - Run advice after a class loads only if class loads successfully.

B - Run advice after a method execution only if method completes successfully.

C - Run advice after http response is returned only if http response is success.

D - Run advice after http request is processed with no exception.

B

Q 8 - How after-throwing advice works?

A - Run advice after a method execution only if method exits by throwing an exception.

B - Run advice after a class loads only if class throws exception during load time.

C - Run advice after http response is returned with error status.

D - Run advice after http request is processed and an exception occurred.

A

Q 9 - How around advice works?

A - Run advice before and after the advised method is invoked.

B - Run advice before and after a class is loaded.

C - Run advice before and after http response is returned.

D - Run advice before and after http request is processed.

A

Q 10 - Which of the following aspect implementation spring supports?

A - XML Schema based aspect implementation

B - @AspectJ based aspect implementation

C - Both of above.

D - None of above.

C

Q 11 - What are the types of the transaction management Spring supports?

A - Programmatic transaction management

B - Declarative transaction management

C - Both of above.

D - None of above.

C

Q 12 - What is Spring MVC framework?

A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready

components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

B

Q 13 - What is @Controller annotation?

A - The @Controller annotation indicates that a particular class serves the role of a controller.

B - The @Controller annotation indicates how to control the transaction management.

C - The @Controller annotation indicates how to control the dependency injection.

D - The @Controller annotation indicates how to control the aspect programming.

A

Q 14 - What are the ways to access Hibernate by using Spring?

A - Inversion of Control with a Hibernate Template and Callback.

B - Extending HibernateDAOSupport and Applying an AOP Interceptor node.

C - Both of above.

D - None of above.

C

Q 15 - Which ORM Spring supports ?

A - Hibernate

B - iBatis

C - JPA

D - All of above.

E - None of above.

D

Q 16 - Which of the following database is not supported using jdbcTemplate?

A - MySql

B - PostgreSQL

C - NoSql

D - Oracle

C

Q 17 - How to get object of a service in spring framework?

A - Using new keyword

B - Using dependency injection

B

Q 18 - Which of the following is part of Data Access layer in Spring framework?

A - Beans

- B - Aspects
- C - JMS
- D - Context

C

Q 19 - How to use ref keyword in beans.xml?

- A - Using setter method only.
- B - Using constructor argument only.
- C - Using setter method and constructor argument both.
- D - None of the above.

C

Q 20 - Core container has AOP as one of its module.

- A - True
- B - False

B

Q 21 - SpEL is part of core container.

- A - False
- B - True

B

Q 22 - Which class acts as IoC Container?

- A - ServletContext
- B - DispatcherServlet
- C - ApplicationContext
- D - None of the above

C

Q 23 - What stands true for spring framework?

- A - Spring framework is a light weight framework.
- B - Spring framework is a heavy weight framework.

A

Q 24 - Expression Language/ SpEL was introduced in which version of spring framework.

- A - 1.0
- B - 2.0
- C - 3.0
- D - 4.0

C

Q 25 - Can we integrate Struts with Spring.

- A - Yes

B - No

A

Q 26 - By default a bean is lazily loaded.

A - True

B - False

B

Q 27 - By default a bean is eagerly loaded.

A - False

B - True

B

Q 28 - If a bean is scoped to HTTP request, scope is

A - session

B - global-session

C - prototype

D - request

D

Q 29 - If a bean is created once per Ioc Container, scope is

A - singleton

B - global-session

C - prototype

D - request

A

Q 30 - Thread scoped bean is introduced in which version of spring framework.

A - 1.0

B - 2.0

C - 3.0

D - 4.0

C

Q 31 - If a bean can be created any number of times, scope is

A - session

B - global-session

C - prototype

D - request

C

Q 32 - What is the scope of stateless bean?

A - global-session

B - singleton

C - prototype

D - request

B

Q 33 - What is the scope of stateful bean?

A - session

B - global-session

C - prototype

D - request

C

Q 34 - If a bean is scoped to HTTP session, scope is

A - global-session

B - session

C - prototype

D - request

B

Q 35 - How to handle shut down of IoC container?

A - Using shutdownHook

B - Using shutdownHandler

C - Using registerHook

D - Using registerShutdownHook

D

Q 36 - How bean life cycle can be controlled?

A - Using init only

B - Using InitializingBean class only

C - Using DisposableBean class only

D - Using All of above

D

Q 37 - What is the scope of bean in portlet context?

A - session

B - global-session

C - prototype

D - request

B

Q 38 - How after advice works?

A - Run advice after a method execution regardless of its outcome.

B - Run advice after a class loads.

C - Run advice after http response is returned.

D - Run advice after http request is processed.

A

Q 39 - Which class is used to map a database row to a java object in spring?

- A - ResultSet
- B - RowMapper
- C - RowSetMapper
- D - ResultSetMapper

B

Q 40 A bean must have id attribute in beans configuration file.

- A - True
- B - False

B

Q 41 - Which of the following class can be used to execute Sql queries in spring?

- B - JDBCHelper
- A - JdbcTemplate
- C - DBHelper
- D - DBTemplate

A

Q 42 - Which of the following class can be used to call Stored Procedures in spring?

- A - SPHelper
- B - JdbcTemplateCall
- C - JdbcTemplate
- D - SimpleJdbcCall

D

Q 43 - What is a DispatcherServlet?

- A - DispatcherServlet is used for transaction management.
- B - DispatcherServlet is used for AOP.
- C - DispatcherServlet handles all the HTTP requests and responses.
- D - DispatcherServlet is used for Dependency injection.

C

Q 44 - What is ACID in transactional management?

- A - Accurate, Controlled, Isolation, Durability
- B - Atomicity, Consistency, Independent, Done
- C - Atomicity, Consistency, Isolation, Durability
- D - Accurate, Controlled, Independent, Done

C

Q 45 - Where do you define DispatcherServlet?

- A - In Beans configuration file.
- B - Web.xml file
- C - Meta-inf/dispatcher.xml
- D - Web-inf/ dispatcher.xml

B

Q 46 - What is true about BeanPostProcessor?

- A - It is a concrete class.
- B - It is an interface.
- C - It is an abstract class.
- D - None of the above.

B

Q 47 - What BeanPostProcessor does?

- A - It processes beans once a bean is initialized.
- B - It defines callback methods that you can implement to provide your own instantiation logic, dependency-resolution logic etc.
- C - It processes beans once a bean is loaded.
- D - It processes beans once a bean exits.

B

Q 48 - Can be bean be configured to have an inner bean?

- A - True
- B - False

A

Q 49 - Can we inject value and ref both together in a bean?

- A - True
- B - False

A

Q 50 - Following class can be extended to create custom event in spring.

- A - SpringEvent
- B - Event
- C - ApplicationEvent
- D - None of above

C



1. Does Spring provide programmatic transaction management? Select a unique answer.

- a. Yes using the `@Transactional` bean post processor
- b. Yes with the `TransactionTemplate` class
- c. Yes with the `TransactionService` class
- d. Yes using the `@Transactional` bean post processor

ans: b

Explanation

- 1. The `TransactionTemplate` class provides an `execute(TransactionCallback)` method
- 2. The `TransactionService` class does not exist
- 3. The `@Transactional` annotation is for declarative transaction management

2. How could you externalize constants from a Spring configuration file or a Spring annotation into a `.properties` file? Select one or more

- a. By using the `<context:property-placeholder />` tag
- b. By using the `<util:constant />` tag
- c. By declaring the `ConstantPlaceholderConfigurer` bean post processor
- d. By using the `c:` namespace

ans: a

3. To validate Java beans in a web application using annotations.

- a) XML
- b) Java Based
- c) JAR-303 standard
- d) All of the mentioned

ans: d

4. Annotation used to indicate a field has to have a minimum of 2 characters.

- a) `@NotNull`
- b) `@Size`
- c) `@MaxSize`
- d) `@size`

Answer: b

Explanation: `@Size` annotation used to indicate a field has to have a minimum of 2 characters.

5. Spring MVC supports generating Excel files using which of the following libraries.

- a) Apache POI library
- b) JExcelAPI library
- c) All of the mentioned

d) None of the mentioned

ans: C

6. Interface for Dispatcher Servlet to auto detect view resolver beans.

- a) localeResolver
- b) tiles
- c) ViewResolver
- d) none of the mentioned.

ans: C

Explanation: Spring MVC comes with several ViewResolver implementations for you to resolve views using different strategies.

7.By default, XmlViewResolver loads view beans from /WEB-INF/views.xml which is final.

- a) True
- b) False

ans: b

8.Views that can't be resolved byInternalResourceViewResolver.

- a) redirect
- b) redirect:prefix
- c) redirect:suffix
- d) all of the mentioned

ans: a

9. Annotation which allows a controller's field to be assigned using Spring Expression Language(SpEL)

- a) @Value
- b) @After
- c) @Default
- d) @None

ans: a

10. SpEL statements are recognizable.

- a) True
- b) False

View Answer

Answer: a

Explanation: They use a notation in the form “#{ SpEL statement }”.

11. To publish a REST service with Spring.

- a) publishing an application's data as a REST service
- b) accessing data from third-party REST services
- c) none of the mentioned
- d) all of the mentioned

ans: a b

12. Publishing an application's data as a REST service requires.

- a) @RequestMapping
- b) @PathVariable
- c) All of the mentioned
- d) None of the mentioned

ans: a b

13. Annotation added as an input parameter to the handler method.

- a) @PathVariable
- b) @Path
- c) @PathLocale
- d) None of the mentioned

ans: a

14. Which of the following statements is true regarding the @ResponseStatus annotation?

- a. @ResponseStatus is detected on nested exceptions
- b. TheExceptionHandlerExceptionResolver uses the @ResponseStatus annotation to map exception to HTTP status code
- c. A controller handler is annotated with the @ResponseStatus, the response status set by RedirectView takes precedence over the annotation value.
- d. The @ResponseStatus annotation can go on a @RequestMapping method or a @RestController class or a

business exception class.

ans: a

15. Default localeResolver used by Spring.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) AcceptLocaleResolver

ans: c

16. Alternative way to resolve locales.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) SessionLocaleResolver

ans: d

17. General-purpose class that allows a response to be rendered using a marshaller.

- a) MarshallingView
  - b) Marshalling
  - c) View
  - d) All of the mentioned
- View Answer

Answer: a

Explanation: The ModelAndView type is defined as a MarshallingView type, which is a general-purpose class that allows a response to be rendered using a marshaller.

18. Annotation which allows the Jaxb2Marshaller marshaller to detect a class's (i.e., object's) fields.

- a) @XmlRootElement
- b) @XmlRoot
- c) @NotNull
- d) None of the mentioned

ans: a

19. How to auto-inject into a field a Spring bean by its name? Select one or more answer choices.

- a. By using both the `@Autowired` and the `@Qualifier` Spring annotations
- b. By using the `@Autowired` annotation and naming the field with the bean name
- c. With the name attribute of the `@Autowired` annotation
- d. By using the single `@Qualifier` annotation

ans : a

20. Considering 2 classes `AccountServiceImpl` and `ClientServiceImpl`. Any of these 2 classes inherits from each other. What is the result of the following pointcut expression?

```
execution(* *..AccountServiceImpl.update(..))  
&& execution(* *..ClientServiceImpl.update(..))
```

- a. No joint point is defined
- b. Matches public update methods of the 2 classes, whatever the arguments
- c. Matches any update methods of the 2 classes, whatever the arguments and method visibility
- d. Matches any update methods of the 2 classes, with one more argument and whatever method visibility

ans: a

21. Select the right statement about referring a Spring configuration file inside the package `com.example.myapp` in the below example?

```
ApplicationContext context = new  
ClassPathXmlApplicationContext("classpath:/com.example.myapp.config.xml");
```

- a. all of the above
- b. The `classpath:` prefix could be omitted
- c. Package name using the dot character is not well formatted
- d. The slash character preceding `com.example` could be omitted

ans: a

22. Using the Spring AOP framework, what is the visibility of the method matches by the following join point?

```
@Pointcut("execution(* *(..))")  
private void anyOperation() {};
```

- a. Public methods
- b. All methods, whatever their visibility
- c. All methods, except private method

d. Protected and public methods

ans: a

23. What the name of the bean defined in the following configuration class? Select a single answer.

@Configuration

```
public class ApplicationConfig {
```

@Autowired

```
private DataSource dataSource;
```

@Bean

```
ClientRepository clientRepository() {
```

```
ClientRepository accountRepository = new JpaClientRepository();
```

```
accountRepository.setDataSource(dataSource);
```

```
return accountRepository;
```

```
}
```

```
}
```

a. clientRepository

b. JpaClientRepository

c. jpaClientRepository

d. Two beans are defined: a data source and a repository

ans: a

24. Which of the following is true regarding the below Spring controller?

@RestController

```
public class OwnerController {
```

@RequestMapping(value = "/owner/{ownerId}", method = RequestMethod.POST)

@ResponseBody

```
public Owner findOwner(@PathVariable("ownerId") int ownerId) {
```

```
return new Owner();
```

```
}
```

```
}
```

a. RequestMethod.GET method is more accurate than POST

b. @ResponseBody could be removed

c. @PathVariable should be replaced with the @PathParam annotation

d. Returning the 201 HTTP status code is better

ans: a

25. What is an advice? Select a unique answer.

- a. An action taken by an aspect at a particular join point
- b. A point during the execution of a program
- c. An aspect and a pointcut
- d. A predicate that matches join points

ans: a

26. What is the easiest method to write a unit test?

- a. `@RequestMapping("/displayAccount")`  
`String displayAccount(@RequestParam("accountId") int id, Model model)`
- b. `void displayAccount(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException`
- c. `void displayAccount(HttpServletRequest req, HttpSession Session)`  
`throws ServletException, IOException`
- d. `@RequestMapping("/displayAccount")`  
`String displayAccount(@PathVariable("accountId") int id, Model model)`

ans: a

27. Select method's signatures that match with the following pointcut:  
`execution(* com.test.service..*.*(*))`

- a. `void com.test.service.MyServiceImpl#transfert(Money amount)`
- b. `void com.test.service.account.MyServiceImpl#transfert(Money amount)`
- c. `void com.test.service.MyServiceImpl#transfert(Account account, Money amount)`
- d. `void com.test.service.account.MyServiceImpl#transfert(Account account, Money amount)`
- e. None of the above

ans: a

28. Given the Spring configuration file, which are the correct statements?

```
<bean class="com.spring.service.BankServiceImpl"  
p:bankName="NationalBank">  
</bean>
```

- a. The p namespace has to be declared
- b. NationalBank is a scalar value
- c. Bean id is bankServiceImpl
- d. The BankServiceImpl references a NationalBank bean

ans: a

29. Given the following configuration class, what are the correct affirmations? Select one or more answers.

```
public class ApplicationConfig {
```

```
    private DataSource dataSource;
```

```
    @Autowired
```

```
    public ApplicationConfig(DataSource dataSource) {
```

```
        this.dataSource = dataSource;
```

```
    }
```

```
    @Bean(name="clientRepository")
```

```
    ClientRepository jpaClientRepository() {
```

```
        return new JpaClientRepository();
```

```
    }
```

```
}
```

- a. Configuration annotation is missing
- b. Default or no-arg constructor is missing
- c. @Bean name is ambiguous
- d. @Bean scope is prototype

ans: a

30. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object? Select a unique answer.

- a. ResultSetExtractor
- b. RowMapper
- c. RowCallbackHandler
- d. ResultSetMapper

ans: a



1. Class used to have full access to the Spring context's life cycle machinery and dependency injection.

- A. DelegatingFilterProxy
- B. WebApplicationContextUtils.getRequiredWeb()
- C. WebApplicationUtils.getRequiredWebApplicationContext()
- D. None of the mentioned

2. Element used to specify access attributes for bean's methods.

- A. security:protect
- B. security:intercept
- C. security:intercept-security
- D. none of the mentioned

3. Stated related to Spring Web flows

- A. view
- B. action
- C. decision
- D. all of the mentioned

4. To enable Spring security for web flows.

- A. SecurityFlowExecutionListener
- B. SecurityFlowExecution
- C. FlowSecurity
- D. SecurityFlowListener

5. To load root application context at the start up.

- A. ContextListener
- B. ContextLoader
- C. ContextLoaderListener
- D. ContextEventListener

6. ActionSupport class provides a method to access spring's application context.

- A. getWebApplicationContext()
- B. getWebApplication()
- C. getApplicationContext()
- D. contextEventListener()

7. XML element to specify the property values for scripting bean

- A. lang:property
- B. bean:property
- C. script:property
- D. None of the mentioned

8. Annotation used to declare an introduction

A. Before

B. After

C. @DeclareParents

D. None of the mentioned

9. Is the following pointcut expression correct?

execution(\* ArithmeticCalculator.\*(..))

A. Yes

B. No

C. If every target class is in same package

D. Depends where target class is located

10. DAO methods require access to the session factory, which can be injected:-

A. a setter method

B. constructor argument

C. none of the mentioned

D. all of the mentioned

11. Which of the following Java Collection types can you inject in a Spring application?

1. None of these.

2. Using list, set, map or collection tag.

3. Using list, set, props or collection tag.

4. Using list, set, map or props tag.

5. Using list, collection, map or props tag.

12. What are the different types of AutoProxying?

a. BeanNameAutoProxyCreator

b. DefaultAdvisorAutoProxyCreator

c. Metadata autoproxyming

d. All of the above

e. none of the above

13. What are the various editors used in spring work?

a. PropertyEditor, URLEditor

b. ClassEditor, CustomDateEditor, StringTrimmerEditor

c. FileEditor, LocaleEditor, StringArrayPropertyEditor

d. All of the above

14. 1. How to use <ref> in spring framework?

a.<ref> is used with bean id.

b.<ref> is used with string values.

15. How to define Inner bean in spring?

a.Use <property/> only

b.Use <property/> or <constructor-arg/>.

16. Which property is replaced by c-namespace in spring?

a.<property/>

b.<constructor-arg>

17. By default a bean is eagerly initialized in spring?

a.True

b.False

18. By default a bean is not autowired?

a.True

b.False

19. Can the class which spring container makes subclass be final?

a.No

b.Yes

20. What is the scope of statefull bean in spring?

a.Singleton scope

b.Prototype scope

21. In which spring scope is a single bean instance created per IoC container?

a.Singleton scope

b.Request scope

22. In which spring scope can any number of instances of bean be created?

a.Request scope

b. **Prototype scope**

23. When a bean has scope limited to only HTTP request, that is called

a. **Request scope**

b. Session scope

24. When a bean has scope limited to HTTP session, that is called

a. Request scope

b. **Session scope**

25. Thread scope bean has been introduced in spring from the version

a. Spring 2.5

b. **Spring 3.0**

26. InitializingBean and DisposableBean is used to customize the nature of bean in spring.

a. **Yes**

b. No

27. How to shutdown the Spring IoC container gracefully in non-web applications?

a. Using shutdownNow()

b. **registerShutdownHook()**

28. Which of the following is correct about dependency injection?

A - **It helps in decoupling application objects from each other.**

B - It helps in deciding the dependencies of objects.

C - It stores objects states in database.

D - It stores object states in file system.

29. Which are the correct implementation classes of ApplicationContext?

A - **FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext**

B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext

C - AdvancedApplicationContext, FileApplicationContext

D - FileSystemApplicationContext, ClassPathApplicationContext

30. What is Spring MVC framework?

A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

31. How to reload my changes on Spring Boot without having to restart server?

a. `<dependency>`  
`<groupId>org.springframework.boot</groupId>`  
`<artifactId>spring-boot-devtools</artifactId>`  
`<optional>true</optional>`  
`</dependency>`

b. `<dependency>`  
`<groupId>org.springframework.boot</groupId>`  
`<artifactId>spring-boot-test</artifactId>`  
`<optional>true</optional>`  
`</dependency>`

c. `<dependency>`  
`<groupId>org.springframework.boot</groupId>`  
`<artifactId>spring-boot-web</artifactId>`  
`<optional>true</optional>`  
`</dependency>`

d. `<dependency>`  
`<groupId>org.springframework.boot</groupId>`  
`<artifactId>spring-boot-jasper</artifactId>`  
`<optional>true</optional>`  
`</dependency>`

32. Annotation to add security to your spring boot app?

a. `@EnableWebSecurity`

b. `@EnableSecurity`

c. `@EnableSpringSecurity`

33. What are the different implementations of Spring AOP ?

A: AspectJ

B: Spring AOP

C: JBoss AOP

D: All

E: None

34. How Spring Batch works?

a. step, ItemReader, ItemProcessor, ItemStreamWriter

b. step, ItemProcessor, ItemStreamWriter

c. step, ItemReader, ItemStreamWriter

35. LoggerInterceptor which implements the HandlerInterceptor. The LoggerInterceptor will override the which following methods

a. preHandlerBean(), postHandlerBean(), afterCompletion()

b. PreProcessHandler(), PostProcessHandler(), afterCompletion()

c. PreHandler(), PostHandler(), afterCompletion()

d. none



1. We need to tell AngularJS what part of our HTML page contains the AngularJS app. You do so by adding the . . . . attribute to the root HTML element of the AngularJS app.

- A) ng-app
- B) ag-app
- C) js-app
- D) aj-app

Ans: A

2. AngularJS can show or hide HTML depending on the state of data in the model. You do so using a set of AngularJS directives such as . . . . . which are created specifically for that purpose.

- A) ng-shown, ng-hidden
- B) ng-show, ng-hide
- C) nt-show, nt-hide
- D) ng-shows, ng-hides

Ans: B

3. EventEmitter class's simple interface, which basically encompass two methods . . . . . can therefore be used to trigger custom events and listen to events as well, both synchronously or asynchronously.

- A) exit()
- B) superscript()
- C) subscribe()
- D) emit()

Ans: C

4. The . . . . . directive substitutes the normal href property and makes it easier to work with route links in Angular 2. Moreover if we want to navigate between routes, we use the . . . . . directive.

- A) RouterLinks
- B) RouterLinking
- C) RouterLink
- D) RouterLinq

Ans: C

5. Three main components of Routing are . . . . .

- A) RouteObject
- B) RouterOutlet
- C) RouterState
- D) RouterLink

Ans: A,B,D

6. Route . . . . . allow us to pass values in our url to our component so we can dynamically change our view content.



- A) pipes
- B) modules
- C) variables
- D) parameters

Ans: D

7. To use observables, Angular uses a third-party library called . . . . .

- A) ng2-bootstrap
- B) angular-cli
- C) numeral
- D) Reactive Extensions (RxJS)

Ans: D

8. Observables help you manage . . . . . data.

- A) synchronous
- B) asynchronous
- C) Both asynchronous & synchronous
- D) None of above

Ans: B

9. We subscribe to the observable ourselves using the actual . . . . . method. The downside is that you have to manage the subscription yourself.

- A) subscribe()
- B) subs()
- C) subscribed()
- D) None of above

Ans: A

10. A promise represents the final result of an . . . . . operation.

- A) asynchronous
- B) synchronous

Ans: A

11. The promise becomes . . . . . after resolving or rejecting.

- A) mutable
- B) ignored
- C) immutable
- D) recurring

Ans: C

12. Which of the following is not built-in pipe in Angular

- A) DatePipe

- B) CurrencyPipe
- C) DataPipe
- D) PercentPipe

Ans: C

13. If you chain multiple pipes together, they are executed . . . . .

- A) in parallel
- B) LIFO order
- C) in the order in which you specify them
- D) none of above

Ans: C

14.The . . . . . filter waits for a promise and returns the most recent value received. It then updates the view.

- A) sync
- B) async
- C) promise
- D) custom

Ans: B

15.The . . . . . decorator allows us to define the pipe name that is globally available for use in any template in the across application.

- A) @pipeName
- B) @pipeDeco
- C) \$Pipe
- D) @Pipe

Ans: D

16. Data bindings are applied as . . . . . or as special sequences of characters in strings.

- A) expressions
- B) attributes on HTML elements
- C) elements itself
- D) none of above

Ans: B

17. A data binding has these four parts:

- A) host element, curly brackets, target, expression
- B) host element, square brackets, target, evaluation
- C) host element, square brackets, target, expression
- D) host element, parentheses, target, expression

Ans: C

18. Attribute binding is similar to property binding but is tied to the . . . . . rather than the DOM property.

- A) Page attribute
- B) DOM attribute
- C) Root attribute
- D) HTML attribute

Ans: D

19. To make your applications ready for animations, you must include the AngularJS Animate library and then you must refer to the . . . . . module in your application.

- A) ng-repeat
- B) ngAnimation
- C) ngAnimate
- D) None of above

Ans: C

20. Using the . . . . . animation function, the parent animation can allow the child animation to run at the exact correct time.

- A) animateChild
- B) childAnimate
- C) animateInner
- D) innerChild

Ans: A

21. Animations are fired using animation property bindings which are prefixed with an . . . . symbol.

- A) &
- B) #
- C) \$
- D) @

Ans: D

22.. . . . . is an animation-specific function that is designed to be used inside of Angular's animation DSL language. It is used to kick off a reusable animation that is created using animation().

- A) useAnimation
- B) reuseAnimation
- C) in-useAnimation
- D) All of above

Ans: A

23. It's always possible to make use of animation input parameters by setting even more data via the . . . . . property.

- A) options.param
- B) option.params

- C) option.param
- D) options.params

Ans: D

24. Which of the following bootstrap styles are used to create a vertical pills navigation?

- A. .nav, .nav-tabs
- B. .nav, .nav-pills
- C. .nav, .nav-pills, .nav-stacked
- D. .nav, .nav-tabs, .nav-justified

Ans: C

97). Which of the following bootstrap styles are used to create a justified tabs navigation?

- A.nav, .nav-tabs
- B.nav, .nav-pills
- C.nav, .nav-pills, .nav-stacked
- D.nav, .nav-tabs, .nav-justified

Answer : D

25.Which class indicates a dropdown menu?

- a) .dropdown-list
- b) .select
- c) .dropdown

Ans: C

26. Ques. Bootstrap's grid system allows up to

- a. 6 columns across the page
- b. 12 columns across the page
- c. columns across the page
- d. columns across the page

Ans: B

27. Which of the following class makes round corner image for automatically adjust to fit the size of the screen?

- a. .img-res-image
- b. .img-responsive-image \* img-fluid (for full width responsible)
- c. .img-responsive
- d. .img-res

Ans: C

28. Which class should be used to indicate a button group?

- a. btn-group-buttons
- b. btn-group
- c. btn-grp
- d. btn-buttons

Ans: B

29. On the opposite side of event bindings (()) lie Angular's square-bracket syntax ([[]]) which signify a . . . . .

- A) property binding
- B) class binding
- C) style binding
- D) both A & B

Ans: A

30. Which object oriented terms are supported by Typescript?

- A. Modules B. Classes C. Interfaces D. All of these

Ans: D

31. What are the types of access modifiers supported by TypeScript?

- A. Public B. Private C. Protected D. All of these

Ans: D

---

31) Which of the following code will not throw an error?

- A. `JSON.parse("");`
- B. `JSON.parse(null);`
- C. `JSON.parse();`
- D. `JSON.parse({});`

Ans: B

32) What does JSONP stand for?

- A. JSON Procedures
- B. JSON Parsing
- C. JSON with padding
- D. JSON Programming

Ans: C

33) What is the value of json in the following code?

```
var obj = { fruit: 'apple', toJSON: function () { return 'orange'; } }; var json = JSON.stringify({x: obj});
```

- A. `{"x": "orange"}`
- B. `{"fruit": "apple"}`
- C. `{"x": "apple"}`
- D. `{"fruit": "orange"}`

Ans: A

34) What is the MIME type of JSON?

- A. `application/x-json`
- B. `text/json`
- C. `application/json`
- D. `application/javascript`

Ans: C

35)What function will convert a JavaScript object to a JSON string?

- A. JSON.text()
- B. JSON.serialize()
- C. JSON.toString()
- D. JSON.stringify()

Ans: D

36)In the below notation, Student is of type { “Student”: [ “Amy”, “Bob”, “John” ] }

- A. Not a valid JSON string
- B. Array
- C. Class
- D. Object

Ans:B

37)JSON-RPC is a remote procedure call protocol

38) To use observables, Angular uses a third-party library called . . . . .

- A) ng2-bootstrap
- B) angular-cli
- C) numeral
- D) Reactive Extensions (RxJS)

Ans:D

39) We subscribe to the observable ourselves using the actual . . . . . method. The downside is that you have to manage the subscription yourself.

- A) subscribe()
- B) subs()
- C) subscribed()
- D) None of above

Ans: A

40)Which of the following is not built-in pipe in Angular

- A) DatePipe
- B) CurrencyPipe
- C) DataPipe
- D) PercentPipe

Ans: C

41)The . . . . . filter waits for a promise and returns the most recent value received. It then updates the view.

- A) sync
- B) async
- C) promise

D) custom

Ans: B

42)If you chain multiple pipes together, they are executed . . . . .

- A) in parallel
- B) LIFO order
- C) in the order in which you specify them
- D) none of above

Ans:D

43)Observables help you manage . . . . . data.

- A. synchronous
- B. asynchronous
- C. Both asynchronous & synchronous
- D. None of above

Ans: B

44) The . . . . . decorator allows us to define the pipe name that is globally available for use in any template in the across application.

- A. pipeName
- B. pipeDeco
- C. Pipe
- D. None

Ans: C

45)console.log(x);  
var x = 5;

Ans: 5

46) RPC:-Remote Procedure call

47). Three main components of Routing are . . . . .

- A) RouteObject
- B) RouterOutlet
- C) RouterState
- D) RouterLink

Ans: A,B,D

48). Route . . . . . allow us to pass values in our url to our component so we can dynamically change our view content.

- A) pipes
- B) modules
- C) variables
- D) parameters

Ans: D

49). Promises are more advanced than Observables.

- A) True
- B) False

Ans: B

50). Which class is used to create a black navigation bar?

- a. .navbar-default
- b. .navbar-black
- c. .navbar-dark
- d. .navbar-inverse

Ans: D

51). The .container-fluid class provides

- a. Fixed width container
- b. Table format
- c. To create a Form
- d. Full width container

Ans: D

52).  $1+2+"3" = 33$  (Ans)

53.) CSS validator: -- csslint

54.) Create responsive images by adding an .img-responsive class to the <img> tag.

55.) img-responsive was in Bootstrap 3, img-fluid is in Bootstrap 4 since beta version.

56.) Bootstrap provides the .img-fluid class to make an image scale appropriately across devices. This class applies `max-width: 100%` and `height: auto` to the image. ... Resize your browser to see the image scale as the viewport becomes smaller.

57.) `<script src="myscripts.js">`

58.) Typescript uses which datatype?

- a) boolean
- b) string
- c) Number
- d) void
- e) All the above

Ans: e

59.) typescript support function overloading

60. css preprocessor

- less
- sass
- both (ans)

61. which prefix is used to add animation



@(ans)

62. default access modifier of typescript

-public(ans)

63. POP()

-Decrement by 1

64. how to implement inheritance in typescript

-extends(ans)

65. What keywords are reserved in JSON and cannot be used as keys?

- A. Value
- B. Object
- C. There are none.
- D. Key

Ans: C

66.) How to get information about browser?

- a) Browser
- b) version
- c) navigator

Ans: C

67.) which event will trigger if button will lost its focus?

:- onblur

68.) Display property?

- A) None
- B) inline
- c) inline-block
- d) block

Ans: All

69.) Shorthand method to represent background of (common)

Ans: background

70.) ----- is an HTML specification used to add more information to HTML tags.

- A) Macrodata
- B) Microdata
- c) Minidata
- d) Modifydata

Ans: B

71.) What is the purpose of the mimeType property of a plug-in entry?

- a) Contains MIME properties
- b) Contains MIME sizes
- c) Contains MIME types
- d) Contains MIME methods

Ans: C

72.) Which of the following can be implemented using animation?

- A) Fireworks
- B) fade Effect

- C) Roll-in or Roll-out
- D) All of the mentioned

Ans: D

- 73.) Which is the property that represents the content displayed in the window?
- A) document
  - B) content
  - C) window
  - D) frame

Ans: A

- 74.) How can we make methods available on all objects?
- A) Object.add(methods)
  - B) Object.methods(add)
  - C) Object.add.methods(...)
  - D) Object.prototype

Ans: D

- 75.) The four kinds of class members are\_\_\_\_\_

- a) Instance methods, Instance fields, Static method, Dynamic method
- b) Instance fields, Instance methods, Class fields, class methods
- c) Instance fields, non-Instance fields, Dynamic methods, Global methods
- d) Global methods, Local methods, Dynamic methods, Static methods

Ans: B

- 76.) Video tag :-- <video>

- 77.) Full Form of Angular CLI?

Ans: command-line interface.

- 78.) TYpe of binding:-

Ans: 4 Type: 1) Property Binding []  
2) Interpolation {{}}  
3) Event Binding ()  
4) Two way Binding {[]}

- 79.) Javascript is used for:- Interactivity

- 80.) How to declare variable in Typescript?

- A) Var
- B) Let
- C) Both

Ans: C

- 81.) Typescript uses which datatype?

- A) String
- B) Number
- C) Boolean
- D) All above option

Ans: D

82.) Default access modifier typescript?

- A) public
- B) private
- c)protected

Ans: A

83.) Super() in typescript?

Ans: Super Constructor

84.)Which field is required in <input>

- A)required
- B) minlength
- c)Both

Ans: A

84.)How to implements inheritance in typescript?

- A) implements
- B) extends

Ans: B

85.) Which keyword we can't use in JSON?

- A) Object
- B) key
- c) value
- D)None

Ans: D

86.) How can we add/ delete in array and returns the removed item.

- A)slice
- B) shift
- c)reverse
- D)splice

Ans: D.splice

87.) How to get information about browser

- A)Browser
- B)Version
- C)navigator

Ans:C

88.)Display property:

- 1)None
- 2)inline
- 3)inline-block
- 4)block

Ans: all 4

89.) Which event will trigger if (button..) will lost its focus:

Ans: onblur

90.) ans: prototype

91.) Drawback of typescript?

ans: all of the above

92.) Ans: 1row-2col

93.) Ans: Carosec

94.) Typescript is co.....

a) It supports overloading

b) partial

c) No

Ans: A

95.) Ans: HTML Attribute

96.) Ans::: n:child (nth)

97.) option: A)owner B)admin c)Both Ans: C

---

### **\*\*NOTES\*\* Pipe**

1.) You use your custom pipe the same way you use built-in pipes.

2.) You must include your pipe in the declarations array of the AppModule.

3.) You must manually register custom pipes. If you don't, Angular reports an error.

4.) "Power Boost Calculator" that combines your pipe and two-way data binding with ngModel.

5.) There are two categories of pipes: pure and impure. Pipes are pure by default.

pipe impure by setting its pure flag to false.

6.) The Angular AsyncPipe is an interesting example of an impure pipe. The AsyncPipe accepts a Promise or Observable as input and subscribes to the input automatically, eventually returning the emitted values.

7.) The JsonPipe provides an easy way to diagnosis a mysteriously failing data binding or inspect an object for future binding.

---

### **\*\* Animation**

1.) Angular animations are built on top of the standard Web Animations API and run natively on browsers that support it.

For other browsers, a polyfill is required.

2.) Animations are defined inside @Component metadata. Before you can add animations, you need to import a few animation-specific functions:

- 3.) It uses animations to transition between two states: active and inactive. When a hero is active, the element appears in a slightly larger size and lighter color.
- 4.) Angular animations are defined as logical states and transitions between states.
- 5.) When both directions of a transition have the same timing, you can use the shorthand syntax `<=>`:
- 6.) a) The `active => *` transition applies when the element's state changes from active to anything else.  
b) The `* => *` transition applies when any change between two states takes place.
- 7.) There are three timing properties you can tune for every animated transition: the duration, the delay, and the easing function. They are all combined into a single transition timing string.

Q. `JSON.parse()` function was developed as a safer alternative to

`Go function()`  
`eval()`  
`Function() eval`  
`call function()`

Ans: `eval()`

- \* fixed width `.container`
- \* full width `.container-fluid`

1. What is the basis for equality state in Java Identity HashMap?

- a) Hash Code and equals
- b) Name
- c) Reference

Answer: c) Reference

2. In Java, which of the following can be marked static?

- a) Methods, Variables, Initialization Blocks, Outer Classes, and Nested Classes
- b) Methods, Variables, Initialization Blocks, and Nested Classes
- c) Methods, Variables, Initialization Blocks, and Outer Classes
- d) Methods, Variables, and Initialization Block

Answer: b) Methods, Variables, Initialization Blocks, and Nested Classes

3. What is the output of the following Java code snippet?

```
public class home implements Runnable {  
    public void run() {  
        System.out.printf("%d",3);  
    }  
  
    public static void main(String[] args) throws InterruptedException {  
        Thread thread = new Thread(new home());  
        thread.start();  
        System.out.printf("%d",1);  
        thread.join();  
        System.out.printf("%d",2);  
    }  
}
```

- a) 132
- b) 123
- c) 213

d) 312

Answer: a) 132

4. Which of the following statements is false about Java main method?

- a) We can override main method.
- b) Main method should be declared public and static.
- c) Main method should have only 1 argument of type String array.
- d) We can overload main method.

Answer: a) We can override main method.

5. HTTP2 Client API is introduced in Java 9.

- a) True
- b) False

Answer: a) True

6. Pick the right statement about creating a custom Java annotation.

- a) Annotations are created by using @interface, followed by annotation name.
- b) An annotation can have elements as well. They look like methods. We should not provide implementation for these elements.
- c) All annotations extend java.lang.annotation.Annotation interface. Annotations cannot include any extends clause.
- d) All of the above

Answer: d) All of the above

7. What is the max capacity of a Java Blocking Queue?

- a) Integer.INFINITY
- b) Integer.MAX\_VALUE
- c) BigDecimal.MAX\_VALUE

Answer: b) Integer.MAX\_VALUE

8. Which of the following is the immutable Java List static factory method?

- a) List.of
- b) List.immutable
- c) List.create

Answer: a) List.of

9. What is the command line tool introduced for JS engine in Java 8?

- a) jjs
- b) jss
- c) jfs
- d) jbd

Answer: a) jjs

10. Which of these Java code snippets is a correct way of defining generic method?

- a) `public <T1, T2, ..., Tn> methodName() { }`
- b) `<T1, T2, ..., Tn> methodName(T1, T2, ..., Tn) { }`
- c) All of the above

Answer: a) `public <T1, T2, ..., Tn> methodName() { }`

11. In Java 9, private methods are allowed in Interfaces.

- a) True
- b) False

Answer: a) True

12. Java system properties can be set or modified at runtime.

- a) True
- b) False

Answer: a) True



13. Identify the Java class used to create a new Process.

- a) ProcessBuilder
- b) ProcessFactory
- c) Process
- d) Thread

Answer: a) ProcessBuilder

14. Pick the correct Java method reference usages.

- a) object::instanceMethod
- b) Class::staticMethod
- c) Class::new
- d) All of the above

Answer: d) All of the above

15. Which is the correct Java code snippet to be used to create a Functional Interface?

- a) interface MyFunctionalInterface implements Functional {
- b) @FunctionalInterface interface MyFunctionalInterface {
- c) @Functional interface MyFunctionalInterface {

Answer: b) @FunctionalInterface interface MyFunctionalInterface {

16. Which one will allow traversing a Java collection in each direction?

- a) ListIterator
- b) Iterator
- c) MapIterator
- d) SetIterator

Answer: a) ListIterator

17. In Java, which method is used to inform the observer of the change in observed object?

- a) Check

b) Update

c) Inform

d) notify

Answer: b) Update

18. In Java, it is NOT possible to override default serialization.

a) True

b) False

Answer: b) False

19. Pick the class that is NOT part of Java collection framework.

a) Map

b) List

c) Queue

d) Struct

Answer: d) Struct

20. Identify the parts of lambda expression in Java.

a) Name

b) Parameter list

c) Body

d) Return Type

Options:

1) i

2) iv

3) i and iv

4) ii and iii

Answer: 4) ii and iii

21. Java Functional Interface can be defined as \_\_\_\_\_.

- a) Simple Abstract Markup
- b) Simple Active Markup
- c) Single Ambivalue Method
- d) Single Abstract Method

Answer: d) Single Abstract Method

22. Which is the correct API to manage reactive streams in Java?

- a) `java.util.concurrent.Flow`
- b) `java.util.concurrent.Flow.Publisher`
- c) `java.util.concurrent.Flow.Processor`
- d) All of the above

Answer: d) All of the above

23. In Java, which of the following data structure classes CANNOT be type parameterized?

- a) List
- b) Set
- c) Array
- d) Map

Answer: c) Array

24. Which of these OOPS concept is enabled by method overriding in Java?

- a) Polymorphism
- b) Abstraction
- c) Inheritance
- d) Encapsulation

Answer: a) Polymorphism

25. What is the output of the following Java code snippet?

```

class First {
    public First() {
        System.out.println("F");
    }
}

class Second extends First {
    public Second() {
        System.out.println("S");
    }
}

class Third extends Second {
    public Third() {
        System.out.println("T");
    }
}

public class TestClass {
    public static void main(String[] args) {
        Third c = new Third();
    }
}

```

a) TSF

b) T

c) F

d) FST

Answer: d) FST

26. How does one assign system property values to variable using Java class fields?

a) `@Autowired("${aProperty}") private String aSysProp;`

b) `@Environment("${aProperty}") private String aSysProp;`

c) `@Property("${aProperty}") private String aSysProp;`

d) `@Value("${aProperty}") private String aSysProp;`

Answer: d) `@Value("${aProperty}") private String aSysProp;`

27. Which of the following types of Java methods CANNOT be Type parameterized?

a) Final methods

b) Overriding methods

c) Overloaded methods

Answer: b) Overriding methods

28. Pick the correct Java Stream API's?

a) `Stream.allMatch`

b) `Stream.noneMatch`

c) `Stream.anyMatch`

d) All the above

Answer: d) All the above

29. Which class is introduced in Java 8 to get the current milliseconds?

a) `Clock`

b) `Current Time`

c) `Date Time`

d) `Time`

Answer: a) `Clock`

30. In Java, which of these class types supports sequential and parallel processing of data?

a) `Queue`

b) `Lambda`

c) `ExecutorService`

d) `Streams`

Answer: d) `Streams`

31. Which Java Set implementation is sorted and synchronized?

- a) TreeSet
- b) LinkedHashSet
- c) HashSet
- d) ConcurrentSkipListSet

Answer: d) ConcurrentSkipListSet

32. Which of the following statements is NOT correct for JVM?

- a) JVM acts as a translator that translates different Machine code (on the basis of Host Machine) for a common Byte Code.
- b) JVM is a Virtual Machine that acts as an intermediary between Java Application and Host Operating System.
- c) JVM reads Byte Code and generates Machine Code.
- d) JVM reads Source Code and generates Byte Code.

Answer: d) JVM reads Source Code and generates Byte Code.

33. Is the following a valid Java code snippet?

```
Optional<String> aValue = Optional.of("Not Empty");  
if (aValue.isPresent()) {  
    System.out.println("Got Not Empty");  
}
```

- a) True
- b) False

Answer: b) False

34. Java Stream does NOT store the elements. It simply performs the aggregate operations to get the desired stream of data.

- a) True

b) False

Answer: a) True

35. Which Java class is used to apply reduction or aggregate functions on streams?

a) Filters

b) Reduction

c) Collectors

Answer: c) Collectors

36. Which of the following statements are true about Java finally block?

i. A finally block is executed, only after the catch block is executed.

ii. A finally block is executed whether an exception is thrown or not.

iii. A finally block is NOT mandatory.

iv. A finally block is executed, only if an exception occurs.

a) i

b) iv

c) i and iv

d) ii and iii

Answer: d) ii and iii

37. Which of these is a standard Java Interface?

a) Serializable

b) Long

c) System

d) Collections

Answer: a) Serializable

38. What is the replacement for PermGen space in Java 8?

a) FixedGen

b) Metaspace

c) MetaGenspace

Answer: b) Metaspace

39. What is the output of the following Java code snippet?

```
StringJoiner astring = new StringJoiner("-", "{", "}");
```

```
astring.add("A");
```

```
astring.add("B");
```

```
astring.add("C");
```

a) {A-B-C}

b) {A}-{B}-{C}

c) {-A-}{-B-}{-C-}

Answer: a) {A-B-C}

40. What is the return type of Java Lambda expressions?

a) Void

b) Lambda

c) Function

d) Regex

Answer: c) Function

41. Name the two types of streams in Java.

a) Sync and async

b) Sequential and parallel

c) Random and ordered

Answer: b) Sequential and parallel

42. What is the difference between peek() and dequeue()?

a) peek() returns the next item in line but dequeue() removes and returns the next item in line



b) peek() and dequeue() return the next item in line

c) peek() and dequeue() remove and return the next item in line

Answer: a) peek() returns the next item in line but dequeue() removes and returns the next item in line

43. Which one of the following statements on Java String is correct?

a) replace() method replaces only first occurrences of a character in invoking string with the given character.

b) replace() method replaces last occurrence of a character in invoking string with the given character.

c) replace() method replaces all occurrences of one character in invoking string with the given character.

d) replace() method replaces all the characters in invoking string with the given character.

Answer: c) replace() method replaces all occurrences of one character in invoking string with the given character.

44. What is the result of the following Java code snippet?

```
class Animal {  
    public String noise() {  
        return "peep";  
    }  
}  
  
class Dog extends Animal {  
    public String noise() {  
        return "bark";  
    }  
}  
  
class Cat extends Animal {  
    public String noise() {  
        return "meow";  
    }  
}
```

```
}  
Animal animal = new Dog();  
Cat cat = (Cat) animal;  
System.out.println(cat.noise());  
// Exception in thread "main" java.lang.ClassCastException
```

- a) peep
- b) bark
- c) meow
- d) Compilation fails
- e) An exception is thrown at runtime

Answer: e) An exception is thrown at runtime

45. What is the output of the following Java code snippet?

```
class A {  
    void foo() throws Exception {  
        throw new Exception();  
    }  
}  
  
class SubB2 extends A {  
    void foo() {  
        System.out.println("B ");  
    }  
}  
  
class HelloWorld {  
    public static void main(String[] args) {  
        A a = new SubB2();  
        a.foo(); // line 15  
    }  
}
```

- A) B
- B) B, followed by an Exception.
- C) Compilation fails due to an error on line 9.
- D) Compilation fails due to an error on line 15.
- E) An Exception is thrown with no other output

Answer: D) Compilation fails due to an error on line 15

46. What is the output of the following Java code snippet?

```
class multithreaded_programming {  
    public static void main(String args[]) {  
        Thread t = Thread.currentThread();  
        t.setName("new thread");  
        System.out.println(t);  
    }  
}
```

- a) Thread[main,5,main]
- b) Thread[new thread,5,main]
- c) Thread[Thread-0,5,main]
- d) Compilation fails

Answer: b) Thread[new thread,5,main]

47. Observe the code:

```
public class MyRunnable implements Runnable {  
    public void run() {  
        // some code  
    }  
}
```

Which of these will create and start this thread?

- a) new MyRunnable().start()

- b) new Thread(new MyRunnable()).run()
- c) new Thread(new MyRunnable()).start()
- d) MyRunnable.startThread()

Answer: c) new Thread(new MyRunnable()).start()

48. Observe the code below:

```
class Mammal { }
```

```
class Raccoon extends Mammal {  
    Mammal m = new Mammal();  
}
```

```
class BabyRaccoon extends Mammal { }
```

Which four statements are true? (Choose four)

- A) Raccoon is-a Mammal.
- B) Raccoon has-a Mammal.
- C) BabyRaccoon is-a Mammal.
- D) BabyRaccoon is-a Raccoon.
- E) BabyRaccoon has-a Mammal.
- F) BabyRaccoon is-a BabyRaccoon.

Answer: A, B, C, F

49. Observe the following code:

```
import java.io.*;
```

```
public class filesinputoutput {
```

```
    public static void main(String[] args) {
```

```
        String obj = "abc";
```

```
        byte b[] = obj.getBytes();
```

```
        ByteArrayInputStream obj1 = new ByteArrayInputStream(b);
```

```
        for (int i = 0; i < 2; ++i) {
```

```
            int c;
```

```

while ((c = obj1.read()) != -1) {
    if (i == 0) {
        System.out.print(Character.toUpperCase((char)c));
    }
}
}
}
}
}

```

- a) abc
- b) ABC
- c) Abc
- d) Compilation fails

Answer: b) ABC

50. Predict the output of the code:

```

class output {
    public static void main(String args[]) {
        StringBuffer s1 = new StringBuffer("Hello World");
        s1.insert(6, "Good ");
        System.out.println(s1);
    }
}

```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld
- d) Hello Good World

Answer: d) Hello Good World

51. Which of these keywords cannot be used for a class which has been declared final?

- a) abstract
- b) extends
- c) abstract and extends
- d) None of the mentioned

Answer: c) abstract

52. Predict the output:

```
sos('1' + new Integer(2) + 3);
```

- a) 6
- b) 123
- c) 33
- d) Compilation fails

Answer: b) 123

53. Observe the code and predict the output:

```
int a[] = new int[]{1, 2, 6, 4, 5};
```

```
int b[] = new int[5];
```

```
b[2] = a[2];
```

```
a[2] = 10;
```

```
System.out.println(b[2]);
```

- a) 10
- b) 6
- c) 2
- d) Compilation fails

Answer: b) 6

54. What is the output of this program?

```
import java.util.*;
```

```
class vector {
```

```

public static void main(String args[]) {
    Vector obj = new Vector(4, 2);
    obj.addElement(new Integer(3));
    obj.addElement(new Integer(2));
    obj.addElement(new Integer(5));
    obj.removeAll(obj);
    System.out.println(obj.isEmpty());
}
}

```

a) 0

b) 1

c) true

d) false

Answer: c) true

55. What is the output?

```

import java.util.*;

class Maps {
    public static void main(String args[]) {
        TreeMap obj = new TreeMap();
        obj.put("A", new Integer(1));
        obj.put("B", new Integer(2));
        obj.put("C", new Integer(3));
        System.out.println(obj.entrySet());
    }
}

```

a) [A, B, C]

b) [1, 2, 3]

c) {A=1, B=2, C=3}

d) [A=1, B=2, C=3]

Answer: d) [A=1, B=2, C=3]

56. What is the output?

```
import java.util.*;

class Bitset {

    public static void main(String args[]) {

        BitSet obj = new BitSet(5);

        for (int i = 0; i < 5; ++i)

            obj.set(i);

        obj.clear(2);

        System.out.print(obj);

    }

}
```

a) {0, 1, 3, 4}

b) {0, 1, 2, 4}

c) {0, 1, 2, 3, 4}

d) {0, 0, 0, 3, 4}

Answer: a) {0, 1, 3, 4}

57. What is the output?

```
import java.util.*;

class Array {

    public static void main(String args[]) {

        int array[] = new int[5];

        for (int i = 5; i > 0; i--)

            array[5-i] = i;

        Arrays.fill(array, 1, 4, 8);

        for (int i = 0; i < 5; i++)
```



```
        System.out.print(array[i]);  
    }  
}
```

a) 12885

b) 12845

c) 58881

d) 54881

Answer: c) 58881

58. What is the output of the following program?

```
import java.util.*;  
  
class Collection_Algos {  
    public static void main(String args[]) {  
        LinkedList list = new LinkedList();  
        list.add(new Integer(2));  
        list.add(new Integer(8));  
        list.add(new Integer(5));  
        list.add(new Integer(1));  
        Iterator i = list.iterator();  
        Collections.reverse(list);  
        Collections.shuffle(list);  
        while(i.hasNext())  
            System.out.print(i.next() + " ");  
    }  
}
```

a) 2 8 5 1

b) 1 5 8 2

c) 1 2 5 8

d) Any random order

Answer: d) Any random order

What is the output?

```
import java.util.*;

class Collection_Algos {

    public static void main(String args[]) {

        LinkedList list = new LinkedList();

        list.add(new Integer(2));

        list.add(new Integer(8));

        list.add(new Integer(5));

        list.add(new Integer(1));

        Iterator i = list.iterator();

        Collections.reverse(list);

        Collections.sort(list);

        while(i.hasNext())

            System.out.print(i.next() + " ");

    }

}
```

a) 2 8 5 1

b) 1 5 8 2

c) 1 2 5 8

d) 2 1 8 5

answer: c) 1 2 5 8

59. What is the output?

```
import java.util.*;

public class genericstack <E> {

    Stack<E> stk = new Stack<E>();

    public void push(E obj) { stk.push(obj); }
```

```

    public E pop() { return stk.pop(); }
}

class Output {

    public static void main(String args[]) {

        genericstack<String> gs = new genericstack<String>();
        gs.push("Hello");
        System.out.print(gs.pop() + " ");
        genericstack<Integer> gs = new genericstack<Integer>();
        gs.push(36);
        System.out.println(gs.pop());
    }
}

```

a) Error // error because gs String and gs Integer use same variable name

b) Hello

c) 36

d) Hello 36

Answer: a) Hello 36

60. What is the output?

```

class output {

    public static void main(String args[]) {

        StringBuffer c = new StringBuffer("Hello");
        StringBuffer c1 = new StringBuffer(" World");
        c.append(c1);
        System.out.println(c);
    }
}

```

a) Hello

b) World

c) HelloWorld

d) Hello World

Answer: d) Hello World

61. What is the output?

```
class exception_handling {  
    public static void main(String args[]) {  
        try {  
            System.out.print("Hello" + " " + 1 / 0);  
        }  
        catch(ArithmeticException e) {  
            System.out.print("World");  
        }  
    }  
}
```

a) Hello

b) World

c) HelloWorld

d) Hello World

Answer: b) World

62. What is the output?

```
class output {  
    public static void main(String args[]) {  
        char c[] = {'a', '1', 'b', ' ', 'A', '0'};  
        for (int i = 0; i < 5; ++i) {  
            if (Character.isDigit(c[i]))  
                System.out.println(c[i] + " is a digit");  
            if (Character.isWhitespace(c[i]))
```

```

        System.out.println(c[i] + " is a Whitespace char");
    if (Character.isUpperCase(c[i]))
        System.out.println(c[i] + " is an Upper case Letter");
    if (Character.isLowerCase(c[i]))
        System.out.println(c[i] + " is a lower case Letter");
    i = i + 3;
}
}
}

```

a) a is a lower case Letter, is White space character

b) b is a lower case Letter, is White space character

c) a is a lower case Letter, A is an Upper case Letter

d) a is a lower case Letter, 0 is a digit

Answer: c) a is a lower case Letter, A is an Upper case Letter

63. What is the output?

```

import java.io.*;

class filesinputoutput {
    public static void main(String args[]) throws IOException {
        InputStream obj = new FileInputStream("inputoutput.java");
        System.out.print(obj.available());
    }
}

```

Note: inputoutput.java is stored in the disk.

a) true

b) false

c) prints number of bytes in file

d) prints number of characters in the file

Answer: c) prints number of bytes in file

64. Observe the code:

```
import java.io.*;

public class filesinputoutput {

    public static void main(String[] args) throws IOException {

        String obj = "abc";

        byte b[] = obj.getBytes();

        ByteArrayInputStream obj1 = new ByteArrayInputStream(b);

        ByteArrayOutputStream obj2 = new ByteArrayOutputStream();

        for (int i = 0; i < 2; ++i) {

            int c;

            while ((c = obj1.read()) != -1) {

                if (i == 0) {

                    System.out.print(Character.toUpperCase((char)c));

                    obj2.write(c);

                }

            }

        }

        System.out.println("\n" + obj2.toString());

    }

}
```

- a) abc
- b) ABC
- c) ABC abc
- d) Compilation fails

Answer: B) ABC

65. Which of these methods are used to read in from file?

- a) get()
- b) read()
- c) scan()
- d) readfileInput()

Answer: b) read()

66. What is the output of the code?

```
public class Test8 {  
    public static void main(String args[]) {  
        Number numberRef = new Integer(0);  
        Double doubleRef = (Double)numberRef;  
    }  
}
```

- a) 0
- b) 0.0
- c) Compilation fails
- d) Throws exception

Answer: d) Throws exception

67. What is the output?

```
import java.io.*;  
  
class HelloWorld {  
    public static void main(String[] args) {  
        StringBuffer s1 = new StringBuffer("Hello");  
        StringBuffer s2 = s1.reverse();  
        System.out.println(s2);  
    }  
}
```

```
}  
}
```

- a) Hello
- b) olleH
- c) HELLO
- d) Compilation fails

Answer: b) olleH

68. Which pattern is used in the following code?

```
import java.util.*;  
  
class HelloWorld {  
    public static void main(String args[]) {  
        List<String> List = new ArrayList<String>();  
        List.add("cricket");  
        List.add("football");  
        List.add("hockey");  
        Iterator it = List.iterator();  
        while(it.hasNext()) {  
            String s = (String) it.next();  
        }  
    }  
}
```

- a) Observer
- b) Singleton
- c) Iterator
- d) Factory

Answer: c) Iterator

69. java.util.Enumeraation is an example of which pattern?



- a) Observer
- b) Singleton
- c) Iterator
- d) Factory

Answer: c) Iterator

70. Observe the following code:

```
class X implements Runnable {  
    public static void main(String args[]) {  
        X run = new X();  
        Thread t = new Thread(run);  
        t.start();  
    }  
    public void run() {}  
}
```

Answer: X run = new X(); Thread t = new Thread(run); t.start();

71. Predict the output of the code:

```
import java.util.*;  
class HelloWorld {  
    public static void main(String args[]) {  
        try {  
            int a = 5;  
            int b = 0;  
            int c = a / b;  
            System.out.println("World");  
        }  
        catch(Exception e) {  
            System.out.println("hello");  
        }  
    }  
}
```

```
    }  
    }  
}
```

OUTPUT: hello

72. What is the output?

```
int a = 10;  
int b = 0;  
int c = a / b;  
System.out.println("c");
```

OUTPUT: runtime exception

73. Which will contain the body of the thread?

Answer: run()

74. Which two of the following methods are defined in class Thread?

- 1) start()
- 2) wait()
- 3) notify()
- 4) run()

Answer: start(), run()

75. Assume the following method is properly synchronized and called from a thread A on an object B: wait(2000);

After calling this method, when will thread A become a candidate to get another turn at the CPU?

- a) After thread A is notified, or after two seconds.
- b) After the lock on B is released, or after two seconds.
- c) Two seconds after thread A is notified.
- d) Two seconds after lock B is released.

Answer: a) After thread A is notified, or after two seconds.

76. Predict the output:

```
class Test {  
    public static void main(String[] args) {  
        String str = new String(".....");  
        do {  
            str = "Hello Stop World";  
            System.out.println(str);  
        } while(str != "Strong");  
    }  
}
```

Answer: Hello Stop World

77. The code snippet below is an example of which of the following?

```
Long myLong = 21L;
```

- a) Auto boxing
- b) Auto unboxing
- c) Auto casting
- d) Auto instancing

Answer: a) Auto boxing

78. What is the result?

```
interface TestA {  
    String toString();  
}
```

```
public class Test {  
    public static void main(String[] args) {  
        System.out.println(new TestA() {
```

```

    public String toString() {
        return "test";
    }
};
}
}

```

- a) test
- b) null
- c) An exception is thrown at runtime
- d) Compilation fails because of an error in line 1
- e) Compilation fails because of an error in line 4
- f) Compilation fails because of an error in line 5

Answer: a) test

79. Output of GenericTest Program:

```

public class Test {
    public static void main(String[] args) {
        GenericTest<String> arr[] = new GenericTest[5];
        arr[0] = new GenericTest("Java");
        arr[1] = new GenericTest(1);
        arr[2] = (GenericTest<String>) new GenericTest(1);
        arr[3] = (GenericTest<String>) new GenericTest<Integer>(1);
        for(GenericTest o : arr) {
            System.out.println(o);
        }
    }
}

```

Answer: Prints the objects in array (GenericTest@hashcode), last element null.

80. Output of substring program:

```
public class Test {  
    public static void main(String[] args) {  
        String word = "IDEAL";  
        System.out.println(word.substring(0, word.length()-1) + word);  
    }  
}
```

Answer: DEAL

81. How many threads from main class?

Answer: Only 1 thread (main thread)

82. Start same thread two times:

```
t1.start();
```

```
t1.start();
```

Answer: Compilation fails

83. What is the output?

```
class Output {  
    public static void main(String args[]) {  
        Object obj = new Object();  
        System.out.print(obj.getClass());  
    }  
}
```

- a) Object
- b) class Object
- c) class java.lang.Object
- d) java.lang.Object

Answer: c) class java.lang.Object

84. What is the output?

```
class A {  
    int i;  
    int j;  
    A() {  
        i = 1;  
        j = 2;  
    }  
}  
  
class Output {  
    public static void main(String args[]) {  
        A obj1 = new A();  
        System.out.println(obj1.toString());  
    }  
}
```

- a) A
- b) 1 2
- c) String associated with obj1, e.g., A@5e265ba4
- d) Compilation fails

Answer: c) String associated with obj1, e.g., A@5e265ba4

85. Output?

```
class Output {  
    public static void main(String args[]) {  
        // Note: The divide method should be outside main in a valid class  
    }  
  
    public void divide(int a, int b) {  
        try {
```

```

        int c = a / b;
    }
    catch(Exception e) {
        System.out.println("Exception");
    }
    finally {
        System.out.println("finally");
    }
}

```

- a) Only Exception is printed
- b) Only finally is printed
- c) Both Exception and finally are printed if exception occurs
- d) Compilation fails

Answer: c) Both Exception and finally are printed if exception occurs.

✓ Whether the exception occurs or not, the `finally` block is executed.

86. Insert the appropriate code:

```

public class Sprite {
    public int fubar(Foo foo) {
        return foo.bar();
    }
    public void testFoo() {
        fubar(
            // insert code here
        );
    }
}

```

- a) new Foo()

b) new Foo() { public int bar() { return 1; } }

c) new Foo(1)

d) Foo.bar()

Answer: b) new Foo() { public int bar() { return 1; } }

87. What is the output of the following code:

```
class Main {  
    String getValue(String word) {  
        System.out.println(word);  
        if (word.length() == 1) {  
            return "";  
        } else {  
            return getValue(word.substring(0, word.length() - 1)) + word.charAt(word.length() -  
1);  
        }  
    }  
}  
  
public static void main(String[] args) {  
    Main obj = new Main();  
    System.out.println(obj.getValue("DEMOS"));  
}  
}
```

Options:

a) D

b) DEMOS

c) EMOS

d) Empty String

Answer: c) EMOS

88. What sort of compilation error you will get? How to fix the compilation error?

```
abstract class xyz {
```



```
abstract abc(int a, int b) { }  
}
```

Options:

- a) Missing return type
- b) Method cannot have a body
- c) Syntax error
- d) Give return type void. and remove empty parenthesis.

Answer: Give return type void. and remove empty parenthesis.

89. Will this code compile?

```
interface ATOB {  
    String s = "yo";  
    public void meth();  
}  
  
interface BTOC { }  
  
interface C extends ATOB, BTOC {  
    public void meth();  
    public void meth(int x);  
}
```

Options:

- a) Yes, this code will compile without any error
- b) No, compilation fails due to multiple inheritance
- c) No, compilation fails because of variable s
- d) No, compilation fails because of method overloading

Answer: a) Yes this code will compile without any error.

90. How can I inherit both the interface and abstract class?

```
interface Data {  
    public void load();
```

```
}  
  
abstract class Info {  
    public abstract void load();  
}
```

Options:

- a) class Xyz extends Info implements Data { void load() { } }
- b) class Xyz implements Data, Info { void load() { } }
- c) class Xyz extends Info, Data { void load() { } }
- d) Not possible in Java

Answer: a) class Xyz extends Info implements Data { void load() { } }

91. Will this code compile?

```
class ATOB {  
    static public String str = "abc";  
}  
  
class BTOC extends ATOB {  
    static public String str = "bcr";  
}  
  
class Main {  
    public static void main(String[] args) {  
        BTOC obj = new BTOC();  
        System.out.println(obj.str);  
    }  
}
```

Options:

- a) Yes, will compile fine and print "abc"
- b) Yes. will compile fine, but str is being overwritten in B class.
- c) No, compilation fails
- d) Runtime exception

Answer: Yes. will compile fine, but str is being overwritten in B class.

92. Which of these handles the exception when no catch is used?

- a) Default handler
- b) Finally
- c) Throw handler
- d) Java run time system

Answer: a) Default handler

93. What is the output of the code?

```
class Main {  
    static boolean foo(char c) {  
        System.out.print(c);  
        return true;  
    }  
    public static void main(String[] args) {  
        int i = 0;  
        for(foo('A'); foo('B') && (i < 2); foo('C')) {  
            i++;  
            foo('D');  
        }  
    }  
}
```

Options:

- a) ABCD
- b) ABDCBDCB
- c) ABABAB
- d) ADBCC

Answer: b) ABDCBDCB

94. What is the OUTPUT of the code?

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        // Creating and adding elements

        TreeSet<Integer> al = new TreeSet<Integer>();

        al.add("7"); // line 5

        al.add(9);

        // Traversing elements

        Iterator<Integer> itr = al.iterator();

        while(itr.hasNext()) {

            System.out.println(itr.next());

        }

    }

}
```

Options:

- a) 7 9
- b) 9 7
- c) error: incompatible types: String cannot be converted to Integer (line 5)
- d) Runtime Exception

Answer: c) error: incompatible types: String cannot be converted to Integer (line 5)

95. Which implementation of Set would you choose if you want the iterator of Set to give objects in the order they were inserted?

- a) LinkedHashSet
- b) TreeSet
- c) HashSet

Answer: a) LinkedHashSet

96. What is the output?

```
import java.util.*;

class TestHashMaps {

    public static void main(String args[]) {

        HashMap<Integer, String> hm = new HashMap<Integer, String>();

        hm.put(100, "John");

        hm.put(101, "Paul");

        hm.put(102, "George");

        hm.put(103, "Ringo");

        for (Map.Entry m : hm.entrySet()) {

            System.out.println(m.getKey() + " " + m.getValue());

        }

    }

}
```

Options:

- a) 100 John 101 Paul 102 George 103 Ringo
- b) 100 John 101 Paul 102 George
- c) 103 Ringo 102 George 101 Paul 100 John
- d) Compilation error

Answer: a) 100 John 101 Paul 102 George 103 Ringo

97. What is the output?

```
import java.util.*;

public class SortOf {

    public static void main(String[] args) {

        ArrayList<Integer> a = new ArrayList<Integer>();

        a.add(1);

        a.add(5);

        a.add(3);

    }

}
```

```
    Collections.sort(a);  
    a.add(2);  
    Collections.reverse(a);  
    System.out.println(a);  
}  
}
```

Options:

- a) [1, 2, 3, 5]
- b) [2, 1, 3, 5]
- c) [2, 5, 3, 1]
- d) [5, 3, 2, 1]
- e) [1, 3, 5, 2]
- f) Compilation fails
- g) An exception is thrown at runtime

Answer: d) [2, 5, 3, 1]

98. Which Man class properly represents the relationship "Man has a best friend who is a Dog"?

Options:

- a) class Man extends Dog { }
- b) class Man implements Dog { }
- c) class Man { private BestFriend dog; }
- d) class Man { private Dog bestFriend; }
- e) class Man { private Dog<bestFriend>; }
- f) class Man { private BestFriend<dog>; }

Answer: d) class Man { private Dog bestFriend; }

99. What is the output?

```
class Alpha {  
    public void foo() {
```

```

        System.out.print("Afoo ");
    }
} class Beta extends Alpha {
    public void foo() {
        System.out.print("Bfoo ");
    }
}

public class Main {
    public static void main(String[] args) {
        Alpha a = new Beta();
        Beta b = (Beta)a;
        a.foo();
        b.foo();
    }
}

```

Options:

- a) Afoo Afoo
- b) Afoo Bfoo
- c) Bfoo Afoo
- d) Bfoo Bfoo
- e) Compilation fails

Answer: d) Bfoo Bfoo

100. Which code fragment, inserted at line 23, allows the code to compile?

```

import java.util.Date;

import java.text.DateFormat;

DateFormat df;

Date date = new Date();

// insert code here

```

```
String s = df.format(date);
```

Options:

1) df = new DateFormat();

2) df = Date.getFormat();

3) df = date.getFormat();

4) df = DateFormat.getFormat();

5) df = DateFormat.getInstance();

Answer: 5) df = DateFormat.getInstance()

101. Output?

```
public class Base {  
    public static final String FOO = "foo";  
    public static void main(String[] args) {  
        Base b = new Base();  
        Sub s = new Sub();  
        System.out.print(Base.FOO);  
        System.out.print(Sub.FOO);  
        System.out.print(b.FOO);  
        System.out.print(s.FOO);  
        System.out.print(((Base)s).FOO);  
    } }  
class Sub extends Base {  
    public static final String FOO = "bar";  
}
```

Options:

a) foofoofoofoofoo

b) foobarfoobarbar

c) foobarfoofoofoo

d) foobarfoobarfoo



e) barbarbarbarbar

f) foofoofoobarbar

g) foofoofoobarfoo

Answer: d) foobarfoobarfoo

102. A company has a business application that provides its users with many different reports: receivables reports, payables reports, revenue projects, and so on. The company has just purchased some new, state-of-the-art, wireless printers, and a programmer has been assigned the task of enhancing all of the reports to use not only the company's old printers, but the new wireless printers as well. When the programmer starts looking into the application, the programmer discovers that because of the design of the application, it is necessary to make changes to each report to support the new printers. Which two design concepts most likely explain this situation? (Choose two.)

Options:

a) Inheritance

b) Low cohesion

c) Tight coupling

d) High cohesion

e) Loose coupling

f) Object immutability

Answer: b) Low cohesion, c) Tight coupling

103. A team of programmers is reviewing a proposed API for a new utility class. After some discussion, they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?

Options:

a) Looser coupling

b) Tighter coupling

c) Lower cohesion

d) Higher cohesion

e) Weaker encapsulation

f) Stronger encapsulation

Answer: a) Looser coupling

104. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself. What design issue has the team discovered?

Options:

- a) Tight coupling
- b) Low cohesion
- c) High cohesion
- d) Loose coupling
- e) Weak encapsulation
- f) Strong encapsulation

Answer: e) Weak encapsulation

105. Output?

```
public class TestString1 {  
    public static void main(String[] args) {  
        String str = "420";  
        str += 42;  
        System.out.print(str);  
    }  
}
```

Options:

- a) 42
- b) 420
- c) 462
- d) 42042
- e) Compilation fails
- f) An exception is thrown at runtime

Answer: d) 42042

106. Which three are valid on line?

```
public interface Status {  
    /* insert code here */ int MY_VALUE = 10;  
}
```

Options:

- a) final
- b) static
- c) native
- d) public
- e) private
- f) abstract
- g) protected

Answer: a) final, b) static, d) public

107. A programmer has an algorithm that requires a `java.util.List` that provides an efficient implementation of `add(0, object)`, but does NOT need to support quick random access. What supports these requirements?

Options:

- a) `java.util.Queue`
- b) `java.util.ArrayList`
- c) `java.util.LinearList`
- d) `java.util.LinkedList`

Answer: d) `java.util.LinkedList`

108. Which three are true? (Choose Three)

```
interface Jumper {  
    public void jump();  
}  
  
class Animal {}
```

```

class Dog extends Animal {
    Tail tail;
}
class Beagle extends Dog implements Jumper {
    public void jump() {}
}
class Cat implements Jumper {
    public void jump() {}
}

```

Options:

- a) Cat is-a Animal
- b) Cat is-a Jumper
- c) Dog is-a Animal
- d) Dog is-a Jumper
- e) Cat has-a Animal
- f) Beagle has-a Tail
- g) Beagle is-a Jumper

Answer: b) Cat is-a Jumper, c) Dog is-a Animal, f) Beagle has-a Tail

109. Which class correctly uses the Data interface and Info abstract class?

```

interface Data { public void load(); }
abstract class Info { public abstract void load(); }

```

Options:

- a) public class Employee extends Info implements Data {
 public void load() { /\* do something \*/ }
 }
- b) public class Employee implements Info extends Data {
 public void load() { /\* do something \*/ }
 }

c) public class Employee extends Info implements Data {  
    public void load() { /\* do something \*/ }  
    public void Info.load() { /\* do something \*/ }  
}

d) public class Employee implements Info extends Data {  
    public void Data.load() { /\* do something \*/ }  
    public void load() { /\* do something \*/ }  
}

e) public class Employee implements Info extends Data {  
    public void load() { /\* do something \*/ }  
    public void Info.load() { /\* do something \*/ }  
}

f) public class Employee extends Info implements Data {  
    public void Data.load() { /\* do something \*/ }  
    public void Info.load() { /\* do something \*/ }  
}

Answer: a) public class Employee extends Info implements Data { public void load() { /\* do something \*/ } }

110. Output?

```
class TestA {  
    public void start() {  
        System.out.println("TestA");  
    }  
}  
  
class TestB extends TestA {  
    public void start() {  
        System.out.println("TestB");  
    }  
  
    public static void main(String[] args) {
```

```
((TestA)new TestB()).start();  
}  
}
```

Options:

- a) TestA
- b) TestB
- c) Compilation fails
- d) An exception is thrown at runtime

Answer: b) TestB

111. Output?

```
import java.io.*;  
  
class files {  
    public static void main(String args[]) {  
        File obj = new File("/java/system");  
        System.out.print(obj.getName());  
    }  
}
```

Options:

- a) java
- b) system
- c) java/system
- d) /java/system

Answer: b) system

112. Output?

```
import java.io.*;  
  
class files {  
    public static void main(String args[]) {
```

```
File obj = new File("/java/system");  
System.out.print(obj.getAbsolutePath());  
}  
}
```

Note: file is made in C drive.

Options:

- a) java
- b) system
- c) java/system
- d) /java/system

Answer: d) /java/system

113. Output?

```
import java.lang.*;  
class exception_handling {  
    public static void main(String args[]) {  
        try {  
            int a = args.length;  
            int b = 10 / a;  
            System.out.print(a);  
            try {  
                if (a == 1)  
                    a = a / a - a;  
                if (a == 2) {  
                    int[] c = {1};  
                    c[8] = 9;  
                }  
            }  
        }  
        catch(ArrayIndexOutOfBoundsException e) {
```

```
        System.out.println("TypeA");
    }
}
catch (ArithmeticException e) {
    System.out.println("TypeB");
}
}
```

Options:

- a) TypeA
- b) TypeB
- c) 0TypeA

Answer: b) TypeB



1. Spring-WS provides various abstract endpoint classes for you to process the request.

a) org.springframework.ws.server.endpoint

b) org.springframework.ws.server

c) org.springframework.\*

d) none of the mentioned

Answer: a) org.springframework.ws.server.endpoint

2. Which method is used to gracefully shutdown all the bean processes after closing the spring container?

A. shutdownHook

B. destroy method

C. none of the mentioned

D. all of the mentioned

Answer: A. shutdownHook

3. Which one is the default scope of the beans?

A. Prototype

B. Session

C. Request

D. Singleton

Answer: D. Singleton

4. Declaring bean from object properties can be done using:-

A. PropertyPathFactoryBean

B. util:constant

C. None of the mentioned

D. All of the mentioned

Answer: PropertyPathFactoryBean

5. What are the ways to create custom Property Editors?

A. implement PropertyEditor interface

B. extend PropertyEditorSupport Class

C. none of the mentioned

D. all of the mentioned

Answer: A. implement PropertyEditor interface

B. extend PropertyEditorSupport Class

6. Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.

A.true

B.false

Answer: True

7. Which of the among is not a special bean in WebApplicationContext?

a.Locale resolver

b.Theme resolver

c.multipart file

d.resolver

e.Handler exception resolver

f.Handler mappings

g. Non of the above -Ans

Answer: Non of the above

8. How the Spring DispatcherServlet supports for returning the last-modification-date, as specified by the Servlet API ?

a.By implementing the LastModified interface

b.By implementing the LastSessionModified interface

c.By implementing the LastDateModified interface

d.By implementing the LastModifiedDate interface

Answer: a.By implementing the LastModified interface

9. @RequestMapping annotation is used to map a HTTP request method (GET or POST) to a specific class or method in the controller which will handle the respective request?

1. false

2. true

Answer: true

10. Which of the following exception classes is related to all the exceptions thrown in spring applications?

1. DataAccessException.

2. ArrayIndexOutOfBounds.

3. NullPointerException.

4. SpringException.

5. None of these.

Answer: DataAccessException.

11. Which of the following is true for Controllers in Spring?

1. An abstract class.
2. An interface.
3. Concrete class.
4. Final class.
5. None of these.

Answer: An interface.

12. Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. false
2. true

Answer: true

13. Which of the following specify the points where weaving can be applied?

1. Compile Time , Classload Time , load time.
2. Compile Time , Classload Time , Runtime
3. Compile Time and Runtime.
4. Compile Time and Classload Time.
5. None of these.

Answer: 2. Compile Time , Classload Time , Runtime

14. Which of the following layers the @Controller annotation is used in?

1. Business layer
2. Service layer.
3. Presentation layer..
4. Session layer.
5. None of these.

Answer: Presentation layer..

15. Which Of The Following Is Not A Valid Difference Between @Inject And @Autowired Annotations?

1. @Inject annotation is part of Java EE 7's Context and Dependency Injection framework while @Autowired is the Spring Framework's own implementation
2. @Autowired is a legacy annotation whereas @Inject is its latest version..
3. There is no difference and both can be used interchangeably.
4. None of these.

Answer: @Autowired is a legacy annotation whereas @Inject is its latest version..

16. Which of the following correctly mentions the Autowire types?

1. None of these.
2. byName , byMethod, constructor and autodetect.
3. byName , byType, destructor and autodetect.

4. byName , byType, constructor and autodetect.
  5. byName , byType, constructor and autocorrect.
- Answer: byName , byType, constructor and autodetect.

17. Which of the following statements is true?

1. BeanFactory extends ApplicationContext.
2. ApplicationContext extends BeanFactory.
3. ApplicationContext implements BeanFactory.
4. BeanFactory implements ApplicationContext
5. None of these.

Answer: ApplicationContext extends BeanFactory.

18. Which of the following is advice supported by Aspect Annotation?

- A. @Before B. @After C. @AfterReturning
- D. All of the mentioned

Answer: D. All of the mentioned

19. Which advice is executed only when joint point returns or throws an exception?

- A. @Before
- B. @After
- C. @AfterReturning
- D. @AfterThrowing

Answer: @AfterReturning

20. Special compiler used during weaving

- A. jvm B. gcc
- C. ajc
- D. none of the mentioned

Answer: ajc

21. How to inject Spring bean into domain objects

- A. AOP
- B. XML C. AspectJ
- D. Java Based

Answer: java Based

22. Method used to process bean before initialization callback

- A. scope
- B. `postProcessAfterInitialization()`
- C. `postProcessBeforeInitialization()`
- D. it's own constructor

Answer: `postProcessBeforeInitialization()`

23. Which attribute is used to set the scope of the bean?

- A. `setScope`
- B. `scope`
- C. `getScope`
- D. none of the mentioned

Answer: `Scope`

24. Which attribute is used to specify classname of the bean?

- A. `name`
- B. `id`
- C. `class`
- D. `constructor-args`

Answer: `Class`

25. Which tag is also allowed by static field?

- A. `util:constant`
- B. `list`
- C. `set`
- D. `constructor-args`

Answer: `util:constant`

26. Class used to have full access to the Spring context's life cycle machinery and dependency injection.

- A. `DelegatingFilterProxy`
- B. `WebApplicationContextUtils.getRequiredWeb()`
- C. `WebApplicationContextUtils.getRequiredWebApplicationContext()`
- D. None of the mentioned

Answer: `DelegatingFilterProxy`

27. Element used to specify access attributes for beans methods.

- A. security:protect
- B. security:intercept
- C. security:intercept-security
- D. none of the mentioned

Answer: A. security:protect

28. Stated related to Spring Web flows

- A. view
- B. action
- C. decision
- D. all of the mentioned

Answer: D. all of the mentioned

29. To enable Spring security for web flows.

- A. SecurityFlowExecutionListener
- B. SecurityFlowExecution
- C. FlowSecurity
- D. SecurityFlowListener

Answer: A. SecurityFlowExecutionListener

1. Does Spring provide programmatic transaction management? Select a unique answer.

- a. Yes using the @Transactional bean post processor
- b. Yes with the TransactionTemplate class
- 3. Yes with the TransactionService class
- 4. Yes using the @Transactional bean post processor

Answer: b Yes with the TransactionTemplate class

30. How could you externalize constants from a Spring configuration file or a Spring annotation into a .properties file? Select one or more

- a. By using the <context:property-placeholder /> tag
- b. By using the <util:constant /> tag
- c. By declaring the ConstantPlaceholderConfigurer bean post processor
- d. By using the c: namespace

Answer: a. By using the <context:property-placeholder /> tag

31. To validate Java beans in a web application using annotations.

- a) XML
- b) Java Based
- c) JAR-303 standard
- d) All of the mentioned

Answer: d) All of the mentioned

32. Annotation used to indicate a field has to have a minimum of 2 characters.

- a) @NotNull
- b) @Size
- c) @MaxSize
- d) @size

Answer: @Size

33. Spring MVC supports generating Excel files using which of the following libraries.

- a) Apache POI library
- b) JExcelAPI library
- c) All of the mentioned
- d) None of the mentioned

Answer: All of the mentioned

34. Interface for Dispatcher Servlet to auto detect view resolver beans.

- a) localeResolver
- b) tiles
- c) ViewResolver
- d) none of the mentioned.

Answer: ViewResolver

35. By default, XmlViewResolver loads view beans from /WEB-INF/views.xml which is final.

- a) True
  - b) False
- ans: False

36. Views that can't be resolved by InternalResourceViewResolver.

- a) redirect
- b) redirect:prefix
- c) redirect:suffix
- d) all of the mentioned

ans: redirect

9. Annotation which allows a controller's field to be assigned using Spring Expression Language (SpEL)

- a) @Value
- b) @After
- c) @Default
- d) @None

ans: @Value

10. SpEL statements are recognizable.

- a) True
- b) False

Answer: True

11.To publish a REST service with Spring.

- a) publishing an applications data as a REST service
- b) accessing data from third-party REST services
- c) none of the mentioned
- d) all of the mentioned

ans: a b

12. Publishing an applicationís data as a REST service requires.

- a) @RequestMapping
- b) @PathVariable
- c) All of the mentioned
- d) None of the mentioned

ans: a b

13.Annotation added as an input parameter to the handler method.

- a) @PathVariable
- b) @Path
- c) @PathLocale
- d) None of the mentioned

ans: a

14. Which of the following statements is true regarding the @ResponseStatus annotation?

- a. @ResponseStatus is detected on nested exceptions
- b.The ExceptionHandlerExceptionHandlerResolver uses the @ResponseStatus annotation to map exception to HTTP status code
- c. A controller handler is annotated with the @ResponseStatus, the response status set by RedirectView takes precedence over the annotation value.
- d. The @ResponseStatus annotation can go on a @RequestMapping method or a @RestController class or a business exception class.

ans: a

15. Default localeResolver used by Spring.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) AcceptLocaleResolver

ans: c

16. Alternative way to resolve locales.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver



d) SessionLocaleResolver

ans: d

17. General-purpose class that allows a response to be rendered using a marshaller.

- a) MarshallingView
- b) Marshalling
- c) View
- d) All of the mentioned

Answer: a

18. Annotation which allows the Jaxb2Marshaller marshaller to detect a class's (i.e., object's) fields.

- a) @XmlRootElement
- b) @XmlRoot
- c) @NotNull
- d) None of the mentioned

ans: a

19. How to auto-inject into a field a Spring bean by its name? Select one or more answer choices.

- a. By using both the @Autowired and the @Qualifier Spring annotations
- b. By using the @Autowired annotation and naming the field with the bean name
- c. With the name attribute of the @Autowired annotation
- d. By using the single @Qualifier annotation

ans : a

20. Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the following pointcut expression?

```
execution(* *..AccountServiceImpl.update(..))  
&& execution(* *..ClientServiceImpl.update(..))
```

- a. No joint point is defined
- b. Matches public update methods of the 2 classes, whatever the arguments
- c. Matches any update methods of the 2 classes, whatever the arguments and method visibility
- d. Matches any update methods of the 2 classes, with one more argument and whatever method visibility

ans: a

21. Select the right statement about referring a Spring configuration file inside the package com.example.myapp in the below example?

ApplicationContext context = new

ClassPathXmlApplicationContext("classpath:/com.example.myapp.config.xml");

- a. all of the above
- b. The classpath: prefix could be omitted
- c. Package name using the dot character is not well formatted
- d. The slash character preceding com.example could be omitted

ans: a

22. Using the Spring AOP framework, what is the visibility of the method matches by the following join point?

```
@Pointcut("execution(* *(..))")
private void anyOperation() {};
```

- a. Public methods
- b. All methods, whereas their visibility
- c. All methods, except private method
- d. Protected and public methods

ans: a

23. What the name of the bean defined in the following configuration class? Select a single answer.

```
@Configuration
public class ApplicationConfig {
    @Autowired
    private DataSource dataSource;
    @Bean
    ClientRepository clientRepository() {
        ClientRepository accountRepository = new JpaClientRepository();
        accountRepository.setDataSource(dataSource);
        return accountRepository;
    }
}
```

- a. clientRepository
- b. JpaClientRepository
- c. jpaClientRepository
- d. Two beans are defined: a data source and a repository

ans: a

24. Which of the following is true regarding the below Spring controller?

```
@RestController
public class OwnerController {
    @RequestMapping(value = "/owner/{ownerId}", method = RequestMethod.POST)
    @ResponseBody
    public Owner findOwner(@PathVariable("ownerId") int ownerId) {
        return new Owner();
    }
}
```

- a. RequestMethod.GET method is more accurate than POST
- b. @ResponseBody could be removed
- c. @PathVariable should be replaced with the @PathParam annotation
- d. Returning the 201 HTTP status code is better

ans: a

25. What is an advice? Select a unique answer.

- a. An action taken by an aspect at a particular join point
- b. A point during the execution of a program
- c. An aspect and a pointcut
- d. A predicate that matches join points

ans: a

26. What is the easiest method to write a unit test?

- a. `@RequestMapping("/displayAccount")`  
`String displayAccount(@RequestParam("accountId") int id, Model model)`
- b. `void displayAccount(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException`
- c. `void displayAccount(HttpServletRequest req, HttpSession Session)`  
`throws ServletException, IOException`
- d. `@RequestMapping("/displayAccount")`  
`String displayAccount(@PathVariable("accountId") int id, Model model)`

ans: a

27. Select method's signatures that match with the following pointcut:  
`execution(* com.test.service..*.*(*))`

- a. `void com.test.service.MyServiceImpl#transfert(Money amount)`
- b. `void com.test.service.account.MyServiceImpl#transfert(Money amount)`
- c. `void com.test.service.MyServiceImpl#transfert(Account account, Money amount)`
- d. `void com.test.service.account.MyServiceImpl#transfert(Account account, Money amount)`
- e. None of the above

ans: a

28. Given the Spring configuration file, which are the correct statements?

```
<bean class="com.spring.service.BankServiceImpl"
p:bankName="NationalBank">
</bean>
```

- a. The p namespace has to be declared
- b. NationalBank is a scalar value
- c. Bean id is bankServiceImpl
- d. The BankServiceImpl references a NationalBank bean

ans: a

29. Given the following configuration class, what are the correct affirmations? Select one or more answers.

```
public class ApplicationConfig {
```

```
private DataSource dataSource;
```

```

@Autowired
public ApplicationConfig(DataSource dataSource) {
    this.dataSource = dataSource;
}
@Bean(name="clientRepository")
ClientRepository jpaClientRepository() {
    return new JpaClientRepository();
}
}

```

- a. Configuration annotation is missing
- b. Default or no-arg constructor is missing
- c. @Bean name is ambiguous
- d. @Bean scope is prototype

ans: a

30. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object? Select a unique answer.

- a. ResultSetExtractor
- b. RowMapper
- c. RowCallbackHandler
- d. ResultSetMapper

ans: a

5. To load root application context at the start up.

- A. ContextListener
- B. ContextLoader
- C. ContextLoaderListener
- D. ContextEventListener

Answer: C. ContextLoaderListener

6. ActionSupport class provides a method to access spring's application context.

- A. getWebApplicationContext()
- B. getWebApplication()
- C. getApplicationContext()
- D. contextEventListener()

Answer: A. getWebApplicationContext()

7. XML element to specify the property values for scripting bean

- A. lang:property
- B. bean:property
- C. script:property
- D. None of the mentioned

Answer: A. lang:property

8. Annotation used to declare an introduction

- A. Before
- B. After
- C. @DeclareParents
- D. None of the mentioned

Answer: C. @DeclareParents

9. Is the following pointcut expression correct?

execution(\* ArithmeticCalculator.\*(..))

- A. Yes
- B. No
- C. If every target class is in same package
- D. Depends where target class is located

Answer: C. If every target class is in same package

10. DAO methods require access to the session factory, which can be injected:-

- A. a setter method
- B. constructor argument
- C. none of the mentioned
- D. all of the mentioned

Answer: D. all of the mentioned

11. Which of the following Java Collection types can you inject in a Spring application?

- 1. None of these.
- 2. Using list, set, map or collection tag.
- 3. Using list, set, props or collection tag.
- 4. Using list, set, map or props tag.
- 5. Using list, collection, map or props tag.

Answer: 4. Using list, set, map or props tag.

12.What are the different types of AutoProxying?

- a.BeanNameAutoProxyCreator
- b.DefaultAdvisorAutoProxyCreator
- c.Metadata autoproxing
- d. All of the above
- e. none of the above

Answer: //d. All of the above

13. What are the various editors used in spring work?

- a. PropertyEditor ,URLEditor
- b. ClassEditor, CustomDateEditor, StringTrimmerEditor
- c. FileEditor, LocaleEditor, StringArrayPropertyEditor
- d. All of the above

Answer: d. All of the above

14. 1. How to use <ref> in spring framework?

- a.<ref> is used with bean id.
- b.<ref> is used with string values.

Answer: a.<ref> is used with bean id.

15. How to define Inner bean in spring?

- a.Use <property/> only
- b.Use <property/> or <constructor-arg/>.

Answer: b.Use <property/> or <constructor-arg/>.

16. Which property is replaced by c-namespace in spring?

- a.<property/>
- b.<constructor-arg>

Answer: b.<constructor-arg>

17. By default a bean is eagerly initialized in spring?

- a. True

b. False  
Answer: True

18. By default a bean is not autowired?

- a. True
- b. False

Answer: False

19. Can the class which spring container makes subclass be final?

- a. No
- b. Yes

Answer: No

20. What is the scope of statefull bean in spring?

- a. Singleton scope
- b. Prototype scope

Answer: Prototype Scope

21. In which spring scope is a single bean instance created per IoC container?

- a. Singleton scope
- b. Request scope

Answer: Singleton Scope

22. In which spring scope can any number of instances of bean be created?

- a. Request scope
- b. Prototype scope

Answer: b. Prototype scope

23. When a bean has scope limited to only HTTP request, that is called

- a. Request scope
- b. Session scope

Answer: a.Request scope

24. When a bean has scope limited to HTTP session, that is called

- a. Request scope
- b. Session scope

Answer: b. Session scope

25. Thread scope bean has been introduced in spring from the version

- a. Spring 2.5
- b. Spring 3.0

Answer: b. Spring 3.0

26. InitializingBean and DisposableBean is used to customize the nature of bean in spring.

- a. Yes
- b. No

Answer: Yes

27. How to shutdown the Spring IoC container gracefully in non-web applications?

- a. Using shutdownNow()
- b. registerShutdownHook()

Answer: b. registerShutdownHook()

28. Which of the following is correct about dependency injection?

A - It helps in decoupling application objects from each other.

B - It helps in deciding the dependencies of objects.

C - It stores objects states in database.

D - It stores object states in file system.

Answer: A - It helps in decoupling application objects from each other

29. Which are the correct implementation classes of ApplicationContext?

A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext

B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext

C - AdvancedApplicationContext, FileApplicationContext

D - FileSystemApplicationContext, ClassPathApplicationContext

Answer: A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext

30. What is Spring MVC framework?

A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

Answer: B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.



31. How to reload my changes on Spring Boot without having to restart server?

a. <dependency>

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
<optional>true</optional>
</dependency>
```

b. <dependency>

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-test</artifactId>
<optional>true</optional>
</dependency>
```

c. <dependency>

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-web</artifactId>
<optional>true</optional>
</dependency>
```

d. <dependency>

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-jasper</artifactId>
<optional>true</optional>
</dependency>
```

Answer: a. <dependency>

```
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
<optional>true</optional>
</dependency>
```

32. Annotation to add security to your spring boot app?

a. @EnableWebSecurity

b. @EnableSecurity

c. @EnableSpringSecurity

Answer: a. @EnableWebSecurity

33. What are the different implementations of Spring AOP ?

A: AspectJ

B: Spring AOP

C: JBoss AOP

D: All

E: None

Answer: All

34. How Spring Batch works?

- a. step, ItemReader, ItemProcessor, ItemStreamWriter
- b. step, ItemProcessor, ItemStreamWriter
- c. step, ItemReader, ItemStreamWriter

Answer: a. step, ItemReader, ItemProcessor, ItemStreamWriter

35. LoggerInterceptor which implements the HandlerInterceptor. The LoggerInterceptor will override the which following methods

- a. preHandlerBean(), postHandlerBean(), afterCompletion()
- b. PreProcessHandler(), PostProcessHandler(), afterCompletion()
- c. PreHandler(), PostHandler(), afterCompletion()
- d. none

Answer: c. PreHandler(), PostHandler(), afterCompletion()

1. arithmetic exception and io exception are of which type?? runtime error

2. what is the sequence of java servlet deployment descriptor??

ans: coding, compiling, deploying

3. which is used to submit query?? options: resultset, statement, connection

4. abstract method doesn't follow which statement??

options: it instantiates its object

abstract class should have at least one abstract method

none of the above

5. what are the implicit objects in jsp??

ans: page, this

6. valid status codes ans: 200-299

7. super class of all classes ans: Object

8. which thread method defines all threads? ans: daemon

9. jdbc belongs to which package? ans: java.sql\*

10. which is not a primitive datatype? ans: String

11. println belongs to class PrintStream

12. void method does not return value

13. which method reverses the string?? reverse()

14. session belongs to which class?? HttpServletRequest

15. frame has default border type and plain has default flow type

16. which interface is used to connect database to application?

17. list of files returns what? ans: an array of String is returned

18. which is a valid declaration??

ans: public void int A() - false

```

public int test2(int o);
public void test int(int i,intj);-false
19. public class B()
static int x;
psvm()
{
sop (B.x);
}
ans: prints 0
20.which is not a directive??
21.wat jsp syntax are following scriptlet,expression,directive?
22.<jsp:usebean id="name" class="java.lang.string"><%=name%>
options: prints 0
prints null
compile time error
none of the above
23.which invoke method will print base class constructor?
ans: super(10)
24.q18 in previous paper
25.how session object can be destroyed?
26.public class test()
{static int i,static double d,static char c,static long l,static short s,
static string s;
psvm{sop(i+d+c+l+s+S);
}
}
27.<%!int i=8%>
<%int i=10%>
<%=i>
options: prints 8, prints 10
28. class.forName() throws class not found exception
29.seek method is used in random access file to specify the new position
30. which is not a valid declaration??
options: char c=a;
char d =\u00432;
char c= \uae262;
31.public class test ()
static int i=012;
psvm(srting args[])
{sop (i);
}
ans: prints 10
32.which is not valid?
ans:int i[] = new int []
int i [][] = new int[] {1,2}
33.public class B()
boolean a,b,c;
boolean =a||(b=true)&&(c=true)

```

sop(a+", "+b+", "+c);

options: false,false,false

false,true,true

34.inputstreamreader i=new inputstreamreader (vds cv);

bufferedReader br = new bufferedreader (i);

35. how ll u create the copy of the same object?? ans: clone()

36.all servlet object implements 1.servlet context 2.web context 3. both 1&2

37.submitting a query is done by ans:statement object

38.which math object is used for pointing 4 while x=4.8

options: math.ceil,math.floor

39.wat is feature of java?? OOPS

40.how does applet communicate with browser?? ans: init and database document

statement = connection.createStatement(

java.sql.ResultSet.TYPE\_FORWARD\_ONL

Y,

java.sql.ResultSet.CONCUR\_READ\_ONLY

);

statement.setFetchSize(Integer.MIN\_VALUE);

Which isolation level prevents dirty read in JDBC,

connection class.

(A) TRANSACTION\_READ\_COMMITTED

(B) TRANSACTION\_UNREAD\_COMMITTED

Isolation Level Table-Level Locking

Row-Level Locking

TRANSACTION\_READ\_UNCOMMITTED Dirty reads, nonre

peatable reads, and phantom reads possible Dirt

y reads, nonrepeatable reads, and phantom reads po

ssible

TRANSACTION\_READ\_COMMITTED Nonrepeatable reads an

d phantom reads possible Nonrepeatable re

ads and phantom reads possible

TRANSACTION\_REPEATABLE\_READ Phantom reads not poss

ible because entire table is locked Phantom read

s possible

TRANSACTION\_SERIALIZABLE None

None

TYPE\_SCROLL\_INSENSITIVE:

The result can be scrolled;

its cursor can move both forward and backward relat

ive to the current position,

and it can move to an absolute position. The result

set is insensitive to changes made

to the underlying data source while it is open. It

contains the rows that satisfy the query

at either the time the query is executed or as the

rows are retrieved.

TYPE\_SCROLL\_SENSITIVE:

The result can be scrolled; its cursor can move both forward and backward relative to the current position, and it can move to an absolute position. The result set reflects changes made to the underlying data source while the result set remains open.

What is the effect of executing the following JSP statement, assuming a class with name Employee exists in classes package.

The code does not compile as there is no property attribute of setProperty tag.

The code does not compile as property attribute cannot take \* as a value.

The code sets value of all properties of employee bean to "".

Ans) The code sets the values of all properties of employee bean to matching parameters in request object.

bookmark is in get.

```
1. class Animal {  
    public String noise() {  
        return "peep";  
    }  
}  
class Dog extends Animal {  
    public String noise() {  
        return "bark";  
    }  
}  
class Cat extends Animal {  
    public String noise() {  
        return "meow";  
    }  
}
```

...

```
30. Animal animal = new Dog();  
31. Cat cat = (Cat)animal;  
32. System.out.println(cat.noise());
```

What is the result?

- A. peep
- B. bark
- C. meow
- D. Compilation fails.
- E. An exception is thrown at runtime.

Answer: E

```
1. class A {  
2. void foo() throws Exception {
```

```

3. throw new Exception();
4. }
5. }
6.
7. class SubB2 extends A {
8. void foo() {
9. System.out.println("B ");
10. }
11. }
12. class Tester {
13. public static void main(String[] args) {
14. A a = new SubB2();
15. a.foo();
16. }
17. }

```

What is the result?

- A. B
- B. B, followed by an Exception.
- C. Compilation fails due to an error on line 9.
- D. Compilation fails due to an error on line 15.
- E. An Exception is thrown with no other output

Answer: D

Section: All

Explanation/Reference:

Unhandled exception type Exception

2. legal  $s3=s2+s1$

3. when we are sending data in URL in get method how to separate

parameters? ANs) &

when we are sending data in URL in get method how to separate

parameter and separate path? ANs) ?

```

4. class multithreaded_programing{
public static void main(String args[]){
Thread t = Thread.currentThread();
t.setName("new thread");
System.out.println(t);
}
}

```

ans- thread[new thread,5,main]

5. current thread

```

public class MyRunnable implements Runnable{
public void run(){
//some code
}}

```

which of these will create and start this thread?

Ans) new Thread(new MyRunnable()).start();

Description: Because the class implements Runnable,

an instance of it has to be passed to the Thread constructor  
ed.

and then the instance of the Thread has to be start

6. class Mammal { }

```
class Raccoon extends Mammal {  
Mammal m = new Mammal();  
}
```

```
class BabyRaccoon extends Mammal { }
```

Which four statements are true? (Choose four.)

- A. Raccoon is-a Mammal.
- B. Raccoon has-a Mammal.
- C. BabyRaccoon is-a Mammal.
- D. BabyRaccoon is-a Raccoon.
- E. BabyRaccoon has-a Mammal.
- F. BabyRaccoon is-a BabyRaccoon.

Answer: A,B,C,F

Explanation :Check is-A and Has-A relationship between both classes.

7. What is the output of this program?

```
import java.io.*;  
public class filesinputoutput {  
public static void main(String[] args)  
{  
String obj = "abc";  
byte b[] = obj.getBytes();  
ByteArrayInputStream obj1 = new ByteArrayInputStream(b);  
for (int i = 0; i < 2; ++ i) {  
int c;  
while ((c = obj1.read()) != -1)  
{  
if (i == 0) {  
System.out.print(Character.toUpperCase((char)c)); }  
} } } }  
ans) ABC
```

```
8. class output {  
public static void main(String args[])  
{  
StringBuffer s1 = new StringBuffer("Hello World");  
s1.insert(6, "Good ");  
System.out.println(s1);  
}  
}
```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld

d) Hello Good World

Answer: d

Explanation: The insert() method inserts one string into another. It is overloaded to accept values of all simple types, plus String and Objects. String is inserted into invoking object at specified position. "Good " is inserted in "Hello World" at index 6 giving "Hello Good World".

output:

```
$ javac output.java
```

```
$ java output
```

```
Hello Good World
```

9. string index(i)

10. Which of these keywords cannot be used for a class which has been declared final?

a) abstract

b) extends

c) abstract and extends

d) None of the mentioned

View Answer

Answer: a

Explanation: A abstract class is incomplete by itself and relies upon its subclasses to provide complete implementation. If we declare a class final then no class can inherit that class, an abstract class needs its subclasses hence both final and abstract cannot be used for a same class.

11. import all- import .\*

12. abstract class abc{  
abstract ... (int a, int b, int c)

Q= method declared right?

13. sos('1'+new Integer(2)+3);

ANS-123

14. int a[]=new int[]{1,2,6,4,5};

int b[]=new int[5];

b[2]=a[2];

a[2]=10;

System.out.println(b[2]);

ans: 6

17. ideal

removing loop

-operations

a) idea

b) deal

18. The sendError method defined in the HttpServlet



class is  
equivalent to invoking the setStatus method with the following  
parameter. Select the one correct answer.

4.

- A. SC\_OK
- B. SC\_MOVED\_TEMPORARILY
- C. SC\_NOT\_FOUND Ans) C
- D. SC\_INTERNAL\_SERVER\_ERROR
- E. ESC\_BAD\_REQUEST

5. The sendRedirect method defined in the HttpServlet class is  
equivalent to invoking the setStatus method with the following  
parameter and a Location header in the URL. Select the one  
correct answer.

- A. SC\_OK
- B. SC\_MOVED\_TEMPORARILY Ans) B
- C. SC\_NOT\_FOUND
- D. SC\_INTERNAL\_SERVER\_ERROR
- E. ESC\_BAD\_REQUEST

19. doGet is in which? a) request, b) response, c) request, d) response, e) both request and response.

20. JSP attributes: differentiate a), b) & c)?

21. independently vary = bridge

22. factory set

23. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself.

What design issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

24. instanceof

25. man dog best friend ans: class man { private dog best friend; }

26. push pop stack

27. blue mix deploy cloud foundry ans) cf push

28. 1..\*At least one instance
29. window if class=underline
30. abstract class- italics
31. slant line for delay
32. hollow diamond on collection side for aggregation, filled for composition and hollow triangle for generalization.
33. jstl
34. arrow towards mammal
35. iterator question
36. hasNext(), traverse
37. junit testing in which order
38. scroll insensitive
39. What is the output of this program?

```
import java.util.*;
class vector {
public static void main(String args[])
{
Vector obj = new Vector(4,2);
obj.addElement(new Integer(3));
obj.addElement(new Integer(2));
obj.addElement(new Integer(5));
obj.removeAll(obj);
System.out.println(obj.isEmpty());
}
}
```

- a) 0
- b) 1
- c) true
- d) false

View Answer

Answer: c

Explanation: firstly elements 3, 2, 5 are entered in the vector obj, but when obj.removeAll(obj); is executed all the elements are deleted and vector is empty, hence obj.isEmpty() returns true.

Output:

```
$ javac vector.java
```

```
$ java vector
```

```
true
```

```
40. import java.util.*;
class Array {
public static void main(String args[])
{
int array[] = new int [5];
for (int i = 5; i > 0; i--)
array[5 - i] = i;
Arrays.sort(array);
```

```

for (int i = 0; i < 5; ++i)
System.out.print(array[i]);
}
} Ans) 12345

```

41. What is the output of this program?

```

import java.util.*;
class Maps {
public static void main(String args[])
{
    TreeMap obj = new TreeMap();
    obj.put("A", new Integer(1));
    obj.put("B", new Integer(2));
    obj.put("C", new Integer(3));
    System.out.println(obj.entrySet());
}
}

```

- a) [A, B, C]
- b) [1, 2, 3]
- c) {A=1, B=2, C=3}
- d) [A=1, B=2, C=3]

Answer: d

Explanation: obj.entrySet() method is used to obtain a set that contains the entries in the map. This method provides set view of the invoking map.

Output:

```

$ javac Maps.java
$ java Maps
[A=1, B=2, C=3]

```

42. What is the output of this program?

```

import java.util.*;
class Bitset {
public static void main(String args[])
{
    BitSet obj = new BitSet(5);
    for (int i = 0; i < 5; ++i)
    obj.set(i);
    obj.clear(2);
    System.out.print(obj);
}
}

```

- a) {0, 1, 3, 4}
- b) {0, 1, 2, 4}
- c) {0, 1, 2, 3, 4}
- d) {0, 0, 0, 3, 4}

View Answer

Answer: a

Explanation: None.

Output:

```
$ javac Bitset.java
```

```
$ java Bitset
```

```
{0, 1, 3, 4}
```

43. What is the output of this program?

```
import java.util.*;
```

```
class Array {
```

```
public static void main(String args[])
```

```
{
```

```
int array[] = new int [5];
```

```
for (int i = 5; i > 0; i--)
```

```
array[5-i] = i;
```

```
Arrays.fill(array, 1, 4, 8);
```

```
for (int i = 0; i < 5 ; i++)
```

```
System.out.print(array[i]);
```

```
}
```

```
}
```

advertisements

a) 12885

b) 12845

c) 58881

d) 54881

[View Answer](#)

Answer: c

Explanation: array was containing 5,4,3,2,1 but when method Arrays.fill(array, 1, 4, 8) is called it fills the index location starting with 1 to 4 by value 8 hence array becomes 5,8,8,8,1.

Output:

```
$ javac Array.java
```

```
$ java Array
```

```
58881
```

44. prefix uri

45. last; or;last

46. What is the output of this program?

```
import java.util.*;
```

```
class Collection_Algos {
```

```
public static void main(String args[])
```

```
{
```

```
LinkedList list = new LinkedList();
```

```
list.add(new Integer(2));
```

```
list.add(new Integer(8));
```

```
list.add(new Integer(5));
```

```
list.add(new Integer(1));
```

```
Iterator i = list.iterator();
```

```
Collections.reverse(list);
```

```
Collections.shuffle(list);
```

```
while(i.hasNext())
```

```
System.out.print(i.next() + " ");
```

```
}  
}
```

a) 2 8 5 1

b) 1 5 8 2

c) 1 2 5 8

d) Any random order

Answer: d

Explanation: shuffle – randomizes all the elements in a list.

Output:

```
$ javac Collection_Algos.java
```

```
$ java Collection_Algos
```

```
1 5 2 8
```

(output will be different on your system)

47.What is the output of this program?

```
import java.util.*;
```

```
class Collection_Algos {
```

```
public static void main(String args[])
```

```
{
```

```
    LinkedList list = new LinkedList();
```

```
    list.add(new Integer(2));
```

```
    list.add(new Integer(8));
```

```
    list.add(new Integer(5));
```

```
    list.add(new Integer(1));
```

```
    Iterator i = list.iterator();
```

```
    Collections.reverse(list);
```

```
    Collections.sort(list);
```

```
    while(i.hasNext())
```

```
        System.out.print(i.next() + " ");
```

```
}
```

```
}
```

a) 2 8 5 1

b) 1 5 8 2

c) 1 2 5 8

d) 2 1 8 5

Answer: c

Explanation: Collections.sort(list) sorts the given list, the list was 2->8->5->1 after sorting it became 1->2->5->8.

Output:

when reverse 1 5 8 2

but sorting 1 2 5 8 ans is C

51.What is the output of this program?

```
import java.util.*;
```

```
public class genericstack <E> {
```

```
    Stack <E> stk = new Stack <E>();
```

```
    public void push(E obj) {
```

```

stk.push(obj);
}
public E pop() {
E obj = stk.pop();
return obj;
}
}
class Output {
public static void main(String args[])
{
genericstack <String> gs = new gene
ricstack<String>();
gs.push("Hello");
System.out.print(gs.pop() + " ");
genericstack <Integer> gs = new gen
ericstack<Integer>();
gs.push(36);
System.out.println(gs.pop());
}
}

```

a) Error

b) Hello

c) 36

d) Hello 36

Answer: d

Explanation: None.

Output:

\$ javac Output.java

\$ java Output

Hello 36

52. class output {

```

{
public static void main(String args[])
StringBuffer c = new StringBuffer(
"Hello");
StringBuffer c1 = new StringBuffer
(" World");
c.append(c1);
System.out.println(c);
}
}

```

a) Hello

b) World

c) Helloworld

d) Hello World Ans-d.

53.class output {

```

{
public static void main(String args[])

```

```

StringBuffer s1 = new StringBuffer("
Hello");
s1.setCharAt(1,'x');
System.out.println(s1);
}
}

```

- a) xello
- b) xxxxx
- c) Hxllo
- d) Hexlo

View Answer

Answer: c

Explanation: None.

Output:

```
$ javac output.java
```

```
$ java output
```

```
Hxllo
```

54. switch case

55. array of string

56. "welcome.trim()"= when compiled ans) equals

57. builder pattern

58. Question on this:

```

Calendar calendar = new GregorianCalendar();
int year = calendar.get(Calendar.YEAR);
int month = calendar.get(Calendar.MONTH);
int dayOfMonth = calendar.get(Calendar.DAY_OF_MONTH
); // Jan = 0, not 1
int dayOfWeek = calendar.get(Calendar.DAY_OF_WEEK)
;
);
H);
int weekOfYear = calendar.get(Calendar.WEEK_OF_YEAR
int weekOfMonth= calendar.get(Calendar.WEEK_OF_MONT
int hour = calendar.get(Calendar.HOUR);
// 12 hour clock
int hourOfDay = calendar.get(Calendar.HOUR_OF_DAY)
; // 24 hour clock
int minute = calendar.get(Calendar.MINUTE);
int second = calendar.get(Calendar.SECOND);
int millisecond= calendar.get(Calendar.MILLISECOND)
;
59. class output {
{
public static void main(String args[])
String s1 = "one";
String s2 = s1 + " two";
System.out.println(s2);
}
}

```

}

a) one

b) two

c) one two

d) compilation error

View Answer

Answer: c

60. Total number of users using listners - ans. ht  
tpsessionlistener

```
61. class exception_handling {  
public static void main(String args[])  
{  
    try {  
        System.out.print("Hello" + " "  
+ 1 / 0);  
    }  
    catch(ArithmeticException e) {  
        System.out.print("World");  
    }  
}  
}
```

a) Hello

b) World

c) HelloWorld

d) Hello World

Answer: b

```
62. class output {  
{  
    public static void main(String args[])  
    char c[]={'a', '1', 'b', ' ', 'A',  
'0'};  
    for (int i = 0; i < 5; ++i)  
    {  
        if(Character.isDigit(c[i]))  
        System.out.println(c[i]+  
" is a digit");  
        if(Character.isWhitespace(c[  
i]))  
        " is a Whitespace character");  
        System.out.println(c[i]+  
        if(Character.isUpperCase(c[i  
]))  
        System.out.println(c[i]+  
" is an Upper case Letter");  
        if(Character.isLowerCase(c[i  
]))  
        System.out.println(c[i]+  
" is a lower case Letter");
```



```
i=i+3;  
}  
}  
}
```

- a) a is a lower case Letter
- is White space character
- b) b is a lower case Letter
- is White space character
- c) a is a lower case Letter
- A is a upper case Letter
- d) a is a lower case Letter
- 0 is a digit

View Answer

Answer:c

62. Two questions on these.

Interfaces for Event Listeners-

Session Serialization--HttpSession

Request Lifecycle Event--ServletRequestListener

Session Attribute Binding--HttpSessionAttribute  
Listener

Session LifecycleEvent--HttpsessionListener

Web Application Lifecycle Event--ServletContext  
Listener

63. Which of the following describes the correct  
sequence of the steps involved in making a connect  
ion with a database.

- 1. Loading the driver
- 2. Process the results.
- 3. Making the connection with the database.
- 4. Executing the SQL statements.

(A) 1,2,3,4

(B) 1,3,4,2

(C) 2,1,3,4

(D) 4,1,2,3

Answer : 1,3,4,2

64. import java.io.\*;

```
class filesinputoutput {  
public static void main(String args[])  
InputStream obj = new FileInputStre  
am("inputoutput.java");  
System.out.print(obj.available());  
}  
}
```

Note: inputoutput.java is stored in the disk.

- a) true
- b) false
- c) prints number of bytes in file
- d) prints number of characters in the file

View Answer

Answer: c

Explanation: obj.available() returns the number of bytes.

Output:

```
$ javac filesinputoutput.java
```

```
$ java filesinputoutput
```

```
1422
```

(Output will be different in your case

```
65.import java.io.*;
```

```
public class filesinputoutput {
```

```
public static void main(String[] args)
```

```
{
```

```
String obj = "abc";
```

```
byte b[] = obj.getBytes();
```

```
ByteArrayInputStream obj1 = new Byte
```

```
ArrayInputStream(b);
```

```
for (int i = 0; i < 2; ++ i) {
```

```
int c;
```

```
while((c = obj1.read()) != -1) {
```

```
if(i == 0) {
```

```
System.out.print(Charact
```

```
er.toUpperCase((char)c));
```

```
obj2.write(1);
```

```
}
```

```
}
```

```
System.out.print(obj2);
```

```
}
```

```
}
```

```
}
```

a) AaBaCa

b) ABCaaa

c) AaaBaaCaa

d) AaBaaCaaa

Answer: d

Explanation: None.

Output:

```
$ javac filesinputoutput.java
```

```
$ java filesinputoutput
```

```
AaBaaCaaa
```

66. Which of these methods are used to read in from file?

a) get()

b) read()

c) scan()

d) readFileInput()

View Answer

Answer: b

67. toString question

68. SC\_OK

69. Which of the following allows non repeatable read in JDBC Connection?

- a. TRANSACTION\_READ\_UNCOMMITTED
- b. TRANSACTION\_READ\_COMMITTED
- c. TRANSACTION\_SERIALIZABLE
- d. TRANSACTION\_REPEATABLE\_READ Ans. D

70. dirty unread- ans)read.

71.package pkg;

class output {

public static void main(String args[])

{

StringBuffer s1 = new StringBuffer(

"Hello");

s1.setCharAt(1, x);

System.out.println(s1);

}

}

a) xello

b) xxxxx

c) Hxlllo

d) Hexllo

Answer: c

72.public class Test8 {

public static void main (String args[]) {

Number numberRef = new Integer(0);

Double doubleRef = (Double)numberRef;

}

}

It shows me exception at runtime:

Exception in thread "main" java.lang.ClassCastException:

ble

java.lang.Integer cannot be cast to java.lang.Dou

73. command design pattern ans)d (history...)

74. Which of these is not a valid value for dispatcher element of filtermapping.

Select the one correct answer.

A. FORWARD

B. RESPONSE Ans) B

C. INCLUDE

D. ERROR

79. Which of the following is not a valid HTTP/1.1 method. Select the one correct answer.

15.

- A. CONNECT method
- B. COMPARE method
- C. OPTIONS method
- D. TRACE method

Name the http method used

answer: B

80. Name the location of compiled class files within a war file? Select the one correct answer.

19.

- A. /META-INF/classes
- B. /classes
- C. /WEB-INF/classes
- D. /root/classes

Answers- web inf

## Docker, Kubernetes

1. You have just executed "docker swarm leave" on a node. What command can be run on the same node to confirm it has left the cluster?

Answer: docker system info

2. The server is running low on disk space. What command can be used to check disk usage of images, containers, and volumes?

Answer: docker system df

3. Which Dockerfile instruction initializes a new build stage and sets the base image?

Answer: FROM

4. Which command automatically creates a volume when a container is started?

Answer: docker container run --name nginx-test --volume /app nginx

5. What is the Docker command to add or update a published port?

Answer: docker service update --publish-add

6. What service mode is used to deploy a single task of a service to each node?

Answer: Global

7. Which description best fits a DaemonSet?

Answer: A way to run a copy of a pod on all or some nodes

8. Docker command to connect a running container to an existing user-defined bridge?

Answer: docker network connect

10. Docker command to display information details on one or more networks?

Answer: docker network inspect

11. In Kubernetes, a node is:

Answer: A worker machine

12. Default format of docker inspect output?

Answer: JSON

13. What is a label in Kubernetes?

Answer: A way to group related things using key/value pairs

14. Docker command to find the current logged driver for a running container?

Answer: docker inspect

15. Docker command to see the storage driver Docker is currently using?

Answer: docker info

16. What does docker image prune do?

Answer: Removes unused images

18. What manages assigning nodes to pods depending on resource availability?

Answer: Scheduler

19. What can you deploy on Kubernetes?

Answer: Containers

20. Which command is used to create a service?

Answer: kubectl expose

21. What is the function of a node?

Answer: To run pods according to master components

22. OpenShift is based on which container orchestration platform?

Answer: Kubernetes

23. What is a Pod?

Answer: A group of one or more application containers with shared volumes and IP address

24. A \_\_\_\_\_ ensures that all nodes run a copy of a pod.

Answer: DaemonSet

25. What is a Kubernetes volume?

Answer: A directory for data accessible to containers in a pod

26. Which command removes the cluster-admin role from a user?

Answer: oc adm policy remove-cluster-role-from-user

27. What is a kubelet?

Answer: A node agent that manages pods and their containers

28. What is the intended use of etcd?

Answer: To store all cluster data and maintain cluster state

29. Docker command to increase the number of replicas?

Answer: docker service scale

30. Docker command to display layers of a Docker image?

Answer: docker history

31. What is the result of "kubectl exec -ti my-pod-name bash"?

Answer: Opens a console on the first container inside the pod

34. Which Docker commands can be used to find all port mappings?

Answer: docker inspect, docker ps, docker port

35. What does Dockerfile LABEL do?

Answer: Adds metadata to an image

36. At its core Kubernetes is a platform for:

Answer: Running and scheduling container applications on a cluster

39. Which command shows volumes for a specific container?

Answer: docker inspect

40. Docker command to update an existing service?

Answer: docker service update

41. What runs on each node to ensure containers are running in a pod?

Answer: kubelet

43. What does Dockerfile EXPOSE do?

Answer: Informs Docker that the container listens on the specified port

44. What is true about using -P option?

Answer: Docker binds each exposed container port to a random host port

45. Difference between ADD and COPY?

Answer: ADD supports remote URL handling, COPY does not

46. What is a cluster?

Answer: A single system made up of several other resources

47. Command to display Docker config on host?

Answer: docker info

48. Function of a node?

Answer: To run pods according to master components

49. Which command provides users edit access to developers group?

Answer: oc adm policy add-role-to-group edit developers -n develop

50. Default port to access WebSphere Application Server console?

Answer: 9080

51. What is proxy service used for?

Answer: To act as an intermediary between endpoint servers

52. Which servers run Kubernetes control components in OpenShift?

Answer: Masters

53. Which resource sets default request/limit for containers?

Answer: LimitRange

54. Docker command to create a bridge network?

Answer: docker network create

55. Purpose of ReplicaSet?

Answer: To duplicate pods



56. Not a valid way to tag a Docker image?

Answer: Tag by image ID (invalid)

57. Mission of control plane?

Answer: Regulate communication between Kubernetes clusters  
an

59. Service mode used to deploy a single task to each node?

Answer: Global

## **CLOUD FSD DUMPS**

| Section Name                                       | No. of Questions | Time Limit (Mins) |
|----------------------------------------------------|------------------|-------------------|
| 1.IBM Cloud Microservices-Java                     | 8                | Untimed*          |
| 10.IBM Cloud Microservices-Microservices           | 3                | Untimed*          |
| 11.IBM Cloud Microservices-DevOps                  | 2                | Untimed*          |
| 2.IBM Cloud Microservices-HTML5 CSS and Bootstrap  | 6                | Untimed*          |
| 3.IBM Cloud Microservices-JavaScript               | 7                | Untimed*          |
| 4.IBM Cloud Microservices-Angular                  | 8                | Untimed*          |
| 5.IBM Cloud Microservices-React                    | 7                | Untimed*          |
| 6.IBM Cloud Microservices-MongoDB                  | 7                | Untimed*          |
| 7.IBM Cloud Microservices-SpringCore AOP and Testi | 5                | Untimed*          |
| 8.IBM Cloud Microservices-Spring DAO Hibernate and | 3                | Untimed*          |
| 9.IBM Cloud Microservices-Spring Boot and Rest Ser | 4                | Untimed*          |

\*Untimed: These sections are without any specific time limit. You can answer these sections within the total assessment time limit.

i.e Total Time of Untimed Sections = Total Time of Test - Total Time of Timed Sections

**Total Test Duration:** 60 Mins

## **HTML & CSS :**

1.Whenever you are assigning two classes to a tag , you must separate them with a

- A.Space
- B.Comma
- C.Dash

ANS : A

2. \_\_\_\_\_ contains the navigation menu, or other navigation functionality for the page

- A.section
- B.header
- C.nav
- D.aside

ANS : C

3.The main purpose of the placeholder attribute in input controls is:

- A.Allow the end-user to remember the location in form
- B.There is no such attribute known as "placeholder"
- C.Allow the end-user to replace the data control with another
- D.Display "watermark" data to assist the end-user

ANS : D

4.Which of the following values are accepted by the float Property?a) leftb) rightc) noned) all of the mentioned

- A.left
- B.right
- C.none
- D.all of the above

ANS : D

5.Which of the following CSS Property controls how an element is positioned?

- A.position
- B.set
- C.static
- D.fix

ANS : A

6.If You are designing a column to occupy 3 of the 12 columns in your desktop designWhat is the width in CSS

- A.Width:25%
- B.Width:33.3333%
- C.Width:21%

ANS : A

7.In what direction does float will work? `img { float: right; }`

- left
- top
- bottom
- right

ANS : D

8.Which of the following CSS Property sets the stacking order of positioned elements?

- A.x-index
- B.y-index
- C.z-index

ANS : C

9. You indicate a class selector with

- A.the hash (#) symbol
- B.the dot (.)
- C.the tag

ANS : B

10.Which of the following media queries would apply to a tabletreporting a screen width of 768px?

- A.@media only screen and (min-width: 1140px){}
- B.@media only screen and (min-width: 641px){}
- C.@media only screen and (max-width: 1140px){}

ANS :- B

11) To get the first or nth child you use

- A.the nested selector
- B.A pseudo-class

ANS : B

12) Which of the following rules allows users to import style rules from other style sheets?

- A.@media
- B.@important
- C.@import
- D.@style

ANS : C

13) I have a <p> which is the child of a <div> which is the child of a <section>, does the style in the section affect the p tag?

- A.Yes, inheritance accumulates
- B.No, only the immediate parent's style is inherited

ANS : A

14) Which HTML tag allows styling from an external .css file?

A.link  
B.css  
C.style  
ANS : A

15) Which of the following properties sets the distance between an element's right border and the rightmost edge of its content?

A.auto  
B.padding-left  
C.padding-right  
D.Padding-top  
ANS : C

## JAVASCRIPT :-

1) What built-in functions allows you to convert a string into an integer?

A.toInt()  
B.parseInt()  
C.toInteger()  
D.parseString()  
ANS : B

2) What is the catch method for a promise?

A.To compose two promises  
B.To listen to the result of a promise  
C.To listen only for a rejected promise  
D.There is no catch method on a promise  
ANS : C

3) What is the effect of using await?

A.The awaited function runs in parallel  
B.A promise is returned from a function instead of a value  
C.A value is returned from a function instead of a promise  
ANS : C

4) What is a function in JavaScript?

A.An instance of a class  
B.A statement that evaluates an expression  
C.A block of organized, reusable code that performs a single action  
D.Code that allows us to write loops  
ANS : C

5) What does an async function return?

A. A promise in a promise  
B. The value returned from the function  
C. A promise  
ANS : C

6) What is dependency injection?

A. A coding technique in which all dependencies are defined with import statements  
B. A coding technique for registering services  
C. A coding pattern in which a class receives the instances of objects it needs from an external source  
ANS : C

7) Which is the proper syntax to make sessionTemplate available by name from a JavaScript module?

A.export function sessionTemplate(sessions) {...}  
B.export default function sessionTemplate(sessions) {...}  
C.module.exports = function sessionTemplate(sessions) {...}  
D.require function sessionTemplate(sessions) {...}

ANS : A

8) What is a callback?

- A.A function that is called when a parallel operation is done
- B.A boolean that is set to true when a parallel operation is done
- C.A function that can be used with the 'await' keyword

ANS : A

9) There is no catch method on a promise

- A.true
- B.false

ANS : B

10) What is blog.name?var blog = { name: 'Ski Utah'};var updatedBlog = blog;updatedBlog.name = 'Rocky Mountain Skiing' }

- A.Ski Utah
- B.undefined
- C.blog
- D.Rocky Mountain Skiing

ANS : D

11) What is the output of the following?greet = function(...months){ months.forEach(month => console.log('Month: ' + month));};greet(['Jan', 'Feb']);

- A. Jan Feb
- B. ReferenceError
- C. Month: Jan, Feb
- D. Month: Jan
- E. Month: Feb

ANS : C

12) What is the relationship between JavaScript and ECMAScript?

- A.ECMAScript is a standard that JavaScript implements.
- B.They are different terms for the same thing.
- C.JavaScript is a standard that ECMAScript implements.
- D.JavaScript is an old term for what is now called ECMAScript.

ANS: A

13) What will show in the console when this code runs?var value = "no value";console.log(!value );

- A.false
- B.(an empty string)
- C.true
- D.no value

ANS: C

Answered changed

Explanation : The expression !value is a shorthand way of converting a value to its boolean equivalent.

Here's how it works:

1. The first ! operator negates the value, so if value is truthy, it becomes false, and if value is falsy, it becomes true.
2. The second ! operator negates the result of the first ! operation, effectively converting it back to its original truthiness.

14) What is the output of the following?function display(num1, num2) { console.log(num1, num2);}let nums = "12345";display(...nums);

- A.1 2
- B.21
- C.12345
- D.45

ANS : A

15) What is the console output of the following code block: function printName() { name = "Mary"; name = "John"; console.log(name);}

A.Empty  
B.None  
C.John  
D.Mary  
ANS: C

16) Where is the correct place to insert a Javascript ?  
A. The <head> section  
B. Both the <head> section and the <body> section are correct.  
C. The <body> section.  
D. The <footer> section.  
ANS : B.

17) .What syntax allows for a variable number of arguments to a function where the arguments are passed as an array?  
1. function([]numbers) { }  
2. function(\*numbers) { }  
3. function(...numbers) { }  
4. function(numbers[]) { }  
ANS : C

18) What statement will load a module from a script named employee.js?  
1. reference "employee";  
2. import "employee";  
3. import "employee.js";  
4. reference "employee.js";  
ANS : C

19) Which flags are required to run nodes with ES6?  
1. harmony and strict-mode  
2. experimental and strict-mode  
3. harmony and block-binding  
4. harmony and experimental  
ANS :- D

20) What contextual keyword is used in a class definition to identify a function containing initialization logic for an object?  
1. ctor  
2. new  
3. constructor  
4. class  
ANS : C

21) What syntax will create the array [1, 2, 3, 4, 5, 6] ?  
1. [1, 2, [3, 4]..., 5, 6];  
2. [1, 2, ...[3, 4], 5, 6];  
3. [1, 2, [3, 4], 5, 6];  
4. [1, 2, {3, 4}, 5, 6];  
ANS :- B

22) Q 6. What symbol identifies a generator function?  
1. &  
2. @  
3. \$  
4. \*  
ANS : D

23) By default, if you load a module that's not a JavaScript file, what will SystemJS do?  
1. Raise an error when loading the module.  
2. Attempt to load the module as if it were a JavaScript file.

3. Look for a plugin with the name of the file extension to load the file.

ANS :- C

24) Babel compiles the JavaScript code you write into

1. ECMAScript 5
2. ECMAScript 7
3. ECMAScript 3
4. ECMAScript 6

ANS :- A

25) . Which of these phrases describes Babel?

- A. JavaScript compiler
- B. Requires a runtime
- B. A programming language
- C. Experimental

ANS : A.

## REACT :-

1.The string "api/route" in Express is an example of a...

- A. Middleware
- B .Route
- C. Error Handler

ANS : B

2. Which library allows the use of the "require" keyword in front-end code?

- A. guid
- B. Browserify
- C. React
- D. jQuery

ANS : B

3. The \_\_\_\_\_ attribute of a React form is called when the main action button inside the form is pressed.

- A. onSubmit
- B. onChange
- C. onComplete

ANS : A

4. Which of the following are automation tools?

- A. Gulp
- B. JSX
- C. Express

ANS : A

5. Which of the following is not a method of a dispatcher?

- A. register
- B. vector
- C. dispatch

ANS : B

6. What React function is used to define a new element tag?

- A. renderComponent
- B. createClass
- C. setInnerHTML

ANS : B

7. Isomorphic means to focus exclusively on an app's back-end.

A. True  
B. False  
ANS : B

8. React's abstraction of the document object model is often called...

A. Virtual DOM  
B. Meta DOM  
C. Virtualized Model  
ANS : A

9. True or false : The only thing that can change the data inside a store is the store itself.

A. false  
B. true  
ANS : B

10.

The Gulp task which takes a stream and outputs a file is...

A. src  
B. task  
C. dest  
ANS : C

11. The best way to detect when a user has completed a form is

A. to handle the onClick event on the form's submit element  
B. to handle the onBlur event of the last form element  
C. to handle the form's onSubmit event  
D. to accumulate the onChange events for each form element  
ANS : C

12. React is

A A library that represents asynchronous data streams with observables  
B A comprehensive client-side application framework  
C An optimizing compiler for JavaScript  
D A client-side library providing rendering and event handling  
ANS : D

13. Routing can be added to a React application by

A. adding an HTML5 or hash fragment based routing library  
B. rendering the application on the server using react-page  
C. handling the window.onLocationChanged event  
D. using the React.route function  
ANS : A

14. Behavior is shared between components using

A. propTypes  
B. state  
C. mixins  
D. props  
ANS : C

15. An advantage of React is

A. jQuery syntax  
B. static type checking  
C. speed  
D. easy integration with other JavaScript libraries  
ANS : C

16. The JSX pre-processor is packaged as a



- A. dynamic link library
- B. Nuget package
- C. Gem
- D. npm module

ANS : D

17. The just-in-time JSX transformer is good for

- A. integrating with JavaScript tools
- B. simple development workflow
- C. reusing behavior
- D. performance

ANS : B

18. The function that creates React components is

- A. React.component.extend
- B. React.createComponent
- C. React.renderComponent
- D. React.createClass

ANS : D

19. How does React Router make URL params accessible to components?

- A. parseUrlParams
- B. State
- C. getUrlParams
- D. Props

ANS : D

20. Why is the Link component useful?

- A. It allows you to hyperlink to routes by name
- B. It enhances page load performance
- C. It removes invalid characters from the URL
- D. It provides faster navigation than hyperlinks

ANS : A

21. How do Flux stores find out about Flux actions?

- A. They query the server via AJAX
- B. They poll HTML5 localStorage
- C. They register a callback with the dispatcher.
- D. They call action creators directly

ANS : C

22. Where should mutable data for React components be stored?

- A. props
- B. reactData
- C. state
- D. React.cookie

ANS : C

23. What mechanism is recommended for passing data down to child components in React?

- A. props
- B. sendChildData
- C. getDefaultProps
- D. state

ANS : A

24. What is JSX?

- A. A language that looks like JavaScript that compiles to HTML.
- B. A styling language that compiles down to CSS.

C. A language for querying RESTful web services.  
D. A language that looks like HTML that compiles down to JavaScript.  
ANS : D

25. What function should all your JSX be placed in?  
A. render  
B. compile  
C. jsx  
D. emit  
ANS : A

26. What does the "webpack" command do ?  
A. Runs React Local Development Server.  
B. Transpiles all the Javascript down into one file.  
C. Both A and B.  
ANS : C

27. Which of the following API is a MUST for every React Component ?  
A. getInitialState.  
B. render  
C. renderComponent.  
ANS : C.

28. What are the advantages of React ?  
A. React can be used on client as well as server side.  
B. Using React increases readability and makes maintainability easier. Component and Data patterns improve readability and thus make it easier for maintaining larger props.  
C. React can be used with any other framework (Backbone.js, Angular.js) as it is just a View Layer.  
D. All of the above.  
ANS : D

29. How does React handle the Web Accessibility Initiative - Accessible Rich Internet Applications (WAI-ARIA) standard ?  
A. aria-\* HTML attributes are fully supported in JSX where most DOM properties and attributes in React are camelCased , these attributes should be lowercase.  
B. React processes aria - \* attributes separately and updates the DOM for accessibility if the user requires it.  
C. aria-\* attributes should be converted to camelCase like other attributes; For example: className, onChange and so on.  
D. React is yet to support the WAI-ARIA standard.  
ANS : D.

30. What is React in MVC ?  
A. Controller  
B. Middleware  
C. Model  
D. Router  
ANS :- None of these. Updated ans is Controller. Option A

31) What is React?  
Just a server-side framework  
Just a user-interface framework  
Both a server-side framework as well as a user-interface framework  
ANS : C

32) What are the limitations of React?  
React is only for the view layer of the app so we still need the help of other technologies to get a complete tooling set for development.  
React uses inline templating and JSX. This can seem awkward to some developers.  
The library of React is too large.  
All of the above

ANS :- D

33) How can you access the state of a component from inside of a member function in React?

this.getState()  
this.prototype.stateValue  
this.state  
this.values

ANS : D

34) You can also use an ES6 class to define a component in React.

True  
False  
ANS : A

35) Which of the following API is a MUST for every React component?

getInitialState  
render  
renderComponent  
ANS : C

36) At the highest level, React components have lifecycle events that fall into \_\_\_\_\_.

Initialization  
State/Property updates  
Destruction  
All of the above  
ANS : D

37) What are the advantages of React?

React can be used on client as well as server side.  
Using React increases readability and makes maintainability easier. Component and Data patterns improve readability and thus make it easier for maintaining larger apps.  
React can be used with any other framework (Backbone.js, Angular.js), as it is just a view layer.  
All of the above  
ANS : D

38) How does React handle the Web Accessibility Initiative - Accessible Rich Internet Applications (WAI-ARIA) standard?

aria-\* HTML attributes are fully supported in JSX where most DOM properties and attributes in React are camelCased, these attributes should be lowercase.  
React processes aria-\* attributes separately and updates the DOM for accessibility if the user requires it.  
aria-\* attributes should be converted to camelCase like other attributes. For example: className, onChange, and so on  
React is yet to support the WAI-ARIA standard.  
ANS :- D

39) What does the "webpack" command do?

Transpiles all the Javascript down into one file  
Runs react local development server  
Both A and B  
ANS : C

40) What is React in MVC?

Controller  
Middleware  
Model  
Router  
ANS : C

## TYPESCRIPT

Q 1 Which features does TypeScript support natively?

- A.Support for standard JavaScript code
- B.Classes and modules
- C.Static typing
- D.Interfaces
- E.Constructors
- F. All of the above

ANS : F

Q 2. Which of the following is not a type keyword in TypeScript?

- 1.float
- 2.bool
- 3.all are valid types in TypeScript
- 4.number
- 5.string

ANS : float

Q 3. Interfaces defined in TypeScript are included in the JavaScript that is generated?

- True
- False

ANS : B

Q 4.How do you extend an interface in TypeScript?

- A.interface ITruckOptions extends IAutoOptions
- B.interface ITruckOptions => IAutoOptions
- C.interface ITruckOptions : IAutoOptions
- D.interface ITruckOptions implements IAutoOptions

ANS : A

Q 5. Benefits offered by TypeScript include:

- A.Code encapsulation
- B.All of these
- C.More maintainable code
- D.Strongly-typed code

ANS :B

Q 6. How do you automatically generate fields in a TypeScript class?

- A.Add the public keyword to a constructor parameter
- B.None of these
- C.Use the auto keyword
- D.TypeScript doesn't support automatically generated fields

ANS : A

Q 7.Which TypeScript function will accept 2 parameters that must be numbers and return their product?

- A.var func = (x: number, y: number) => x \* y;
- B.var func = (x, y) => x \* y;
- C.None of these
- D.var func = function(x, y) => x \* y;
- E.All of these

ANS : A

Q 8. Which statement will allow your code to reference an internal module in a file named shapes.ts using TypeScript?

- A./// <reference="shapes.ts" />
- B./// <reference path="shapes.ts" />
- C.import "shapes"
- D./// shapes.ts

ANS : C

Q 9. Which TypeScript statement will import an external module in a file named viewmodels.ts that is in the same folder as your code?

- A. `import vm = module(viewmodels);`
- B. `import vm = module('../viewmodels');`
- C. `import module('viewmodels');`
- D. `import vm = module('viewmodels');`

ANS : DC

Q 10. Which TypeScript compiler parameter will emit JavaScript that is AMD compliant?

- AMD
- module AMD
- Neither of these

ANS : B

Q 11. What keyword is used to apply all of the properties on one interface to another interface?

- implements
- extends
- class
- further
- references

ANS B

Q 12. What value will be output at the end of the following code block?

```
enum Color { Red = 5, Green = 10, Blue = 15 };
```

- A. `console.log(Color.Green);`
- B. 2
- C. Color.Green
- D. 1
- E. Green

F. 10

ANS : F

Q 13. Which of the following best defines a "class"?

- A. Objects with no implementation details
- B. Object with no properties or methods
- C. JSON objects with methods
- D. Collection of private, reusable functions
- E. Template for creating objects

ANS : E

Q 14. What term describes the concept of treating an object as if it were a particular type, even if it weren't declared as that type?

- A. Extension Typing
- B. Duck Typing
- C. Interface Typing
- D. Generic Typing
- E. Ruby Typing

ANS : B

Q 15. What access modifier limits the accessibility of a class member to its own class and subclasses?

- A. export
- B. internal
- C. protected
- D. private
- E. sensitive

ANS : C

Q 16.What is the primary difference between a generic class and a non-generic class?

- A.Generic classes do not accept constructor parameters.
- B.Generic classes are always declared abstract.
- C.Generic classes may not implement interfaces.
- D.Generic classes accept a type parameter.
- E.Generic classes must inherit from a base class.

ANS : D

Q17.Which TypeScript compiler options lets you specify the location of the tsconfig.json file to be used for the compilation?

- A.--project
- B.--json
- C.--outDir
- D.--input
- E.--source

ANS : A

Q 18.Which of the following symbols is used to denote that a function parameter is a rest parameter?

- A. <>
- B. >>>
- C. !!
- D. ...
- E. ||

ANS : D

Q 19.Which of the following will correctly import the items exported by this export statement?

`export { GetUser as GetLibraryUser, GetBook as GetLibraryBook };`

- A.`import TypeScript from './library';`
- B.`import namespace from './library';`
- C.`import { GetLibraryUser, GetLibraryBook } from './library';`
- D.`import { GetUser, GetBook } from './library';`
- E.`import all from './library';`

ANS : C

1. Which are the different Data Types supported by Typescript?

- A. `Boolean var bValue: boolean = false`
- B. `Number var age: number = 16`
- C. `String var name: string = "join"`
- D. All of the above mentioned

ANS : D

2. What are the variable scopes available in TypeScript?

- A. Global Scope
- B. Class Scope
- C. Local Scope
- D. All of the above

ANS : D

3. Which object oriented terms are supported by Typescript?

- A. Modules
- B. Classes
- C. Interfaces
- D. All of these

ANS : D

4. In typescript, it calls an javascript internally and then performs validations

A. True

B. False

ANS : A

5. To change the datatype from one type to another, we require

A. TypeOperator

B. TypeAssertion

C. InferredTyping

ANS : B

6. For a variable, Without Specifying any data type is

A. InferredTyping

B. TypeOperator

C. TypeAssertion

ANS : A

7. Which of the following is valid command to compile typescript file?

A. ts abc.ts

B. t abc.ts

C. tsc abc.ts

D. tst abc.ts

ANS : C

8. What will be output of : var a:string=47;

console.log("Value of a="+a);

A. Value of a=47

B. Value of a=0

C. Value of a=

D. None of the above

Ans : D

9. Internal Modules are known as namespaces in Typescript.

A. True

B. False

Ans : A

10. Which of the following Arrow Functions are true

A. var reflect = value => value;

B. var sum = (num1, num2) => num1 + num2;

C. var getName = () => "Nicholas";

D. var sum = (num1, num2) => { return num1 + num2; };

E. All

ANS : E

```
11. function fun1(...params) {  
    console.log(params.length);  
}  
  
    fun1();  
    fun1(5);  
    fun1(5, 6, 7);
```

Which concept resemble the above program

A. String Interpolation

B. Tagged Templates

C. Spread Operator/Rest Parameters

D.Object Destructuring

ANS : C

12.TypeScript is a

A.Strongly typed

B.Object oriented

C.Compiled Language

D.All the above

ANS : D

13.TypeScript is

A.Language

B.set of tools

C.Both

D.None

ANS : C

14.Select user defined data types in typescript

A.arrays

B.String

C.Boolean

D.Void

ANS : A

15.Which of the following are true

A.we can use printf in ts(typescript)

B.we can use console.log in ts(typescript)

ANS : B

16.Which of the following are array methods

A.Unshift()

B.Sort()

C.Pop()

D.All the above

ANS: D

17.Supertype of all datatypes in typescript

A.number

B.Enum

C.Void

D.Any

ANS : D

18.Which of the following are true

A.var a=new String(hii)

B.var b=new array[1]

C.var a:String=hii;

D.All the above

ANS : B

19. List some features of TypeScript?

A. TypeScript can be compiled to all major versions of Javascript(ES3,ES5,ES6,ES7)

B. TypeScript can be used for cross –browser development and is an open source project

C. TypeScript is a superset of JavaScript that provides typed nature to your code

D. All of the above

ANS : D



20.What are the components of TypeScript?

- A. TypeScript Language
- B. TypeScript Compiler
- C. TypeScript Language Service
- D. All of these

ANS : D

21) How to declare variables in Typescript?

- A) Var
- B) Let
- C) Both

ANS: C

22) Typescript uses which data type?

- A) String
- B) Number
- C) Boolean
- D)All above option

ANS: D

23.) Default access modifier typescript?

- A) public
- B) private
- c) protected

ANS: A

24) What is async/await?

- A.New syntax to write JavaScript promises
- B.New syntax to consume JavaScript promises
- C.New syntax to consume asynchronous callbacks

ANS: B

25)Which of the following is an attribute css selector?

- A.[selector]
- B. .selector
- C. #selector

ANS: A

26) When is it recommended to pass this.setState a function instead of an object

- A.When the new state should completely replace the old state .
- B.When the new state does not depend on the old state.
- C.When the new state depends on the old state.

ANS: C

27) A promise represents what?

- A. A single value in the future
- B. Multiple values in the present
- C. Multiple values in the future
- D. A single value in the present

ANS : A

28) Can a component be styled with a global styled sheet?

- A.Only function components can be styled with a global styled sheet
- B.Only if the component returns HTML elements
- C.Yes
- D.No

ANS : C

29) When binding to data in templates, how do you avoid issues with binding to properties of null or undefined objects?

- A. You don't need to worry about null objects, Angular will handle them gracefully.
- B. Use the \* operator on any object that could be null.
- C. Use the ? operator on any object that could be null.
- D. In the component, ensure the objects are never null.

ANS : C

30) Where should you sort and filter data?

- A. In a directive
- B. In a pipe
- C. In your component
- D. In a filter

ANS: C

31) What's the right syntax to assign the function doSomething as the handler of a click event?

- A. `onClick={doSomething}`
- B. `onClick={() => doSomething}`
- C. `onClick={doSomething()}`

ANS : A

32) Assuming you define the statement: "var person: string;" TypeScript will alert you that there is an error if you enter which additional code?

- A. `person = { name: 'Colleen', age: 25; }`
- B. `person = 0`
- C. `person = ['Colleen', 'John'];`
- D. all of the above

ANS : D

33) When bootstrapping an Angular app, how do you specify the main, top-level app component to be used?

- A. You add the component as a package in your SystemJs config.
- B. You load the component directly in your index.html file.
- C. You add the component to the bootstrap array in your main app module.
- D. You load the component directly from your main SystemJs package (e.g. main.ts).

ANS : C

34) What is the format of the data returned by the JSON Server API?

- YAML
- XML
- JSON

ANS : C

35) What syntax do you use to render the App component into an HTML element with id "root"?

- A. `ReactDOM.render(App, document.getElementById("root"));`
- B. `React.render(App, document.getElementById("root"));`
- C. `ReactDOM.mountNode(<App />, document.getElementById("root"));`
- D. `ReactDOM.render(<App />, document.getElementById("root"));`

ANS : D

36) What function can be used to change the state of a React component?

- A. `this.state = {}`
- B. `this.setState`
- C. `this.changeState`

ANS : B

37) What can we do if components A and B are siblings and they require access to the same state element?

- A. Restructure the application to have A and B be related differently
- B. Place the state in either A or B
- C. Place the state in a third component C that is a sibling to A and B

D.Place the state in the parent component of A and B

ANS : D

38) What is the name of the file that manages TypeScript projects?

- A.tsproj.config
- B.tsmeta.json
- C.tsconfig.json
- D.package.json

ANS : A

39) When dealing with asynchronous functions, what type of generic must your return type be?

- A.Async
- B.Promise
- C.Array
- D.AsyncAwait

ANS : B

40) What tool installs type declaration files?

- A.Duck
- B.npm
- C.Ruby
- D.NuGet
- E.Async

ANS : C updated to d=b -> npm

41) What should appear to the left of the arrow in an arrow function?

- A.Function name
- B.Function parameters
- C.Curly braces
- D.Function return value
- E.Number of function parameters.

ANS : B

42) What character separates a variable from its type annotation in a declaration?

- A. #
- B.\$
- C.%
- D. :

ANS : D

43) How would you define a string type variable with the value bar in TypeScript?

- A.let string: foo = 'bar';
- B.let foo:string = 'bar';
- C.let string foo = 'bar';
- D.let foo (string) = 'bar';

ANS : B

44) What keyword requires a variable to be declared before it's used?

- A.let
- B.dim
- C.declare
- D.set

ANS : A

45) What keyword declares a class member in the class rather than an instance of the class?

- A.enum
- B.static

C.intersection  
D.function  
E.async  
ANS : B

46) Within which function scope would a variable that you declare using the keyword var cover?  
A.Within the entire function scope  
B.Within an if statement scope  
C.Within a for loop scope  
D.Within a while loop scope  
ANS : A

47) What keyword lets you derive a subclass from a superclass?  
stretches  
extends  
sub  
follows  
buildsUpon  
ANS : B

## ANGULAR :

1. We need to tell AngularJS what part of our HTML page contains the AngularJS app. You do so by adding the . . . . attribute to the root HTML element of the AngularJS app.  
A) ng-app  
B) ag-app  
C) js-app  
D) aj-app  
Ans: A

2. AngularJS can show or hide HTML depending on the state of data in the model. You do so using a set of AngularJS directives such as . . . . . which are created specifically for that purpose.  
A) ng-shown, ng-hidden  
B) ng-show, ng-hide  
C) nt-show, nt-hide  
D) ng-shows, ng-hides  
Ans: B

3. EventEmitter class's simple interface, which basically encompass two methods . . . . . can therefore be used to trigger custom events and listen to events as well, both synchronously or asynchronously.  
A) exit()  
B) superscript()  
C) subscribe()  
D) emit()  
Ans: C

4. The . . . . . directive substitutes the normal href property and makes it easier to work with route links in Angular 2. Moreover if we want to navigate between routes, we use the . . . . . directive.  
A) RouterLinks  
B) RouterLinking  
C) RouterLink  
D) RouterLinq  
Ans: C

5. Three main components of Routing are . . . . .  
A) RouteObject  
B) RouterOutlet

C) RouterState  
D) RouterLink  
Ans: A,B,D

6. Route . . . . . allow us to pass values in our url to our component so we can dynamically change our view content.  
A) pipes  
B) modules  
C) variables  
D) parameters  
Ans: D

7. To use observables, Angular uses a third-party library called . . . . .  
A) ng2-bootstrap  
B) angular-cli  
C) numeral  
D) Reactive Extensions (RxJS)  
Ans: D

8. Observables help you manage . . . . . data.  
A) synchronous  
B) asynchronous  
C) Both asynchronous & synchronous  
D) None of above  
Ans: B

9. We subscribe to the observable ourselves using the actual . . . . . method. The downside is that you have to manage the subscription yourself.  
A) subscribe()  
B) subs()  
C) subscribed()  
D) None of above  
Ans: A

10. A promise represents the final result of an . . . . . operation.  
A) asynchronous  
B) synchronous  
Ans: A

11. The promise becomes . . . . . after resolving or rejecting.  
A) mutable  
B) ignored  
C) immutable  
D) recurring  
Ans: C

12. Which of the following is not built-in pipe in Angular  
A) DatePipe  
B) CurrencyPipe  
C) DataPipe  
D) PercentPipe  
Ans: C

13. If you chain multiple pipes together, they are executed . . . . .  
A) in parallel  
B) LIFO order  
C) in the order in which you specify them  
D) none of above  
Ans: C

14. The . . . . . filter waits for a promise and returns the most recent value received. It then updates the view.

- A) sync
- B) async
- C) promise
- D) custom

Ans: B

15. The . . . . . decorator allows us to define the pipe name that is globally available for use in any template in the application.

- A) @pipeName
- B) @pipeDeco
- C) \$Pipe
- D) @Pipe

Ans: D

16. Data bindings are applied as . . . . . or as special sequences of characters in strings.

- A) expressions
- B) attributes on HTML elements
- C) elements itself
- D) none of above

Ans: B

17. A data binding has these four parts:

- A) host element, curly brackets, target, expression
- B) host element, square brackets, target, evaluation
- C) host element, square brackets, target, expression
- D) host element, parentheses, target, expression

Ans: C

18. Attribute binding is similar to property binding but is tied to the . . . . . rather than the DOM property.

- A) Page attribute
- B) DOM attribute
- C) Root attribute
- D) HTML attribute

Ans: D

19. To make your applications ready for animations, you must include the AngularJS Animate library and then you must refer to the . . . . . module in your application.

- A) ng-repeat
- B) ngAnimation
- C) ngAnimate
- D) None of above

Ans: C

20. Using the . . . . . animation function, the parent animation can allow the child animation to run at the exact correct time.

- A) animateChild
- B) childAnimate
- C) animateInner
- D) innerChild

Ans: A

21. Animations are fired using animation property bindings which are prefixed with an . . . . . symbol.

- A) &
- B) #
- C) \$
- D) @

Ans: D

22. . . . . is an animation-specific function that is designed to be used inside of Angular's animation DSL language. It is used to kick off a reusable animation that is created using animation().

- A) useAnimation
- B) reuseAnimation
- C) in-useAnimation
- D) All of above

Ans: A

23. It's always possible to make use of animation input parameters by setting even more data via the . . . .property.

- A) options.param
- B) option.params
- C) option.param
- D) options.params

Ans: D

24. Which of the following bootstrap styles are used to create a vertical pills navigation?

- A. .nav, .nav-tabs
- B. .nav, .nav-pills
- C. .nav, .nav-pills, .nav-stacked
- D. .nav, .nav-tabs, .nav-justified

Ans: C

26). Which of the following bootstrap styles are used to create a justified tabs navigation?

- A.nav, .nav-tabs
- B.nav, .nav-pills
- C.nav, .nav-pills, .nav-stacked
- D.nav, .nav-tabs, .nav-justified

Answer : D

25.Which class indicates a dropdown menu?

- a) .dropdown-list
- b) .select
- c) .dropdown

Ans: C

26. Ques. Bootstrap's grid system allows up to

- a. 6 columns across the page
- b. 12 columns across the page
- c. columns across the page
- d. columns across the page

Ans: B

27. Which of the following classes makes a round corner image automatically adjust to fit the size of the screen?

- a. .img-res-image
- b. .img-responsive-image \* img-fluid (for full width responsive)
- c. .img-responsive
- d. .img-res

Ans: C

28. Which class should be used to indicate a button group?

- a. btn-group-buttons
- b. btn-group
- c. btn-grp
- d. btn-buttons

Ans: B

29. On the opposite side of event bindings (()) lie Angular's square-bracket syntax ([[]]) which signify a . . . .

- A) property binding
- B) class binding
- C) style binding
- D) both A & B

Ans: A

30. What are the types of access modifiers supported by TypeScript?

- A. Public B. Private C. Protected D. All of these

Ans: D

31. The purpose of the @Output decorator in a nested component is to:

- A. Expose an output channel for logging
- B. Expose an event property that a container can listen for using event binding
- C. Expose a property that a container can set using property binding

ANS: B

32. A function that adds metadata to a class, its members, or its method arguments is a(n):

- A. Attribute
- B. Directive
- C. Annotation
- D. Decorator

ANS: D

33. What is the current recommended way to register a service with the root Angular injector?

- A. By setting the providedIn property of the Injectable decorator for the service.
- B. By setting the providedIn property of the Component decorator for the component that needs it
- C. By declaring the service in an Angular module
- D. By setting the providedIn property of the Component decorator for the App component.

ANS : A

34. What does the Angular CLI do when you execute npm start/ng serve?

- A. It builds the application for production and starts the deployment process
- B. It compiles the application and starts a web server to serve the application to the browser
- C. It starts the default editor for the application so you can make changes
- D. It installs the packages defined in the package.json file

ANS : B

35. What is the purpose of the \*ngIf directive?

- A. To use style classes to hide or show elements based on an expression
- B. To repeat a set of elements based on an expression
- C. To optionally navigate to another view based on an expression
- D. To add or remove elements from the DOM based on an expression

ANS : D

36. Route definitions are configured in an array passed to:

- A. A method of the AppModule
- B. A method of the routed component
- C. A method of the AppComponent
- D. A method of the RouterModule

ANS : D

37. What is the purpose of the subscribe method on an observable?

- A. To map observable responses to alternate formats
- B. To transform the Observable stream
- C. To request logging of observable actions
- D. To request notifications and data from the observable

ANS: D



38.What is the purpose of the Angular CLI?

- A.Installing and running Angular applications.
- B.Building, executing, testing, and deploying Angular applications
- C.Testing and deploying Angular applications
- D.Executing, testing, and debugging Angular applications.

ANS: B

39.What is the purpose of a feature module?

- A.To define all the logic for your application all in one place
- B.To separate out the responsibilities for a specific feature set into its own Angular module
- C.To consolidate Angular system modules, such as BrowserModule and FormsModule, into a logical group
- D.To provide services specifically for one set of application features

ANS: B

40.Which of the following is the correct syntax for two-way binding?

- A. [(ngModel)]='listFilter'
- B. [(ngModel)]='listFilter'
- C. {{listFilter}}
- D.ngModel='listFilter'

ANS : B

41. In Angular / RxJs , when should you unsubscribe from a Subscription ?

- A. There is no need to unsubscribe . Angular routing handles it.
- B. Unsubscribe them in ngOnDestroy.

ANS :- B.

42. When you apply an Angular 'pipe' , it changes the value of the underlying component's member variable as well.

- A . TRUE.
- B. FALSE.

ANS :- B.

43. What's the best way to inject one service into another in Angular ?

- A . A service cannot be injected in another service.
- B. Through constructor of depending service.
- C. By instantiating using a new operator.

ANS : B

44. In Angular, how can you set a header in every http request ?

- A. Use HttpClientInterceptor.
- B. Extend HttpClient class.
- C. Set as a request header parameter.

ANS : A.

45. In Angular, you can pass data from parent component to child component using \_\_\_\_\_.

- A : '@Output()'
- B . '@Input()'
- C. Input.
- D. Output

ANS : B

46. In Angular, you can pass data from child component to parent component using \_\_\_\_\_

- A : '@Output'
- B . '@Input'
- C. Input
- D. Output

ANS : A

47. In Angular , one can create a local HTML reference of an HTML tag using a variable , which starts with character \_\_\_\_ .

- A. @
- B. #
- C. \*
- D. &

48. If you provide a service in two Angular components, in the "providers" section of @Component decorator, how many instances of service shall get created?

- A. 4
- B. 2
- C. 3

49. In Angular routing , which of these tags is used to show the selected route component dynamically ?

- A. <router></router>
- B. <router-output></router-output>
- C. <router-outlet></router-outlet>
- D. <router-input></router-input>

ANS : C

50. Which method of Angular RouterModule should be called for providing all routes in AppModule?

- A. RouterModule.forChild
- B. RouterModule.forRoot
- C. RouterModule
- D. RouterModule.all

ANS : B

## JAVA SPRING :

Question: 1 Expression Language is the part of Core Container in Spring Framework.

Your Answer: None

Correct Answer: True

Description: None

Question: 2 AOP is the part of Core Container in Spring Framework.

Your Answer: None

Correct Answer: False

Description: None

Question: 3 In which spring version has Spring Expression Language been supported?

Your Answer: None

Correct Answer: Spring 3.0

Description: None

Question: 4 Which class represents the IoC container?

Your Answer: None

Correct Answer: ApplicationContext

Description: None

Question: 5 How to get the object of DAO in spring framework?

Your Answer: None

Correct Answer: Using Spring Dependency injection

Description: None

Question: 6 Which is correct for Spring Framework?

Your Answer: None

Correct Answer: Spring framework is a light-weight solution.

Description: None

Question: 7 In which spring version, java 5 features has been introduced?

Your Answer: None

Correct Answer: Spring 3.0

Description: None

Question: 8 Can we integrate spring with struts?

Your Answer: None

Correct Answer: Yes

Description: None

Question: 9 How to use idref in spring framework?

Your Answer: None

Correct Answer: With setter method and constructor argument both

Description: None

Question: 10 Which is the part of the Data Access layer in Spring Framework?

Your Answer: None

Correct Answer: JMS

Description: None

1. Does Spring provide programmatic transaction management? Select a unique answer.

- a. Yes using the @Transactional bean post processor
- b. Yes with the TransactionTemplate class
- c. Yes with the TransactionService class
- d. Yes using the @Transactional bean post processor

ans: b

Explanation

- 1. The TransactionTemplate class provides an execute(TransactionCallback) method
- 2. The TransactionService class does not exist
- 3. The @Transactional annotation is for declarative transaction management

2. How could you externalize constants from a Spring configuration file or a Spring annotation into a .properties file? Select one or more

- a. By using the <context:property-placeholder /> tag
- b. By using the <util:constant /> tag
- c. By declaring the ConstantPlaceholderConfigurer bean post processor
- d. By using the c: namespace

ans: a

3. To validate Java beans in a web application using annotations.

- a) XML
- b) Java Based
- c) JAR-303 standard
- d) All of the mentioned

ans: d

4. Annotation used to indicate a field has to have a minimum of 2 characters.

- a) @NotNull
- b) @Size
- c) @MaxSize
- d) @size

Answer: b

Explanation: @Size annotation used to indicate a field has to have a minimum of 2 characters.

5. Spring MVC supports generating Excel files using which of the following libraries.

- a) Apache POI library
- b) JExcelAPI library
- c) All of the mentioned
- d) None of the mentioned

ans: C

6. Interface for Dispatcher Servlet to auto detect view resolver beans.

- a) localeResolver
- b) tiles
- c) ViewResolver
- d) none of the mentioned.

ans: C

Explanation: Spring MVC comes with several ViewResolver implementations for you to resolve views using different strategies.

7. By default, XmlViewResolver loads view beans from /WEB-INF/views.xml which is final.

- a) True
- b) False

ans: b

8. Views that cannot be resolved by InternalResourceViewResolver.

- a) redirect
- b) redirect:prefix
- c) redirect:suffix
- d) all of the mentioned

ans: a

9. Annotation which allows a controller's field to be assigned using Spring Expression Language(SpEL)

- a) @Value
- b) @After
- c) @Default
- d) @None

ans: a

10. SpEL statements are recognizable.

- a) True
- b) False

View Answer

Answer: a

Explanation: They use a notation in the form "{ SpEL statement }".

11. To publish a REST service with Spring.

- a) publishing an application's data as a REST service
- b) accessing data from third-party REST services
- c) none of the mentioned
- d) all of the mentioned

ans: a b

12. Publishing an application's data as a REST service requires.

- a) @RequestMapping
- b) @PathVariable
- c) All of the mentioned
- d) None of the mentioned

ans: a b

13. Annotation added as an input parameter to the handler method.

- a) @PathVariable
- b) @Path
- c) @PathLocale
- d) None of the mentioned

ans: a

14. Which of the following statements is true regarding the @ResponseStatus annotation?

- a. @ResponseStatus is detected on nested exceptions
- b. TheExceptionHandlerResolver uses the @ResponseStatus annotation to map exception to HTTP status code
- c. A controller handler is annotated with the @ResponseStatus, the response status set by RedirectView takes precedence over the annotation value.
- d. The @ResponseStatus annotation can go on a @RequestMapping method or a @RestController class or a business exception class.

ans: a

15. Default localeResolver used by Spring.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) AcceptLocaleResolver

ans: c

16. Alternative way to resolve locales.

- a) AcceptHeaderLocale
- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) SessionLocaleResolver

ans: d

17. General-purpose class that allows a response to be rendered using a marshaller.

- a) MarshallingView
- b) Marshaling
- c) View
- d) All of the mentioned

View Answer

Answer: a

Explanation: The member template view is defined as a MarshallingView type, which is a general-purpose class that allows a response to be rendered using a marshaller.

18. Annotation which allows the Jaxb2Marshaller marshaller to detect a class's (i.e., object's) fields.

- a) @XmlElement
- b) @XmlRoot
- c) @NotNull
- d) None of the mentioned

ans: a

19. How to auto-inject into a field a Spring bean by its name? Select one or more answer choices.

- a. By using both the @Autowired and the @Qualifier Spring annotations
- b. By using the @Autowired annotation and naming the field with the bean name
- c. With the name attribute of the @Autowired annotation
- d. By using the single @Qualifier annotation

ans : a

20. Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the following pointcut expression? execution(\* \*..AccountServiceImpl.update(..)) && execution(\* \*..ClientServiceImpl.update(..))

- a. No joint point is defined
- b. Matches public update methods of the 2 classes, whatever the arguments
- c. Matches any update methods of the 2 classes, whatever the arguments and method visibility
- d. Matches any update methods of the 2 classes, with one more arguments and whatever method visibility

ans: a

21. Select the right statement about referring a Spring configuration file inside the package com.example.myapp in the below example?

ApplicationContext context = new ClassPathXmlApplicationContext("classpath:/com.example.myapp.config.xml");

- a. all of the above
- b. The classpath: prefix could be omitted
- c. Package name using the dot character is not well formatted
- d. The slash character preceding com.example could be omitted

ans: a

22. Using the Spring AOP framework, what is the visibility of the method matched by the following join point?

```
@Pointcut("execution(* *(..))")
private void anyOperation() {};
```

- a. Public methods
- b. All methods, regardless of their visibility
- c. All methods, except private method
- d. Protected and public methods

ans: a

23. What is the name of the bean defined in the following configuration class? Select a single answer.

```

@Configuration
public class ApplicationConfig {
    @Autowired
    private DataSource dataSource;
    @Bean
    ClientRepository clientRepository() {
        ClientRepository accountRepository = new JpaClientRepository();
        accountRepository.setDataSource(dataSource);
        return accountRepository;
    }
}

```

- a. clientRepository
  - b. JpaClientRepository
  - c. jpaClientRepository
  - d. Two beans are defined: a data source and a repository
- ans: a

24. Which of the following is true regarding the below Spring controller?

```

@RestController
public class OwnerController {
    @RequestMapping(value = "/owner/{ownerId}", method = RequestMethod.POST)
    @ResponseBody
    public Owner findOwner(@PathVariable("ownerId") int ownerId) {
        return new Owner();
    }
}

```

- a. RequestMethod.GET method is more accurate than POST
  - b. @ResponseBody could be removed
  - c. @PathVariable should be replaced with the @PathParam annotation
  - d. Returning the 201 HTTP status code is better
- ans: a

25. What is advice? Select a unique answer.

- a. An action taken by an aspect at a particular join point
  - b. A point during the execution of a program
  - c. An aspect and a pointcut
  - d. A predicate that matches join points
- ans: a

26. What is the easiest method to write a unit test?

- a. @RequestMapping("/displayAccount")  
String displayAccount(@RequestParam("accountId") int id, Model model)
  - b. void displayAccount(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException
  - c. void displayAccount(HttpServletRequest req, HttpSession session) throws ServletException, IOException
  - d. @RequestMapping("/displayAccount")  
String displayAccount(@PathVariable("accountId") int id, Model model)
- ans: a

27. Select method's signatures that match with the following pointcut:

- ```
execution(* com.test.service..*.*(*))
```
- a. void com.test.service.MyServiceImpl#transfert(Money amount)
  - b. void com.test.service.account.MyServiceImpl#transfert(Money amount)
  - c. void com.test.service.MyServiceImpl#transfert(Account account, Money amount)
  - d. void com.test.service.account.MyServiceImpl#transfert(Account account, Money amount)
  - e. None of the above
- ans: a

28. Given the Spring configuration file, which are the correct statements?

- ```

<bean class="com.spring.service.BankServiceImpl"
    p:bankName="NationalBank">
</bean>

```
- a. The p namespace has to be declared
  - b. NationalBank is a scalar value
  - c. Bean id is bankServiceImpl
  - d. The BankServiceImpl references a NationalBank bean

ans: a

29. Given the following configuration class, what are the correct affirmations? Select one or more answers.

```
public class ApplicationConfig {  
    private DataSource dataSource;  
    @Autowired  
    public ApplicationConfig(DataSource dataSource) {  
        this.dataSource = dataSource;  
    }  
    @Bean(name="clientRepository")  
    ClientRepository jpaClientRepository() {  
        return new JpaClientRepository();  
    }  
}
```

- a. Configuration annotation is missing
- b. Default or no-arg constructor is missing
- c. @Bean name is ambiguous
- d. @Bean scope is prototype

ans: a

30. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object? Select a unique answer.

- a. ResultSetExtractor
- b. RowMapper
- c. RowCallbackHandler
- d. ResultSetMapper

ans: B

Q 1 - Which of the following is correct about dependency injection?

- A - It helps in decoupling application objects from each other.
- B - It helps in deciding the dependencies of objects.
- C - It stores objects states in a database.
- D - It stores object states in the file system.

Answer : A

Q 2 - Which are the correct implementation classes of ApplicationContext?

- A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext
- B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext
- C - AdvancedApplicationContext, FileApplicationContext
- D - FileSystemApplicationContext, ClassPathApplicationContext

Answer : A

Q 3 - What is true about <set> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

Answer : B

Q 4 - What is the constructor mode of autowiring?

- A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.
- B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.
- C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the bean name in the configuration file.
- D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

Answer : D

Q 5 - What is true about @Autowired annotation?

- A - The @Autowired annotation can be used to autowire bean on the setter method.
- B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.
- C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.
- D - All of the above.

Answer : B

Q 6 - What are the different points where weaving can be applied?

A - Compile time, load time  
B - Compile time, run time  
C - Run time  
D - Compile time, load Time, Run time  
Answer : D

Q 7 - What is Spring MVC framework?  
A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.  
B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.  
C - Spring MVC framework is used for Transaction management for Web Applications.  
D - Spring MVC framework is used for AOP for Web Applications.  
Answer : B

Q 8 - By default a bean is lazily loaded.  
A - True  
B - False  
Answer : B

Q 9 - What is the scope of bean in portlet context?  
A - session  
B - global-session  
C - prototype  
D - request  
Answer : B

Q 10 - Which of the following classes can be used to call Stored Procedures in spring?  
A - SPHelper  
B - JdbcTemplateCall  
C - JdbcTemplate  
D - SimpleJdbcCall  
Answer : D

Q 42 - What is a ContextRefreshedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the HTTP Response is returned.  
D - This event is published when the ApplicationContext is either initialized or refreshed.  
D

Q 43 - What is a ContextStartedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the ApplicationContext is started using the start method on the ConfigurableApplicationContext interface.  
D - This event is published when the HTTP Response is returned.  
C

Q 44 - What is ContextStoppedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the ApplicationContext is stopped using the stop method on the ConfigurableApplicationContext interface.  
C - This event is published when the HTTP Request is received.  
D - This event is published when the HTTP Response is returned.  
B

Q 45 - What is ContextClosedEvent event?  
A - This event is published when the Servlet Context is either initialized or refreshed.  
B - This event is published when the HTTP Request is received.  
C - This event is published when the HTTP Response is returned.  
D - This event is published when the ApplicationContext is closed using the close method on the ConfigurableApplicationContext interface.  
D

Q 46 - What is RequestHandledEvent:event?



- A - This event is published when the Servlet Context is either initialized or refreshed.
  - B - This event is published when the HTTP Request is received.
  - C - This event is published when the HTTP session is initialized or refreshed.
  - D - This event is published when the HTTP Request is serviced.
- C

Q 2 - What is Introduction?

- A - An introduction represents a point in your application where you can plug-in the AOP aspect.
- B - This is used to inject values in objects.
- C - This is not invoked during program execution by Spring AOP framework.
- D - An introduction allows you to add new methods or attributes to existing classes.

Answer: D

Q 3 - What is @Controller annotation?

- A - The @Controller annotation indicates that a particular class serves the role of a controller.
- B - The @Controller annotation indicates how to control the transaction management.
- C - The @Controller annotation indicates how to control the dependency injection.
- D - The @Controller annotation indicates how to control the aspect programming.

Answer: A

Q 4 - What is the prototype scope?

- A - This scopes a single bean definition to have any number of object instances.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

Answer: A

Q 5 - What is true about <props> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings

Answer: D

Q 6 - How to handle shut down of IoC containers?

- A - Using shutdownHook()
- B - Using shutdownHandler()
- C - Using registerHook()
- D - Using registerShutdownHook()

Answer: D

Q 7 - By default a bean is eagerly loaded.

- A - False
- B - True

Answer: B

Q 8 - What is true about BeanPostProcessor?

- A - It is a concrete class.
- B - It is an interface.
- C - It is an abstract class.
- D - None of the above.

Answer: B

Q 9 - How can you inject Java Collection in Spring?

- A - Using list, set, map or props tag.
- B - Using list, set, map or collection tag.
- C - Using list, set, props or collection tag.
- D - Using list, collection, map or props tag.

Answer: A

Q 10 - Which class is used to map a database row to a java object in spring?

- A - ResultSet
- B - RowMapper
- C - RowSetMapper
- D - ResultSetMapper

Answer: B

Q 11 - How can the bean life cycle be controlled?

- A - Using init() only
- B - Using InitializingBean class only
- C - Using DisposableBean class only
- D - Using All of above

Answer:D

Q 12 - What is the scope of bean in portlet context?

- A - session
- B - global-session
- C - prototype
- D - request

Answer:B

Q 13 - What is a ContextStoppedEvent event?

- A - This event is published when the Servlet Context is either initialized or refreshed.
- B - This event is published when the ApplicationContext is stopped using the stop() method on the ConfigurableApplicationContext interface.
- C - This event is published when the HTTP Request is received.
- D - This event is published when the HTTP Response is returned.

Answer:B

Q 14 - Which ORM Spring supports ?

- A - Hibernate
- B - iBatis
- C - JPA
- D - All of the above.
- E - None of the above.

Answer:D

Q 15 - What is true about <map> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This tag is not supported.

Answer:C

Q 16 - Core container has AOP as one of its modules.

- A - True
- B - False

Answer:B

Q 17-Can you inject null and empty string values in Spring?

- A - Yes
- B - No

Answer:A

Q 18 - What stands true for the spring framework?

- A - Spring framework is a lightweight framework.
- B - Spring framework is a heavy weight framework.

Answer:A

Q 19 - Which class acts as IoC Container?

- A - ServletContext
- B - DispatcherServlet
- C - ApplicationContext
- D - None of the above

Answer:C

Q 20 - What is the scope of stateless bean?

- A - global-session
- B - singleton
- C - prototype

D - request  
Answer:B

Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. true
2. false

ANS : A

Which of the following layers the @Controller annotation is used in?

1. Service layer.
2. Presentation layer.
3. Session layer.
4. Business layer.
5. None of these.

ANS : B

Which of the following statements is true?

1. ApplicationContext extends BeanFactory.
2. None of these.
3. ApplicationContext implements BeanFactory.
4. BeanFactory extends ApplicationContext.
5. BeanFactory implements ApplicationContext

ANS : A

Q 20 - What is singleton scope?

- A - This scopes the bean definition to a single instance per Spring IoC container.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 21 - What is the prototype scope?

- A - This scopes a single bean definition to have any number of object instances.
- B - This scopes the bean definition to a single instance per HTTP Request.
- C - This scopes the bean definition to a single instance per HTTP Session.
- D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

A

Q 22 - What is the request scope?

- A - This scopes a bean definition to an HTTP request.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP Session.
- D - This scopes the bean definition HTTP Application/ Global session.

A

Q 23 - What is session scope?

- A - This scopes a bean definition to an HTTP session.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Application/ Global session.

A

Q 24 - What is global-session scope?

- A - This scopes a bean definition to an HTTP Application/ Global session.
- B - This scopes the bean definition to Spring IoC container.
- C - This scopes the bean definition to HTTP request.
- D - This scopes the bean definition to HTTP Session.

A

Q 30 - What is true about collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

ANS : C

Q 30 - What is true about <props> collection configuration elements?

- A - This helps in wiring a list of values, allowing duplicates.
- B - This helps in wiring a list of values but without any duplicates.
- C - This can be used to inject a collection of name-value pairs where name and value can be of any type.
- D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

ANS : D

Q 48 - What is the Join point?

- A - This represents a point in your application which joins two objects.
- B - This represents a point in your object where you join values.
- C - This represents a point in your object where you join injected values.
- D - This represents a point in your application where you can plug-in the AOP aspect.

ANS :- D

Q 49 - What is Advice?

- A - This is the way to instruct an object to behave in a certain manner.
- B - This is used to inject values in objects.
- C - This is the actual action to be taken either before or after the method execution.
- D - This is not invoked during program execution by Spring AOP framework.

ANS :- C

Q 50 - What is Pointcut?

- A - This represents a point in your application where you can plug-in the AOP aspect.
- B - This is a set of one or more join points where an advice should be executed.
- C - This is used to inject values in objects.
- D - This is invoked during program execution by Spring AOP framework.

ANS :- B

Q 2 - What is the Target object?

- A - A represents an object in your application where you can plug-in AOP aspect.
- B - The object being advised by one or more aspects, this object will always be a proxy object, also referred to as the advised object.
- C - This is used to inject values in objects.
- D - This is not invoked during program execution by Spring AOP framework.

B

Q 3 - What is Weaving?

- A - Weaving is the process of injecting values in objects to create an advised object.
- B - Weaving is the process of linking aspects with other application types or objects to create an advised object.
- C - This is used to inject values in objects.
- D - Weaving is used to check object dependencies.

B

Q 10 - Which of the following aspects implementation spring supports?

- A - XML Schema based aspect implementation
- B - @AspectJ based aspect implementation
- C - Both of the above.
- D - None of the above.

C

Q 11 - What are the types of the transaction management Spring supports?

- A - Programmatic transaction management
- B - Declarative transaction management
- C - Both of the above.
- D - None of the above.

C

Q 12 - What is Spring MVC framework?

- A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.
- B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.
- C - Spring MVC framework is used for Transaction management for Web Applications.
- D - Spring MVC framework is used for AOP for Web Applications.

B

Q 43 - What is a DispatcherServlet?

- A - DispatcherServlet is used for transaction management.

B - DispatcherServlet is used for AOP.  
C - DispatcherServlet handles all the HTTP requests and responses.  
D - DispatcherServlet is used for Dependency injection.  
C

Q 44 - What is ACID in transactional management?

A - Accurate, Controlled, Isolation, Durability  
B - Atomicity, Consistency, Independent, Done  
C - Atomicity, Consistency, Isolation, Durability  
D - Accurate, Controlled, Independent, Done  
C

Q 45 - Where do you define DispatcherServlet?

A - In Beans configuration file.  
B - Web.xml file  
C - Meta-inf/dispatcher.xml  
D - Web-inf/ dispatcher.xml  
B

Q 46 - What is true about BeanPostProcessor?

A - It is a concrete class.  
B - It is an interface.  
C - It is an abstract class.  
D - None of the above.  
B

Q 47 - What BeanPostProcessor does?

A - It processes beans once a bean is initialized.  
B - It defines callback methods that you can implement to provide your own instantiation logic, dependency-resolution logic etc.  
C - It processes beans once a bean is loaded.  
D - It processes beans once a bean exits.  
B

Q 48 - Can a bean be configured to have an inner bean?

A - True  
B - False  
A

Q 49 - Can we inject value and ref both together in a bean?

A - True  
B - False  
A

Q 50 - Following class can be extended to create custom events in spring.

A - SpringEvent  
B - Event  
C - ApplicationEvent  
D - None of above  
C

Q 2 - Which of the following is the correct assertion about spring?

A - Spring enables developers to develop enterprise-class applications using POJOs.  
B - Spring is organized in a modular fashion.  
C - Testing an application written with spring is simple because environment-dependent code is moved into this framework.  
D - All of the above.  
D

Q 5 - What AOP stands for?

A - Aspect Oriented Programming  
B - Any Object Programming  
C - Asset Oriented Programming  
D - Asset Oriented Protocol  
A

Q 6 - What is true about cross-cutting concerns?

A - The functions that span multiple points of an application are called cross cutting concerns.  
B - Cross-cutting concerns are conceptually separate from the application's business logic.

C - Logging is one of the examples of cross cutting concerns.

D - All of the above.

A

Q 7 - Which are the modules of the core container?

A - Beans, Core, Context, SpEL

B - Core, Context, ORM, Web

C - Core, Context, Aspects, Test

D - Bean, Core, Context, Test

A

Q 8 - Which are the modules of the Data Access/ integration layer?

A - JDBC, ORM, OXM, JMS, Transactions

B - JDBC, ORM, OXM, JMS

C - JDBC, ORM, Web, Beans

D - JDBC, ORM, OXM, JMS

A

Q 9 - Which are the modules of the Web layer?

A - WebSocket, Servlet, Web, Portlet

B - WebSocket, Servlet, Web-MVC, Web

C - HTML, JSP, WEB, Portlet

D - HTML, Servlet, WEB, Portlet

A

Q 11 - Which of the statements is correct?

A - The JDBC module provides a JDBC-abstraction layer that removes the need to do tedious JDBC related coding.

B - The ORM module provides integration layers for popular object relational mapping APIs, including JPA, JDO, Hibernate, and iBatis.

C - The Java Messaging Service JMS module contains features for producing and consuming messages.

D - All of the above.

D

Q 13 - What types of Dependency injection does spring supports?

A - Constructor based, Setter based

B - Constructor based, Setter based, Getter Based

C - Setter based, Getter based, Properties based

D - Constructor based, Setter based, Properties based

A

Q 14 - Which are the IoC containers in Spring?

A - BeanFactory, ApplicationContext

B - BeanFactory, ApplicationContext, locContextFactory

C - BeanFactory, BeanContext, locContextFactory

D - BeanFactory, ApplicationContext, BeanContext

A

Q 17 - Which of the following stands true for spring beans?

A - Spring beans are managed by the Spring IoC container.

B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.

C - Spring beans are simple POJOs.

D - All of the above.

D

Q 18 - Which is the way to provide configuration metadata to spring?

A - XML Based configuration file.

B - Annotation based configuration.

C - Java based configuration.

D - All of the above.

D

Q 19 - What is bean scope?

A - Bean scope forces Spring to produce a new bean instance as per the scope defined.

B - Bean scope defines the accessibility of bean in a java class.

C - Bean scope defines the accessibility of bean in a java package.

D - Bean scope defines the accessibility of bean in a web application.

A

Q 33 - What is no mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Similar to byType, but type applies to constructor arguments.

A

Q 34 - What is byName mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Similar to byType, but type applies to constructor arguments.

B

Q 35 - What is byType mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in the configuration file.

D

Q 36 - What is the constructor mode of autowiring?

A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the bean names in the configuration file.

D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

D

Q 37 - What is autodetect mode of autowiring?

A - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in the configuration file.

C

Q 15 - Which is the correct implementation class of BeanFactory?

A - XmlBeanFactory

B - ClassPathBeanFactory

C - FileSystemBeanFactory

D - AdvancedBeanFactory

A

Q 40 - What does @Required annotation mean?

A - This annotation indicates that bean property must be populated by the user.

B - This annotation indicates that bean property is required while saving the bean data to the database.

C - This annotation simply indicates that the affected bean property must be populated at configuration time, through an explicit property value in a bean definition or through autowiring.

D - This annotation indicates that bean property is required while serializing the bean.

C

Q 41 - What is true about @Autowired annotation?

A - The @Autowired annotation can be used to autowire bean on the setter method.

B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.

C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.

D - All of the above.

B

Q 47 - What is the aspect?

- A - Aspect is a way to do the dependency injection.
  - B - A module which has a set of APIs providing cross-cutting requirements.
  - C - Aspect is used to log information about applications.
  - D - Aspect represents properties of spring based application.
- B

Q 48 - What is the Join point?

- A - This represents a point in your application which joins two objects.
  - B - This represents a point in your object where you join values.
  - C - This represents a point in your object where you join injected values.
  - D - This represents a point in your application where you can plug-in AOP aspect.
- D

Q 50 - What is Pointcut?

- A - This represents a point in your application where you can plug-in AOP aspect.
  - B - This is a set of one or more join points where an advice should be executed.
  - C - This is used to inject values in objects.
  - D - This is invoked during program execution by Spring AOP framework.
- B

Which of these operations instruct the IOC container to create a bean instance?

- The initialization method
- The destruction method
- The lazy-initialization mode
- The auto-wiring mode
- None of the above

ANS : E

Which interface in Spring MVC ensures that implementation is detected?

- WebApplicationInitializer
- WebInitializer
- XmlWebApplicationContext
- AbstractAnnotationConfigDispatcherServletInitializer
- None of the above

ANS : D

The getModeMap() method of ModelAndView returns a(n): \_\_\_\_\_ instance.

- Empty
- Http
- ModelMap
- Map

ANS : C

## SPRING AOP :-

Question: 1 What is an after returning advice? Select a unique answer.

Your Answer: None

Correct Answer: Advice to be executed after a join point completes without throwing an exception

Description: None

Question: 2 Select method's signatures that match with the following pointcut:

execution(\* com.test.service..\*.\*(\*))

Your Answer: None

Correct Answer: void com.test.service.MyServiceImpl#transfert(Money amount)

Description: None

Question: 3 Using the Spring AOP framework, what is the visibility of the method matches by the following join point?



```
@Pointcut("execution(* *(..))")
private void anyOperation() {};
```

Your Answer: None

Correct Answer: Public methods

Description: Due to the proxy-based nature of Spring's AOP framework, protected methods are by definition not intercepted, neither for JDK proxy nor for CGLIB proxies. As a consequence, any given pointcut will be matched against public methods only! To intercept private and protected methods, AspectJ weaving should be used instead of the Spring's proxy-based AOP framework.

Question: 4 Which one is the correct statement about AOP proxy?

Your Answer: None

Correct Answer: AOP proxies are created by Spring in order to implement the aspect contracts

Description: 1. An object created by the AOP framework in order to implement the aspect contracts

2. If the target object does not implement any interfaces then a CGLIB proxy will be created.

You could also use CGLIB proxy instead of JDK dynamic proxy

3. If the target object does not implement any interfaces then a CGLIB proxy will be created.

4. When CGLIB proxy is used, final methods cannot be advised, as they cannot be overridden.

Question: 5 What is a pointcut?

Your Answer: None

Correct Answer: An expression to identify join points

Description: None

Q 6 - How before advice works?

A - Run advice before a class loads.

B - Run advice before a method execution.

C - Run advice before http response is to be returned.

D - Run advice before http request is to be processed.

B

Q 7 - How after-returning advice works?

A - Run advice after a class loads only if class loads successfully.

B - Run advice after a method execution only if the method completes successfully.

C - Run advice after http response is returned only if http response is success.

D - Run advice after http request is processed with no exception.

B

Q 8 - How after-throwing advice works?

A - Run advice after a method execution only if the method exits by throwing an exception.

B - Run advice after a class loads only if the class throws an exception during load time.

C - Run advice after http response is returned with error status.

D - Run advice after http request is processed and an exception occurred.

A

Q 9 - How around advice works?

A - Run advice before and after the advised method is invoked.

B - Run advice before and after a class is loaded.

C - Run advice before and after http response is returned.

D - Run advice before and after http request is processed.

A

Q 12 - What is Spring MVC framework?

A - Spring MVC framework is Model-View-Controller architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

B

Question: 7 What are the unique correct answers about Spring AOP support?

Your Answer: None

Correct Answer: A point cut could select methods that have a custom annotation

Description: The @annotation designator enables to select methods that are annotated by a given annotation

Question: 8 What is advice? Select a unique answer.

Your Answer: None

Correct Answer: An action taken by an aspect at a particular join point

Description: None

Question: 9 Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the pointcut expressions?

execution(\* \*..AccountServiceImpl.update(..))

&& execution(\* \*..ClientServiceImpl.update(..))

Your Answer: None

Correct Answer: No joint point is defined

Description: Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the pointcut expressions?

execution(\* \*..AccountServiceImpl.update(..))

&& execution(\* \*..ClientServiceImpl.update(..))

Pointcut expression could not satisfy both first and second execution points. Do not confuse the && operator and || operator.

Question: 10 Using the Spring AOP framework, what are the joinpoint methods of the following pointcut expressions?

execution(public \* \*(..))

Your Answer: None

Correct Answer: The execution of all public method

Description: The execution of all public method

Question: 10 How many types of Dynamic proxy are available in Spring?

Your Answer: None

Correct Answer: two

Description: 1. JDK Dynamic Proxy

2. CGLIB Dynamic Proxy

Question: 11 What is the Type of Proxy in Spring Framework?

Your Answer: None

Correct Answer: two

Description: 1. static

2. Dynamic

Question: 6 How many types of the injection method in Spring?

Your Answer: None

Correct Answer: two

Which of the following is not a valid difference between @Inject and @Autowired annotations?

1. There is no difference and both can be used interchangeably.

2. @Autowired is a legacy annotation whereas @Inject is its latest version.

3. @Inject annotation is part of Java EE 7's Context and Dependency Injection framework while @Autowired is the Spring Frameworks own implementation.

4. None of these.

ANS :- B

Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. true

2. false

ANS : A

Question: 10 What is the role of ApplicationContextAware in spring?

Your Answer: None

Correct Answer: Make a bean aware to the container.

Description: None

Question: 9 Can the class which spring container makes subclass be final?

Your Answer: None

Correct Answer: No

Description: None

Question: 10 When a bean has scope limited to HTTP session, that is called

Your Answer: None

Correct Answer: Session scope

Description: None

Q 14 - What are the ways to access Hibernate by using Spring?

- A - Inversion of Control with a Hibernate Template and Callback.
  - B - Extending HibernateDAOSupport and Applying an AOP Interceptor node.
  - C - Both of above.
  - D - None of above.
- C

Q 15 - Which ORM Spring supports ?

- A - Hibernate
  - B - iBatis
  - C - JPA
  - D - All of above.
  - E - None of above.
- D

Q 16 - Which of the following database is not supported using jdbcTemplate?

- A - MySql
  - B - PostgresSql
  - C - NoSql
  - D - Oracle
- C

Q 24 - Expression Language/ SpEL was introduced in which version of spring framework.

- A - 1.0
  - B - 2.0
  - C - 3.0
  - D - 4.0
- C

Q 25 - Can we integrate Struts with Spring.

- A - Yes
  - B - No
- A

Q 26 - By default a bean is lazily loaded.

- A - True
  - B - False
- B

Q 27 - By default a bean is eagerly loaded.

- A - False
  - B - True
- B

Q 31 - If a bean can be created any number of times, scope is

- A - session
  - B - global-session
  - C - prototype
  - D - request
- C

Q 32 - What is the scope of stateless beans?

- A - global-session
  - B - singleton
  - C - prototype
  - D - request
- B

Q 33 - What is the scope of stateful beans?

- A - session
  - B - global-session
  - C - prototype
  - D - request
- C

Q 34 - If a bean is scoped to HTTP session, scope is

- A - global-session
  - B - session
  - C - prototype
  - D - request
- B

Q 41 - Which of the following classes can be used to execute Sql queries in spring?

- B - JDBCHelper
  - A - JdbcTemplate
  - C - DBHelper
  - D - DBTemplate
- A

Q 42 - Which of the following classes can be used to call Stored Procedures in spring?

- A - SPHelper
  - B - JdbcTemplateCall
  - C - JdbcTemplate
  - D - SimpleJdbcCall
- D

Q 50 - Following class can be extended to create custom events in spring.

- A - SpringEvent
  - B - Event
  - C - ApplicationEvent
  - D - None of above
- C

32. Annotation to add security to your spring boot app?

- a. @EnableWebSecurity
- b. @EnableSecurity
- c. @EnableSpringSecurity

ANS : A

2.Element used to specify access attributes for bean's methods.

- A. security:protect
- B. security:intercept
- C. security:intercept-security
- D. none of the mentioned

ANS : A

31. How to reload my changes on Spring Boot without having to restart the server?

- a. org.springframework.boot spring-boot-devtools true
- b. org.springframework.boot spring-boot-test true
- c. org.springframework.boot spring-boot-web true
- d. org.springframework.boot spring-boot-jasper true

ANS : A.

35. LoggerInterceptor which implements the HandlerInterceptor.The LoggerInterceptor will override the which following methods

- a. preHandlerBean(), postHandlerBean(), afterCompletion()
- b. PreProcessHandler(), PostProcessHandler(), afterCompletion()
- c. PreHandler(), PostHandler(), afterCompletion()
- d. none

ANS : C

34. How does the Spring Batch work?

- a. step, ItemReader, ItemProcessor, ItemStreamWriter
- b. step, ItemProcessor, ItemStreamWriter

c. step, ItemReader, ItemStreamWriter  
ANS : A.

Q - Which of the following stands true for spring beans?  
A - Spring beans are managed by the Spring IoC container.  
B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.  
C - Spring beans are simple POJOs.  
D - All of the above.  
Answer : D

## JSON :

31) Which of the following codes will not throw an error?

- A. JSON.parse("");
- B. JSON.parse(null);
- C. JSON.parse();
- D. JSON.parse({});

Ans: B

32) What does JSONP stand for?

- A. JSON Procedures
- B. JSON Parsing
- C. JSON with padding
- D. JSON Programming

Ans: C

33) What is the value of json in the following code?

```
var obj = { fruit: 'apple', toJSON: function () { return 'orange'; } }; var json = JSON.stringify(obj);
```

- A. {"x": "orange"}
- B. {"fruit": "apple"}
- C. {"x": "apple"}
- D. {"fruit": "orange"}

Ans: A

34) What is the MIME type of JSON?

- A. application/x-json
- B. text/json
- C. application/json
- D. application/javascript

Ans: C

35) What function will convert a JavaScript object to a JSON string?

- A. JSON.text()
- B. JSON.serialize()
- C. JSON.toString()
- D. JSON.stringify()

Ans: D

36) In the below notation, Student is of type { "Student": [ "Amy", "Bob", "John" ] }

- A. Not a valid JSON string
- B. Array
- C. Class
- D. Object

Ans: B

37) JSON-RPC is a remote procedure call protocol

38) To use observables, Angular uses a third-party library called . . . . .

- A) ng2-bootstrap
- B) angular-cli
- C) numeral
- D) Reactive Extensions (RxJS)

Ans: D

39) We subscribe to the observable ourselves using the actual . . . . . method. The downside is that you have to manage the subscription yourself.

- A) subscribe()
- B) subs()
- C) subscribed()
- D) None of above

Ans: A

41) The . . . . . filter waits for a promise and returns the most recent value received. It then updates the view.

- A) sync
- B) async
- C) promise
- D) custom

Ans: B

42) If you chain multiple pipes together, they are executed . . . . .

- A) in parallel
- B) LIFO order
- C) in the order in which you specify them
- D) none of above

Ans: D

43) Observables help you manage . . . . . data.

- A. synchronous
- B. asynchronous
- C. Both asynchronous & synchronous
- D. None of above

Ans: B

45) console.log(x);

var x = 5;

Ans: 5

49). Promises are more advanced than Observables.

- A) True
- B) False

Ans: B

50). Which class is used to create a black navigation bar?

- a. .navbar-default
- b. .navbar-black
- c. .navbar-dark
- d. .navbar-inverse

Ans: D

51). The .container-fluid class provides

- a. Fixed width container
- b. Table format
- c. To create a Form
- d. Full width container

Ans: D

65. What keywords are reserved in JSON and cannot be used as keys?

- A. Value
- B. Object
- C. There are none.
- D. Key

Ans: C

66.) How to get information about browsers?

- a) Browser
- b) version
- c) navigator

Ans: C

67.) Which event will trigger if button will lose its focus?

:- onBlur

68.) Display property?

- A) None
- B) inline
- c) inline-block
- d) block

Ans: All

69.) Shorthand method to represent background of (common)

Ans: background

70.) ----- is an HTML specification used to add more information to HTML tags.

- A) Macrodata
- B) Microdata
- c) Minidata
- d) Modifydata

Ans: B

71.) What is the purpose of the mimeType property of a plug-in entry?

- a) Contains MIME properties
- b) Contains MIME sizes
- c) Contains MIME types
- d) Contains MIME methods

Ans: C

72.) Which of the following can be implemented using animation?

- A) Fireworks
- B) fade Effect
- C) Roll-in or Roll-out
- D) All of the mentioned

Ans: D

73.) Which is the property that represents the content displayed in the window?

- A) document
- B) content
- C) window
- D) frame

Ans: A

74.) How can we make methods available on all objects?

- A) Object.add(methods)
- B) Object.methods(add)
- C) Object.add.methods(...)

D) Object.prototype

Ans: D

75.) The four kinds of class members are\_\_\_\_\_

- a) Instance methods, Instance fields, Static method, Dynamic method
- b) Instance fields, Instance methods, Class fields, class methods
- c) Instance fields, non-Instance fields, Dynamic methods, Global methods
- d) Global methods, Local methods, Dynamic methods, Static methods

Ans: B

84.) Which field is required in <input>

- A) required
- B) minlength
- c) Both

Ans: A

86.) How can we add/ delete in an array and return the removed item.

- A) slice
- B) shift
- c) reverse
- D) splice

Ans: D.splice

87.) How to get information about browser

- A) Browser
- B) Version
- C) navigator

Ans: C

## MONGODB :-

1) In MongoDB, what is the equivalent of SQL term row?

- A. Primary Key
- B. Index
- C. Field
- D. Document

ANS : D

2) What is the equivalent of the following SQL command in MongoDB?

```
UPDATE user  
SET email = 'NA'  
WHERE age < 18
```

- A. db.user.updateMany({ gt: { \$age: 18 } }, { \$set: { email: "NA" } })
- B. db.user.updateMany({ age: { \$lt: 18 } }, { \$set: { email: "NA" } })
- C. db.user.updateMany({ gt: { \$age: 18 } }, { \$set: { email: "NA" } })
- D. db.user.updateMany({ age: { \$gt: 18 } }, { \$set: { email: "NA" } })

ANS : B

3) How would you limit the results of a find to 10 items?

```
adb.collection.find({}, { limit: 10 })  
db.collection.find().limit(10)  
db.collection.limit(query, 10)
```

This is not possible in the node.js driver

ANS : B

4) Which of the following code will give an error on the MongoDB command shell?

- A. use databasename
- B. show dbs



C.USE databasename  
D.Db  
ANS : C

5) What theorem or principal MongoDB follows?  
A.APAC  
B.Always-Sync  
C.ASCII  
D.CAP  
ANS : D

6) What is NoSQL?  
A.No Object SQL  
B.No SQL Server  
C.Not Only SQL  
D.No Relational Database  
ANS : C

7) When there is no match of the document in the collection what do upsert: true do?  
A.Delete  
B.Nothing  
C.Insert  
D.Update  
ANS : C

8) Upon running the following command, what all the fields will be returned in the output?  
`db.collection.find( { qty: { $gt: 6 } }, { name: 1, _id: 0 } )`  
A.\_id  
B.name  
C.\_id, name  
D.name1, \_id0  
ANS : B

9) What method is used to remove a single item from a MongoDB collection?  
`db.collection.deleteOne()`  
`db.collection.delete()`  
`db.collection.removeOne()`  
`db.collection.remove()`  
ANS : A

10) In the following command, what does the second parameter represent?  
`db.collection.find( { qty: { $gt: 6 } }, { name: 1, wins: 1 } )`  
Read Concern  
Projection  
Write Concern  
Query  
ANS : B

## **NODE / EXPRESS :**

1) What is the command used to create a new item when using mongoose?  
A.Book.Save()  
B.Book.New()  
C.Book.Create()  
ANS : A

2)What is the HTTP verb used to update a resource in REST?  
A.DELETE

B.GET  
C.POST  
D.PUT  
ANS : D

3) Google's V8 Engine, also used in Chrome, provides what capability to node.js?  
A.JavaScript support  
B. 3rd party module repository  
C.Support for HTML rendering and DOM manipulation  
D.Asynchronous File and Network access  
ANS : A

4) What is the package you use to connect to MongoDB?  
A.Mongod  
B.MongoConnect  
C.Schema  
D.Mongoose  
ANS : D

5) Assigning a value to "module.exports" in a node.js JavaScript file  
A.allows that value to be imported into other JavaScript files in your project.  
B.publishes the exported value to npm, node's package repository.  
C.persists that value to disk, allowing you to retrieve it later.  
ANS : A

6) Running "npm init" will...  
A.initialize your project, creating a project.json file.  
B.initialize the npm cli on your system.  
C.download and add the "init" module to your node.js project.  
AND : A

7) What is the module for Node that allows you to run a web server?  
A.Sublime  
B.REST  
C.FS  
D.HTTP  
ANS : D

8) What is the mongoose command used to delete an item?  
A.delete()  
B.trash()  
C.remove()  
ANS : C

9) What verb do you use to do a partial update on an item?  
A.Post  
B.Patch  
C.Put  
ANS : B

10) What is the object that you use to pull query string params?  
A.res.query  
B.res.params  
C.req.query  
D.req.params  
ANS :- C

**JAVA:**

1) Given the variable `List<String> strings`, which of the following is valid?

- A. `strings.put("a");`
- B. `List.add(strings, "a");`
- C. `strings[0] = "a";`
- D. `strings.add("a");`

ANS : D

2) When is a method required to include a return statement?

- A. When the method has a non-void return type
- B. When the method contains more than one statement
- C. Never
- D. Always

ANS : A

3) What is the type of the class literal `Foo.class`?

- A. `Class<Foo>`
- B. `Class`
- C. `Class<?>`
- D. `Class<Class>`

ANS : A

4) How would you declare and instantiate a variable, given the generic class `Bag<T>`?

- A. `Bag bag = new Bag();`
- B. `Bag<T> bag = new Bag<T>();`
- C. `Bag bag<T> = new Bag<T>();`
- D. `Bag<T> bag = new Bag();`

ANS : B

5) What is the keyword used to transfer control to another part of the program?

- A. `catch`
- B. `throw`
- C. `try`
- D. `finally`

ANS : B

6) What indicates the end of a Java statement?

- A. Both a semicolon and a newline
- B. A newline
- C. A double-slash ( `//` )
- D. A semicolon ( `;` )

ANS : D

7) Which of the following will initialize the array `myVals` to contain two values: 1 and 2

- A. `int[] myVals = new int[2] {1, 2};`
- B. `int[] myVals = {1, 2};`
- C. `int[2] myVals = {1, 2};`
- D. `int[] myVals = [1, 2];`

ANS : B

8) Given the following code: `float f = 7.0f/2.0f; long l = 7L/2L;` Which statement is true?

- A. The values of `f` and `l` are equivalent
- B. The attempt to assign `l` the result of `7L/2L` will create an error
- C. The value of `f` is less than the value of `l`
- D. The value of `f` is greater than the value of `l`

ANS : D

9) Which of the following can be a valid exception constructor?

- A. Exception(Object cause)
- B. Exception(String message)
- C. Exception(int cause)
- D. Exception(String message, Object cause)

ANS : B

10) Which of these classes is a checked exception?

- A. ClassNotFoundException
- B. NumberFormatException
- C. ThreadDeath
- D. NoClassDefFoundError

ANS : A

11) Given the following code: String s1 = "hi"; String s2 = "h"; s2 += "i"; boolean b1 = s1 == s2; boolean b2 = s1.equals(s2); Which variable(s) is/are assured to be set to true?

- A. Both b1 and b2
- B. b1
- C. Neither b1 or b2
- D. b2

12) Given a type parameter T, what would assign to T[] ts;?

- A. ts = (T[]) new Object[SIZE];
- B. Its impossible.
- C. ts = new T[SIZE];
- D. ts = new Object[SIZE];

ANS : A

13) How would you declare a generic factory method that creates a new generic list with a given size?

- A. public static List<T> <T> newList(int size)
- B. <T> public static List<T> newList(int size)
- C. public <T> static List<T> newList(int size)
- D. public static <T> List<T> newList(int size)

ANS : D

14) Which instances of this class are treated as unchecked exceptions?

- A. Error
- B. Throwable
- C. Exception

ANS : A

15) In which of the following situations will the compiler generate an error?

- A. Catching an checked exception that is not thrown in a try block
- B. Mixing subclasses and superclasses in a throws clause
- C. Don't having at least one catch block after a try block
- D. Downcasting an instance of Exception to an incorrect type (e.g. RuntimeException re = (RuntimeException)e;

ANS : A

Q.1) Select the correct statement for the given code based on Polymorphism in Java.

```
class Bike{
    void run(){System.out.println("running");}
}
class Splendor extends Bike{
    void run(){System.out.println("running safely with 60km");}

    public static void main(String args[]){
```

```
Bike b = new Splendor();  
b.run();  
}  
}
```

- A.Compile time polymorphism
- B.Encapsulation Runtime
- C.Java Runtime Polymorphism
- D.None of the above

**ANS :- A**

Q.2) In the Running phase of a program, JVM executes the \_\_\_\_\_ generated by the compiler.

- classcode
- bytecode
- plugin code
- Bitcode

**ANS :- B**

What is the premise of equality for IdentityHashMap?

- Name equality
- Hashcode equality
- Length equality
- Reference equality

**ANS :- D**

Which of the following options is incorrect in terms of the given statement?

If a class inherits from some other class, it should \_\_\_\_\_.

- Over-ride or add the minimum to accomplish the derived class' purpose
- Override all the methods of its parent class
- Make sure the result "contains" its base class
- Make use of the parent class's capabilities.

**ANS : B**

Find the output of the following program. Choose the correct option.

```
public class Solution{  
    public static void main(String[] args){  
        byte x = 127;  
        x++;  
        x++;  
        System.out.print(x);  
    }  
}
```

- A. -128
- B. 128
- C. 129
- D. 2

**ANS :- A**

(Reason , the range is from [-128,127] )

How many classes of the java.lang package are known as wrapper classes in Java?

- 6
- 5
- 8
- 3

**ANS : C**

Rohini uses static keywords in Java.

Find the option in which she cannot apply a static keyword.

variables  
methods  
blocks  
nested classes  
class  
ANS : E

What is the main use of the keyword "this" in the given options?

To invoke current class method (implicitly)  
To return the current class instance from the method  
To refer to the current class instance variable  
All of the above.  
ANS : D

Predict the output of the code given below and select the correct option.

```
int i =20;  
int j =30;  
int k = j += i/5;  
System.out.println(i+":"+j+":"+k);  
A.0.19067129629629628  
B.0.857175925925926  
C.1.4406712962962962  
D.0.8573379629629629  
ANS : D
```

What is the output of the following code snippet?

```
long choice=2;  
switch(choice) {  
case 1: System.out.print("Hello");  
break;  
case 2: System.out.print(" Java");  
default:  
System.out.print("Try Again");  
}  
A.JavaTry Again  
B.Try Again  
C.Compilation Error  
D.Java  
ANS : A
```

What will be the output of the following code?

```
import java.util.*;  
class Student  
{  
    int id;  
    String name;  
    private Student(int i,String n)  
    {  
        id = i;  
        name = n;  
    }  
  
    void display(){System.out.println(id+" "+name);}  
}  
public class Example{  
  
    public static void main(String args[]){
```

```

Student s1 = new Student(111,"Karan");
Student s2 = new Student(222,"Aryan");

s1.display();
s2.display();
}
}

```

- A. 111 Karan  
222 Aryan
  - B. 111 "Karan"  
222 "Aryan"
  - C. 111  
Karan  
222  
Aryan
  - D. Error: Student4 (int,String) has private access in Student4
- ANS :- D

Which option is correct about the below code?

```

import java.util.*;
class ThreadSample extends Thread{
public void run(){
for(int time=1;time<5;time++){
try{Thread.sleep(1000);}catch(InterruptedException e)
{System.out.println(e);}
System.out.println(time);
}
}
}

public class ThreadExample{

public static void main(String args[]){
ThreadSample t1=new ThreadSample();
ThreadSample t2=new ThreadSample();

t1.run();
t2.run();
System.out.println("hello");
}
}

```

- A. 123412341234.....infinite
  - B. Hello will be printed after the run method of both the thread objects are executed.
  - C. It will give an error from try catch block
  - D. None of the above
- ANS : B

Which are the two classes that have high-level streams containing the methods for serializing and deserializing an object?

- A. ObjectInputStream and ObjectOutputStream
  - B. ObjectInputStream and OutputStream
  - C. InputStream and OutputStream
  - D. ObjectOutputStream and OutputStream
- ANS :- B

```

class output {
public static void main(String args[])

```

```
{
StringBuffer s1 = new StringBuffer("
Hello World");
s1.insert(6 , "Good ");
System.out.println(s1);
}
}
```

- a) HelloGoodWorld
- b) HellGoodoWorld
- c) HellGood oWorld
- d) Hello Good World

Answer: D

Explanation: The insert() method inserts one string into another. It is overloaded to accept values of all simple types, plus String and Objects. Sting is inserted into the invoking object at specified position. "Good " is inserted in "Hello World" T index 6 giving "Hello Good World".

10. Which of these keywords cannot be used for a class which has been declared final?

- a) abstract
- b) extends
- c) abstract and extends
- d) None of the mentioned

View Answer

Answer: a

23. A team of programmers is involved in reviewing a proposed design for a new utility class. After some discussion, they realize that the current design allows other classes to access methods in the utility class that should be accessible only to methods within the utility class itself. What design issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

```
import java.util.*;
class vector {
public static void main(String args[])
{
Vector obj = new Vector(4,2);
obj.addElement(new Integer(3));
obj.addElement(new Integer(2));
obj.addElement(new Integer(5));
obj.removeAll(obj);
System.out.println(obj.isEmpty());
}
```

- ```
}
```
- a) 0
  - b) 1
  - c) true
  - d) false

Answer: C

```
import java.util.*;
class Array {
public static void main(String args[])
```



```

{
    int array[] = new int [5];
    for (int i = 5; i > 0; i--)
        array[5 - i] = i;
    Arrays.sort(array);
    for (int i = 0; i < 5; ++i)
        System.out.print(array[i]);
}

```

Ans) 12345

What is the output of this program?

```

import java.util.*;
class Maps {
    public static void main(String args[])
    {
        TreeMap obj = new TreeMap();
        obj.put("A", new Integer(1));
        obj.put("B", new Integer(2));
        obj.put("C", new Integer(3));
        System.out.println(obj.entrySet());
    }
}

```

- a) [A, B, C]
  - b) [1, 2, 3]
  - c) {A=1, B=2, C=3}
  - d) [A=1, B=2, C=3]
- ANS :- D

42. What is the output of this program?

```

import java.util.*;
class Bitset {

    public static void main(String args[])
    {
        BitSet obj = new BitSet(5);
        for (int i = 0; i < 5; ++i)
            obj.set(i);
        obj.clear(2);
        System.out.print(obj);
    }
}

```

- a) {0, 1, 3, 4}
  - b) {0, 1, 2, 4}
  - c) {0, 1, 2, 3, 4}
  - d) {0, 0, 0, 3, 4}
- View Answer  
Answer: a

43. What is the output of this program?

```

import java.util.*;
class Array {
    public static void main(String args[])
    {
        int array[] = new int [5];
    }
}

```

```
for (int i = 5; i > 0; i--)
```

```
array[5-i] = i;  
Arrays.fill(array, 1, 4, 8);  
for (int i = 0; i < 5 ; i++)  
System.out.print(array[i]);  
}  
}
```

advertisements

- a) 12885
- b) 12845
- c) 58881
- d) 54881

Answer: C , fill(start\_idx,end\_idx,value).

46. What is the output of this program?

```
import java.util.*;  
class Collection_Algos {  
    public static void main(String args[])  
    {  
        LinkedList list = new LinkedList();
```

```
        list.add(new Integer(2));  
        list.add(new Integer(8));  
        list.add(new Integer(5));  
        list.add(new Integer(1));  
        Iterator i = list.iterator();  
        Collections.reverse(list);  
        Collections.shuffle(list);  
        while(i.hasNext())  
            System.out.print(i.next() + " ");  
    }  
}
```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8
- d) Any random order

Answer: d

Explanation: shuffle – randomizes all the elements in a list.

47.What is the output of this program?

```
import java.util.*;  
class Collection_Algos {  
    public static void main(String args[])  
    {  
        LinkedList list = new LinkedList();  
  
        list.add(new Integer(2));  
        list.add(new Integer(8));  
        list.add(new Integer(5));  
        list.add(new Integer(1));  
        Iterator i = list.iterator();  
        Collections.reverse(list);  
        Collections.sort(list);  
        while(i.hasNext())  
            System.out.print(i.next() + " ");  
    }  
}
```

- a) 2 8 5 1
- b) 1 5 8 2
- c) 1 2 5 8
- d) 2 1 8 5

ANS : C , Sorting the list in the ascending order.

```
52. class output {
    public static void main(String args[])
    {
        StringBuffer c = new StringBuffer("Hello");
        StringBuffer c1 = new StringBuffer("World");
        c.append(c1);
        System.out.println(c);
    }
}
```

- a) Hello
- b) World
- c) Helloworld
- d) Hello World

ANS : D

```
53. class output {
    public static void main(String args[])
    {

        StringBuffer s1 = new StringBuffer("Hello");
        s1.setCharAt(1,'x');
        System.out.println(s1);
    }
}
```

- a) xello
- b) xxxxx
- c) Hxlllo
- d) Hexlo

ANS : C

```
59. class output {
    public static void main(String args[])
    {
        String s1 = "one";
        String s2 = s1 + " two";
        System.out.println(s2);
    }
}
```

- a) one
- b) two
- c) one two
- d) compilation error

View Answer

Answer: c

```
61. class exception_handling {
    public static void main(String args[])
    {
```

```
        try {
            System.out.print("Hello" + " "
                + 1 / 0);
```

```

    }
    catch(ArithmeticException e) {
        System.out.print("World");
    }
}
}

```

- a) Hello
- b) World
- c) HelloWorld
- d) Hello World

Answer: b

63. Which of the following describes the correct sequence of the steps involved in making a connection with a database.

1. Loading the driver
2. Process the results.
3. Making the connection with the database.
4. Executing the SQL statements.

(A) 1,2,3,4

(B) 1,3,4,2

(C) 2,1,3,4

(D) 4,1,2,3

Answer : 1,3,4,2

```

64. import java.io.*;
class fileinputoutput {
    public static void main(String args[]) {
        InputStream obj = new FileInputStream("inputoutput.java");
        System.out.print(obj.available());
    }
}

```

Note: inputoutput.java is stored in the disk.

- a) true
- b) false
- c) prints number of bytes in file
- d) prints number of characters in the file

View Answer

Answer: C

66. Which of these methods are used to read in from a file?

- a) get()
- b) read()
- c) scan()
- d) readfileinput()

Answer: B

67. Which of the following allows non repeatable read in JDBC Connection?

- a. TRANSACTION\_READ\_UNCOMMITTED
- b. TRANSACTION\_READ\_COMMITTED
- c. TRANSACTION\_SERIALIZABLE
- d. TRANSACTION\_REPEATABLE\_READ

Ans. D

## DOCKER & DEVOPS :-

1) After you initialize a new Git repository and create a file named git-quiz.html, which of the following commands will NOT work if issued?

git add git-quiz.html

git status  
git add .  
git commit -m "git quiz web file added"  
ANS : D

2) Which of the following are the benefits of using Gradle over Maven? Select all that apply.

- A. Gradle supports more languages than Maven.
- B. Gradle incorporates Maven features as well as additional functionality.
- C. Gradle builds are successful more often than Maven builds.
- D. A Gradle build is 1000 times faster than a Maven build.

ANS : A , B

3) State whether the statement is true or false.

Adopting Agile is necessary prior to embracing DevOps practices.

True

False

ANS : B

4) What is continuous testing?

Testing continuously for 24 hours

Manually testing each build, end-to-end

Automated testing

Automated testing on each continuous deployment

ANS : D

5) Adopting DevOps requires planning that spans which of the following? Select all that apply.

People

Efforts

Process

Tools/Technology

ANS : A,C,D

6) Which DevOps principle focuses on product and service thinking?

Customer-centric action

Continuous improvement

Create with the end in mind

Automate everything you can

ANS : C

7) Shift-left concept refers to moving everything to the earlier phase in the \_\_\_\_\_ cycle.

Requirement

Development

Design

Testing

ANS : B

8) In order to minimize risks to the business within DevOps, what is the main goal of version control?

Ensure the ability to alert when configuration changes from the desired state.

Ensure the ability to re-create the previous state of the test environment.

Ensure the ability to re-create the production environment and build processes.

Ensure the ability to share the source code between different developers teams.

ANS :- B

9) What is a continuous build? Select all that apply.

A build that runs non-stop, over and over again.

A build that has not failed in 10 continuous attempts.

A build that is executed every hour/day by a continuous integration server.

A build that is executed on every commit by a continuous integration server.

ANS : C , D

10) DevOps is primarily about:

Tools and their usage

Processes and adherence

Governance and audit processes

Tools, processes, and culture

ANS : D

11) Which statement best describes the relationship between DevOps and Continuous Delivery?

DevOps and Continuous Delivery are the same thing.

DevOps and Continuous Delivery share a background in Agile methods and LEAN thinking.

DevOps and Continuous Delivery share common processes.

DevOps and Continuous Delivery are not related and are mutually exclusive.

ANS : B

12) How does a DevOps organization act in principle when it comes to financing its work?

It finances special projects to serve its clients.

It finances products and services to serve its clients.

It finances teams in matrix organizations and these teams are responsible for handling their own budgets.

It finances development and operations teams separately, so they take care of their own business.

ANS : B

13) In Kubernetes, a node is:

A worker machine

A tool for starting a kubernetes cluster on a local machine

A machine that coordinates the scheduling and management of application containers on the cluster

A virtual machine

ANS : A

14) GIT can be integrated with Jenkins.

True

False

ANS : A

15) POM in Maven stands for \_\_\_\_\_.

Project Object Model

Process Object Model

Project Object Method

All of the above

ANS : A

16) What is a widely used reusable asset to reinforce information security of deliverables from your DevOps team?

Data storage systems

Handling the logging of sensitive client information

Data transfer between clients and software

All of the above

ANS : D

17) DevOps is NOT a set of \_\_\_\_\_ steps.

Development

Fixed

Testing

Design

ANS : B

18) What is the sequence in which Maven looks for the resources?

Local -> Remote - Maven Central

Remote → Maven Central - → Local

Remote → Local - → Maven Central

Maven Central - → Local - → Remote

ANS : A

19) Why is it recommended that production and development environments should be similar?

A. It is easier to manage for the ops team.

B. It improves the accuracy of testing, eases the task of deployment of software, and reduces chances of major problems when deploying software into production.

C. People can fix defects directly in the production environment.

D. Development environments are cheaper, hence it makes sense to make the production environment similar (and cheaper) to the development environments.

ANS : B

20) DevOps can be implemented in any \_\_\_\_\_.

Process

Environment

Defects

None of the above

ANS : B

21) What is continuous deployment? Select all that apply.

Deployment on every code push/commit

Deployment every hour/day

Local deployment on the developer's laptop

Deployment on IBM Cloud

ANS : A.

22) In DevOps, one of the significant ROI areas is:

Zero defects

Faster time to value

Less feedback

Less customer experience

ANS : B

23) How do you remove local (untracked) files and directories from the current Git working tree?

i. git clean -f

ii. git delete -f

iii. git remove -f -d

iv. git add --all

v. git reset --hard HEAD

A : Only i

B : Only ii and iii

C : Only i and iv

D : All of the above.

ANS : A

## PREV BATCH FSD QUESTIONS :-

1) Which event occurs when the user clicks on the HTML element ?

A. onmouseover

B. onchange

C. onmouseclick

D. onclick

ANS : D

2) The correct HTML tag to left align the content inside a table cell is :-

A. <td align = "left">

B.<td align = "right">  
C.<td leftAlign>  
D.<table align="left">  
ANS : A.

3) What is the output of the Java Code Snippet ?  
StringJoiner astring = new StringJoiner("-", " ( ", " ) ");  
astring.add("A");  
astring.add("B");  
astring.add("C");

- A. (A-B-C)
- B. (A)-(B)-(C)
- C. (-A-)(-B-)(-C-)

ANS : A

4) What are the Different Modes of Autowiring ?  
A. byName,byType,constructor,autodetect.  
B. byName,byType,constructor,autocorrect.  
C.byName,byContent,constructor,autodetect.  
D.byName,byContent,setter,autodetect.

ANS :- A

5) Which Annotation is used to refer to Pointcuts.  
A. @PointCut.  
B. @PointCutExecution.  
C. @PointCutBefore.

ANS :- A

6) Which of the below API Calls provides you the security information ?  
1) HttpServletRequest.isUserInRole()  
2) HttpServletRequest.getUserPrincipal()

- A. Only 1
- B. Only 2
- C. Both i and ii

ANS:- C

7) What does it mean by publishing a REST service in Spring ?  
i. Publishing an application's data as Json over a REST service.  
ii.Publishing an application's data as XML over a REST service.  
lii. Sharing Database within two microservices using Spring Data Access Object (DAO).  
A. Only 1  
B. Only 1 and 2  
C. Only 2  
D. All of the above.

ANS : B -> Chatgpt

8) Which element is used to restrict the access to a particular URL within Spring Security ?  
i. restrict-url  
ii.intercept-url  
iii.intercept-restrict.

ANS : B

9) What is used to dispatch requests to the WEB Application ?  
i. DispatchherServlet.  
li. ViewHandler.



iii. ViewResolver.

ANS:A

10) Which annotation is used to create Primary KEY : Foreign KEY relation between two tables ?

- i. ForeignKey.
- ii. JoinedKey.
- iii. JoinColumn.

ANS: C

11) You can only use Hibernate in ORM in Spring Boot Applications.

- i. TRUE
- II.FALSE

ANS : B

12) In your Data Access Object (DAO) method , you neither need to surround your code with a try/catch block nor declare throwing an exception in the method signature.

- i. TRUE
- ii. FALSE.

ANS : B

13) Which annotation is used as a substitute for the initialization method ?

- 1. @PostConstruct.
- 2. @PreDestroy.
- 3. @PreConstruct.

ANS : 1

14) A logger location where logging information is sent is called

- 1. appender.
- 2. logger
- 3. component.

ANS : appender A

15) By default, a bean is lazily loaded in Spring.

- 1. True
- 2. False

ANS : B

16) Sharding a MongoDB database across many server instances can be achieved with \_\_\_\_

- A.LAN
- B.SAN.
- C.MAN

ANS : LAN A

17) In MongoDB , which is the operation that uses an index often , having better performance than those who do NOT use an index ?

- A. SELECT.
- B. UPDATE
- C. DELETE.
- D. SORT.

ANS : D.

18) Which one is NOT a spring data MongoDB annotation ?

- A. @Document
- B. @Field
- C. @Entity
- D. @Query.

ANS : C @Entity

19) What type of management does "Capped Collections" provide for inserted documents in MongoDB ?

- A. FIFO.
- B. LIFO
- C. LRU
- D. MRU

ANS :- A FIFO updated

20) Identify the incorrect statement on MongoDB ?

- A. Secondary Indices are not available in MongoDB.
- B. MongoDB supports search by field, range queries and regular expression matches.
- C. MongoDB can store the business subject in the minimal number of documents.

ANS :- A. -> it can create

21) In MongoDB, which of the following methods return one document ?

- A. findOne()
- B. findOne1()
- C. selectOne()

ANS : A

22) Redux is an exclusive feature of the React store ?

- A. TRUE
- B. FALSE

ANS : B

23) Identify the correct ES6 statement.

- i. `var React = require('react');`
- ii. `import React from react;`
- iii. `module.exports = Component.`
- iv. `export default Component.`

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only ii and iv

ANS : D

24) You can also use an ES6 class to define a component in React.

- A. True.
- B. False.

ANS: A

25) Which of the below methods are NOT a part of React Component's lifecycle ?

- A. Initial Rendering.
- B. Render / Re-Render.
- C. unMounting.
- D. Remote URL calls.

ANS : D

26) Identify the correct statement in MongoDB.

- A. Queries specify criteria, or conditions , which identify the documents that MongoDB returns to the clients.
- B. Write Operations , or queries , retrieve data stored in the database.
- C. The selection limits the amount of data that MongoDB returns to the client over the network.

ANS : A -> mostly

27) Identify the incorrect statement on MongoDB.

- A. Secondary indices are not available in MongoDB.
- B. MongoDB supports search by field, range queries, and regular expression searches.
- C. MongoDB can store the business subject in the minimal number of documents.

ANS : A

28) Which of the following is not a callback method of React ?

- A. render()
- B. componentDidMount()
- C. componentFailedToMount()
- D. componentWillunMount()

ANS : C -> failedtomount

29) How can we render JSX in the browser ?

- i. Using Babel
- ii. Using Typescript.
- iii. Using Flux.
- iv. Using Redux.

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only iii and iv.

ANS : A

30) What is the command line tool introduced for the JS engine in Java 8 ?

- A. jjs
- B. jss
- C. jfs
- D. jbd

ANS : A

31) Which of the following is the immutable Java List static factory method ?

- A. List.of
- B. List.immutable.
- C. List.create

ANS : - A

32) Which of the following statements are true about Java ?

- i. A finally block is executed, only after the catch block is executed.
- ii. A finally block is executed whether an exception is thrown or not.
- iii. A finally block is NOT Mandatory.
- iv. A finally block is executed, only if an exception occurs.

- A. i
- B. iv
- C. i and iv
- D. ii and iii

ANs :- D

33) Which Java Class has natural sorting implemented ?

- A. LinkedHashSet.
- B. TreeSet
- C. HashSet

ANS :- B -> TreeSet

34) Is the following a valid Java code snippet ?

```
Optional< String > aValue = Optional.of("Not Empty");
if(aValue.isPresent()) {
    System.out.println("Got Not Empty");
}
```

- A. True
- B. False

ANS :- True its an valid code

35) Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

- A. True
- B. False

ANS :- A -> True

36) In Java Optional.ofNullable() method of the Optional Class, returns a Non-empty Optional if the given object has a value, otherwise it returns an empty Optional.

- A. true
- B.false

ANS :- A->True

37) Which of the following types of Java methods cannot be type parameterized ?

- A. FINAL methods.
- B. Overriding methods.
- C. Overloading methods.

Ans:- C-> chatgpt updated by sir -> B Overriding

38) Which of the following statements is true about microservices architecture ?

- i. Microservices applications are required to be deployed on the same physical host.
- ii. Microservices architecture supports high availability of individual microservices.
- iii. Microservices cannot be used if a company is embracing DevOps.
- iv. Microservices are designed using a bounded context that can communicate with other bounded contexts.

- A. Only i
- B. Only ii
- C. Only ii and iv
- D. All of the above.

Ans -> D

39) Load Balancing improves the distribution of workloads across multiple computing resources , such as computers or a computer cluster.

- A. True.
- B. False.

ANS:- A

40) What are the core operations of DevOps with applications development and with infrastructure.

- A. Provisioning, Configuration and Orchestration.
- B. Code Building, Coverage and Unit Testing.
- C. Packaging and Deployment.
- D. All of the above.

ANS : D

41) Which CSS property is used to change the font of an element ?

- A. font
- B. font-family
- C. font-style
- D.font-weight

ANS :- B

42) The Bootstrap grid system is based on how many columns ?

- A. 12

- B. 3
  - C. 6
  - D. 9
- ANS:- A

43) After you initialize a new GIT repository and create a file named git-quiz.html , which of the following commands will NOT work if needed?

- A. git add git-quiz.html
- B. git status
- C. git add
- D. git commit -m "gut quiz web file added"

ANS: C git add (without argument it won't add anything)

44) Choose the correct HTML element to define important text.

- A. <strong>
- B. <important>
- C. <b>
- D. <pre>

ANS :- A

45) Which is the correct CSS Syntax ?

- A. body color-black
- B. {body color-black}
- C. body {color:black;}
- D. {body;color:black}

ANS : C

46) Which Bootstrap class adds zebra-stripes to a table ?

- A. table-bordered.
- B. table-zebra.
- C. table-striped
- D. even and odd

ANS :- C

47) The \_\_ Javascript method is like the mathematical "for all" quantifier , which resumes true if and only if the function returns true for all elements in the array.

- A. every()
- B. some()
- C. all()
- D. forall()

ANS : B Updated ans is A

48) How can you detect the client's browser name in Javascript ?

- A. class.navName
- B. navigator.appName
- C. browser.name
- D. window.browser.

ANS : B

49) Which of the following methods CANNOT be used to add array elements in Javascript ?

- A. push()
- B. splice()
- C. shift()
- D. All can be used.

ANS : D

50) In Javascript \_\_ \_\_ is a comma separated list of colon separated name value pairs enclosed within curly braces.

- A. An object prototype.

- B. A map object.
  - C. An class
  - D. An object literal.
- Ans. D

51) In Javascript , what is the observation made of the statement if (!a[i]) continue ?

- i. Skips the undefined elements.
- ii. Skips the null elements.
- iii. Skips the non existent elements.

ANS :-

- A. i and ii only.
- B. ii and iii only.
- C. i and iii only.
- D. i,ii and iii

ANS :- D

52) Which of these methods is the ultimate element selection method and a very powerful technique by which client side Javascript programs can select the document elements that they are going to manipulate ?

- A. querySelectAll()
- B. querySelector()
- C. querySelect()
- D. querySelectorAll()

ANS :- D

53) To use the Angular HttpClient component, you must import the \_\_\_\_ module.

- A. HttpClientModule.
- B. HttpModule.
- C. Http

Ans :- A

54) In Angular , one can create local HTML reference or HTML tag using variable which starts with character \_\_\_\_

- A. @
- B. #
- C. "
- D. &

ANS :- B

55) Which of these Angular services can be injected in your component to the dynamic mode navigation ?

- A. Routing.
- B. RouterService.
- C. RoutingService
- D. Router

ANS :- D

56) In Angular , which of the following wild card route paths will you use to define a 404 route ?

- A. \*\*
- B. /
- C. ##
- D. default

ANS :- A

1. What annotation does Spring Boot provide to conditionally wire a bean based on an environment variable?

A. `@ConditionalOnValue`

B. `@ConditionalOnProperty`  
correct

C. `@ConditionalOnSystemProperty`

D. `@ConditionalOnEnvironmentVariable`

2. You have a number of external properties that you want to use in your application. How can you configure your Spring Boot application to source them from a file?

Add your `.properties` files to `src/main/resources`, where Spring Boot loads them automatically.

A. Place the `.properties` file in the `src/main/resources` directory. In the code, load it with `java.util.Properties`.

B. Place the properties in `my.properties` and add the annotation `@PropertySource("my.properties")` to the Spring Boot application main class.

C. Create a properties file named `application.properties` in the `src/main/resources` of your project.  
Correct

3. The following method only queries or retrieves data from the data source:

```
@Transactionalpublic Customer retrieveCustomer(String customerId) { ... }
```

It does not perform any write operations. What optimizations can you make so the provider can process the query more efficiently?

A. Add the `readOnly` option to the `@Transactional` annotation. This provides a hint to the provider to optimize the transaction for read-only operations.  
correct

B. Add the `timeout` option to the `@Transactional` annotation. This limits the time the transaction can run.

- C. Add the readOnly option to the @Transactional annotation. This prevents any write operations in the transaction.
- D. Set the appropriate transactionManager option in the @Transactional annotation. This allows you to customize the underlying provider.

4. How do you configure a Spring Boot application to run as a command-line application?

A. Implement CommandLineRunner and publish it to the application context.  
correct

B. Call SpringApplication#setHeadless(true).

C. Annotate a configuration class with @CommandLineApplication.

D. Use SpringConsoleApplication.run instead of SpringApplication.run

5. You write a Java Server Pages (JSP) error page that should deal with HTTP errors. You also create a controller to populate the error page:

@Controller

public class ErrorPageController {

    @RequestMapping(value = "errors", method = RequestMethod.GET)

    public ModelAndView showError(HttpServletRequest request) {

        var errorPage = new ModelAndView("errorPage");

        var httpCode = getHttpCode(request);

        var message = switch (httpCode) {

            case 400 -> "Http Code: 400";

            case 404 -> "Http Code: 404";

            case 500 -> "Http Code: 500";

        }

        errorPage.addObject("message", message);

        return errorPage;

    }

    private int getHttpCode(HttpServletRequest request) {

        // implementation omitted

    }

}

The controller is not dealing with the errors. How can you make it work?

A. Add the following to the web.xml:

<error-page>



```
<location>/errors</location>
</error-page>
```

Correct

B. Store the error page in the folder errors where Spring will look for it. If that is not an option, configure explicitly where the application can find the view.

C. Add the following bean:

```
<bean id="viewResolver"
class="org.springframework.web.servlet.view.InternalResourceViewResolver">
  <property name="prefix" value="/WEB-INF/errors/" />
  <property name="suffix" value=".jsp" />
</bean>
```

D. Annotate the controller with `@ExceptionHandler("errorPage")`.

6. How would you configure the default configuration and dependencies that a Spring Boot application requires in Maven?

A. In the Project Object Model (POM) file, add the following parent configuration:

```
<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>X.Y.Z</version>
</parent>
correct
```

B. In the Project Object Model (POM) file, add the following parent configuration:

```
<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-dependencies</artifactId>
  <version>X.Y.Z</version>
  <relativePath>../../spring-boot-dependencies</relativePath></parent>
```

C. In the Project Object Model (POM) file, add the following dependency starter project:

```
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-parent</artifactId>
</dependency>
</dependencies>
```

D. In the Project Object Model (POM) file, add the following dependency starter project:

```
<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter</artifactId>
</dependency>
</dependencies>
```

7. You write the following code, but the test is not working. What is the most likely solution?

```
@ExtendWith(SpringExtension.class)
@ContextConfiguration(classes = EmployeeService.class)
class EmployeeServiceTest {

    @Autowired
    private ApplicationEvents applicationEvents;

    @Autowired
    private EmployeeService employeeService;

    @Test
    void employeeCreationEvent() {
        this.employeeService.createEmployee("nathan");
        assertEquals(1, applicationEvents
            .stream(EmployeeCreationEvent.class)
            .filter(event -> event.getName().equals("nathan"))
            .count());
        applicationEvents.stream().forEach(System.out::println);
    }
}
```

A. The ContextConfiguration is too minimalistic. Extend it for integration testing.

B. You must annotate the creation method with @EventListener(EmployeeCreationEvent.class).

C. You must add @RecordApplicationEvents to the top of the class.

correct

D. You must make sure that the application is registering a bean of type ApplicationEvents

8. Which Spring class can you configure to function as a special environment with custom set properties for testing purposes?

A. ConfigurableEnvironment

B. PropertySource

C. TestEnv

D. MockEnvironment

Correct

9. You write a test class. The test class should only run when the operating system is Apple macOS. Which built-in annotation could you add on top of the class to achieve this?

A. @SpringJUnitWebConfig(system = "mac")

B. @EnabledOnMac

C. @EnabledIf("#{systemProperties['os.name'].toLowerCase().contains('mac')}")

correct

D. @DetectOS(category = "mac")

10. You want to package your application into a native image. You correctly configure Maven and Docker. What command do you need to build the image?

A. mvn compile

B. mvn -Pnative spring-boot:build-image

correct

C. mvn spring-boot:build-native-image

D. mvn package

11. How would you change the following code snippets so the list of names in the model contains four items?

```
@Controller
public class NameController {
    @ModelAttribute("names")
    public List <String> getNames() {
        List <String> names = new ArrayList<>();
        names.add("User1");
        names.add("User2");
        names.add("User3");
        return names;
    }

    @GetMapping("/names")
    public String allNames(@ModelAttribute("list") List <String> list, Model model) {
        // Add another name
        list.add("User4");
        return "names";
    }
}
```

A. Change the signature of getNames to:

```
public List <String> getNames(Model model)
```

B. Change the signature of getNames to:

```
public List <String> getNames(Model model, @ModelAttribute("names") List <String> names)
```

C. Change the signature of allNames to:

```
public String allNames(Model model, @ModelAttribute("names") List <String> names)
```

D. Change the signature of allNames to:

```
public String allNames(@ModelAttribute("names") List <String> names, Model model)
```

Correct

12. The ExceptionHandler that should pick up ArithmeticException is not working. The only problem is in the following code snippet. What is the explanation for this?

A.

```
@ExceptionHandler(ArithmeticException.class)
@ResponseBody
Map<String, Object> handleException(Exception e, HttpServletRequest request) {
    // implementation omitted
}
```

}

- A. The method must have the name of the specific exception, so you must rename it `handleArithmeticException`.
- B. Spring cannot find the view associated with this method because it is not specified in the annotation.
- C. The return type is incorrect. It can only return views and strings.
- D. The annotation should be `@ExceptionHandler(exceptions={ArithmeticException.class})`.

13. Which statement about the `TestTransaction` class is true?

A. The static methods on the `TestTransaction` class interact with the current transaction of the test method.

Correct

B. `TestTransaction` only works on classes that you annotate with the `@EnableTestTransaction` and `@Transactional` annotations.

C. You must write your custom class to implement the abstract `TestTransaction` class to interact with the transaction of the test method.

D. You must autowire `TestTransaction`, then call the instance method to interact with the transactions of the entire test class.

14. You create the following auto-configuration class:

```
@AutoConfiguration
public class MyAutoConfiguration {
    @Bean
    @ConditionalOnMissingBean(MyComponent.class)
    MyComponent myComponent() {
        return new MyComponent();
    }
}
```

A. It is on your classpath, but it is not loading. What configuration step is missing?

B. You must reference the auto-configuration class in each application's `@SpringBootConfiguration` annotation.

C. You must add the @Configuration annotation.

D. You must list the auto-configuration class in a META-INF/spring/org.springframework.boot.autoconfigure.AutoConfiguration.imports file on the classpath.

Correct

E. You must list the auto-configuration class name in an application.properties file under the property spring.autoconfigure.imports.

15. Which is an example of the XML for mapping a NullPointerException to a page called npe?

A.

```
<bean class="org.springframework.web.servlet.handler.SimpleMappingExceptionResolver">
  <property name="exceptionMappings">
    <props>
      <prop key="java.lang.NullPointerException">npe</prop>
    </props>
  </property>
</bean>
```

Correct

B.

```
<exceptionHandler
class="org.springframework.web.servlet.handler.SimpleMappingExceptionResolver">
  <property name="exceptionMappings">
    <props>
      <prop key="java.lang.NullPointerException">npe</prop>
    </props>
  </property>
</exceptionHandler>
```

C.

```
<bean class="org.springframework.web.servlet.handler.SimpleMappingExceptionResolver">
  <property name="exceptionMappings">
    <props>
      <prop view="npe">java.lang.NullPointerException</prop>
    </props>
  </property>
</bean>
```

D.

```
<exceptionHandler
class="org.springframework.web.servlet.handler.SimpleMappingExceptionResolver">
  <property name="exceptionMappings">
    <props>
      <prop view="npe">java.lang.NullPointerException</prop>
    </props>
  </property>
</exceptionHandler>
```

16. You configure your Maven-based Spring Boot application to be packaged as a jar. How would you create the jar from the command line?

A. mvn install followed by jar cvf build/classes

B. mvn spring-boot:jar

C. mvn package

Correct

D. mvn compile

17. Why should a developer remove the @ExtendWith annotation in this JUnit 5 test?

@SpringBootTest

@ExtendWith(SpringExtension.class)

public class MyTest { ...

A. @SpringBootTest implies @ExtendWith(SpringExtension.class).

correct -

B. Only JUnit 4 Spring Boot tests recognize @ExtendWith.

C. @SpringBootTest contradicts @ExtendWith(SpringExtension.class).

D. @ExtendWith(SpringExtension.class) overrides @SpringBootTest.

18. What is the default behavior of Spring's TestContext for handling database persistence?

A. The framework creates a transaction for every test and does an automatic rollback. The database is the same as before the test.

correct

B. TestContext configures a new database for every test that runs. The database persistence of the test does not influence the next test.

C. The framework stores the changes in the test database that is being configured by default. This influences the tests to follow. If you do not want this, you can apply @NoCommit.

D. You can mark tests with @Transactional, specifying to rollback for a certain custom exception. You will throw the exception at the end of the test to rollback.

19. Which class meets the conditions for having methods with the @ModelAttribute annotation?

A.

@Repository

```
public class DataController {}
```

B.

@Controller

```
public class DataController {}
```

correct

C.

@Model

```
public class DataController {}
```

D.

@Attributed

```
public class DataController {}
```

20. How can you disable an auto configuration class in Spring Boot?

A. Add the @ContextConfiguration annotation to your configuration class and set the classes property as the auto config class.

B. Add the @Conditional annotation to your configuration class and set the value property as the auto config class.

C. Add the @EnableAutoConfiguration annotation to your configuration class and set the exclude property as the auto config class.

D. Add the auto config class to the exclude property in the @SpringBootApplication annotation.  
Correct



# 3. Spring

## First

### Spring AOP

**1. What is advice?**

- a. A concern that cuts across multiple classes
- b. The process of linking aspects with other application types
- c. Action taken by an aspect at a particular join point
- d. A method to implement transaction management

▼ Answer

- c. Action taken by an aspect at a particular join point

▼ Extra

AOP:- Aspect Oriented Programming

**2. Which annotation represents an advice to be executed after the method execution only if method exits by throwing an exception?**

- a. @Before
- b. @After
- c. @Around
- d. @AfterThrowing

▼ Answer

- d. @AfterThowring

**3. Which of the following terminologies are typically used with regards to AOP?**

- a. **Concerns**
- b. **Advice**
- c. **Float**

d. **Pointcut**

- i. A, B, and D
- ii. A, C, and D
- iii. A, B, and C
- iv. B, C, and D

▼ Answer

- i. A, B and D

## Spring Auto-Wiring

1. **What attribute is added in the bean.xml to reduce or eliminate the need of <property> and <constructor-arg>?**

- a. Autodiscovery
- b. Autowire
- c. Scope
- d. Auto-detect

▼ Answer

- b. Autowrie

## Spring Core Concepts and Life Cycle

1. **Which of the following examples summarizes Program to an interface rather than an implementation?**

- a. Map m = new HashMap();
- b. ArrayList list = new ArrayList();
- c. ArrayList<Integer> = new ArrayList<Integer>();
- d. Strings = "HelloWorld";

▼ Answer

- c. ArrayList<Integer> = new ArrayList();

## Spring Dependency Check, Inner beans and Scopes

**1. If any of the beans implements the BeanPostProcessor interface, Spring will call which of these methods?**

- a. processBeforeInitialization()
- b. BeanPostProcessor()
- c. postProcessInterface()
- d. postProcessBeforeInitialization()

▼ Answer

- d. postProcessBeforeInitializatoin()

**2. Which Collections elements does Spring support?**

- a. List, hashmap, arraylist, andset
- b. List, props, hashmaps, and arraylist
- c. List, set, map, and props
- d. Hashmaps, arraylist, list, and treemap

▼ Answer

- c. List, set, map, and props

**3. What will be the result if you use Web-aware Spring Application Context scopes in a regular Spring IoC container?**

- a. It will return nothing.
- b. It will return a NullPointerException
- c. It will return a IllegalStateException
- d. It will work correctly.

▼ Answer

- a. It will return nothing.

**4. Which Bean scopes are valid only in the context of a web-aware Spring Application Context?**

- a. Singleton and Prototype
- b. Session, Global Session, and Request
- c. Singleton, Prototype, Request, and Session
- d. Session, Singleton, and Global Session

▼ Answer

b. Session, Global Session, and Request

5. **To use `java.util.List`, what is the Spring Collections element that you need?**

- a. `<list>`
- b. `<List>`
- c. `<lists>`
- d. `util:list`

▼ Answer

a. `<list>`

6. **Which of the below are benefits of using DI?**

- a. Tight coupling
- b. Loose coupling
- c. Easy testing
- d. Removing run time error

▼ Answer

b. Loose coupling and c. Easy testing

7. **What is the difference between `ClassPathXMLApplicationContext` and `FilePathXMLApplicationContext`?**

- a. First one reads from application class path and second one reads from absolute file path.
- b. First one reads from application class path and second one reads from relative file path.
- c. First one reads from relative application class path and second one reads from absolute file path.
- d. None of the above.

▼ Answer

c. First one reads from relative application class path and second one reads from absolute file path.

8. The Dependency Look up is a methodology where we get the resource after demand. There can be various ways to get the resource. Types of Dependency Look up include:

- a. Factory Method
- b. JNDI Method
- c. Key Word Method
- d. All of the above

▼ Answer

- d. All of the above

9. Which of the following declarations of inner Beans are correct

```
A.<beanid="outerBean"class="...">
    <propertyname="target">
        <beanid="innerBean"class="..."/>
    </property>
</bean>
```

```
B.<beanid="outerBean"class="...">
    <beanid="innerBean"class="...">
        <propertyname="target">
            </property>
        </bean>
    </bean>
```

```
C.<beanid="outerBean"class="...">
    <beanid="innerBean"class="...">
        <propertyname="target">
            </property>
        </bean>
    </bean>
```

```
D.<beanid="outerBean"class="...">
    <propertyname="target">
        <beanclass="...">
            <propertyname="name"value="HelloWorld"/>
        </bean>
```

```
        </property>
    </bean>
```

- a. A and B
- b. B and C
- c. B and D
- d. A and D

10. **When do you use Dependency Injection?**

- A. DI is very useful for decoupling your system.**
- B. DI is most useful when you are aiming for code reuse.**
- C. DI helps in designing tightly coupled code.**
- D. DI does NOT help in unit testing the code.**

- a. A and C
- b. B and C
- c. A and B
- d. A and D

▼ Answer

- c. A and B

## Spring Java Configuration

1. **For Java-based configuration, which is the class that sets the Spring Context?**

- a. Annotationconfig
- b. ComponentScan
- c. AnnotationConfigApplicationContext
- d. AnnotationConfigApplication

▼ Answer

- c. AnnotationConfigApplicationContext

2. **Which of the following is an example of java-based configuration?**

- a. context:annotation-config/

- b. @Configuration public class HelloWorldConfig{..}
- c. @Autowired
- d. ApplicationContext context=new  
ClassPathXmlApplicationContext("Beans.xml");

▼ Answer

- b. @Configuration public class HelloWorldConfig{..}

## Spring ORM and MVC

### 1. Which is the main component in Spring Web MVC?

- a. Faces Servlet
- b. Action Servlet
- c. View Renderer Servlet
- d. Dispatcher Servlet

▼ Answer

- d. Dispatcher Servlet

### 2. Which element has to be used in order to configure Spring MVC?

- a. <mvc:annotation-driven/>
- b. <aop:aspect/>
- c. <tx:advice/>
- d. <oxm:jaxb2-marshaller/>

▼ Answer

- a. <mvc:annotation-driven/>

## Spring Stereotype Annotations

### 1. What annotation wiring element is added to scan packages to find and register beans with in the Application Context?

- a. <context:annotation-config/>
- b. <context:annotation-scan/>
- c. <context:component-config/>

d. <context:component-scan>

▼ Answer

d. <context:component-scan>

**2. In the following example, what type of autowire will be used?**

```
@Autowired(required=false)
public Employee AutowiredByConstructorService(@Qualifier("employee")
    Employee emp) {
    this.employee=emp;
}
```

- a. ByName
- b. ById
- c. @Autowired
- d. Constructor

▼ Answer

d. Constructor

**3. What are different valid types of Spring bean autowiring?**

- a. byName, byId, and constructor
- b. byName, byType, constructor, @Autowired, and @Qualifier annotations
- c. by Scope, by Constructor, and @Autowired
- d. ById, byNameScope, constructor, and @Autowire

▼ Answer

b. byName, byType, constructor, @Autowire, @Qualifier annotations

**4. Spring handles object creation in order to be non-invasive. Can you identify which definition creates a spring-managed object?**

- a. String myObj;
- b. @Autowired private String myObj;
- c. private String myObj = new String("spring");
- d. @Spring String myObj=null;



▼ Answer

c. `private String myObj = new String("spring");`

**5. How many types of autowiring modes are supported in Spring?**

- a. Three
- b. Five
- c. Six
- d. Four

▼ Answer

b. Five

## Other

**1. The use of Autodiscovery and Autowire will reduce the amount of \_\_\_\_\_.**

- a. Java code generated
- b. Methods generated
- c. XML Spring configurations
- d. <bean> elements

▼ Answer

d. <bean> elements

**2. Dependency Injection (DI) or Inversion of Control (IOC) is based on which Hollywood principle ?**

- a. Do not call us. We will call you.
- b. Understand the problem first.
- c. The customer is always right.
- d. What you see is what you get.

▼ Answer

a. Do not call us. We will call you.

**3. From where does the Spring container gets its instructions?**

- a. Properties File

- b. Spring JARS
- c. XML File
- d. Annotations

▼ Answer

c and d

**4. The use of the prototype scope is recommended for \_\_\_\_\_ type of beans.**

- a. Stateless
- b. Stateful
- c. Restless
- d. Restful

▼ Answer

b. Stateful

**5. Which is the alternative to create, load, and manage beans in a Spring Application?**

- a. Application Factory
- b. Context Factory
- c. Bean Factory
- d. Spring Bean Factory

▼ Answer

c. Bean Factory

**6. Java- based configuration allows bean definition and Spring configuration to move out of \_\_\_\_\_.**

- a. XML files
- b. Java code
- c. Property files
- d. All of the above

▼ Answer

a. XML files

**7. Which one of these statements about XML declaration of transaction manager bean is NOT true?**

- a. The tx namespace provides JTA transaction manager declaration shortcut syntax.
- b. The Id of the bean has to be transactionManager.
- c. Depending on the application persistence technology, the HibernateTransactionManager or the DataSourceTransactionManager could be used as bean class.
- d. A default transaction timeout could be given.

▼ Answer

- b. The Id of the bean has to be transactionManager.

**8. Which one of the following is one of the challenge of Traditional Transaction Management?**

- a. Transaction Management at Database Connection Level
- b. Programmatic Management of Transactions
- c. Error-Prone Connection Management
- d. All of the above

▼ Answer

- d. All of the above

**9. What does the JDBC Template users to connect to the database?**

- a. A simple java bean with connection properties
- b. A data source bean configuration
- c. An XML with DB connection parameters
- d. A JDBC connection object

▼ Answer

- b. A data source bean configuration

**10. What is the use of Autodiscovery?**

- a. Eliminate the use of the <Property> element
- b. Reduce the use of the <Constructor-arg> element

- c. Eliminate the use of the <Props> element
- d. Reduce the use of the <bean> element

▼ Answer

- d. Reduce the use of the <bean> element

**11. The ----- packages are the basis of spring Framework's IOC container**

- a. org.springframework.beans and org.springframework.Application
- b. org.springframework.properties and org.springframework.context
- c. org.springframework.beans and org.springframework.properties
- d. org.springframework.beans and org.springframework.context

▼ Answer

- d. org.springframework.beans and org.springframework.context

**12. Which of the following attributes are valid in a Spring bean definition?**

- A. ID**
- B. Map**
- C. Set**
- D. Class**

▼ Answer

- A.ID and D.Class

**13. What Spring Collection can be used to inject a collection of name-value pairs, where name and value can be of any type?**

- a. <set>
- b. <List>
- c. <map>
- d. <props>

▼ Answer

- c. <map>

**14. Which attributes are applicable for an inner bean?**

- a. Context, Parent

- b. Id, Class
- c. Class, Inner
- d. Interface, Id

▼ Answer

c. Class, Inner

**15. To use bean definition template, you should add what attribute in the bean?**

- a. abstract="true"
- b. interface="true"
- c. class ="com.example.helloworld"
- d. public ="true"

▼ Answer

a. abstract= "true"

**16. A collection value of the type java.util.Properties is defined by what element?**

- a. <props>
- b. <prop>
- c. <properties>
- d. <property>

▼ Answer

a. <props>

**17. Which of the following is an annotation for Spring Java-based configuration?**

- a. @Component
- b. @SpringConfiguration
- c. @SpringXml
- d. @Configuration

▼ Answer

@Configuration

**18. What does the JDBC Template uses to connect to the database?**

- a. A simple java bean with connection properties
- b. A data source bean configuration
- c. An XML with DB connection parameters
- d. A JDBC connection object

▼ Answer

- b. A data source bean configuration

**19. Which of the following tasks are performed by a JDBCTemplate?**

- a. **Execute SQL query.**
- b. **Execute Update statements.**
- c. **Execute a Java program.**
- d. **Execute stored procedure calls.**

▼ Answer

- a, b and d

**20. Does Spring provide programmatic transaction management?**

- a. Yes, with the TransactionTemplate class
- b. Yes, with the TransactionService class
- c. Yes, using the @Transactional bean post processor
- d. No, Spring does not provide transactional management

▼ Answer

- a. Yes, with the Transaction Template class

**21. Which of the following are MVC – Specific annotations:**

- a. **@Controller**
- b. **@RequestMapping**
- c. **@PathVariable**
- d. **@Autowired**

▼ Answer

a, b and c only

22. To annotate all your Data Access Object (DAO) classes, use the -----

- a. @Service
- b. @Classes
- c. @Repository
- d. @Component

▼ Answer

c. @Repository

## Second

### Imp4

1. Which of the following statement is false ?

- a. The Corepackage is the most fundamental part of the framework and provides the loc an dDependency Injection feature.
- b. TheDAO package provides a JDBC-abstraction layer that removes the need to do tedious JDBC coding and parsing of database-vendor specific error code
- c. The ORM package provides integration layers for popular object-relational mapping APIs, including JPA,JDO, Hibernate, iBatis
- d. AOP takes out the direct dependency of crosscutting tasks from classes which, we can achieve through normal object oriented programming model. or example, method-interceptors and pointcuts to cleanly decouple code implementing functionality that
- e. Spring's Web package provides basic web-oriented integration features, such as multipart file-upload functionality
- f. Spring's MVC package provides a Model-View-Controller implementation for web-application MVC package provides a Model-

View-Controller implementation for web-application

g. None of these

▼ Answer

g. None of these

**2. Which of the following is false?**

- a. A BeanFactory pretty much just instantiates and configures beans.
- b. An ApplicationContext also instantiates bean, and it provides the supporting infrastructure to enable lots of enterprise- specific features such as transactions and AOP, message resource handling (for use in internationalization), event propagation.
- c. Application-layer specific contexts such as the WebApplicationContext use in web application
- d. none of the above

▼ Answer

d. none of the above

**3. Which of the following is actual representation of the Spring loc container?**

- a. Application Context
- b. Xml Bean Factory
- c. Bean Factory
- d. Web Application Context

▼ Answer

c. Bean Factory

**4. A bean is simply an object that is instantiated, assembled and otherwise managed by a Spring loc container.**

- a. true
- b. false

▼ Answer

a. true



**5. How container load configuration metadata from a variety of external resources such as the local file system from the Java CLASSPATH?**

- a. Application Context constructor
- b. Init()
- c. Context Contructor
- d. XmlClassPath Contructor

▼ Answer

Application Context constructor

**6. Which of following is correct way to load bean definitions from another file?**

- a. <bean id="bean1" class="..." import="something.xml"/>
- b. <resources name="something.xml" import="true"/>
- c. <import resource="something.xml"/>
- d. all the above

▼ Answer

a. <import resource="something.xml"/>

**7. What is the correct way to write constructor injection?**

```
package com.bullraider;

public class TestBean {

    private int year;

    private String happy;

    public TestBean(String happy,int year) {

        this.year=yearj

        this.happy = happy;
    }
}
```

```
}  
}
```

a.

```
<bean name="testClass" class="com.bullraider. TestBean">  
  
<constructor-arg type="java.lang.String" value="Happy new  
  
<constructor-arg type="int" value="2015"/>  
  
</bean>
```

b.

```
<bean name="testClass" class="com.bullraider.TestBean">  
  
<constructor-arg index="0" value="Happy new year"/>  
  
<constructor-arg index="1" value="2015"/>
```

c.

```
<bean name="testClass" class="com.bullraider. TestBean">  
  
<constructor-arg index=" 1 " value="Happy new year"/>  
  
<constructor-arg index=" 0 " value =^ prime prime 2015 ^ p  
  
</bean>
```

d.

```
<bean name="testClass" class="com.bullraider TestBean">  
  
<constructor-arg type="int" value="2015"/>  
  
<constructor-arg type="java.lang.String" value="Happy new  
  
</bean>
```

e.All of the above

▼ Answer

e. All of the above

**8. Which of the following are correct statements?**

- a. Form Controllers handle form interaction
- b. Multipart resolver to handle file uploads
- c. Dynamic binding of data to domain model
- d. Interaction with velocity.jsp, thymeleaf,pdf, Excel
- e. Sending email remote accessing via Soap, Burlap

▼ Answer

All of the above

**9. What do you mean by 'bean'?**

- a. component
- b. object
- c. class
- d. container
- e. xml file

▼ Answer

a. component and b. object

**10. What are responsibilities included for bean factory do ?**

- a. Instantiating Instantiating or sourcing application objects
- b. configuring such objects
- c. Assembling the dependencies between these objects
- d. Manage lifecycle of spring

▼ Answer

a, b and c

**11. What are the responsibilities of XmlBeanFactory included ?**

- a. This implementation allows you to express the objects that compose your application
- b. Resolve inter-dependencies between such objects, in terms of XML
- c. The XmlBeanFactory takes this XML configuration metadata and uses it to create a fully configured system or application
- d. None of this above

▼ Answer

a, b, c(if it is multi option or else a)

12. Which of the following are the correct way to nstantiating a Spring loc container?

```
ApplicationContext context = new ClassPathXmlApplicationCo
```

```
XmlBeanFactory beanFactory=new XmlBeanFactory(new ClassPat
```

```
BeanFactory fac=new ClassPathXmlApplicationContext("beans-
```

```
Resource res=new Classpathresource("beans-definitions.xml"
```

▼ Answer

all of the above

13. **What are the correct ways to instantiate bean?**

- a. Using default (empty) constructor
- b. Instantiation using a static factory method
- c. Instantiation using an instance factor method
- d. Using the container
- e. All these above are true

▼ Answer

e. All these above are true

14. **What is Bean Factory?**

- a. A BeanFactory is essentially maintains a registry of different beans and their dependencies.
- b. A BeanFactory enables you to read bean definitions and access them using the bean factory.
- c. Using just the BeanFactory we can create and read in some bean definitions in the XML format as following
  - i. `Resource definition1 = new FileSystemResource("beans-def.xml");`
  - ii. `BeanFactory factory = new XmlBeanFactory(definition1)`
- d. We can get ApplicationContext and WebApplicationContext object using BeanFactory

▼ Answer

a, b and c

#### 15. Which statements are true about DI(Dependency Injection)?

**DI says your components and services should tightly coupled in code.**

- a. DI says that no need create your objects but describe how they should be created.
- b. We should not directly connect your components and services together in code but describe which services are needed by which components in a configuration file
- c. Inversion of control relies on dependency injection
- d. We should not directly connect your components and services together in code but describe which services are needed by Which components in a configuration file

▼ Answer

a, c, d

#### 16. Which statements are true about IOC(Inversion of Control) ?

- a. Decoupling of the execution of a certain task from implementation.
- b. Each module can focus on what it is designed for.
- c. Replacing modules should not affect to other modules.
- d. Modules should observe about what other systems do

e. All the above

▼ Answer

a, b, c

**17. What do you mean by inner bean?**

- a. A <bean/> element inside the <bean/> elements is used to define a so-called Inner bean.
- b. A <bean/> element inside the <property/> elements is used to define a so-called inner bean.
- c. A <bean/> element inside the <constructor-arg/> elements is used to define a so-called inner bean.
- d. A <bean/> element inside the property or <constructor-arg/> elements is used to define a so-called inner bean.

▼ Answer

d. A <bean/> element inside the property or <constructor-arg/> elements is used to define a so-called inner bean.

**18. Which of the following are basic techniques to implement inversion of control?**

- a. using a factory pattern
- b. using a service locator pattern
- c. a constructor injection
- d. a setter injection
- e. an interface injection
- f. all the above

▼ Answer

f. all of the above

**19. Which of the following statements are false?**

**Constructor-based DI is effected by invoking Setter Injection each representing a dependency**

**Constructor-based DI is accomplished when the container invokes a class constructor with a number of arguments**

**@Autowire annotation annotation on constructor for constructor based autowiring.**

- a. Constructor-based DI is effected by invoking a constructor with a number of arguments, each representing a dependency
- b. We can aslo use @Autowire annotation on constructor for constructor based autowiring.
- c. Constructor-based DI is accomplished when the container invokes a class constructor with a number of arguments
- d. Constructor-based DI is effected by invoking Setter Injection each representing a dependency

▼ Answer

- d. Constructor-based DI is effected by invoking Setter Injection each representing a dependency

**20. What are the correct ways to define constructor injection?**

- a.

```
<bean d ="salute" class=^ prime prime com . com.bullraider
  <constructor-arg>
    <value>Salute</value>
  </constructor-arg>
</bean>
```
- b.

```
<bean id="salute" class="com.bullraider.app.beans.Salutati
  <constructor-arg le ="Salute">
  </constructor-arg>
</bean>
```
- c.

```
<bean id="salute" class="com.bullraider.app.beans.Salutati
  <constructor-arg value="Slute"/>
</bean>
```
- d. All of the above

▼ Answer

d. all of the above

## SS

1. **Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.**

a. true

b. false

▼ Answer

a. true

2. **Which of the among is not a special bean in WebApplicationContext?**

a. Locale resolver

b. Theme resolver

c. multipart file

d. resolver

e. Handler exception resolver

f. Handler mappings

g. None of the above

▼ Answer

d. resolver

3. **How the Spring DispatcherServlet supports for returning the last-modification-date, as specified by the Servlet API?**

a. By implementing the LastModified interface

b. By implementing the LastSessionModified interface

c. By implementing the LastDateModified interface

d. By implementing the LastModifiedDate interface

▼ Answer



4. **WebApplicationContext has more scopes like request, session, globalSession to the standard Application context (which only provides Singleton and Prototype scopes)**

a. true

b. false

▼ Answer

a. true

5. **All Spring's various controller inherit from AbstractController?**

a. true

b. false

▼ Answer

a. true

6. **You can write your own controller by using**

**handleRequestInternal(HttpServletRequest, HttpServletResponse) method from abstract controller and a ModelAndView object.**

a. true

b. false

▼ Answer

a. true

7. **Which of the following controller inspects the URL and retrieves the filename of the file request and uses that as a viewname ? For example, the filename of <http://www.springframework.org/welcome.html> request is welcome.**

a. UrlFilenameViewController

b. MultiActionMultiActionController

c. CommandController

d. AbstractController

▼ Answer

a. UrlFilenameViewController

**8. Which statements are true about DispatcherServlet?**

- a. DispatcherServlet is an expression of the "Front Controller" design pattern.
- b. The DispatcherServlet is the actual Servlet.
- c. DispatcherServlet is declared in the web.xml of your web application.
- d. The Spring DispatcherServlet has a couple of special beans it uses in order to be able to process requests and render the appropriate views. These beans can be configured in WebApplicationContext
- e. All of the above

▼ Answer

- e. All of the above

**9. Which statements are true about WebApplicationContext?**

- a. The WebApplicationContext is an extension of the plain ApplicationContext
- b. The WebApplicationContext is bound in the ServletContext, and by using static methods on the RequestContextUtils class.
- c. WebApplicationContext differs from a normal ApplicationContext in that it is capable of resolving themes
- d. We can get WebApplicationContext reference by using ApplicationContext.
- e. All the above

▼ Answer

- a, c, d (doubt)

**10. Which of the following controller combines multiple request-handling methods into one controller?**

- a. UrlFilenameViewController
- b. MultiActionController
- c. CommandController
- d. MultiRequestController

▼ Answer

11. **'https://localhost:8080/testing.view' will result in the method `testing(HttpServletRequest, HttpServletResponse)` being invoked.**

Which is the correct view resolver for this?

- a. ParameterMethodName Resolver
- b. Properties MethodName Resolver
- c. InternalViewResolver
- d. InternalPathMethodName Resolve

▼ Answer

- d. InternalPathMethodName Resolve

12. **'/index/welcome.html=dolt' and a request to /index/welcome.html comes in, the `dolt(HttpServletRequest, HttpServletResponse)` method will be invoked.**

**Which is the correct view resolver for this?**

- a. InternalPathMethodName Resolver
- b. ParameterMethodName Resolver
- c. Properties MethodName Resolver

▼ Answer

13. **Which of the following controller provide a way to interact with data objects and dynamically bind parameters from the `HttpServletRequest` to the data object specified?**

- a. MultiActionController
- b. AbstractController
- c. FormActionControllor
- d. CommandController

▼ Answer

- d. CommandController

14. **Which of the following is not a view resolver?**

- a. BeanNameViewResolver
- b. VelocityViewResolver
- c. JasperReportsViewResolver
- d. UriBasedViewResolver
- e. Resour Resource BundleViewResolver
- f. None of above

▼ Answer

- c. JasperReportsViewResolver

## **HTML & CSS :**

1. Whenever you are assigning two classes to a tag , you must separate them with a

A.Space

B.Comma

C.Dash

ANS : A

2. \_\_\_\_\_ contains the navigation menu, or other navigation functionality for the page

A.section

B.header

C.nav

D.aside

ANS : C

15) Which of the following properties sets the distance between an element's right border and the rightmost edge of its content?

A.auto

B.padding-left

C.padding-right

D.Padding-top

ANS : C

## **JAVASCRIPT :-**

1) What built-in functions allows you to convert a string into an integer?

A.toInt()

B.parseInt()

C.toInteger()

D.parseString()

ANS : B

4) What is a function in JavaScript?

- A. An instance of a class
- B. A statement that evaluates an expression
- C. A block of organized, reusable code that performs a single action
- D. Code that allows us to write loops

ANS : C

11) What is the output of the following?

```
greet = function(...months){ months.forEach(month => console.log('Month: ' + month));}greet(['Jan', 'Feb']);
```

- A. Jan Feb
- B. ReferenceError
- C. Month: Jan, Feb
- D. Month: Jan E. Month: Feb

ANS : C

14) What is the output of the following?

```
function display(num1, num2) { console.log(num1, num2);}let nums = "12345";display(...nums);
```

- A. 1 2
- B. 21
- C. 12345
- D. 45

ANS : A

15) What is the console output of the following code block: `function printName() { name = "Mary"; name = "John"; console.log(name); }`

- A. Empty

B.None

C.John

D.Mary

ANS: C

24) Babel compiles the JavaScript code you write into

1. ECMAScript 5

2. ECMAScript 7

3. ECMAScript 3

4. ECMAScript 6

ANS :- A

### **REACT :-**

1.The string "api/route" in Express is an example of a...

A. Middleware

B .Route

C. Error Handler

ANS : B

2. Which library allows the use of the "require" keyword in front-end code?

A. guid

B. Browserify

C. React

D. jQuery

3. The \_\_\_\_\_ attribute of a React form is called when the main action button inside the form is pressed.

A. onSubmit

- B. onChange
- C. onComplete

ANS : A

4. Which of the following are automation tools?

- A. Gulp
- B. JSX
- C. Express

ANS : A

6. What React function is used to define a new element tag?

- A. renderComponent
- B. createClass
- C. setInnerHTML

ANS : B

9. True or false : The only thing that can change the data inside a store is the store itself.

- A. false
- B. true

ANS : B

10. The Gulp task which takes a stream and outputs a file is...

- A. src
- B. task
- C. dest

ANS : C

11. The best way to detect when a user has completed a form is



- A. to handle the onClick event on the form's submit element
- B. to handle the onBlur event of the last form element
- C. to handle the form's onSubmit event
- D. to accumulate the onChange events for each form element

ANS : C

14. Behavior is shared between components using

- A. propTypes
- B. state
- C. mixins
- D. props

ANS : C

27. Which of the following API is a MUST for every React Component ?

- A. getInitialState.
- B. render
- C. renderComponent.

ANS : C.

28. What are the advantages of React ?

- A. React can be used on client as well as server side.
- B. Using React increases readability and makes maintainability easier. Component and Data patterns improve readability and thus make it easier for maintaining larger props.
- C. React can be used with any other framework (Backbone.js, Angular.js) as it is just a View Layer.
- D. All of the above.

ANS : D

30. What is React in MVC ?

- A. Controller
- B. Middleware
- C. Model
- D. Router

ANS :- None of these. Updated ans is Controller. Option A

31) What is React?

Just a server-side framework

Just a user-interface framework

Both a server-side framework as well as a user-interface framework

ANS : C

32) What are the limitations of React?

React is only for the view layer of the app so we still need the help of other technologies to get a complete tooling set for development.

React uses inline templating and JSX. This can seem awkward to some developers. The library of React is too large.

All of the above

### **ANGULAR :**

3. Event Emitter class's simple interface, which basically encompass two methods . . . . . can therefore be used to trigger custom events and listen to events as well, both synchronously or asynchronously.

- A) exit()
- B) superscript()
- C) subscribe()
- D) emit()

Ans: C

4. The . . . . . directive substitutes the normal href property and makes it easier to work with route links in Angular 2. Moreover if we want to navigate between routes, we use the . . . . . directive.

- A) RouterLinks
- B) RouterLinking
- C) RouterLink
- D) RouterLink

Ans: C

6. Route . . . . . allow us to pass values in our url to our component so we can dynamically change our view content.

- A) pipes
- B) modules
- C) variables
- D) parameters

Ans: D

8. Observables help you manage . . . . . data.

- A) synchronous
- B) asynchronous
- C) Both asynchronous & synchronous
- D) None of above

Ans: B

11. The promise becomes . . . . . after resolving or rejecting.

- A) mutable
- B) ignored
- C) immutable
- D) recurring

Ans: C

12. Which of the following is not built-in pipe in Angular

- A) DatePipe
- B) CurrencyPipe
- C) DataPipe
- D) PercentPipe

Ans: C

15. The . . . . . decorator allows us to define the pipe name that is globally available for use in any template in the across application.

- A) @pipeName
- B) @pipeDeco
- C) \$Pipe
- D) @Pipe

Ans: D

18. Attribute binding is similar to property binding but is tied to the . . . . . rather than the DOM property.

- A) Page attribute
- B) DOM attribute
- C) Root attribute
- D) HTML attribute

Ans: D

23. It's always possible to make use of animation input parameters by setting even more data via the . . . . .property.

- A) options.param
- B) option.params

- C) option.param
- D) options.params

Ans: D

26. Ques. Bootstrap's grid system allows up to

- a. 6 columns across the page
- b. 12 columns across the page
- c. columns across the page
- d. columns across the page

Ans: B

27. Which of the following classes makes a round corner image automatically adjust to fit the size of the screen?

- a. .img-res-image
- b. .img-responsive-image \* img-fluid (for full width responsible)
- c. .img-responsive
- d. .img-res

Ans: C

30. What are the types of access modifiers supported by TypeScript?

- A. Public
- B. Private
- C. Protected
- D. All of these

Ans: D

35.What is the purpose of the \*ngIf directive?

- A.To use style classes to hide or show elements based on an expression

- B.To repeat a set of elements based on an expression
- C.To optionally navigate to another view based on an expression
- D.To add or remove elements from the DOM based on an expression

ANS : D

47. In Angular , one can create a local HTML reference of an HTML tag using a variable , which starts with character \_\_\_\_ .

- A. @
- B. #
- C. \*
- D. &

ANS : B

49. In Angular routing , which of these tags is used to show the selected route component dynamically ?

- A. <router></router>
- B.<router-output></router-output>
- C.<router-outlet></router-outlet>
- D.<router-input></router-input>

ANS : C

### **SPRING AOP :-**

Q 12 - What is Spring MVC framework?

A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

ANS : B

13. Which of the following is not a valid difference between @Inject and @Autowired annotations?

1. There is no difference and both can be used interchangeably.

2. @Autowired is a legacy annotation whereas @Inject is its latest version.

3. @Inject annotation is part of Java EE 7's Context and Dependency Injection framework while @Autowired is the Spring Frameworks own implementation.

4. None of these.

ANS :- B

Q 14 - What are the ways to access Hibernate by using Spring?

A - Inversion of Control with a Hibernate Template and Callback.

B - Extending HibernateDAOSupport and Applying an AOP Interceptor node.

C - Both of above.

D - None of above.

ANS : C

Q 15 - Which ORM Spring supports ?

A – Hibernate

B – iBatis

C – JPA

D - All of above.

E - None of above.

ANS : D

Q 16 - Which of the following database is not supported using jdbcTemplate?

A - MySql

B – PostgresSql

C – NoSql

D – Oracle

ANS : C

Q 25 - Can we integrate Struts with Spring.

A – Yes

B - No

ANS : A

Q 26 - By default a bean is lazily loaded.

A - True

B - False

ANS : B

Q 32 - What is the scope of stateless beans?

A - global-session

B - singleton

C – prototype

D - request

ANS : B

32. Annotation to add security to your spring boot app?

a. @EnableWebSecurity

b. @EnableSecurity

c. @EnableSpringSecurity

ANS : A



35. LoggerInterceptor which implements the HandlerInterceptor. The LoggerInterceptor will override the which following methods

- a. preHandlerBean(), postHandlerBean(), afterCompletion()
- b. PreProcessHandler(), PostProcessHandler(), afterCompletion()
- c. PreHandler(), PostHandler(), afterCompletion()
- d. none

ANS : C

### **MONGODB :-**

1) In MongoDB, what is the equivalent of SQL term row?

- A. Primary Key
- B. Index
- C. Field
- D. Document

ANS : D

4) Which of the following code will give an error on the MongoDB command shell?

- A. use databasename
- B. show dbs
- C. USE databasename
- D. Db

ANS : C

5) What theorem or principal MongoDB follows?

- A. APAC
- B. Always-Sync
- C. ASCII

D.CAP

ANS : D

9) What method is used to remove a single item from a MongoDB collection?

`db.collection.deleteOne()`

`db.collection.delete()`

`db.collection.removeOne()`

`db.collection.remove()`

ANS : A

### **DOCKER & DEVOPS :-**

4) What is continuous testing?

Testing continuously for 24 hours

Manually testing each build, end-to-end Automated testing

Automated testing

on each continuous deployment

ANS : D

7) Shift-left concept refers to moving everything to the earlier phase in the \_\_\_\_\_ cycle.

Requirement

Development

Design

Testing

ANS : B

13) In Kubernetes, a node is:

A worker machine

A tool for starting a kubernetes cluster on a local machine

A machine that coordinates the scheduling and management of application containers on the cluster A virtual machine

ANS : A

14) GIT can be integrated with Jenkins.

True

False

ANS : A

### **PREV BATCH FSD QUESTIONS :-**

5) Which Annotation is used to refer to Pointcuts

A. @PointCut.

B. @PointCutExecution.

C. @PointCutBefore.

ANS :- A

8) Which element is used to restrict the access to a particular URL within Spring Security ?

i. restrict-url

ii.intercept-url

iii.intercept-restrict.

ANS : B

10) Which annotation is used to create Primary KEY :

Foreign KEY relation between two tables ?

i. ForeignKey.

ii. JoinedKey.

iii. JoinColumn.

iv. ANS: C

11) You can only use Hibernate in ORM in Spring Boot Applications.

i.TRUE

II.FALSE

ANS : B

14) A logger location where logging information is sent is called

1. appender.

2. logger

3. component.

ANS : appender A

15) By default, a bean is lazily loaded in Spring.

1. True

2. False

ANS : B

20) Identify the incorrect statement on MongoDB ?

A. Secondary Indices are not available in MongoDB.

B. MongoDB supports search by field, range queries and regular expression matches.

C .MongoDB can store the business subject in the minimal number of documents.

ANS :- A. -> it can create

26) Identify the correct statement in MongoDB.

A. Queries specify criteria, or conditions , which identify the documents that MongoDB returns to the clients.

B. Write Operations , or queries , retrieve data stored in the database.

C. The selection limits the amount of data that MongoDB returns to the client over the network.

ANS : A -> mostly

30) What is the command line tool introduced for the JS engine in Java 8 ?

A. jjjs

B. jss

C.jfs

D.jbd

ANS : A

38) Which of the following statements is true about microservices architecture ?

i. Microservices applications are required to be deployed on the same physical host.

ii. Microservices architecture supports high availability of individual microservices.

iii. Microservices cannot be used if a company is embracing DevOps.

iv. Microservices are designed using a bounded context that can communicate with other bounded contexts.

A. Only i

B. Only

ii C. Only ii and iv

D. All of the above.

Ans -> D

39) Load Balancing improves the distribution of workloads across multiple computing resources , such as computers or a computer cluster.

A. True.

B. False.

ANS:- A

41) Which CSS property is used to change the font of an element ?

A. font

B. font-family

- C. font-style
- D. font-weight

ANS :- B

42) The Bootstrap grid system is based on how many columns ?

- A. 12
- B. 3
- C. 6
- D. 9

ANS:- A

44) Choose the correct HTML element to define important text.

- A. <strong>
- B. <important>
- C. <b>
- D. <pre>

ANS :- A

45) Which is the correct CSS Syntax ?

- A. body color-black
- B. {body color-black}
- C. body {color:black;}
- D. {body;color;black}

ANS : C

48) How can you detect the client's browser name in Javascript ?

- A. class.navName
- B. navigator.appName

- C. browser.name
- D. window.browser.

ANS : B

53) To use the Angular HttpClient component, you must import the \_\_\_\_ module.

- A. HttpClientModule.
- B. HttpModule.
- C. Http

Ans :- A

54) In Angular , one can create local HTML reference or HTML tag using variable which starts with character \_\_\_\_

- A. @
- B. #
- C. “
- D. &

ANS :- B

55) Which of these Angular services can be injected in your component to the dynamic mode navigation ?

- A. Routing.
- B. RouterService.
- C. RoutingService
- D. Router

ANS :- D

56) In Angular , which of the following wild card route paths will you use to define a 404 route ?

- A. \*\*
- B. /

C. ##

D. default

ANS :- A

### Core Java:

#### 3. What is the output of the following Java code snippet?

```
public class home implements Runnable {  
    public void run() {  
        System.out.printf("%d", 3);  
    }  
    public static void main(String[] args) throws InterruptedException {  
        Thread thread = new Thread(new home());  
        thread.start();  
        System.out.printf("%d", 1);  
        thread.join();  
        System.out.printf("%d", 2);  
    }  
}
```

- a. 132
- b. 123
- c. 213
- d. 312



8. Which of the following is the immutable Java List static factory method?

- a. **List.of**
- b. List.immutable
- c. List.create

▼ Explanation:

`List.of` is a standard method introduced in Java 9 specifically for creating immutable lists. It's important to note that while there are other ways to create immutable lists in Java, such as using `Collections.unmodifiableList`, `List.of` provides a more concise and readable way to achieve immutability without the need for explicit instantiation or copying.

9. What is the command line tool introduced for JS engine in Java8?

- a. **jjs**
- b. jss
- c. jfs
- d. jbd

▼ Explanation:

`jjs` is the command-line tool provided in Java 8 for running JavaScript code via the Nashorn JavaScript engine. Nashorn was introduced in Java 8 as a modern JavaScript engine, allowing Java developers to execute JavaScript code seamlessly within their Java applications.

13. Identify the Java class used to create a new Process.

- a. **ProcessBuilder**
- b. ProcessFactory
- c. Process
- d. Thread

▼ Explanation:

The Java class used to create a new process is `java.lang.ProcessBuilder`.

14. **Pick the correct Java method reference usages.**

- a. object::instanceMethod
- b. Class::staticMethod
- c. Class::new
- d. **All of the above**

▼ Explanation:

The correct Java method reference usages are:

16. **Which one will allow traversing a Java collection in each direction?**

- a. **ListIterator**
- b. Iterator
- c. MapIterator
- d. SetIterator

▼ Explanation:

The question is asking about Java interfaces that allow traversal of collections in both forward and backward directions.

The correct answer is 1. ListIterator.

19. **Pick the class that is NOT part of Java collection framework.**

- a. Map
- b. List
- c. Queue
- d. **Struct**

▼ Explanation:

21. **Java Functional Interface can be defined as\_\_\_\_\_.**

- a. Simple Abstract Markup
- b. Simple Active Markup
- c. Single Ambivalue Method
- d. **Single Abstract Method**

22. **Which is the correct API to manage reactive streams in Java?**

- a. java.util.concurrent.Flow
- b. java.util.concurrent.Flow.Publisher
- c. java.util.concurrent.Flow.Processor
- d. **All of the above**

23. **In Java, which of the following data structure classes CANNOT be type parameterized?**

- a. List
- b. Set
- c. **Array**
- d. Map

24.

```

    }
    classSecondextendsFirst {
        publicSecond() {
            System.out.println("S");
        }
    }
    classThirdextendsSecond {
        publicThird() {
            System.out.println("T");
        }
    }
    publicclassTestClass {
        publicstaticvoidmain(String[]args) {
            Thirde=newThird();
        }
    }

```

- a. TSF
- b. T
- c. F
- d. **FST**

29. Which class is introduced in Java 8 to get the current milliseconds?

- a. **Clock**
- b. Current Time
- c. Date Time
- d. Time

▼ Explanation:

The class introduced in Java 8 to get the current milliseconds is:

1. Clock

30. In Java, Which of these class types supports sequential and parallel processing of data?

- a. Queue
- b. Lambda
- c. ExecutorService
- d. **Streams**

▼ Explanation:

38. What is the replacement for PermGen space in java8?

- a. FixedGen
- b. **Metaspace**
- c. MetaGenspace

▼ Explanation:

In Java 8, the Permanent Generation (PermGen) was replaced by Metaspace. Metaspace is a memory space where class metadata is stored, including information about classes, methods, fields, annotations, etc. Unlike PermGen, Metaspace is not a fixed size and can dynamically expand or contract depending on the application's needs. It helps mitigate issues

42. What is the difference between peak() and dequeue()?

- a. **Peek() returns the next item in line but dequeue() removes and returns the next item in line**
- b. peek() and dequeue() return the next item in line
- c. peek() and dequeue remove and return the next item in line

3.MongoDB database can be accessed using Spring Data.

True

False

ANS : A

13.How to handle shutdown of IoC container?

Using shutdownHook()

Using shutdownHandler()

Using registerHook()

Using registerShutdownHook()

ANS : D

14.What is default scope of bean in Spring framework?

singleton

prototype

request

session

ANS : A

16. Which endpoints are provided by Spring Boot to monitor metrics of individual microservices? /actuator /health

/actuator

/rest

/logger

/health

ANS : A

20.REST service endpoint comprises of which address? Starts with http:// and may end with?

Starts with http:// and may end with?

No certain URL is specified

Depends upon the platform used

ANS : A

23. Microservices is most closely related to which other established acronym?

API

AWS

GCP

PAAS

ANS : A

25. What are the core operations of DevOps with application development and with infrastructure?

Provisioning, Configuration, and Orchestration

Code building, coverage, and Unit testing

Packaging and Deployment

All of the above

ANS : D

27. What is the return type of Java lambda expressions?

void

Lambda

Function

Regex

ANS : C

29. In Java, which of these class types supports sequential and parallel processing of data?

Queue

Lambda

ExecutorService

Streams

ANS : D

31.What is the replacement for PermGen space in Java 8?

FixedGen

Metaspace

MetaGenspace

ANS : B

36.Where in an HTML document is the correct place to refer to an external style sheet?

In the <head> section

In the <body> section

In the <head> section

At the end of the document

At the start of the document

ANS : B

42.Where is the correct place to insert a JavaScript? Both the <head> section and the <body> section are correct

The <head> section

Both the <head> section and the <body> section are correct

The <body> section

The <footer> section

ANS : B



45.How can you detect the client's browser name in JavaScript?

client.navName

navigator.appName

browser.name

window.browser

ANS : B

48.In JavaScript, var declares a block-scoped variable.? False

True

False

ANS : B

50.What is React?

Just a server-side framework

Just a user-interface framework

Both a server-side framework as well as a user-interface framework

ANS : B

51.What are the limitations of React?

React is only for view layer of the app so we still need the help of other technologies to get a complete tooling set for development.

React uses inline templating and JSX. This can seem awkward to some developers.

The library of React is too large.

All of the above

ANS : D

52.How can you access the state of a component from inside of a member function in React?

this.getState()

this prototype stateValue

this.state

this values

ANS : C

54.Which of the following API is a MUST for every React component? render

getInitialState

render

renderComponent

ANS : B

58.What does the "webpack" command do?

Transpiles all the Javascript down into one file

Runs react local development server

ANS : A



**1.**

Which tag is used to define a clickable image map in HTML5?

- a) <map>
- b) <area>
- c) <img>
- d) <figure>

**Answer: a) <map>**

---

**2.**

What is the effect of display: flex; justify-content: space-between;?

- a) Aligns children vertically with equal spacing
- b) Aligns children horizontally with space between them
- c) Centers children horizontally
- d) Stacks children in a column

**Answer: b) Aligns children horizontally with space between them**

---

**3.**

What is the output?

```
console.log(5 + "5");
```

- a) 10
- b) "55"
- c) 25
- d) Error

**Answer: b) "55"**

---

**4.**

Which method is used to add an event listener?

- a) addEvent()
- b) attachEvent()
- c) addEventListener()
- d) onEvent()

**Answer: c) addEventListener()**

---

5.

Which property keeps a background image fixed while scrolling?

- a) background-repeat: no-repeat
- b) background-attachment: fixed
- c) background-position: center
- d) background-size: cover

**Answer: b) background-attachment: fixed**

---

## React

6.

What is the purpose of the useState hook?

- a) Fetch data from API
- b) Manage state in functional components
- c) Handle side effects
- d) Create context

**Answer: b) Manage state in functional components**

---

7.

What does this render?

```
return <h1>{undefined}</h1>;
```

- a) Nothing (empty string)
- b) "undefined"
- c) Error
- d) null

**Answer: a) Nothing (empty string)**

---

8.

Which hook performs side effects?

- a) useEffect
- b) useReducer

- c) useContext
- d) useCallback

**Answer: a) useEffect**

---

**9.**

How do you pass data from parent to child?

- a) Using state
- b) Using props
- c) Using context
- d) Using hooks

**Answer: b) Using props**

---

**10.**

Correct way to update state?

```
const [user, setUser] = useState({ name:"John", age:30 });
```

- a) setUser({ name:"Jane" })
- b) setUser({ ...user, name:"Jane" })
- c) user.name="Jane"
- d) setUser(user.name="Jane")

**Answer: b) setUser({ ...user, name:"Jane" })**

---

**11.**

Purpose of key prop in a list?

- a) Styling
- b) Identifying unique elements for efficient rendering
- c) Handling events
- d) Managing state

**Answer: b) Identifying unique elements for efficient rendering**

---

**12.**

Lifecycle method replaced by useEffect?

- a) componentDidMount
- b) render
- c) constructor
- d) setState

**Answer: a) componentDidMount**

---

**13.**

What does JSX stand for?

- a) JavaScript XML
- b) JavaScript Extension
- c) JSON XML
- d) Java Syntax

**Answer: a) JavaScript XML**

---

**14.**

How to create a ref in functional component?

- a) `const ref = useRef();`
- b) `const ref = createRef();`
- c) `const ref = useState();`
- d) `const ref = useEffect();`

**Answer: a) `const ref = useRef();`**

---

**15.**

If count = 0, output?

```
useEffect(() => {  
  setCount(count + 1);  
}, []);
```

- a) 0
- b) 1
- c) 2
- d) Infinite loop

**Answer: b) 1**

---

## MongoDB

**16.**

Primary key in MongoDB document?

- a) \_id
- b) id
- c) key
- d) index

**Answer: a) \_id**

---

**17.**

Find users where age > 25?

- a) `db.users.find({ age: { $gt: 25 } })`
- b) `db.users.find({ age: { $gte: 25 } })`
- c) `db.users.find({ age: { $lt: 25 } })`
- d) `db.users.find({ age: 25 })`

**Answer: a) `db.users.find({ age: { $gt: 25 } })`**

---

**18.**

What does \$push do?

- a) Replace field
- b) Add element to array
- c) Remove element
- d) Update field

**Answer: b) Add element to an array**

---

**19.**

Which index improves query performance?

- a) Compound index
- b) Text index
- c) Unique index
- d) All of the above



**Answer: d) All of the above**

---

## **Angular**

**20.**

Purpose of \*ngFor?

- a) Conditional display
- b) Iterate list
- c) Data binding
- d) Form submission

**Answer: b) Iterate over a list**

---

**21.**

Decorator defining component?

- a) @Component
- b) @Directive
- c) @NgModule
- d) @Injectable

**Answer: a) @Component**

---

**22.**

Output:

```
<div *ngIf="false">Hello</div>
```

- a) Hello
- b) Nothing
- c) Error
- d) false

**Answer: b) Nothing**

---

**23.**

How to inject service?

- a) Using @Inject
- b) In constructor
- c) Using @Service
- d) In template

**Answer: b) In constructor**

---

**24.**

What does FormBuilder simplify?

- a) HTTP requests
- b) Reactive forms creation
- c) Template rendering
- d) Dependency injection

**Answer: b) Reactive form creation**

---

**25.**

Module for HTTP requests?

- a) HttpClientModule
- b) FormsModule
- c) RouterModule
- d) CommonModule

**Answer: a) HttpClientModule**

---

### **Core Java Code Snippets**

**26.**

Output?

```
int x = 5;  
System.out.println(x++ + ++x);
```

- a) 10
- b) 11
- c) 12
- d) 13

**Answer: c) 12**

---

**27.**

```
String s1="Hello";  
String s2=new String("Hello");  
System.out.println(s1==s2);
```

- a) true
- b) false
- c) Error
- d) null

**Answer: b) false**

---

**28.**

```
List<Integer> list=new ArrayList<>();  
list.add(1);  
list.add(2);  
list.remove(Integer.valueOf(1));  
System.out.println(list);
```

- a) [1,2]
- b) [2]
- c) []
- d) Error

**Answer: b) [2]**

---

**29.**

```
static int x=10;  
Test t=new Test();  
t.x=20;  
System.out.println(Test.x);
```

- a) 10
- b) 20
- c) 30
- d) Error

**Answer: b) 20**

---

**30.**

```
int[] arr={1,2};  
System.out.println(arr[2]);
```

with try-catch.

- a) 1
- b) 2
- c) Error
- d) null

**Answer: c) Error**

**31. Java: What is the purpose of the final keyword?**

- a) Prevents method overriding
- b) Allows dynamic binding
- c) Enables garbage collection
- d) Permits multiple inheritance

**Answer: a) Prevents method overriding**

---

**32. Java: Which collection is synchronized by default?**

- a) ArrayList
- b) HashMap
- c) Vector
- d) LinkedList

**Answer: c) Vector**

---

**33. Java: What is the default access modifier for class members?**

- a) public
- b) private
- c) protected
- d) package-private

**Answer: d) package-private**

---

**34. Java: What does the transient keyword do?**

- a) Prevents serialization
- b) Enables multithreading
- c) Marks a variable as static
- d) Allows method overriding

**Answer: a) Prevents serialization**

---

**35. Java: Which interface is used for sorting objects?**

- a) Comparable
- b) Comparator
- c) Serializable
- d) Cloneable

**Answer: a) Comparable**

---

## **TypeScript**

**36. TypeScript: What is the purpose of interface in TypeScript?**

- a) Define class behavior
- b) Specify object structure
- c) Create modules
- d) Handle exceptions

**Answer: b) Specify object structure**

---

**37. TypeScript: What is the type of let x: any;?**

- a) number
- b) string
- c) any
- d) void

**Answer: c) any**

---

**38. TypeScript: What does this code do?**

```
function greet(name: string): string {  
  return `Hello, ${name}`;  
}
```

- a) Throws a type error
- b) Returns a string
- c) Returns void
- d) Returns any

**Answer: b) Returns a string**

---

**39. TypeScript: Which keyword extends a class?**

- a) implements
- b) extends
- c) super
- d) interface

**Answer: b) extends**

---

**40. TypeScript: How do you define an optional property?**

- a) name: string?
- b) name?: string
- c) name: string | null
- d) name: string = null

**Answer: b) name?: string**

---

**Spring / Spring Boot**

**41. Spring: Which annotation defines a Spring bean?**

- a) @Bean
- b) @Component
- c) @Service
- d) All of the above

**Answer: d) All of the above**

---

**42. Spring Boot: What does @SpringBootApplication include?**

- a) @Configuration, @EnableAutoConfiguration, @ComponentScan
- b) @RestController, @EnableWebMvc
- c) @Repository, @Transactional
- d) @Autowired, @Qualifier

**Answer: a) @Configuration, @EnableAutoConfiguration, @ComponentScan**

---

**43. Spring: What is the output of this code?**

```
@Component
public class MyBean {
    public String getName() {
        return "Hello";
    }
}
```

```
@Autowired
MyBean bean;
```

```
System.out.println(bean.getName());
```

- a) Hello
- b) null
- c) Error
- d) MyBean

**Answer: a) Hello**

---

**44. Spring Boot: Which dependency enables REST APIs?**

- a) spring-boot-starter-web
- b) spring-boot-starter-data-jpa
- c) spring-boot-starter-security
- d) spring-boot-starter-test

**Answer: a) spring-boot-starter-web**

---

**45. Spring: What is Dependency Injection?**

- a) Creating objects manually
- b) Injecting dependencies into objects
- c) Managing database connections
- d) Handling HTTP requests

**Answer: b) Injecting dependencies into objects**

---

**46. Spring Boot: How do you externalize configuration?**

- a) Using application.properties
- b) Using @Configuration
- c) Using @Bean
- d) Using @Autowired

**Answer: a) Using application.properties**

---

**47. Spring: Which annotation maps HTTP GET requests?**

- a) @PostMapping
- b) @GetMapping
- c) @RequestMapping
- d) @PutMapping

**Answer: b) @GetMapping**

---

**48. Spring Boot: What is the default embedded server?**

- a) Jetty
- b) Tomcat
- c) Undertow
- d) Netty

**Answer: b) Tomcat**

---

**49. Spring: What does @Transactional do?**

- a) Manages database transactions
- b) Configures REST endpoints
- c) Injects dependencies
- d) Scans components

**Answer: a) Manages database transactions**

---

**50. Spring Boot: How do you connect to MongoDB?**

- a) spring-boot-starter-data-mongodb
- b) spring-boot-starter-data-jpa
- c) spring-boot-starter-web
- d) spring-boot-starter-security



**Answer: a) spring-boot-starter-data-mongodb**

---

## **DevOps**

**51. DevOps: What is the purpose of Docker?**

- a) Code compilation
- b) Containerization
- c) Version control
- d) Load balancing

**Answer: b) Containerization**

---

**52. DevOps: Which command builds a Docker image?**

- a) docker run
- b) docker build
- c) docker push
- d) docker pull

**Answer: b) docker build**

---

**53. DevOps: What is CI/CD?**

- a) Continuous Integration / Continuous Deployment
- b) Code Inspection / Code Delivery
- c) Continuous Improvement / Continuous Development
- d) Code Integration / Code Deployment

**Answer: a) Continuous Integration / Continuous Deployment**

---

**54. DevOps: Which tool manages container orchestration?**

- a) Jenkins
- b) Kubernetes
- c) Git
- d) Ansible

**Answer: b) Kubernetes**

---

**55. DevOps: What does nginx -g "daemon off;" do in a Docker container?**

- a) Runs NGINX in the background
- b) Runs NGINX in the foreground
- c) Stops NGINX
- d) Restarts NGINX

**Answer: b) Runs NGINX in the foreground**

---

**56. DevOps: Which file defines a Jenkins pipeline?**

- a) Dockerfile
- b) Jenkinsfile
- c) pom.xml
- d) build.gradle

**Answer: b) Jenkinsfile**

---

**57. DevOps: What is Infrastructure as Code?**

- a) Manual server setup
- b) Defining infrastructure using scripts
- c) Database management
- d) Application deployment

**Answer: b) Defining infrastructure using scripts**

---

**58. DevOps: Which AWS service manages containerized apps?**

- a) EC2
- b) ECS
- c) S3
- d) RDS

**Answer: b) ECS**

---

**Microservices**

**59. Microservices: What is a key benefit of microservices architecture?**

- a) Monolithic codebase
- b) Independent scalability
- c) Tight coupling
- d) Single database

**Answer: b) Independent scalability**

---

**60. Microservices: Which pattern ensures loose coupling in microservices?**

- a) Circuit Breaker
- b) Service Registry
- c) Event-Driven
- d) Monolith

**Answer: c) Event-Driven**

## Vaishali

1. How to convert mongodb object in javascript? toString()
2. How to run mongodb in js? - mongo
3. No. of columns in bootstrap grid layout? -12
4. Is @entity is an annotation in mongodb?- False
5. Do we add @Repository annotation on DAO implementation class? T/F - False
6. Function which add or remove elements from array in js? - join(), shift(), splice(), all of
7. the above? Ans- all of the above
8. How to create custom annotations in java?- @interface
9. Microservice closely resembles to? API, PAAS, SEO, AWS. Ans - API
10. States in the redux are immutable? T/F - False
11. Which set is sorted and synchronous in java? – TreeSet

## Suraj

Which one of the following methods are ... functions of react ? (... I am forgetting)

1. componentDidMount
2. componentDidUnmount
3. Render
4. componentFailToMount

Which of the following are valid class constructs in java?

1. class::StaticMethods()
2. Class::new
3. Instance::classMethods()
4. All of the above -> ans

Which function in js is used to convert an array to a string?

1. Splice
2. Slice
3. Join
4. Concat

Join - ans

Which function in js is equivalent to returning true if conditions corresponding to all of the

elements of an array are true?

1. Every - ans (my opinion)
2. All
3. Some
4. any

How many columns are there in bootstrap grid system?(Repeated 4 times)

Ans – 12

### **1** forkJoin

#### **Question:**

Which RxJS operator waits for all observables to complete and then returns the last emitted values?

**Answer:** forkJoin

---

### **2** mergeMap

#### **Question:**

Which RxJS operator maps each value to an observable and merges the results?

**Answer:** mergeMap

---

### **3** merge

#### **Question:**

Which RxJS operator combines multiple observables into one observable?

**Answer:** merge

---

### **4** a.length = 0

#### **Question:**

How can you clear all elements of an array in JavaScript?

**Answer:** a.length = 0

Example

```
let a = [1,2,3];  
a.length = 0;
```

---

## 5 Constructor Chaining

### Question:

What is the process of calling one constructor from another constructor in Java?

**Answer:** Constructor Chaining

---

## 6 Thread Basic Example

### Question:

Give a simple example of creating a thread in Java.

### Answer Example:

```
class MyThread extends Thread {  
    public void run(){  
        System.out.println("Thread running");  
    }  
}
```

---

## 7 @ResponseBody

### Question:

Which annotation is used in Spring MVC to return data directly to the HTTP response body?

**Answer:** @ResponseBody

---

## 8 RxJS Method

### Question:

Which library provides operators like merge, forkJoin, and mergeMap in Angular?

**Answer:** RxJS

---

## 9 Canvas

**Question:**

Which HTML element is used to draw graphics using JavaScript?

**Answer:** <canvas>

---

**10 container****Question:**

Which Bootstrap class is used to create a responsive fixed-width layout container?

**Answer:** .container

---

**1 1 ul****Question:**

Which HTML tag is used to create an unordered list?

**Answer:** <ul>

---

**1 2 ol****Question:**

Which HTML tag is used to create an ordered list?

**Answer:** <ol>

---

**1 3 ContextLoaderListener****Question:**

Which class is used in Spring to load the root application context?

**Answer:** ContextLoaderListener

---

**1 4 Open Tracking API in Microservice****Question:**

Which API specification is used for tracking and documenting microservice APIs?

**Answer:** Open Tracking API (often referring to OpenAPI)

---

## **1 5 Default Scope of Bean**

### **Question:**

What is the default scope of a bean in Spring Framework?

**Answer:** Singleton

---

## **1 6 Flux**

### **Question:**

Which reactive type in Spring WebFlux represents 0 to many asynchronous elements?

**Answer:** Flux

---

## **1 7 Flux is Multidimensional**

### **Question:**

In Spring Reactive Programming, which reactive stream type supports multiple elements?

**Answer:** Flux

---

## **1 8 State and Props**

### **Question:**

Which two data handling concepts are used in React components?

**Answer:** State and Props

---

## **1 9 React Port**

### **Question:**

Which default port does a React development server run on?

**Answer:** 3000

---

## **2 0 Everything in React**

### **Question:**

In React, what is everything built using?

**Answer:** Components



---

## 2 1 RouterModule.forRoot

### Question:

Which Angular method is used to register routes at the root level?

**Answer:** RouterModule.forRoot()

---

## 2 2 @Injectable

### Question:

Which Angular decorator marks a class as available to be injected as a dependency?

**Answer:** @Injectable

---

## 2 3 "5" < 2

### Question:

What will be the result of the JavaScript expression "5" < 2?

**Answer:** False

Because "5" is converted to number **5**, so **5 < 2 → false**

---

## 2 4 Script Tag Location

### Question:

Where can the <script> tag be added in HTML?

Options:

- Head
- Body
- Both

**Answer:** Both head and body

## Sneha

1. how to create zebra striped table in bootstrap? Ans: <table class="table table-striped">
2. Static keyword in java can be used with?
3. Features of devops?
4. Microservices vs Monolithic?
5. three major parts of flux application: Ans: the dispatcher, the stores, and the views
6. Which class is used to create a basic pagination? //HTML w3 school quiz  
a) pagination b) navigation c) page d) .pages
7. Which class is used to create a big box for calling extra attention?  
a) .bigbox b) .jumbotron c) .container
8. Which button class is used to create a large button?  
a) .btn-lg b) .btn-xl c) .btn-l d) .btn-large
9. Which class indicates a dropdown menu?  
a) .dropdown-list b) .dropdown c) .select
10. In Java, static keyword is mainly used for ? Ans: variables, methods, blocks and nested classes

## Srishti

- Mark important text in web pages ----> strong
- Constructor::new , constructor ::(), constructor::method--->> options of questions. (Question I don't remember).
- Relation between dev ops and microservices
- Why to use microservices
- Java 8 has private methods in constructor ->> false
- typeof(null) in javascript

- Form table structure - > table, tr, td
- Microservices closely related to API
- Default scope of bean → singleton
- Natural sort -> TreeSet
- add/remove methods in array -> splice
- Everything in react is component
- Slide show is carousel
- 404 error -> \*\*
- Flux has multidimensional data flow -> false (its unidirectional)
- “5”<2 (false)
- React has multi- binding -> False
- Subscribe a service using @injectable
- W3 school - > html, css, javascript, bootstrap
- \$ - > used to set property

how to capitalize text in css : text-transform: capitalize;-

how to hyperlink local html page- #-

which is the correct table structure - <table><tr><td>

## Priyashi

1) In Java 8 constructor are allowed in interfaces. (True /False)

Ans :False

2) .How to create zebra stripped table in bootstrap?

Ans:<table class="table table-striped">

3) Main method cannot be -(4 options)

ans will be Overriden

4) graphics defined by svg is in which format

Ans: xml

5) What does replace() do?(Java question)

6) You can also only use tomcat as embedded system in spring?(T/F)

Ans:False

7)What is the default scope of bean?

Ans: Singleton

8)What of this has natural sorting?

Ans: TreeSet

9)mongodb identifies

10)flux has multi dimensional data flow?

Ans: False bcz flux has unidirectional data flow

11)What is Babel?

a. A transpiler

c. A compiler

Ans : d

b. An interpreter

d. Both Compiler and transpiler

12)In which directory React Components are saved?

a. Inside js/components/

c. Inside external/components/

b. Inside vendor/components/

d. Inside vendor/

Ans: a

13)What is a react.js in MVC?

a. Middleware

b. Controller

c. Model

Ans: b

d. Router

14)404 error -> \*\*

15)Features of devops?

16)W3 school questions - img.circle,

17). In Angular one can create local html reference of html tag using variable, which starts with character.

Ans: ans-- #

18)Point out the correct statement.

- a) MongoDB is classified as a NoSQL database
- b) MongoDB favours XML format more than JSON
- c) MongoDB is column oriented database store
- d) All of the mentioned

Ans: a

19)How to make table ?

Ans: <table><tr><td>

20).what is the java class used to create new class

ans.process builder

## Ameya

1. A linkage of series of prototype objects is called as \_\_\_\_\_

- a) prototype stack
- b) prototype chain
- c) prototype class
- d) prototypes

View Answer

Answer: b

2. Redux has a Single Store.

Ans: True

3. Which of these are true:

Answers: finally will be executed irrespective of whether

exception occurred or not

Finally is not mandatory

4. Element used to restrict access to a particular URL.

- a) restrict-url
- b) intercept-url
- c) intercept-restrict
- d) none of the mentioned

View Answer

Answer: b

Explanation: Inside the http configuration element, you can restrict access to particular URLs with one or more intercept-url elements.

5. Point out the correct statement.

- a) Documents can contain many different key-value pairs, or key-array pairs, or even nested documents
- b) MongoDB has official drivers for a variety of popular programming languages and development environments
- c) When compared to relational databases, NoSQL databases are more scalable and provide superior performance
- d) All of the mentioned

View Answer

Answer: d

Explanation: There are also a large number of unofficial or community-supported drivers for other programming languages and frameworks.

6. NoSQL databases is used mainly for handling large volumes of \_\_\_\_\_ data.

- a) unstructured
- b) structured

- c) semi-structured
- d) all of the mentioned

View Answer

Answer: a

Explanation: MongoDB is an a typical choice for unstructured data storage.

7. What is the correct HTML for making a checkbox?

<checkbox>

<input type="check">

<input type="checkbox">

<check>

Ans: <input type="checkbox">

. Aashray

1. W3 school questions - img.circle, text bold
2. Interface implements @funtion something
3. How to create object in js - Object.create() method is the ans.
4. Can we define methods in interface in java 9/8? - ans not sure (yes)
5. React returns how many items - multiple
6. Can we define a service/component in more than 1 ngmodule- ans - no
7. Logger need not return explained results .... Ans - true
8. Tell elements of devops - all of the above
9. Mongo- which one is correct- ans - mongo runs something in batches.
10. How to change font of text - font family
11. Load balancing - ans - balances the load by distributing things.
12. Svg is in which format - ans xml - check w3
13. List map props question

1,what annotation is used in spring bean for declaring a document?

1. @Document
2. @entity
3. @collection
4. none of the above

Ans: @Document

2.Spring Properties can be changed at runtime

True or false

Ans : True

3. Everything in react is : \_\_\_\_\_

Ans : Component

4.Flux components

- Dispatcher
- Stores
- Views (React components)
- Action

Ans : All of the above

5.JavaScript is Case-sensitive :True or false

Ans : Tru

6. Beans defined in a Spring are by default : \_\_\_\_\_

Ans: Singleton

7.The <aside> content is often placed as a sidebar in a document.

8. var arr = [1,2,3];

arr.length = 0;

console.log(arr)

Ans: []

9.What are the two ways that data gets handled in React?



state & props

services & components

Answer :state & props

10. How can you access the state of a component from inside of a member function?

this.getState()

this.values

this.prototype.stateValue

this.state

Answer this.values

11. MongoDB is \_\_\_\_\_ oriented database

ANS: Document

Some Reference for

Document-based and Key/Value-based databases?

There are three different, popular types of no-sql databases.

- Key/Value: Redis, Tokyo Cabinet, Memcached

- ColumnFamily: Cassandra, HBase

- Document: MongoDB, CouchDB

12. Wildcard Route for defining error 404 path or undefined route is

Ans: \*\*

13. What are the advices in Aop?

a) After

b) before

c) AfterReturning

d) ALL of the above

ANS: D

14. Syntax for Checkbox in HTML

A)<Check> B)<box> C)<checkbox>

ANS:D

D)<input type= "checkbox">

15.Syntax for large buttons in Bootstrap?

A).large

ANS:C

B).button-large

16.Changing Text size in CSS?

C).btn-lg

D)btn-large

A)font-size B)font-style C)text-style D)text-format

ANS:A

17. \_\_\_\_\_ is a part of the standard MongoDB distribution and provides a full JavaScript environment.

a) mongod

b) mongodb

c) mongo

Answer: c

18. When you query a collection, MongoDB returns a \_\_\_\_\_ object that contains the results of the query.

a) row

b) cursor

c) columns

d) none of the mentioned

Answer: b

19. React is \_\_\_\_\_

A)View

ANS:C

B)Model

C)Controller

20. What is the default port for webpack dev server?

A. 3000

B. 8080

C. 3306

D. 8809

21. The mongodb is written in \_\_\_\_\_ language

A. C++

B. Javascript

C. C

D. All of the above

22. Which of the following statement is wrong for mongodb?

A. MongoDB supports search by field, range queries, regular expression searches

B. MongoDB can store the business subject in the minimal number of documents

C. Secondary indices is not available in MongoDB

23. Which of the following cannot be initialised parameterized?

A. List

B. Set

C. Array

D. Map

24. React.js Covers only the view layer of the app.

A. True

B. False

25. <canvas> is used for?

Ans:- to draw graphics

26. An object's ..... is a reference to another object from which properties are inherited.

A) Characteristics

B) Prototype

C) Class

D) Extensible flag

Ans: B

27. Which one of the following release patterns does not enable you to do low risk DevOps

code deployments in your production systems?

Choice-1:

Canary Deployment Pattern (The Dark Launch).

Choice-2:

Blue-Green Deployment Pattern.

Choice-3:

Cluster Immune System Release Pattern.

Choice-4:

Big bang code deployments of fully tested and validated releases.

Correct Answer:

Choice-4

28.. allows the same event handler function to be registered more than once. When an event of the specified type occurs, the registered function will be invoked as many times as

it was registered.

A. addEvent()

B. addMultipltEvent()

C. attachEvent()

D. reattachEvent()

Ans : C

) associative arrays

29. The constructor prototype refers to the correct prototype for .....

but does not usually do so for objects created with `Object.Create()`.

- A) object prototype
- B) object literal
- C) object class
- D) associative arrays

Ans: B

30. Which routing guard is used to check whether routing can take place or not

- A. CanActivate
- B. Resolve
- C. CanLoad
- D. CanDeactive

ans: A

31. In Angular application there can be at most one `<router-outlet>` can be used for routing

- A. True
- B. False

Ans: B

32. Select type of pipe Angular executes during every change detection cycle of component

- A. Pure Pipe
- B. Impure Pipe

Ans: B

33. Below compilation option create smaller in size and faster in performance angular distributed package

- A. JIT
- B. AOT

ans: B

34. There can be more than one `<router-outlet>` element in angular application

A. True

B. False

ans: A

35.To use HttpClient component you need to import below module

A. HttpClientModule

B. HttpClientModule

C. Http

D. None of the above

ans B

36.When you apply 'pipe', it changes the value of the underlying component's member variable as well.

a. True

b. False

ans: False

37.Select type of pipe Angular executes during every change detection cycle of component

A. Pure Pipepath

B. Impure Pipe

ans B

1. Let has block scope?

- A. True
- B. False

**Ans:- True**

2. what does <aside> tag in html do?

**Ans : aside the content of the page**

3. Which of the following class in Bootstrap is used to provide a responsive fixed width container?

**Ans : .Container**

4. Which of the following is the correct syntax of creating a standard navigation tab?

**Ans : <ul class="nav nav-tabs">**

5. In Bootstrap, the class .img-circle is used to turn an image into a circle.

6. In Java, which of these class types supports sequential and parallel processing of data?

**Ans : Streams**

7. What is the replacement for PermGen space in Java 8?

**Ans : Metaspace**

8. Java Functional Interface can be defined as \_\_\_\_.

**Ans : Single Abstract Method**

9. Where in an HTML document is the correct place to refer to an external style sheet?

**Ans : Head section**

10. Which of the following represents the correct way to implement a functional interface in Java?

**Ans : interface MyFunctionalInterface { }**

11. In javascript const declares block scoped?

**Ans : true**

12. What is React in MVC?

**Ans : Controller**

13. How can you access the state of a component from inside of a member function in React?

**Ans : this.getState()**

14. Which of the following API is a MUST for every React component?

**Ans : renderComponent**

15. MongoDB database can be accessed using Spring Data.

**Ans : True**

16. Which of the following stands true for Spring beans?

- i. Spring beans are managed by the Spring IoC container.
- ii. Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.
- iii. Spring beans are simple POJOs.

**Ans : All of the Above**

17. What is true about `<list>` collection configuration elements?

**Ans : wiring list of values and allowing duplicates.**

18. What is default scope of bean in Spring framework?

**Ans : Singleton**

19. REST service endpoint comprises of which address?

**Ans : Starts with http:// and may end with ?**

20. What tag is used to get a bulleted list?

**Ans: <ul>**

21. What tag is used to get an ordered list?

**Ans: <ol>**

22. React is service side rendering. True or False.

**Ans: not sure**

23. What are the different types of autowire?

**Ans: bytype, byname, constructor, autodetect**

24. After you initialize a new Git repository and create a file named git-quiz.html, which of the following commands will NOT work if issued?

- A. git add git-quiz.html
- B. git status
- C. git add .
- D. git commit -m "git quiz web file added"

**ANS : D**

25. Which event occurs when the user clicks on the HTML element?

- A. onmouseover
- B. onchange
- C. onmouseclick
- D. onclick

**ANS : D**



26. Which of the below API Calls provides you the security information?

1. HttpServletRequest.isUserInRole()
2. HttpServletRequest.getUserPrincipal()
  - A. Only 1
  - B. Only 2
  - C. Both i and ii

**ANS:- C**

27. What is used to dispatch requests to the WEB Application?

- i. DispatcherServlet.
- ii. ViewHandler.
- iii. ViewResolver.

**ANS:A**

28. By default, a bean is lazily loaded in Spring.

29. True

30. False

**ANS : B**

31. Which one is NOT a spring data MongoDB annotation?

- A. @Document
- B. @Field
- C. @Entity
- D. @Query

**ANS : C @Entity**

32. Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

- A. True
- B. False

**ANS :- A -> True**

33. In Angular, one can create local HTML reference or HTML tag using variable which starts with character \_\_\_\_

- A. @
- B. #
- C. “
- D. &

**ANS :- B**

34. How to add search bar using bootstrap?

**Ans Option A - <class= “gly.. Gly..”>**

35. React is two-way binding? True/False

**Ans : (Longest option)**

36. Q )Line can be break using

**Ans : <br>**

37. Q.)Overriding can be done in?

Poly

Abstr

Inher

Encaps

**Ans. polymorphism**

38. What theorem or principal MongoDB follows?

**Ans : capped-->opn D**

39. What method is used to remove a single item from a MongoDB collection?

**Ans : deleteOne()**

40. Which of these Angular services can be injected in your component to the dynamic mode navigation?

A. Routing

B. RouterService

C. RoutingService

D. Router

**ANS :- D**

41. Q) which method is used to add/ remove items?

**Ans. splice()**

42. The \_\_ Javascript method is like the mathematical “for all” quantifier

A. every()

B. some()

C. all()

D. forall()

**Ans. every()**

43. What is the command line tool introduced for the JS engine in Java 8?

A. jjs

B. jss

C. jfs

D. jbd

**ANS : A**

44. In MongoDB, which of the following methods return one document?

A. findOne()

- B. findOne1()
- C. selectOne()

**Ans. opn A**

45. What is the output of the Java Code Snippet?

- (A-B-C)
- (A)-(B)-(C)
- (-A-)(-B-)(-C-)

**ANS : A**

46. Which of the following media queries would apply to a tablet reporting a screen width of 768px?

- A.@media only screen and (min-width: 1140px){}
- B.@media only screen and (min-width: 641px){}
- C.@media only screen and (max-width: 1140px){}

**ANS :- B**

47. Which of the following rules allows users to import style rules from other style sheets?

- A.@media
- B.@important
- C.@import
- D.@style

**ANS : C**

48. Which class indicates a dropdown menu?

- a) .dropdown-list
- b) .select
- c) .dropdown

**Ans: C**

49. What is the relationship between JavaScript and ECMAScript?

- A. ECMAScript is a standard that JavaScript implements.
- B. They are different terms for the same thing.
- C. JavaScript is a standard that ECMAScript implements.
- D. JavaScript is an old term for what is now called ECMAScript.

**ANS: A**

50. What syntax allows for a variable number of arguments to a function where the arguments are passed as an array?

51. function([]numbers) { }

52. function(\*numbers) { }

53. function(...numbers) { }

54. `function(numbers[]) { }`

**ANS : C**

55. The `__` attribute of a React form is called when the main action button inside the form is pressed.

- A. `onSubmit`
- B. `onChange`
- C. `onComplete`

**ANS : A**

56. Behavior is shared between components using

- A. `propTypes`
- B. `state`
- C. `mixins`
- D. `props`

**ANS : C**

57. The function that creates React components is

- A. `React.component.extend`
- B. `React.createComponent`
- C. `React.renderComponent`
- D. `React.createClass`

**ANS : D**

58. correct annotation

`transaction`, `transactions`, `transactional`

**Ans : transaction**

59. Annotation used to indicate a field has to have a minimum of 2 characters.

- a) `@NotNull`
- b) `@Size`
- c) `@MaxSize`
- d) `@size`

**Answer: b**

60. Views that cannot be resolved by `InternalResourceViewResolver`.

- a) `redirect`
- b) `redirect:prefix`
- c) `redirect:suffix`
- d) all of the mentioned

**ans: a**

61. Alternative way to resolve locales.

- a) `AcceptHeaderLocale`

- b) AcceptHeader
- c) AcceptHeaderLocaleResolver
- d) SessionLocaleResolver

**ans: d**

62. General-purpose class that allows a response to be rendered using a marshaller.

- a) MarshallingView
- b) Marshaling
- c) View
- d) All of the mentioned

**Answer: a**

63. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object?

- a. ResultSetExtractor
- b. RowMapper
- c. RowCallbackHandler
- d. ResultSetMapper

**ans: B**

64. Which of the following classes can be used to call Stored Procedures in spring?

- A - SPHelper
- B - JdbcTemplateCall
- C - JdbcTemplate
- D - SimpleJdbcCall

**Answer : D**

65. How to handle shut down of IoC containers?

- A - Using shutdownHook()
- B - Using shutdownHandler()
- C - Using registerHook()
- D - Using registerShutdownHook()

**Answer:D**

1) What is the command line tool introduced for the JS engine in Java 8 ?

A. jjs

B. jss

C. jfs

D. jbd

ANS : A jjs

2) API and MUST component answer component

3) Where is the correct place to insert a Javascript ?

A. The <head> section

B. Both the <head> section and the <body> section are correct.

C. The <body> section.

D. The <footer> section.

4). What does the “webpack” command do ?

A. Runs React Local Development Server.

B. Transpiles all the Javascript down into one file.

C. Both A and B.

ANS : C

5). Which of the following API is a MUST for every React Component ?

A. getInitialState.

B. render

C. renderComponent.

ANS : C.

6). btn-large

7) sequential and parallel-streams

8) which among is right for monolithic

form related question

option A-method

option B-action

Option C-react

9) which is sorted and synchronized

option A-treeset(not sure)

option B-hashmap

linkedlisthashmap

concurrentskip

10) host/target new tab

11) js/component

12)-----?

ans <h1>.....</h1>answer

13) What does it mean by publishing a REST service in Spring ?

- i. Publishing an application's data as Json over a REST service.
  - ii. Publishing an application's data as XML over a REST service.
  - iii. Sharing Database within two microservices using Spring Data Access Object (DAO).
- A. Only 1  
B. Only 1 and 2  
C. Only 2  
D. All of the above.

18) Which of the below API Calls provides you the security information ?

- 1) HttpServletRequest.isUserInRole()
  - 2) HttpServletRequest.getUserPrincipal()
- A. Only 1  
B. Only 2  
C. Both i and ii  
ANS:- C

19) Which Annotation is used to refer to Pointcuts.

- A. @PointCut.  
B. @PointCutExecution.  
C. @PointCutBefore.  
ANS :- A

20) In your Data Access Object (DAO) method , you neither need to surround your code with a try/catch block nor declare throwing an exception in the method signature.

- i. TRUE
  - ii. FALSE.
- ANS : B

21) A logger location where logging information is sent is called

- 1. appender.
  - 2. logger
  - 3. component.
- ANS : appender A

22)-log4J---answers is logger

23)redux is single-binding TRUE/FALSE

24) Identify the correct ES6 statement.

- i. `var React = require('react');`
- ii. `import React from react;`
- iii. `module.exports = Component.`
- iv. `export default Component.`

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only ii and iv

ANS : D

25) You can also use an ES6 class to define a component in React.

- A. True.
- B. False.

ANS: A

26) Identify the correct statement in MongoDB.

- A. Queries specify criteria, or conditions , which identify the documents that MongoDB returns to the clients.
- B. Write Operations , or queries , retrieve data stored in the database.
- C. The selection limits the amount of data that MongoDB returns to the client over the network.

ANS : A -> mostly

27)STREAM API USED IN JAVA9 TRUE/FALSE

28) Which Java Class has natural sorting implemented ?

- A. `LinkedHashSet.`
- B. `TreeSet`
- C. `HashSet`

ANS :- B -> `TreeSet`

29)`var o=newfun`  
`console.log,o===fun`

true  
false

0  
1

30)What is the command line tool introduced for JS engine in Java8?

- a ☐ `jjs`
- b ☐ `jss`
- c ☐ `jfs`
- d ☐ `jbd`

☐ In Java9, private methods are allowed in Interfaces.

- a ☐ True
- b ☐ False



31)typeof null return ans:Object

32)Which Java Set implementation is sorted and synchronized?

- a ☐ TreeSet
- b ☐ LinkedHashSet
- c ☐ HashSet
- d ☐ ConcurrentSkipListSet

33)Name the two types of streams in Java.

- a ☐ Sync and async
- b ☐ Sequential and parallel
- c ☐ Random and ordered

34)RxJs

option-map  
etc

35)host/target listener

option-@HostListener  
option-@TargetListener

36)what does <aside> tag in html do?

Ans : aside the content of the page

37)Which of the following is the correct syntax of creating a standard navigation tab?

Ans : <ul class="nav nav-tabs">

38)Java Functional Interface can be defined as \_\_\_\_\_.A)0,B)1,

39)Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

A. True

B. False

ANS :- A -> True

## CORE JAVA

---

1)What is the max capacity of a Java Blocking Queue?

- a ☐ Integer.INFINITY
- b ☐ Integer.MAX\_VALUE
- c ☐ BigDecimal.MAX\_VALUE

2)Which is the correct Java code snippet to be used to create a Functional Interface?

- a ☐ interface MyFunctionalInterface implements Functional{
- b ☐ " ☐FunctionalInterface interface MyFunctionalInterface{"
- c ☐ " ☐Functional interface MyFunctionalInterface{

3)Which is the correct API to manage reactive streams in Java?

- a ☐ java.util.concurrent.Flow
- b ☐ java.util.concurrent.Flow.Publisher
- c ☐ java.util.concurrent.Flow.Processor
- d ☐ All of the above

4)In Java, Which of these class types supports sequential and parallel processing of data?

- a ☐ Queue
- b ☐ Lambda
- c ☐ ExecutorService
- d ☐ Streams

5)Which Java Set implementation is sorted and synchronized?

- a ☐ TreeSet
- b ☐ LinkedHashSet
- c ☐ HashSet
- d ☐ ConcurrentSkipListSet

6)What is the output of the following Java code snippet?

```
StringJoiner astring = new StringJoiner("-", "{", "}");  
astring.add("A");  
astring.add("B");  
astring.add("C");
```

- a ☐ {A-B-C}
- b ☐ {A}-{B}-{C}
- c ☐ {-A-}{-B-}{-C-}

7)Name the two types of streams in Java.

- a ☐ Sync and async
- b ☐ Sequential and parallel
- c ☐ Random and ordered

## ANGULAR

-----

1)Two Way binding---[(ngmodel)]

2)What's the best way to inject one service into another in Angular ?

- A . A service cannot be injected in another service.
- B. Through constructor of depending service.
- C. By instantiating using a new operator.

ANS : B

2)Which of these Angular services can be injected in your component to the dynamic mode navigation ?

- A. Routing.
- B. RouterService.
- C. RoutingService
- D. Router

ANS :- D

3)Which method of Angular RouterModule should be called for providing all routes in AppModule?

- A. RouteModule.forChild
- B. RouteModule.forRoot
- C. RouterModule
- D. RouterModule.all

ANS : B

4)How can you detect the client's browser name in JavaScript?

- client.navName
- navigator.appName
- browser.name
- window.browser

ANS : B

5)To use the Angular HttpClient component, you must import the \_\_\_\_ module.

- A. HttpClientModule.
- B. HttpModule.
- C. Http

Ans :- A

## React

-----

1)does react support two way binding?(No)

2)How can you access the state of a component from inside of a member function in React?  
this.getState()

this.prototype.stateValue  
this.state  
this.values  
ANS : c

3)What are the advantages of React?

React can be used on client as well as server side.

Using React increases readability and makes maintainability easier. Component and Data patterns improve readability and thus make it easier for maintaining larger apps.

React can be used with any other framework (Backbone.js, Angular.js), as it is just a view layer.  
All of the above

ANS : D

4)How does React handle the Web Accessibility Initiative - Accessible Rich Internet Applications (WAI-ARIA) standard?

aria-\* HTML attributes are fully supported in JSX where most DOM properties and attributes in React are camelCased, these attributes should be lowercase.

React processes aria-\* attributes separately and updates the DOM for accessibility if the user requires it.

aria-\* attributes should be converted to camelCase like other attributes. For example: className, onChange, and so on

React is yet to support the WAI-ARIA standard.

ANS :- D

5)What is React?

Just a server-side framework

Just a user-interface framework

Both a server-side framework as well as a user-interface framework

ANS : c

JS

-----

1)what is babel?

a)compiler

b)interpreter

c)both

2)which keyword is used to check whether it exists or not?

3)which is not js keyword?

a)if

b)with

c)use strict

## HTML,CSS&Bootstrap

1)Which is the correct CSS Syntax ?

- A. body color-black
- B. {body color-black}
- C. body {color:black;}
- D. {body;color:black}

ANS : C

correct syntax for applying background color?

which is used to make text bold?

html syntax for creating link to open new tab?

ans)<a href="https://example.com" target="\_blank"/>

## Spring

1)Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

- A. True
- B. False

ANS :- A-> True

2)By default, a bean is lazily loaded in Spring.

- 1. True
- 2. False

ANS : B

3)Which of the below API Calls provides you the security information ?

- 1) HttpServletRequest.isUserInRole()
- 2) HttpServletRequest.getUserPrincipal()

- A. Only 1
- B. Only 2
- C. Both i and ii

ANS:- C

4)Which event occurs when the user clicks on the HTML element ?

- A. onmouseover
- B. onchange
- C. onmouseclick
- D. onclick

ANS : D

5) Which of the following stands true for Spring beans?

- i. Spring beans are managed by the Spring IoC container.
- ii. Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.
- iii. Spring beans are simple POJOs.

Ans : All of the Above

6) Load Balancing improves the distribution of workloads across multiple computing resources, such as computers or a computer cluster.

A. True.

B. False.

ANS:- A

7) Which of the following statements is true about microservices architecture?

- i. Microservices applications are required to be deployed on the same physical host.
- ii. Microservices architecture supports high availability of individual microservices.
- iii. Microservices cannot be used if a company is embracing DevOps.
- iv. Microservices are designed using a bounded context that can communicate with other bounded contexts.

A. Only i

B. Only

ii C. Only ii and iv

D. All of the above.

Ans -> D

8) You can only use Hibernate in ORM in Spring Boot Applications.

i. TRUE

ii. FALSE

ANS : B

question is like:

9) If the component data is frequently changing what will be pipe

a) pure

b) impure

In Angular, pipes can be Pure or Impure depending on how often the input data changes.

If the component data is frequently changing

→ You should use an Impure Pipe.

10) Publishing an application's data as a REST service requires.

a) @RequestMapping

b) @PathVariable

c) All of the mentioned

d) None of the mentioned

ans: a b

11)Service discovery tools help by reducing the problem of how processes and services in a cluster can find and talk to one another.

A. True

B. False

ANS :- A -> True

MongoDb

-----

1)find is used for filtering

2)which annotation is used to connect a mongoDB

3)whic stores data like name and value pairs ---ans:Map

4)does junit need junit class for running?

# Study Notes (Converted from Notebook Images)

- 1) Is React single-dimensional
- 2) What is React in MVC – Controller
- 3) ES6 – option
- 4) FST
- 5) Delegating filters
- 6) Is var block scope?
- 7) Which acquires properties & behaviours – Inheritance
- 8) ng syntax
- 9) To know existence collection – show collections
- 10) @PathVariable
- 11) @Pointcut
- 12) application.properties
- 13) Which creates on request – Prototype, Session, Request, Singleton
- 14) Does Java support interface?
- 15) Options – { } [ ] ; REST ends with \*, ;, |
- 16) In Java which data structure cannot be type parameterized? List, Set, Array, Map – Answer: Array
- 17) MongoDB operation using index often – SELECT, UPDATE, DELETE, SORT – Answer: SORT
- 18) FIFO
- 19) findOne()
- 20) In Flux, store and create are separated
- 30) Render JSX in browser – Using Babel, Using TypeScript
- 31) Command line tool introduced – JS
- 32) Immutable Java list static factory method – List.of()
- 33) Natural sorting implemented – TreeSet
- 34) Load balancing
- 35) Git commit -m
- 36) body { color: black }
- 37) navigator.appName
- 38) In JS comma separated – Object literal
- 39) ProcessBuilder
- 40) Collectors
- 41) { A - B - C }
- 42) Java 8 get current milliseconds – Clock
- 43) Java Functional Interface – Single Abstract Method



- 44) External stylesheet – head section
- 45) Remove hyperlink underline – `a { text-decoration: none; }`
- 46) Correct place to insert JS – head or body
- 47) In JS var declares block scope – False
- 48) MongoDB related classes – MongoTemplate, MongoRepository, MongoClient
- 49) MongoDB is classified as NoSQL database
- 50) MongoDB format – BSON
- 51) Incorrect statement – Secondary indices not available in MongoDB
- 52) Default bean scope in Spring – Singleton
- 53) CI/CD essential for microservices – True/False
- 54) Git WORKDIR options (notes)
- 55) Run MongoDB in JS – Mongo
- 56) Default React development port – 3000
- 57) Babel – Compiler & Transpiler
- 58) Class – `class::constructor()`, `class::function`
- 59) MongoDB is a document oriented database

\java spring:

---

1) @controller(in ques) ans:Spring MVC

2) class(ques) : ans: Annotationconfig applicationcontext

3) interface rather than implementations: map m = new HashMap();

4) which module provide communication between ORM and core module(ques) :  
options: a) ORM b) Transaction c) Oxm d)JMS

Expected ans for 4rth ques are b or d

5) autowiring by default(ques) ans:"off"

6) how to inherit beans (ques) ans: parent = ""

7) program starting line abstract ="true" ans: country = japan , type= 1 ;

8) three basic to create a simple spring application (ques): 1st option (context, retrieve , work)

9) default(ques) ans:singleton

10) web- aware(ques) ans: session , request session , global session

11) modes of autowiring(ques) ans: bytype , by name , constructor , @ autowiring , @ Qualifier

12) Autowiring is an alternative of(ques) ans: 3rd option with AOP

13) standard collection : serializable

14) dependency look up(ques) ans: all the above(factory method , keyword method , JNDI method)

15) @transactional

16) do we use transaction management in spring (ques) ans: yes, with template class

17) how do define destroy method (ques): destroy- method ="destroy"

18) (ans)<context: component-scan>

19) (ans)<bean id="hell world" class="com.Helloworld" destroy-method="destroy"/>

20) (ans) org.springframework.beans.factory.InitializingBean(option a)

21) how to define datasource (ques) ans: DriverManager DataSource(option a)

22) Spring is a pojo class. which of the following makes it manageable(ques): option a) some extended b) public static

myclass extends Spring Bean implements Manageable.....c) something extends d) public class MyClass(expected ans b or d)

23) by default inner bean is(ques) ans: prototype

24)ans: context.getBean(in question you will have id name this should match with that if it is case sensitive means ans

also case sensitive)

25) in ques they will give programme and asks for finding which of the following represents inner bean(ques) ans: in the

proramme bean id comes after the constructor-arg or property

26)classpath means Relative and file path means absolute

26) (ans)set auto commit(false)

27) by using auto wiring and auto discovery which you can reduce(ques) ans: java code(1st option)

28) how to acess instruction in java configuration (ques)ans : property file and XML file

29) meta data (ques): ans : XML based , Annotation based , java based configuration

30) my bean = "null"

31) some program with constructor(ques) ans: constructor type

32) some exceptions in autowiring modes (ques): b) null pointer exception c) illegal state exception d) none (please check

expected option c)

34) JSR-250 annotation(ques) ans: @postconstruct @ predestroy @ resource

35) advice(def and syntaxes read)

36) aspect (def and syntaxes read)

37) point cut (ques) ans: one or more join points

38) java.util.collection.properties (ques) ans: props

39) <list> allows duplicates

40) <set> doesn't allow duplicates

41) result set(ques) ans: row mapper

42) <map> name value pair of any type

43) <props> name value pair of string type

44) @ transactional

45) same instance one in application (ques) ans: singleton

46) each instance object creatio (ques) ans:prototype

47) spring bean lifecycle(ques) ans: before before after propertieset()

48) JDBC template provides: options a) execute SQL queries b) execute update statements c) execute java programe

d) execute stored callprocedures ans :(abd)

49) add method (amount, -SRC) add method (amount src) in (ques) ans: transfer method();

50) new xml classpath application context(c:/project/config/application.xml) and new Filepathapplicationcontext(config/applict.xml)

these ans will be in "a " and "c" options ans:(a and c)

//////////SSSSPRING DUMPS//////////

## 1. VALID SPRING BEAN DEFINING

ID,CLASS

option a,d

## 2.STANDARD JAVA INTERFACE

A. SERIALIZE

B.LONG

C.SYSTEM

D.COLLECTION

SERIALIZE

## 3. BENIFITS OF DI or ADVANTAGES OF IOC

LOOSE COUPLING,EASY TESTING

4.WHICH OF THE BEAN SCOPE ATTRIBUTE THAT FORCES SPRING TO PRODUCE A NEW BEAN INSTANCE EACH TIME?

PROTOTYPE

5.TO ENABLE PATTERN AUTOWIRE?

AUTOWIRE\_PATTERN="TRUE"

6. SET AUTOWIRE IN A BEAN WHAT ATTRIBUTE DO YOU ADD?

AUTOWIRE="ON"

Default autowire in a bean what attribute do you add?

autowire="off"

7.NOT ADDRESSED BY SPRING JDBC?

if in question DataSourceAccess, then answer DataSourceAccess

otherwise

DAO

8.TO USE JAVA.UTIL.LIST, WHAT IS THE SPRING COLLECTION ELEMENT THAT U NEED?

<list>

9.TO USE BEAN DEFINATION TEMPLATE, YOU SHOULD ADD WHAT ATTRIBUTE IN THE BEAN?

- A. ABSTRACT="TRUE"
- B. INTERFACE="TRUE"
- C. PUBLIC ="TRUE"

A

10.USER MANAGED TRANSACTION HAS TO BE COMMITTED EXPLICITLY?

SET AUTOCOMMIT(FALSE)

11. PROGRAMatic TRANSACTION

C

12. Spring INHERITANCE

to share common values, properties or configurations.

13. SPRING RESOLVE

AT RUNTIME

14. JDBC DAO

@REPOSITORY

15. JAVA BASED WHAT WE USE



@CONFIGURATION

## 16. HOLLYWOOD

DONT CALL US WE WILL CALL YOU

## 17. modes of AUTOWIRING

FIVE

types of autowire

4

## 18. MINIMIZE XML config file

AUTOWIRING

## 19. EACH REQUIREMENT

PROTOTYPE

## 20. CONTRUCTOR IN PROGRAM

CONTRUCTOR TYPE

## 21. SCOPE OF BEAN

SESSION, REQUEST, PROTOTYPE, SINGLETON, GLOBAL SESSION

22. HOW DO YOU DECLARE MVC ANNOTATION IN BEAN IN XML FILE?

<MVC: ANNOTATION-CONFIG/>

23. ALTERNATIVE OF @AUTOWIRED

USING AOP

24. USE OF DEPENDENCY INJECTION

A & C (LOOSE COUPLING AND EASY TESTING)

25. NON INVASIVE

AUTOWIRED

26. Which of the following beans isn't significant for the WebApplicationContext?

1. Locale resolver.
2. Handler exception resolver.
3. Multipart file.
4. None of these.
5. Resolver.

4

27. Which of the following statements is true for @RequestMapping annotation?

1. Can be applied to the controller classes.
2. Can be applied to the controller methods.
3. None of these.
4. Used to map web requests onto specific handler methods.
5. Used to map web requests onto specific handler classes.

3

if multiple ans then 4,5

28. which annotation is used for setter method?

@autowired

29. SOME PROGRAM

@AUTOWIRED SOMETHING OBJECT

30. AUTOWIRING IS AN ALTERNATIVE TO

something with aop

31.ANNOTATION FOR CONFIGURATION

@CONFIGURATION

32. BASIC PACKAGES

org.springframework.context.ApplicationContext;

```
org.springframework.context.support.ClassPathXMLApplicationContext;
```

or

```
//      ORG.SPRINGFRAMEWORK CONTEXT AND  
ORG.SPRINGFRAMEWORK.APPLICATION
```

33. CODE IN QUESTION regarding HelloWorld

OPTION WITH PROTOTYPE

(MAYB OPTION B)

34. PROGRAMTIC TRANSACTION MANAGEment (with the help of normal programming)

Declarative TRANSACTION MANAGEMENT(AOP)

35. JAVA-BASED CONFIGURATION

a) <Context: annotation-config>

b) @Configuration public class HelloWorldConfig{...}

c) @Autowired

d) ApplicationContext context= new  
ClassPathXMLApplicationContext("Bean.XML")

Ans: b)@Configuration public class HelloWorldConfig{...}

36. TRADITIONAL TRANSACTION MANAGEMENT

USING TEMPLATE CLASS

37. GENERATOR GENERATOR....PROGRAM

LOOSE COUPLING

TIGHT COUPLING

DECOUPLING

if new operator used--> tight coupling

if this. used--> loose coupling

38. DAO ACCESS

@REPOSITORY

39. MYBEAN= NULL

A. TRUE

B. FALSE

C. COMPILATION ERROR

D. NULL

D.NULL

40. program with spell checker

SETTER DI

Constructor DI

Setter DI

\*41.what does @ Transactional annotation mean

Transaction = true (option b)

42. current bean for request scope

43. I am object A

null

```
<bean id="helloWorld" class="com.tutorialspoint.HelloWorldId"
scope="prototype"></bean>
```

44. class String MyBeans

```
{
    if(MyBeans.equals(MyBean)
        sop(true);
    else
        sop(false)
}
```

A. TRUE

B. FALSE

C. NULL POINTER

D. CLASS CAST

c

45. types of autowire-4

diff modes of autowiring=5

46. set map id class

id class

47. difference between autowire and autowiring

48. @ autodiscovery

need to eliminate <property> or <constructor tag> (option a)

49. autowire->

50. autowire+auto discovery

-> reduce java code

51. Spellchecker

-> setterDI

52. INNER BEAN?

-> DEFINED INSIDE CONSTRUCTOR ARG OR PROPERTY TAG

53. public class TextEditor {

private SpellChecker spellChecker;

public TextEditor(SpellChecker spellChecker) {

this.spellChecker = spellChecker;

```
}  
}
```

Which DI?

a.CONTRUCTOR DI

b.Getter DI

c. Setter DI

d. NOne

c

54. WHAT BASIC PACKAGES ARE IMPORTED IN JAVA PROG IN SPRING

org.springframework.context.ApplicationContext;

org.springframework.context.support.ClassPathXMLApplicationContext;

55. Mandatory attribute for bean

id & class

56.what mandatory attribute does aop include?

concern

float

advice

pointcut

concern mayb



57.....object myObject;

if(myObject==null)

sop(true)

else

(false)

true

false

null

compilation error

null(mayb)

58. order of inner bean

ans --><constructor-arg> or <property>

59. programmatic transaction management

a. pojo based model

d. aop

60. auto-discovery

1st option eliminate<property>

61. mvc question

<mvc:annotation-driven>

62.abstract layer, init layer, dependency check

long answer

63. default bean and its functionality

singleton and single ioc container

64. @autowired is used in

constructor destructor option

65.What is the purpose of Spring Inheritance?

Ans: to share common values, properties or configurations.

66.How to handle Named Query Parameters?

Ans: NamedParameterJdbcTemplate

67. In the bean configuration file for Hibernate (i.e., beans-hibernate.xml), you have to declare a `HibernateTransactionManager` instance for this application and enable declarative transaction via:-

a) tx:annotation

b) tx:annotationdriven

c) tx:annotation-driven

d) none of the mentioned

Ans: c) tx:annotation-driven

68. Which of the below option is not a valid collection type?

a) list

b)Set

c)props

d)property

Ans: property

-----

1. How to import all package?

Answer : import java.\*

2. Which of the following is correct about junit?

Answer: - It is an open source framework.

- It provides Annotation to identify  
the test methods.

- It provides Assertions for testing  
Expected results

3. Pattern which follows sequential order?

Answer : Iterator

4. class Animal

```

{
    public String noise()
    {
        return "peep";
    }
}

class Dog extends Animal
{
    public String noice()
    {
        return "bark";
    }
}

class cat extends Animal
{
    public String noise()
    {
        return "meow";
    }
}

}.....

Animal animal = new Dog();

Cat cat= (Cat)animal;

SOP(cat.noise());

```

Answer : Exception is thrown at the Runtime

```

5. class output {

public static void main(String args[]){

```

```
String buffer s1 = new StringBuffer("Hello world")
s1.insert(6,"Good");
SOP(s1);
}
}
```

Answer: Hello Goodworld

6. SOP ('1'+new integer (2) +3);

Answer: 123

7. A team of programmer is involved in reviewing a proposed design for a new utility class after some discussion.....

Answer: weak encapsulation

8. Man dog best friend

Answer : class man private dog best friend;

9. Bluemix deploy cloud foundry

Answer: cf push

10. Class vector {

PSVM(String args[])

Vector obj = new vector(4,2);

Obj.add element(new Interger(3));

Obj.add element(new Interger(2));

Obj.add element(new Interger(5));

Obj.removeAll(obj);

SOP(obj.isEmpty());

}

Answer : True

11. `StringBuffer s1 = new StringBuffer("Hello");`

`StringBuffer s2 = reverse(s1);`

`SOP(s2);`

Answer : olleH

12. How to access the resultsetobject?

Answer : by using special getter method.....

13. `X= something`

`While(X<10&&X>24)`

Answer : condition false

14. `RunWith(Suit.class)`

`@SuiteClasses({`

`annotationTest.class,`

`AssertTest.class`

`CalculateTest.class,`

`EmployeeTest.class})`

`Public class SuiteTest{`

`//the class remains empty`

`//used only as a holder for the above annotation`

`}`

What is the sequence of junit test ?

Answer : AnnotationTest, AsserTest, CalculateTest,EmployeeTest

15. `Public static void main(String args[])`

`{`

`List<string> List = new ArrayList<string>();`

`//add string`

```
List.add("cricket");  
List.add("football");  
List.add("hockey");  
Iterator it = List.iterator();  
While(it.hasNext())  
{  
String s = it.next();  
}  
}
```

Which pattern

Answer : Iterator

```
16. Class x implements Runnable {  
Public static void main(String args[])  
{  
/*missing code? */  
}  
Public void run() {  
}}  
}}
```

Which of the following line of code to start thread?

Answer : x run = new x();

```
Thread t = new Thread(run);    t.start();
```

17. PSVM()

```
{  
try{  
int a = 5;  
int b = 0;  
int c = a/b;
```

```
SOP("World");  
}  
Catch(exception e)  
{  
SOP("hello");  
}}
```

Answer : hello

18. Int a = 10;

Int b = 0;

Int c = a/b;

SOP("c");

Output :

Answer: runtime exception not arithmetic exception

19. Which will contain the body of the thread?

Answer : run()

20. Which two of the following methods are defined in class thread?

1. Start() 2. Wait() 3. Notify() 4. Run()

Answer : start() and run()

21. To send binary o/p?

Answer : getOutputStream

22. Unit testing?

Answer : Single Entity

23. Assume the following method is properly synchronized and called from a thread A on an object B:

```
wait(2000);
```

After calling this method, when will the thread A become a candidate to get another turn at the CPU?



- A. After thread A is notified, or after two seconds.
- B. After the lock on B is released, or after two seconds.
- C. Two seconds after thread A is notified.
- D. Two seconds after lock B is released.

24. Invoked only once?

Answer : @BeforeClass

25. Invoked before each test?

Answer : @Before

26. Invoked after each test?

Answer : @After

27. Only once invoked finishing all tests?

Answer : @AfterClass

28. What will be proper order of session?

Answer : loading, instantiation, init, service, destroy

29. Proper order of access modifier

Answer : private default protected public

30. If the options will be like

New class foo{print(.....)},

Answer: new foo(){print(.....)}

31. Insert(6, "Good")

Answer : Hello Good Word

32. Class{

PSVM

String str = new String( ".....");

}

Do{

```
str = "Hello Stop World ";
```

```
SOP(str);
```

```
}
```

```
While(str!=Strong);
```

```
{
```

```
.....
```

```
}
```

Answer : Hello Stop World

33. The code snippet below is an example of which of the following?

```
Long myLong = 21l;
```

A Autoboxing

B Autounboxing

C Autocasting

D Autoinstantancing

Answer: Autoboxing

34. clear(2) in the string of index 2

Answer : 0134

35.

```
interface TestA { String toString(); }
```

```
public class Test {
```

```
public static void main(String[] args) {
```

```
System.out.println(new TestA() {
```

```
public String toString() { return "test"; }
```

```
});
```

```
}
```

```
}
```

What is the result?

- A. test
- B. null
- C. An exception is thrown at runtime.
- D. Compilation fails because of an error in line 1.
- E. Compilation fails because of an error in line 4.
- F. Compilation fails because of an error in line 5.

Answer: A

36.  $2+3*5$

Answer 17 integer

17 byte

25 byte

25 integer

37. Identify the correct signature of the main method?

Answer : public static void main(String args[])

38. public static void main(String[] args)

```
{
GenericTest<String> arr[] = new GenericTest[5];           //line 1
arr[0] = new GenericTest("Java");                       //line 2
arr[1] = new GenericTest(1);                             //line 3
arr[2] = (GenericTest<String>)new GenericTest(1);        //line 4
arr[3] = (GenericTest<String>)new GenericTest<Integer>(1); //line 5
for(GenericTest o:arr)
    {
System.out.println(o);
    }
}
```

Output: Error in Line 1

39. Answer: AaBaaaCaaaa

40. Multiple exception catch block..

Answer : superclass exception can not be caught first

Can we caught either super class or subclass

41. l.add(2)

l.add(5)

l.add(8)

l.add(1)

l.reverse();

Answer: 1852

42. getValue(IDEAL)

SOP(word.substring(0,word.length()-1)+word.charAt(word.length()-1))

Answer: DEAL

43. in javabean reference id define?

- a. Classname
- b. Package
- c. Instance
- d. None

44. Int x=6;

Int y = 12;

If(x>y)

{

X++;

y--;

```
}
```

```
SOP(....);
```

Answer: 2

45. Datatype of the no 9.6352

Answer: float

double

Float

Double

46. How many thread one from exception mainclass

Answer : is only 1 thread(main thread)

47. Start same thread two times

```
t1.start()
```

```
t1.start()
```

Answer: compilation fails, compile time error

48. Decouples in which design pattern?

Answer: Bridge

49. String s1 = abc;

String s2 = def;

String s3 = s1.concat(s2.UpperCase)

Answer:: abcdefabcDEF

50. what is the output of this program

```
class output{
```

```
public static void main(String args[])
```

```
{
```

```
Object obj = new object();
```

```
System.out.print(obj.getClass());  
}}
```

Answer: class java.lang.Object

51. declare array of string

```
String[] s;  
String s[]  
String []s;b
```

Answer: ABD

52. class A{

```
int i;
```

```
int j;
```

```
A(){
```

```
    I =1;
```

```
    J=2;
```

```
}
```

```
}
```

```
Class output{
```

```
    Public static void main(String args[])
```

```
{
```

```
    A obj1 = new A();
```

```
    SOP(obj1.toString());
```

```
}
```

```
}
```

Answer: String associated with obj1

And output will be A@1cde5f

53. public void divide(int a, int )

```
{
```

```
Try{
```

```

Int c = a/b;
}
Catch(Exception e)
{
SOP(Exception);
}
Finally{
SOP("finally")
}
}

```

Output :

Answer if a,b not initialized then error and if b = 0; then compile time error with finally will work.

54. Current thread

```

Public class MyRunnable implements runnable {
Public void run()
{
//some code
}
}

```

Which of these will create and start this thread?

Answer:

```

new Thread(new MyRunnable()).start();

```

55. To execute in the parameterized statement?

Answer: prepared statement

56. Given

```

10 interface Foo{int bar();}

```

```

11 public class sprite{
12 public int fubar(Foo foo){return foo.bar();}
13 public void testFoo(){
14 fubar(
15 //insert code here
16 );
17 }}

```

Answer: new Foo(){public int bar(){return 1;}}

57. A: unit testing is the testing of single entity

B: unit testing is the process of checking the functionality of the application whether it is working as per requirements

Options :

- a. A is true
- b. B is true
- c. Both are true
- d. None of the above

What will be the output of this -

```

58. String river = new String("TRAINING");
    System.out.println(river.length());

```

answer-8

```

59. String getValue(String word)
    {
    if (word.length() == 1)
    return "";
    else
    return getValue( word.substring(0, word.length() - 1) ) + word.charAt(word.length() - 1);
    }

```



What is the value of the string returned by `getValue("DEMOS")`

Answer: EMOS

```
60. public class Question {  
    public static void main(String args[]) {  
        String s1 = "uvw";  
        String s2 = "xyz";  
        String s3 = s1.concat(s2.toUpperCase( ));  
        System.out.println(s1+s2+s3);  
    }  
}
```

answer: uvwxyzuvwXYZ

```
61. int a = 9;  
    int b = 14;  
    while(a<b) {  
        System.out.println("In the loop");  
        a+=2;  
        b-=2;  
    }
```

answer- In the loop

In the loop

```
62. abstract class xyz  
    {  
        abstract abc (int a, int b) { }  
    }
```

- What sort of compilation error you will get ?

- How to fix the compilation error

Answer-- give return type void. and remove empty parentheses.

```
63. interface ATOB {  
    String s = "yo";  
    public void meth();  
}  
interface BTOC { }  
interface C extends ATOB, BTOC {  
    public void meth();  
    public void meth(int x);  
}
```

- Will this compile ?

answer--- interface can also be extended (when inheriting from two interfaces.)

```
64. int i = -1;  
    int b = 10;  
    int val = b / i;
```

answer- -10

```
interface Data { public void load(); }  
abstract class Info { public abstract void load(); }
```

65. How Can i inherit both the interface and abstract class ?

answer= class xyz extends Info implements interface{ void load}

```
66. public class GenericTest<G> {  
    G g;  
    GenericTest(G g){  
        this.g =g;  
    }  
    public static void main(String[] args)
```

```

{
    GenericTest<String> arr[] = new GenericTest[5];           //line 1
    arr[0] = new GenericTest("Java");                       //line 2
    arr[1] = new GenericTest(1);                             //line 3
    arr[2] = (GenericTest<String>)new GenericTest(1);        //line 4
    arr[3] = (GenericTest<String>)new GenericTest<Integer>(1); //line 5
    for(GenericTest o:arr)
    {
        System.out.println(o);
    }
}

```

Answer-- can not cast <Integer> to <String> line5

and default value is null in an array

```

67. class A {
    static String str = "STRING_IN_A";
}
class B extends A {
    static String str = "STRING_IN_B";
}

```

//

in void main

B b1=new B();

sop b1.str

//

Answer--- will compile fine.bt str is being overwritten in B class.

```

68. int j=8;

while( (j < 10) && (j > 24)) {

    System.out.println(" Am here");

}

```

answer-while() condition is always false

```

69. class output {

    public static void main(String args[]){

        StringBuffer c = new StringBuffer("Hello");

        StringBuffer c1 = new StringBuffer("World");

        c.append(c1);

        System.out.print(c);

    }

}

```

answer-helloworld

70. are resultset updateable?

Answer: yes but only if you indicate a concurrency strategy when executing the statement and if the driver and database support this option.

71. How can you start a database transaction in the database

Answer: by setting the autocommit property of the connection to false & execute a statement in the database

72. The object that does not contain the stored procedure itself but contains only a call to the store procedure?

Answer: callableStatement

73. To execute a stored procedure “total stack” in a database server which of the following code snippet is used?

Answer: callablestatement clbstmnt = con.preparecall("{call totoal stack}");  
cs.executeQuery();

74. Relative path

Answer: c:/test/\*\*\*

75. Which type of driver converts jdbc calls into the Network protocol used by the datamanagement system?

Answer: Type 4

76. SOP(index(i)+index(0)+lastIndex(i)+lastIndex(0));

Answer: 6469

77. When a select statement returns only one record with a single column of integer datatype on invoking resultset getinteger(0) causes\_\_\_\_\_

Answer:

78. Which listener class will close() method?

Answer: windowlistner

79. Class exception\_Handling{

Public static void main(String args[]){

Try{

SOP("Hello"+" "+1/0);

}

Catch(ArithmeticException e)

{

SOP("World");

}

}}

Answer: World

80. What is the output of this program?

```
Import java.util.*  
  
Public static void main(String args[]){  
  
    TreeMap obj = new Treemap();  
  
    Obj.put("A", newInteger(1));  
  
    Obj.put("b", newInteger(2));  
  
    Obj.put("c", newInteger(3));  
  
    SOP(obj.entrySet());  
  
}
```

Answer: [A=1,B=2,C=3]

```
Public class genericstack<E>{  
  
    Stack<E>{  
  
    Stack<E>stk = new stack<E>();  
  
    }  
  
}
```

81. Which of the following describes the purpose of JUnit?

- A It is a framework to help with writing unit tests for your code.
- B It is a framework to help with writing code involving unit conversions.
- C It is a framework to help with packaging your code into units for distribution.
- D It is a framework to convert all days/dates in your code to June.

Answer: A

82. Which of these keywords must be used to monitor for exceptions?

- a) try
- b) finally
- c) throw
- d) catch

View Answer

Answer: a

83. Treemap question and options will be like

(7,9)

(9,7)

Answer : 7,9

84. Testing Amit Gupta and Ravi Kumar wala question?

a) Amit Gupta False

b) Ravi Kumar False

c) Amit Gupta False(Expected name as Ravi Kumar)

d) Ravi Kumar False(Expected name as Amit Gupta)

use

85. Fill in the blank

```
Exception _____{
```

```
printstack e;
```

```
}
```

```
Exception_____ {
```

```
printStack e;
```

```
}
```

Answer class not found

Sql exception

86. Get value from resultset

Answer: getBoolean(),getClob()

getBlob()

87. class exception\_handling {

```
public static void main(String args[]) {
```

```
try {
```

```

int a, b;

b = 0;

a = 5 / b;

System.out.print("A");

}

catch(ArithmeticException e) {

System.out.print("B");

}

}

}

```

a) A

b) B

c) Compilation Error

d) Runtime Error

Ans:B

88. class A implements foo {

```

    16 foo bar()

    17 {

    18 Return 1;

}

```

```

}

```

Class A implements foo

```

{

25 foobar()

26 {

27 return 2;

}

```

```

}

```



Answer: return 2;

If eliminate 16,17,18 return 2

If eliminate 25 26 27 return 1

89. Which of these handles the exception when no catch is used?

- a. Default handler
- b. Finally
- c. Throw handler
- d. Java run time system

```
90. class exception_handling {  
    public static void main(String args[]) {  
        try {  
            int a, b;  
            b = 0;  
            a = 5 / b;  
            System.out.print("A");  
        }  
        catch(ArithmeticException e) {  
            System.out.print("B");  
        }  
        finally {  
            System.out.print("C");  
        }  
    }  
}
```

- a) A
- b) B
- c) AC

d) BC

Answer:BC

91. public class Delta

```
{ static boolean foo(char c)
{
    System.out.print(c);
    return true;
}
public static void main( String[] argv )
{
    int i = 0;
    for (foo('A'); foo('B') && (i < 2); foo('C'))
    {
        i++;
        foo('D');
    }
}
}
```

Answer: A.

ABDCBDCB

92. public class TreeSet {

```
public static void main(String args[]){
    //Creating and adding elements
    TreeSet<Integer> al=new TreeSet<Integer>();
    al.add("7"); //line 5
    al.add(9);
}
```

```

        //Traversing elements
        Iterator<Integer> itr=al.iterator();
        while(itr.hasNext()){
            System.out.println(itr.next());
        }
    }
}

```

Answer: compilation error at line 5

93. Which implementation of set would you choose if you want the iterator of set would give you object in the order it were inserted?

LinkedHashSet

TreeSet

HashSet

94. Push(7);

Push(2);

Pop();

Push(9);

Answer: print[7,9]

95. (first,two,three,four;,five;) delimits(,)

Answer: firsttwothreefour;five;

143. Which packages contain the JDBC classes?

(a) java.jdbc and javax.jdbc

(b) java.jdbc and java.jdbc.sql

- (c) java.sql and javax.sql
- (d) java.rdb and javax.rdb

144. Which type of driver provides JDBC access via one or more ODBC drivers?

- (a) Type 1 driver
- (b) Type 2 driver
- (c) Type 3 driver
- (d) Type 4 driver

145. Which type of driver converts JDBC calls into the network protocol used by the database management system directly?

- (a) Type 1 driver
- (b) Type 2 driver
- (c) Type 3 driver
- (d) Type 4 driver

146. How can you retrieve information from a ResultSet?

- (a) By invoking the method get(..., String type) on the ResultSet, where type is the database type
- (b) By invoking the method get(..., Type type) on the ResultSet, where Type is an object which represents a database type
- (c) By invoking the method getValue(...), and cast the result to the desired Java type.
- (d) By invoking the special getter methods on the ResultSet: getString(...), getBoolean(...), getClob(...),...

147. How can you execute DML statements (i.e. insert, delete, update) in the database?

- (a) By making use of the InsertStatement, DeleteStatement or UpdateStatement classes
- (b) By invoking the execute(...) or executeUpdate(...) method of a normal Statement object or a sub-interface object thereof
- (c) By invoking the executeInsert(...), executeDelete(...) or executeUpdate(...) methods of the DataModificationStatement object
- (d) By making use of the execute(...) statement of the DataModificationStatement object

148. How do you know in your Java program that a SQL warning is generated as a result of executing a SQL statement in the database?

- (a) You must catch the checked SQLException which is thrown by the method which executes the statement
- (b) You must catch the unchecked SQLWarningException which is thrown by the method which executes the statement
- (c) You must invoke the getWarnings() method on the Statement object (or a sub interface thereof)
- (d) You must query the ResultSet object about possible warnings generated by the database

149. What is, in terms of JDBC, a DataSource?

- (a) A DataSource is the basic service for managing a set of JDBC drivers
- (b) A DataSource is the Java representation of a physical data source
- (c) A DataSource is a registry point for JNDI-services
- (d) A DataSource is a factory of connections to a physical data source

150. What is the meaning of ResultSet.TYPE\_SCROLL\_INSENSITIVE

- (a) This means that the ResultSet is insensitive to scrolling

(b) This means that the Resultset is sensitive to scrolling, but insensitive to updates, i.e. not updateable

(c) This means that the ResultSet is sensitive to scrolling, but insensitive to changes made by others

(d) The meaning depends on the type of data source, and the type and version of the driver you use with this data source

151. Are ResultSets updateable?

(a) Yes, but only if you call the method `openCursor()` on the ResultSet, and if the driver and database support this option

(b) Yes, but only if you indicate a concurrency strategy when executing the statement, and if the driver and database support this option

(c) Yes, but only if the ResultSet is an object of class `UpdateableResultSet`, and if the driver and database support this option

(d) No, ResultSets are never updateable. You must explicitly execute DML statements (i.e. insert, delete and update) to change the data in the underlying database

152. How can you start a database transaction in the database?

(a) By asking a Transaction object to your Connection, and calling the method `begin()` on it

(b) By asking a Transaction object to your Connection, and setting the `autoCommit` property of the Transaction to false

(c) By calling the method `beginTransaction()` on the Connection object

(d) By setting the `autoCommit` property of the Connection to false, and execute a statement in the database

153. What is the meaning of the transaction isolation level

`TRANSACTION_REPEATABLE_READ`

(a) Dirty reads, non-repeatable reads and phantom reads can occur

(b) Dirty reads are prevented; non-repeatable reads and phantom reads can occur

(c) Dirty reads and non-repeatable reads are prevented; phantom reads can occur

(d) Dirty reads, non-repeatable reads and phantom reads are prevented

154. What statements are correct about positioned updates (i.e. cursor updates) in ResultSets? (2

correct answers)

[a] Using the cursor technique is currently the only possible way to change the data in the current row of a ResultSet

[b] Insert statements are only supported when using scrollable cursors.

[c] Only scrollable updateable ResultSets can use this approach to change the data in the current row of a ResultSet

[d] The name of the cursor is specified by the setCursorName(String name) method the Statement object.

155. How can you execute a stored procedure in the database?

(a) Call method execute() on a CallableStatement object

(b) Call method executeProcedure() on a Statement object

(c) Call method execute() on a StoredProcedure object

(d) Call method run() on a ProcedureCommand object

156. What happens if you call the method close() on a ResultSet object?

(a) the method close() does not exist for a ResultSet. Only Connections can be closed.

(b) the database and JDBC resources are released

(c) you will get a SQLException, because only Statement objects can close ResultSets

(d) the ResultSet, together with the Statement which created it and the Connection from which the Statement was retrieved, will be closed and release all database and JDBC resources

157. What happens if you call deleteRow() on a ResultSet object?

(a) The row you are positioned on is deleted from the ResultSet, but not from the database.

- (b) The row you are positioned on is deleted from the ResultSet and from the database
- (c) The result depends on whether the property `synchronizeWithDataSource` is set to true or false
- (d) You will get a compile error: the method does not exist because you can not delete rows from a ResultSet

158. What statements are correct about batched insert and updates? (2 answers)

[a] To create a batch of insert and update statements, you create an object of type `Batch`, and call the method

`addStatement(String statement)` for each statement you want to execute in the batch

[b] Batch insert and updates are only possible when making use of parameterized queries.

[c] To do a batched update/insert, you call `addBatch(String statement)` on a `Statement` object for each statement you want to execute in the batch

[d] To execute a batched update/insert, you call the `executeBatch()` method on a `Statement` object

159. What is correct about DDL statements (create, grant,...)?

(a) DDL statements are treated as normal SQL statements, and are executed by calling the `execute()` method on a `Statement` (or a sub interface thereof) object

(b) To execute DDL statements, you have to install additional support files

(c) DDL statements can not be executed by making use of JDBC, you should use the native database tools for this.

(d) Support for DDL statements will be a feature of a future release of JDBC

160. The JDBC-ODBC Bridge supports multiple concurrent open statements per connection?

a. True

b. False

162. Which of the following allows non repeatable read in JDBC Connection?



- a. TRANSACTION\_READ\_UNCOMMITTED
- b. TRANSACTION\_READ\_COMMITTED
- c. TRANSACTION\_SERIALIZABLE
- d. TRANSACTION\_REPEATABLE\_READ

163. Which of the following statements is false as far as different type of statements is concern in JDBC?

- a. Regular Statement
- b. Prepared Statement
- c. Callable Statement
- d. Interim Statement

164. Which of the following methods are needed for loading a database driver in JDBC?

- a. registerDriver() method
- b. Class.forName()
- c. Both A and B
- d. getConnection()

165. Which of the following is false as far as type 4 driver is concern?

- a. Type 4 driver is “native protocol, pure java” driver
- b. Type 4 drivers are 100% Java compatible
- c. Type 4 drivers uses Socket class to connect to the database
- d. Type 4 drivers can not be used with Netscape

166. To execute a stored procedure “totalStock” in a database server, which of the following code snippet is used?

- a. Statement stmt = connection.createStatement();stmt.execute("totalStock()");
- b. CallableStatement clbstmnt = con.prepareCall("{call totalStock}");cs.executeQuery();

c. `StoreProcedureStatement`

```
stmt=connection.createStoreProcedure("totalStock()");spstmt.executeQuery();
```

d. `PreparedStatement pstmt =`

```
connection.prepareStatement("totalStock()");pstmt.execute();
```

167. Which driver is efficient and always preferable for using JDBC applications?

a. Type – 4

b. Type – 1

c. Type – 3

d. Type – 2

168. JDBC facilitates to store the java objects by using which of the methods of `PreparedStatement`

`setObject ()` 2. `setBlob()` 3. `setClob()`

a. 1, 2

b. 1,2,3

c. 1,3

d. 2,3

169. Which statement is static and synchronized in JDBC API?

a. `executeQuery()`

b. `executeUpdate()`

c. `getConnection()`

d. `prepareCall()`

170. The JDBC-ODBC bridge is

a. Three tiered

b. Multithreaded

c. Best for any platform

d. All of the above

171. All raw data types (including binary documents or images) should be read and uploaded to the database as an array of

a. byte

b. int

c. boolean

d. char

172. The class `java.sql.Timestamp` has its super class as

a. `java.sql.Time`

b. `java.util.Date`

c. `java.util.Time`

d. None of the above

173. BLOB, CLOB, ARRAY and REF type columns can be updated in

a. JDBC 1.0

b. JDBC 4.0

c. JDBC 2.0

d. JDBC 3.0

174. Which of the following methods finds the maximum number of connections that a specific driver can obtain?

a. `Database.getMaxConnections`

b. `Connection.getMaxConnections`

c. `DatabaseMetaData.getMaxConnections`

d. `ResultSetMetaData.getMaxConnections`

175. Are prepared statements actually compiled?

- a. Yes, they compiled
- b. No, they are bound by the JDBC driver

176. When the message “No Suitable Driver” occurs?

- a. When the driver is not registered by Class.forName() method
- b. When the user name, password and the database does not match
- c. When the JDBC database URL passed is not constructed properly
- d. When the type 4 driver is used

177. Which driver is called as thin-driver in JDBC?

- a. Type-4 driver
- b. Type-1 driver
- c. Type-3 driver
- d. Type-2 driver

178. How many transaction isolation levels are defined in java.sql.Connection interface?

- a. 4
- b. 3
- c. 5
- d. 2

179. Which method is used to perform DML statements in JDBC?

- a. execute()
- b. executeQuery()
- c. executeUpdate()
- d. executeResult()

180. What is the disadvantage of Type-4 Native-Protocol Driver?

- a. At client side, a separate driver is needed for each database.
- b. Type-4 driver is entirely written in Java
- c. The driver converts JDBC calls into vendor-specific database protocol
- d. It does not support to read MySQL data.

```
181. import java.util.*;

class TestHashMaps{

public static void main(String args[]) {

HashMap<Integer,String> hm= new HashMap<Integer,String> ();

hm.put(100, "John");

hm.put(101, "Paul");

hm.put(102, "George");

hm.put(103, "Ringo");

for (Map.Entry m: hm.entrySet()) {

System.out.println(m.getKey() + " " + m.getValue());

}

}

}
```

Answer:

100 John

101 Paul

102 George

103 Ringo

```
182. import java.util.Map;

import java.util.TreeMap;

public class TestTreeMap {
```

```

public static void main(String args[]) {
    TreeMap< Integer, String > hm= new TreeMap< Integer, String > ();
    hm.put(100, "John");
    hm.put(102, "Paul");
    hm.put(101, "George");
    hm.put(103, "Ringo");
    for (Map.Entry m: hm.entrySet()) {
        System.out.println(m.getKey() + " " + m.getValue());
    }
}

```

Answer

100 John

101 George

102 Paul

103 Ringo

183.Given:

```

5. import java.util.*;
6. public class SortOf {
7. public static void main(String[] args) {
8. ArrayList<Integer> a = new ArrayList<Integer>();
9. a.add(1); a.add(5); a.add(3);
11. Collections.sort(a);
12. a.add(2);
13. Collections.reverse(a);
14. System.out.println(a);
15. }

```

16. }

What is the result?

A. [1, 2, 3, 5]

B. [2, 1, 3, 5]

C. [2, 5, 3, 1]

D. [5, 3, 2, 1]

E. [1, 3, 5, 2]

F. Compilation fails.

G. An exception is thrown at runtime.

Answer: C

184. class BabyRaccoon extends Mammal { }

Which four statements are true? (Choose four.)

A. Raccoon is-a Mammal.

B. Raccoon has-a Mammal.

C. BabyRaccoon is-a Mammal.

D. BabyRaccoon is-a Raccoon.

E. BabyRaccoon has-a Mammal.

F. BabyRaccoon is-a BabyRaccoon.

Answer: A,B,C,F

185. Which Man class properly represents the relationship "Man has a best friend who is a Dog"?

A. class Man extends Dog { }

B. class Man implements Dog { }

C. class Man { private BestFriend dog; }

D. class Man { private Dog bestFriend; }

E. class Man { private Dog<bestFriend>; }

F. class Man { private BestFriend<dog>; }

Answer: D

186. Given:

```
11. class Alpha {  
12. public void foo() { System.out.print("Afoo "); }  
13. }  
  
14. public class Beta extends Alpha {  
15. public void foo() { System.out.print("Bfoo "); }  
16. public static void main(String[] args) {  
17. Alpha a = new Beta();  
18. Beta b = (Beta)a;  
19. a.foo();  
20. b.foo();  
21. }  
22. }
```

What is the result?

- A. Afoo Afoo
- B. Afoo Bfoo
- C. Bfoo Afoo
- D. Bfoo Bfoo
- E. Compilation fails.
- F. An exception is thrown at runtime.

Answer: D

187. Given:



```
5. import java.util.Date;
6. import java.text.DateFormat;
21. DateFormat df;
22. Date date = new Date();
23. // insert code here
24. String s = df.format(date);
```

Which code fragment, inserted at line 23, allows the code to compile?

- A. df = new DateFormat();
- B. df = Date.getFormat();
- C. df = date.getFormat();
- D. df = DateFormat.getFormat();
- E. df = DateFormat.getInstance();

Answer: E

188. Given:

```
1. public class Base {
2.     public static final String FOO = "foo";
3.     public static void main(String[] args) {
4.         Base b = new Base();
5.         Sub s = new Sub();
6.         System.out.print(Base.FOO);
7.         System.out.print(Sub.FOO);
8.         System.out.print(b.FOO);
9.         System.out.print(s.FOO);
10.        System.out.print(((Base)s).FOO);
11.    } }
12. class Sub extends Base {public static final String FOO="bar";}
```

What is the result?

- A. foofoofoofoofoo
- B. foobarfoobarbar
- C. foobarfoofoofoo
- D. foobarfoobarfoo
- E. barbarbarbarbar
- F. foofoofoobarbar
- G. foofoofoobarfoo

Answer: D

189 A company has a business application that provides its users with many different reports:

receivables reports, payables reports, revenue projects, and so on. The company has just

purchased some new, state-of-the-art, wireless printers, and a programmer has been assigned the

task of enhancing all of the reports to use not only the company's old printers, but the new

wireless printers as well. When the programmer starts looking into the application, the programmer

discovers that because of the design of the application, it is necessary to make changes to each

report to support the new printers. Which two design concepts most likely explain this situation?

(Choose two.)

- A. Inheritance
- B. Low cohesion
- C. Tight coupling
- D. High cohesion
- E. Loose coupling
- F. Object immutability

Answer: B,C

190. A team of programmers is reviewing a proposed API for a new utility class. After some discussion, they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?

- A. Looser coupling
- B. Tighter coupling
- C. Lower cohesion
- D. Higher cohesion
- E. Weaker encapsulation
- F. Stronger encapsulation

Answer: A

191. A team of programmers is involved in reviewing a proposed design for a new utility class. After

some discussion, they realize that the current design allows other classes to access methods in

the utility class that should be accessible only to methods within the utility class itself. What design

issue has the team discovered?

- A. Tight coupling
- B. Low cohesion
- C. High cohesion
- D. Loose coupling
- E. Weak encapsulation
- F. Strong encapsulation

Answer: E

192. Given:

```
1. public class TestString1 {  
2. public static void main(String[] args) {  
3. String str = "420";  
4. str += 42;  
5. System.out.print(str);  
6. }  
7. }
```

What is the output?

- A. 42
- B. 420
- C. 462
- D. 42042
- E. Compilation fails.
- F. An exception is thrown at runtime.

Answer: D

193. Given

```
11. public interface Status {  
12. /* insert code here */ int MY_VALUE = 10;  
13. } Which three are valid on line  
12?
```

(Choose three.)

- A. final
- B. static
- C. native
- D. public
- E. private

F. abstract

G. protected

Answer: A,B,D

194. Given:

10. interface Foo { int bar(); }

11. public class Sprite {

12. public int fubar( Foo foo ) { return foo.bar(); }

13. public void testFoo() {

14. fubar(

15. // insert code here

16. );

17. }

18. }

Which code, inserted at line 15, allows the class Sprite to compile?

A. Foo { public int bar() { return 1; } }

B. new Foo { public int bar() { return 1; } }

C. new Foo() { public int bar() { return 1; } }

D. new class Foo { public int bar() { return 1; } }

Answer: C

195. Given:

11. class Animal { public String noise() { return "peep"; } }

12. class Dog extends Animal {

13. public String noise() { return "bark"; }

14. }

15. class Cat extends Animal {

16. public String noise() { return "meow"; }

17. } ...

30. Animal animal = new Dog();

31. Cat cat = (Cat)animal;

32. System.out.println(cat.noise());

What is the result?

A. peep

B. bark

C. meow

D. Compilation fails.

E. An exception is thrown at runtime.

Answer: E

196. A programmer has an algorithm that requires a java.util.List that provides an efficient

implementation of add(0, object), but does NOT need to support quick random access. What

supports these requirements?

A. java.util.Queue

B. java.util.ArrayList

C. java.util.LinearList

D. java.util.LinkedList

Answer: D

197. Which Man class properly represents the relationship "Man has a best friend who is a Dog"?

A. class Man extends Dog { }

B. class Man implements Dog { }

C. class Man { private BestFriend dog; }

D. class Man { private Dog bestFriend; }

E. class Man { private Dog<bestFriend>; }

F. class Man { private BestFriend<dog>; }

Answer: D

199. A team of programmers is reviewing a proposed API for a new utility class. After some discussion,

they realize that they can reduce the number of methods in the API without losing any functionality. If they implement the new design, which two OO principles will they be promoting?

- A. Looser coupling
- B. Tighter coupling
- C. Lower cohesion
- D. Higher cohesion
- E. Weaker encapsulation
- F. Stronger encapsulation

Answer: A

200. Given:

10. interface Jumper { public void jump(); } ...

20. class Animal {} ...

30. class Dog extends Animal {

31. Tail tail; 32. } ...

40. class Beagle extends Dog implements Jumper{

41. public void jump() {}

42. } ...

50. class Cat implements Jumper{

51. public void jump() {}

52. }

Which three are true?(Choose Three)

- A. Cat is-a Animal

- B. Cat is-a Jumper
- C. Dog is-a Animal
- D. Dog is-a Jumper
- E. Cat has-a Animal
- F. Beagle has-a Tail
- G. Beagle has-a Jumper

Answer: B,C,F

201. Given:

10. interface Data { public void load(); }

11. abstract class Info { public abstract void load(); }

Which class correctly uses the Data interface and Info class?

A. public class Employee extends Info implements Data {  
public void load() { /\*do something\*/ }  
}

B. public class Employee implements Info extends Data {  
public void load() { /\*do something\*/ }  
}

C. public class Employee extends Info implements Data {  
public void load(){ /\*do something\*/ }  
public void Info.load(){ /\*do something\*/ }  
}

D. public class Employee implements Info extends Data {  
public void Data.load(){ /\*do something\*/ }  
public void load(){ /\*do something\*/ }  
}

E. public class Employee implements Info extends Data {  
public void load(){ /\*do something\*/ }



```
public void Info.load(){ /*do something*/ }  
}
```

```
F. public class Employee extends Info implements Data{  
public void Data.load() { /*do something*/ }  
public void Info.load() { /*do something*/ }  
}
```

Answer: A

202. Given:

```
1. class TestA {  
2. public void start() { System.out.println("TestA"); }  
3. }  
4. public class TestB extends TestA {  
5. public void start() { System.out.println("TestB"); }  
6. public static void main(String[] args) {  
7. ((TestA)new TestB()).start();  
8. }  
9. }
```

What is the result?

- A. TestA
- B. TestB
- C. Compilation fails.
- D. An exception is thrown at runtime.

Answer: B

```
203. import java.io.*;  
  
class files {  
  
public static void main(String args[]) {
```

```
File obj = new File("/java/system");  
System.out.print(obj.getName());  
}  
}
```

- a) java
- b) system
- c) java/system
- d) /java/system

Answer:B

```
204. import java.io.*;  
  
class files {  
  
public static void main(String args[]) {  
File obj = new File("/java/system");  
System.out.print(obj.getAbsolutePath());  
}  
}
```

Note: file is made in c drive.

- a) java
- b) system
- c) java/system
- d) /java/system

View Answer

Answer: d

Explanation: None.

Output:

```
$ javac files.java
```

```
$ java files
```

`\java\system`

205. What is the output of this program?

```
import java.io.*;

class files {

public static void main(String args[]) {

File obj = new File("/java/system");

System.out.

}

}
```

Note: file is made in c drive.

- a) java
- b) system
- c) java/system
- d) /java/system

Answer:D

206. class exception\_handling {

```
public static void main(String args[]) {

try {

int a = args.length;

int b = 10 / a;

System.out.print(a);

try {

if (a == 1)

a = a / a - a;

if (a == 2) {
```

```
int c = {1};  
c[8] = 9;  
}  
}  
catch (ArrayIndexOutOfBoundsException e) {  
System.out.println("TypeA");  
}  
catch (ArithmeticException e) {  
System.out.println("TypeB");  
}}}
```

- a) TypeA
- b) TypeB
- c) 0TypeA

Important link:

<http://javabysri.blogspot.in/2012/06/jdbc-objective-questions.html>

1.arithmetic exception and io exception are of which type??

runtime error

7. super class of all classes

ans: objects

8.which thread method defined all threads?

ans: daemon

9.jdbc belongs to which package?

ans: java.sql\*

10. which is not a primitive datatype?

ans: string

13. which method reverse the string??

reverse()

17. list of file returns wat?

ans:an array of string is returned

18.which is a valid declaration??

ans: public void int A()-false

public int test2(int o);

public void test int(int i,intj);-false

19. public class B()

static int x;

psvm()

{

sop (B.x);

}

ans: prints 0

23.which invoke method will print base class constructor?

ans: super(10)

30. which is not a valid declaration??

options: char c=a;

char d =\u00432;

char c= \uae262;

ju006B

31.public class test ()

static int i=012;

psvm(srting args[])

{sop (i);

}

ans: prints 10

32.which is not valid?

ans:int i[] = new int []

int i [][] = new int[] {1,2}

35. how ll u create the copy of the same object?? ans: clone()

2.what are the sequence of java servletdeployment descriptor??

ans: coding,compiling,deploying

3.which is used to submit query??

options: statement,

4.abstract method doesn follow which statement??

options:it instantiates its object

abstract class shud ve atleast one abstract method

none of the above-----

5. what are the implicit objects in jsp??

ans: page, this

6. valid status codes

ans: 200-299

11. println belongs to class printstream- print writer

12. void method does not return value---true

14. session belongs to which class?? httpServletRequest

15. frame has default border type and plain has default flow type

16. which interface is used to connect database to application?-connection

20. which is not a directive??

21. what jsp syntax are following scriptlet, expression, directive?

22. `<jsp:usebean id="name" class="java.lang.string"><%=name%>`

options: prints 0

prints null

compile time error

none of the above

24. q18 in previous paper

25. how session object can be destroyed?

26. `public class test()`

`{static int i, static double d, static char c, static long l, static short s,`

`static string s;`

`psvm{ sop(i+d+c+l+s+S);`

`}`

`}`

27. `<%!int i=8%>`

`<%int i=10%>`

`<%=i>`

options: prints 8, prints 10

28. `class.forName()` throws class not found exception--true



29.seek method is used in random access file to specify the new position--true

33.public class B()

boolean a,b,c;

boolean =a||(b=true)&&(c=true)

sop(a+","+b+","+c);

options: false,false,false

false,true,true

34.inputstreamreader i=new inputstreamreader (vdscl);

bufferedReader br = new bufferedreader (i);----br.read

36.all servlet object implements 1.servlet context 2.web context 3. both 1&2

37.submitting a query is done by ans:statement object

38.which math object is used for pointing 4 while x=4.8

options: math.ceil,math.floor---

39.what is feature of java?? OOPS

40.how does applet communicate with browser?? ans: init and database document

### 1. Question

10 points

Which of the following correctly mentions the Autowire types?

- ☐ 1. byName , byType, constructor and autodetect.
- ☐ 2. byName , byType, destructor and autodetect.
- ☐ 3. byName , byType, constructor and autocorrect.
- ☐ 4. None of these.
- ☐ 5. byName , byMethod, constructor and autodetect.

Correct

### 2. Question

10 points

Which of the following statements is true?

- ☐ 1. ApplicationContext extends BeanFactory.
- ☐ 2. None of these.
- ☐ 3. ApplicationContext implements BeanFactory.
- ☐ 4. BeanFactory extends ApplicationContext.
- ☐ 5. BeanFactory implements ApplicationContext

Incorrect

Correct 3

### 3. Question

10 points

Which of the following is an IOC or Dependency injection?

- ☐ 1. Java Module
- ☐ 2. Framework
- ☐ 3. ORM Framework
- ☐ 4. None of these.
- ☐ 5. Design Pattern

Incorrect

#### 4. Question

10 points

What is the default nature of the Beans defined in spring framework?

- ☐ 1. Initialized
- ☐ 2. None of these.
- ☐ 3. Abstract
- ☐ 4. Final
- ☐ 5. Singleton

Incorrect

#### 5. Question

10 points

Which of the following is not a component of the Spring framework?

- ☐ 1. AOP
- ☐ 2. HTML/JSP
- ☐ 3. None of these.
- ☐ 4. Spring MVC
- ☐ 5. O/R Integration

Incorrect

#### 6. Question

10 points

Which of the following is a type of bean injection?

- ☐ 1. constructor and setter.
- ☐ 2. setter,getter and constructor.
- ☐ 3. constructor and getter.
- ☐ 4. None of these.
- ☐ 5. getter and setter.

Incorrect

### 7. Question

10 points

Which of the following specify the points where weaving can be applied?

- ☐ 1. Compile Time and Runtime.
- ☐ 2. Compile Time , Classload Time , load time.
- ☐ 3. Compile Time , Classload Time , Runtime
- ☐ 4. None of these.
- ☐ 5. Compile Time and Classload Time.

Incorrect

### 8. Question

10 points

Which of the following correctly defines the Spring?

- ☐ 1. A licensed framework.
- ☐ 2. A free framework.
- ☐ 3. An open source framework.
- ☐ 4. A proprietary framework.
- ☐ 5. None of these.

Incorrect

### 9. Question

10 points

Which of the following exception classes is related to all the exceptions thrown in spring applications?

- ☐ 1. NullPointerException.
- ☐ 2. DataAccessException.
- ☐ 3. ArrayIndexOutOfBounds.
- ☐ 4. SpringException.
- ☐ 5. None of these.

Incorrect

### 10. Question

10 points

Which of the following is true for Controllers in Spring?

- ☐ 1. Concrete class.
- ☐ 2. Final class.
- ☐ 3. An abstract class.
- ☐ 4. An interface.
- ☐ 5. None of these.

Incorrect

### 11. Question

10 points

Which of the following classes can you use for executing the SQL queries in Spring?

- ☐ 1. DBHelper.
- ☐ 2. JdbcTemplate.
- ☐ 3. DBTemplate
- ☐ 4. None of these.
- ☐ 5. JDBCHelper.

Incorrect

### 12. Question

10 points

All of the Spring's Controllers inherit from an AbstractControllor?

- ☐ 1. true
- ☐ 2. false

Incorrect

Ans ? False

### 13. Question

10 points

Which of the following beans isn't significant for the WebApplicationContext?

- ☐ 1. Locale resolver.
- ☐ 2. Handler exception resolver.
- ☐ 3. Multipart file.
- ☐ 4. Resolver.
- ☐ 5. None of these.

Incorrect

#### 14. Question

10 points

Which of the following is a valid scope of the beans in portlet context?

- ☐ 1. session.
- ☐ 2. global-session.
- ☐ 3. None of these.
- ☐ 4. prototype.
- ☐ 5. request.

Incorrect

#### 15. Question

10 points

Which of the following Java Collection types can you inject in a Spring application?

- ☐ 1. Using list, collection, map or props tag.
- ☐ 2. Using list, set, map or props tag.
- ☐ 3. Using list, set, props or collection tag.
- ☐ 4. Using list, set, map or collection tag.
- ☐ 5. None of these.

Incorrect

#### 16. Question

10 points

Which of the following layers the @Controller annotation is used in?

- ☐ 1. Service layer.

- ☐ 2. Presentation layer.
- ☐ 3. Session layer.
- ☐ 4. Business layer.
- ☐ 5. None of these.

Incorrect

#### 17. Question

10 points

Which of the following statements is true for @RequestMapping annotation?

- ☐ 1. Used to map web requests onto specific handler methods.
- ☐ 2. Used to map web requests onto specific handler classes.
- ☐ 3. Can be applied to the controller methods.
- ☐ 4. None of these.
- ☐ 5. Can be applied to the controller classes.

Incorrect

#### 18. Question

10 points

@RequestMapping annotation is used to map a HTTP request method (GET or POST) to a specific class or method in the controller which will handle the respective request?

- ☐ 1. true
- ☐ 2. false

Incorrect

#### 19. Question

10 points

Which of the following is not a valid difference between @Inject and @Autowired annotations?

- ☐ 1. There is no difference and both can be used interchangeably.
- ☐ 2. @Autowired is a legacy annotation whereas @Inject is its latest version.
- ☐ 3. @Inject annotation is part of Java EE 7's Context and Dependency Injection framework while @Autowired is the Spring Frameworks own implementation.

- ☐ 4. None of these.

Incorrect

## 20. Question

10 points

Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

- ☐ 1. true
- ☐ 2. false

## Spring Questions and Answers – JDBC Template

This set of Java Spring Multiple Choice Questions & Answers (MCQs) focuses on “JDBC Template”.

1. Class which declares a number of overloaded update() template methods to control the overall update process.

- a) org.springframework.jdbc.core.JdbcTemplate
- b) org.springframework.jdbc.core.\*
- c) org.springframework.jdbc.\*
- d) none of the mentioned

View Answer

Answer: a

Explanation: The org.springframework.jdbc.core.JdbcTemplate class declares a number of overloaded update() template methods to control the overall update process.

2. You implement this interface to override the statement creation task.

- a) PreparedStatement
- b) PreparedStatementCreator
- c) PreparedCreator
- d) None of the mentioned

View Answer

Answer: b

Explanation: The first callback interface to introduce is PreparedStatementCreator. You implement this interface to override the statement creation task and the parameter binding task of the overall update process.

3. When implementing the PreparedStatementCreator interface, you will get the database connection as the createPreparedStatement() method's argument.



- a) True
- b) False

[View Answer](#)

Answer: a

Explanation: All you have to do in this method is to create a PreparedStatement object on this connection and bind your parameters to this object.

4. It is better to implement the PreparedStatementCreator interface and other callback interfaces as inner classes if they are used within one method only.

- a) True
- b) False

[View Answer](#)

Answer: a

Explanation: This is because you can get access to the local variables and method arguments directly from the inner class, instead of passing them as constructor arguments.

5. PreparedStatementSetter, as its name indicates, create a PreparedStatement object on this connection the parameter as well as binding task of the overall update process.

- a) True
- b) False

[View Answer](#)

Answer: b

Explanation: The second callback interface, PreparedStatementSetter, as its name indicates, performs only the parameter binding task of the overall update process.

6. The JdbcTemplate class offers template method for batch update operations.

- a) batchUpdate()
- b) update()
- c) all of the mentioned
- d) none of the mentioned

[View Answer](#)

Answer: c

Explanation: It requires a SQL statement and a BatchPreparedStatementSetter object as arguments. In this method, the statement is compiled (prepared) only once and executed multiple times.

7. The JdbcTemplate class declares a number of overloaded query() template methods to control the overall query process.

- a) True

b) False

[View Answer](#)

Answer: c

Explanation: You can override the statement creation (task 2) and the parameter binding by implementing the PreparedStatementCreator and PreparedStatementSetter interfaces, just as you did for the update operations.

8. The primary interface that allows you to process the current row of the result set.

- a) PreparedStatementSetter
- b) PreparedStatementCreator
- c) RowCallbackHandler
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: RowCallbackHandler is the is the primary interface that allows you to process the current row of the result set.

9. RowCallbackHandler's purpose is to map a single row of the result set to a customized object.

- a) True
- b) False

[View Answer](#)

Answer: b

Explanation: RowCallbackHandler is the is the primary interface that allows you to process the current row of the result set.

10. Method of RowMapper interface in which, you have to construct the object that represents a row and return it as the method's return value.

- a) mapRow()
- b) query()
- c) update()
- d) none of the mentioned

[View Answer](#)

Answer: a

Explanation: From the viewpoint of reuse, it's better to implement the RowMapper interface as a normal class than as an inner class. In the mapRow() method of this interface, you have to construct the object that represents a row and return it as the method's return value.

11. RowMapper implementation which can automatically map a row to a new instance of the specified class.

- a) BeanPropertyRowMapper
- b) BeanPropertyRow
- c) All of the mentioned
- d) None of the mentioned

[View Answer](#)

Answer: a

Explanation: Note that the specified class must be a top-level class and must have a default or no-argument constructor. It first instantiates this class and then maps each column value to a property by matching their names.

12. Method which provides list of maps.

- a) queryForList()
- b) update
- c) query()
- d) all of the mentioned

[View Answer](#)

Answer: a

Explanation: Without the help of RowMapper, you can still call the queryForList() method and pass in a SQL statement. The returned result will be a list of maps. Each map stores a row of the result set with the column names as the keys.

13. Spring JDBC framework offers a convenient class, to simplify your DAO implementation.

- a) org.springframework.jdbc.core.support
- b) org.springframework.jdbc.core.support.JdbcDaoSupport
- c) all of the mentioned
- d) none of the mentioned

[View Answer](#)

Answer: b

Explanation: Spring JDBC framework offers a convenient class, org.springframework.jdbc.core.support.JdbcDaoSupport, to simplify your DAO implementation. This class declares a jdbcTemplate property, which can be injected from the IoC container or created automatically from a data source.

14. The org.springframework.jdbc.core.support.JdbcDaoSupport class has a setDataSource() method and a setJdbcTemplate() method.

- a) True
- b) False

[View Answer](#)

Answer: a

Explanation: Your DAO class can extend this class to have these methods inherited.

15. Method to retrieve the JDBC template.

- a) setJdbcTemplate()
- b) getTemplate()
- c) getJdbc()
- d) getJdbcTemplate()

[View Answer](#)

Answer: d

Explanation: In your DAO methods, you can simply call the getJdbcTemplate() method to retrieve the JDBC template. You also have to delete the dataSource and jdbcTemplate properties, as well as their setter methods, from your DAO class, because they have already been inherited. Again, for simplicity's sake, only the change to the insert() method is shown.

## Spring Questions and Answers – Web Services with XML Marshalling and service end points

This set of Java Spring online test focuses on “Web Services with XML Marshalling and service end points”.

1. Spring-WS supports using XML marshalling technology to marshal and unmarshal objects to and from XML documents.

- a) True
- b) False

[View Answer](#)

Answer: a

Explanation: This technology is also known as object/XML mapping (OXM), because you are actually mapping objects to and from XML documents.

2. To implement endpoints with an XML marshalling technology, you have to extend the:-

- a) AbstractMarshallingPayload
- b) AbstractMarshalling
- c) AbstractMarshallingPayloadEndpoint
- d) None of the mentioned

[View Answer](#)

Answer: c

Explanation: To implement endpoints with an XML marshalling technology, you have to

extend the AbstractMarshallingPayloadEndpoint class and configure an XML marshaller for it.

3. Marshaller for JAXB 1.0:-

- a) org.springframework.xml.jaxb.Jaxb1Marshaller
- b) org.springframework.xml.jaxb.Jaxb2Marshaller
- c) org.springframework.xml.castor.CastorMarshaller
- d) org.springframework.xml.xmlbeans.XmlBeansMarshaller

[View Answer](#)

Answer: a

Explanation: JAXB 1.0 org.springframework.xml.jaxb.Jaxb1Marshaller.

4. Marshaller for JAXB 2.0:-

- a) org.springframework.xml.jaxb.Jaxb1Marshaller
- b) org.springframework.xml.jaxb.Jaxb2Marshaller
- c) org.springframework.xml.castor.CastorMarshaller
- d) org.springframework.xml.xmlbeans.XmlBeansMarshaller

[View Answer](#)

Answer: b

Explanation: JAXB 2.0 org.springframework.xml.jaxb.Jaxb2Marshaller.

5. Marshaller for Castor:-

- a) org.springframework.xml.jaxb.Jaxb1Marshaller
- b) org.springframework.xml.jaxb.Jaxb2Marshaller
- c) org.springframework.xml.castor.CastorMarshaller
- d) org.springframework.xml.xmlbeans.XmlBeansMarshaller

[View Answer](#)

Answer: c

Explanation: Castor org.springframework.xml.castor.CastorMarshaller.

6. Marshaller for XMLBeans:-

- a) org.springframework.xml.jaxb.Jaxb1Marshaller
- b) org.springframework.xml.jaxb.Jaxb2Marshaller
- c) org.springframework.xml.castor.CastorMarshaller
- d) org.springframework.xml.xmlbeans.XmlBeansMarshaller

[View Answer](#)

Answer: d

Explanation: XMLBeans org.springframework.xml.xmlbeans.XmlBeansMarshaller.

7. Marshaller for JiBX:-

- a) org.springframework.xml.jaxb.Jaxb1Marshaller
- b) org.springframework.xml.jaxb.Jaxb2Marshaller

c) org.springframework.xml.castor.CastorMarshaller

d) org.springframework.xml.jibx.JibxMarshaller

View Answer

Answer: d

Explanation: JiBX org.springframework.xml.jibx.JibxMarshaller.

8. Marshaller for XStream:-

a) org.springframework.xml.jaxb.Jaxb1Marshaller

b) org.springframework.xml.jaxb.Jaxb2Marshaller

c) org.springframework.xml.xstream.XStreamMarshaller

d) org.springframework.xml.jibx.JibxMarshaller

View Answer

Answer: c

Explanation: XStream org.springframework.xml.xstream.XStreamMarshaller.

9. Spring-WS supports various XML marshalling APIs.

a) Castor

b) JAXB 1.0

c) XMLBeans

d) All of the mentioned

View Answer

Answer: d

Explanation: Spring-WS supports various XML marshalling APIs, including JAXB 1.0, JAXB 2.0, Castor, XMLBeans, JiBX, and XStream.

10. For some marshalling APIs, the object model must be generated by them so that they can insert marshalling-specific information.

a) True

b) False

View Answer

Answer: a

Explanation: Because Castor supports marshalling between XML messages and arbitrary Java objects, you can start creating the following classes by yourself.

11. A marshalling endpoint requires:-

a) marshaller

b) unmarshaller

c) all of the mentioned

d) none of the mentioned

View Answer

Answer: c

Explanation: A marshalling endpoint requires both the marshaller and unmarshaller properties to be set. Usually, you can specify a single marshaller for both properties. For Castor, you declare a CastorMarshaller bean as the marshaller.

12. Castor requires a mapping configuration file to know how to map objects to and from XML documents.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: You can create this file in the classpath root and specify it in the mappingLocation property (e.g., mapping.xml).

13. Spring-WS supports annotating an arbitrary class as a service endpoint by :-

a) @Endpoint

b) @End

c) @Point

d) None of the mentioned

[View Answer](#)

Answer: a

Explanation: Spring-WS supports annotating an arbitrary class as a service endpoint by the @Endpoint annotation, without extending a framework-specific class.

14. When you are using XML marshalling, WebServiceTemplate requires both the marshaller and unmarshaller properties to be set.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: You can also set them to WebServiceGatewaySupport if you extend this class to have WebServiceTemplate auto-created.

15. A Spring-WS client can also marshall and unmarshall the request and response objects to and from XML messages.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: As an example, we will create a client using Castor as the marshaller so that you can reuse the object models.

## Spring Questions and Answers – Spring Portlet MVC

This set of Java Spring Multiple Choice Questions & Answers (MCQs) focuses on “Spring Portlet MVC”.

1. DispatcherServlet dispatches portlet requests to appropriate handlers that handle the requests.

- a) True
- b) False

View Answer

Answer: b

Explanation: DispatcherPortlet. It dispatches portlet requests to appropriate handlers that handle the requests. It acts as the front controller of Spring Portlet MVC, and every portlet request must go through it so that it can manage the entire request-handling process.

2. Maps each request to a handler through one or more handler mapping beans.

- a) DispatcherServlet
- b) DispatcherPortlet
- c) All of the mentioned
- d) None of the mentioned

View Answer

Answer: b

Explanation: Once it has picked an appropriate handler, it will invoke this handler to handle the request.

3. In portlets, there are URLs:-

- a) render
- b) action
- c) all of the mentioned
- d) none of the mentioned

View Answer

Answer: c

Explanation: In portlets, there are two types of URLs: render URLs and action URLs.

4. When a user triggers a render URL, the portlet container will ask all the portlets in the same page to handle a render request to render its view.

- a) True
- b) False

View Answer



Answer: a

Explanation: When a user triggers a render URL, the portlet container will ask all the portlets in the same page to handle a render request to render its view, unless the view's content has been cached.

5. When a user triggers an action URL in a portlet, the portlet container will first ask the target portlet to handle an action request.

a) True

b) False

View Answer

Answer: a

Explanation: A controller should return an object in response to a render request. However, when a user triggers an action URL in a portlet, the portlet container will first ask the target portlet to handle an action request. A controller needn't return anything for an action request.

6. After Controller has finished handling a render request, it returns:-

a) model name

b) view name

c) view object

d) all of the mentioned

View Answer

Answer: d

Explanation: After a controller has finished handling a render request, it returns a model and a view name, or sometimes a view object, to DispatcherPortlet.

7. DispatcherPortlet resolves a view name from one or more view resolver beans.

a) True

b) False

View Answer

Answer: a

Explanation: DispatcherPortlet renders the view and passes in the model returned by the controller. Note that an action request doesn't need to render a view.

8. portlet deployment descriptor file is:-

a) portlet.xml

b) portlet.config

c) portlet.xhtml

d) none of the mentioned

View Answer

Answer: a

Explanation: The Java Portlet specification defines the valid structure of a portlet application, which is very similar to that of a web application, but with the addition of a portlet deployment descriptor (i.e., portlet.xml).

9. Servlet Listener to load the root application context at startup.

- a) ContextLoader
- b) ContextLoaderListener
- c) All of the mentioned
- d) None of the mentioned

View Answer

Answer: b

Explanation: In the web deployment descriptor (i.e., web.xml), you have to register the servlet listener ContextLoaderListener to load the root application context at startup.

10. Unlike in a web application, you can't control URLs directly in a portlet.

- a) True
- b) False

View Answer

Answer: a

Explanation: In the web deployment descriptor (i.e., web.xml), you have to register the servlet listener ContextLoaderListener to load the root application context at startup.

11. You can chain multiple handler mapping annotations as required to portlet.

- a) True
- b) False

View Answer

Answer: a

Explanation: Spring Portlet MVC comes with several strategies for you to map portlet requests.

12. The preceding controller handles portlet requests:-

- a) render requests
- b) action requests
- c) all of the mentioned
- d) none of the mentioned

View Answer

Answer: c

Explanation: The preceding controller handles two types of portlet requests: render requests and action requests.

13. When handling a render request, it gets the time zone attribute from the portlet preference.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: When handling a render request, it gets the time zone attribute from the portlet preferences and passes it to the view for editing.

14. When handling an action request, it gets the time zone parameter from the portlet request.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: When handling an action request, it gets the time zone parameter from the portlet request and stores it in the portlet preferences as an attribute, which will override the existing one.

15. The submission URL of this form should be a portlet action URL that will trigger an action request to the current portlet.

a) True

b) False

[View Answer](#)

Answer: a

Explanation: In portlets, you have to construct an action URL with the portlet:actionURL tag defined in the portlet tag library.

Q 1.

**Given:**

```
<beans>
```

```
<bean name="____" class="jbb.MyBean"/>
```

**</beans>**

**What can be inserted in the gap?**

☐ name1 - name2

☐ name1 - name2 - name3

☐ none of the above.

1,2

Q 2.

**Choose equivalent tags from the following :**

☐ `<property name="message">`  
`<value>spring</value>`  
`</property>`

☐

and

`<property name="message" value="spring"/>`

```
<constructor-arg>  
  <value>spring</value>  
</constructor-arg>
```



and

```
<constructor-arg value="spring"/>
```

```
<property name="someBean">  
  <ref bean="myBean">  
</property>
```



and

```
<property name="someBean" ref="myBean"/>
```

```
<property name="someBean">  
  <ref local="myBean">  
</property>
```



and

```
<property name="someBean" ref="myBean"/>
```

1,2,3

Q 3

**What will be the output of the following program?**

**Car.xml:**

```
<beans...>
```

```
<bean id="expensiveCar" class="foo.Car">
```

```
  <property name="name" value="Mercedes"/>
```

```
</bean>
<bean id="luxuryCar" class="foo.Car">
  <property name="name" value="Bentley"/>
</bean>
</beans>
```

**Car.java:**

```
public class Car {
    private String name;
    private static int carCount;

    public Car() {
        carCount++;
    }

    public static int getCount() {
        return carCount;
    }

    public void setName(String name) {
        this.name = name;
    }

    public static void main(String[] args) {
        ApplicationContext appContext =
new    FileSystemXmlApplicationContext("car.xml");
        Car car = (Car) appContext.getBean("expensiveCar");
        System.out.println("Total cars:"+Car.getCount());
    }
}
```



It is not possible to declare two Car objects and get only one - Spring container throws an exception



 2

3

Q 4

**The BeanFactory supports the following variants of dependency injection:**

☐ constructor injection without setter injection

☐ setter injection without constructor injection

☐ first constructor injection, then setter injection

☐ first setter injection, then constructor injection

1,2,3

Q 5.

Which ones of the following XML snippets correctly define singletons?

☐ `<bean id="bean1" class="com.foo.MyBean" prototype="singleton"/>`

☐ `<bean id="bean1" class="com.foo.MyBean" scope="singleton"/>`

☐ `<bean id="bean1" class="com.foo.MyBean" singleton="yes"/>`

☐ `<bean id="bean1" class="com.foo.MyBean"/>`

2,4

Q 6.

Which of the following elements have to be insert in a current configuration file for integrate another spring configuration file:

☐ `<load/>`

☐ `<import/>`

☐ `<include/>`

☐ `<use/>`



**Given the fact that our Java class has a firstName property of the String type, which of the following mappings are valid?**

☐ `<property name="firstName" column="FIRST_NAME" type="java.lang.String" />`

☐ `<property name="firstName" column="FIRST_NAME"/>`

☐ `<property name="firstName" column="FIRST_NAME" type="string" />`

☐ `<property name="firstName" column="FIRST_NAME" type="VARCHAR" />`

Q 8.

**For a typical business application, with a business logic and a DAO layer.  
Check true assertions about the creation of the beans.**



Most beans are singleton, so they are created when the application context is created.



Most bean are accessed via the `getBean(String name)` method from the application context.



Most beans are accessed via getters from other beans.

1,3

Q 9.

**This is the class A.**

```
public class A {  
    B b;  
    public B getB(){  
        return this.b;  
    }  
    public void setB(B newB){  
        this.b = newB;  
    }  
}
```

**A has 1 bean definition:**

```
<bean name="a" autowire="byType" class="A" />
```

**B has 2 bean definitions:**

```
<bean name="b2" class="B" />
```

```
<bean name="b" class="B" />
```

**What will happen when a is created by Spring?**



Bean *a* will be linked with *b*, because when *byType* fails, Spring tries *byName*.



Bean *a* will be linked with *b2*, because of the definition order between *b2* and *b*.



An exception will be thrown by Spring which cannot choose between *b* and *b2*.

3

Q 10

**Given the following xml code snippet :**

```
<bean id="jdbcTemplate" class="org.springframework.jdbc.core.JdbcTemplate"
```

```

scope="singleton">
  <property name="dataSource" ref="dataSource"/>
</bean>
<bean id="bookDao" class="BookDAO">
  <property name="jdbcTemplate" ref="jdbcTemplate"/>
  ...
</bean>
<bean id="authorDao" class="BookDAO">
  <property name="jdbcTemplate" ref="jdbcTemplate"/>
  ...
</bean>
<bean id="publisherDao" class="BookDAO">
  <property name="jdbcTemplate" ref="jdbcTemplate"/>
  ...
</bean>

```

We will have to change the jdbcTemplate bean's scope to *prototype* if we work in a multithreaded environment.

(dataSource bean is e.g. of the BasicDataSource type (from the Jakarta Commons DBCP)).

|                                  |       |
|----------------------------------|-------|
|                                  |       |
| <input checked="" type="radio"/> | True  |
| <input type="radio"/>            | False |

2

Q 11.

Spring JDBC DAO handles for us tedious low-level JDBC details like opening/closing connections and catching exceptions associated with them.



True



False

1

Q 12.

```
public class TestClass {  
    private String name;  
  
    public TestClass(String name) {  
        this.name = name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public void printName() {  
        System.out.println(name);  
    }  
}
```

and

...

```
<bean id="testBean" class="TestClass">  
    <constructor-arg>  
        <value>Constructor</value>
```

```
</constructor-arg>
<property name="name">
  <value>Setter</value>
</property>
</bean>
```

...

What must be done if we want the `printName()` method of the `TestClass` to print **Constructor**?

☐ Nothing has to be done. It already prints **Constructor**.

☐ Delete `<property>` tag from the xml file.

☐ Change the order of the `<property>` and `<constructor-arg>` tags.

2

Q 13.

Consider the following snippet:

```
<bean id="topLevelBean" class="Foo">
  <property name="innerFoo">
    <bean class="Foo"/>
  </property>
</bean>
```

Assume that `Foo` class implements the `DisposableBean` interface and that `topLevelBean` is a top level bean indeed.

During the Spring container shutdown `destroy()` method will be called...

 0 times.

 1 times.

 2 times.

 3 times.

2

**Q 14.**

**Consider the following code:**

**01: package foo;**

**02: ...**

**03: public class Bar {**

**04:**

**05: private String test;**

**06:**

**07: public static void main(String[] args) {**

**08:   BeanFactory bf = new XmlBeanFactory(**

**09:     new FileSystemResource("mapper.xml"));**

**10:   bf.getBean("bar");**

**11: }**

**12:**

```

13: public Bar() {
14:     System.out.println("Bar()");
15: }
16:
17: public Bar init() {
18:     System.out.println("init()");
19:     return this;
20: }
21:
22: public void setTest(String test) {
23:     this.test = test;
24:     System.out.println("setTest(String test)");
25: }
26: }

```

Bar bean is configured with following entry:

```

<bean id="bar" class="foo.Bar" init-method="init" >
  <property name="test" value="test" />
</bean>

```

What will be the output of execution of the main() method?

☐ Bar()  
setTest(String test)

☐ init()

☐ Bar()  
☐ setTest(String test)  
☐ init()



☐ init()  
☐ setTest(String test)

☐ Bar()  
☐ init()  
☐ setTest(String test)

3

Q 15.

**To define a bean to be created using an instance static method you should write the following configuration:**

**<bean id="myBean" factory-bean="myFactory" />**

☐ True

☐ False

2

Q 16.

```
package com;
public class Bar {
    public static void main(String[] args) {
        BeanFactory bf = new XmlBeanFactory(
            new FileSystemResource("mapper.xml"));
        bf.getBean("bar");
        bf.getBean("bar");
        bf.getBean("bar");
    }
    public Bar() {
        System.out.println("new Bar");
    }
}
```

Bar bean is configured with following entry:

```
<bean id="bar" class="foo.Bar"/>
```

What will be the output of execution of the main() method?

How many times "new Bar" will be printed?

☐ 1

☐ 3

☐ 0

☐ Spring container throws an exception.

//////////SSSSPRING DUMPS//////////

## 1. VALID SPRING BEAN DEFINING

ID,CLASS

option a,d

## 2.STANDARD JAVA INTERFACE

A. SERIALIZE

B.LONG

C.SYSTEM

D.COLLECTION

SERIALIZE

## 3. BENIFITS OF DI or ADVANTAGES OF IOC

LOOSE COUPLING,EASY TESTING

## 4.WHICH OF THE BEAN SCOPE ATTRIBUTE THAT FORCES SPRING TO PRODUCE A NEW BEAN INSTANCE EACH TIME?

PROTOTYPE

## 5.TO ENABLE PATTERN AUTOWIRE?

AUTOWIRE\_PATTERN="TRUE"

## 6. SET AUTOWIRE IN A BEAN WHAT ATTRIBUTE DO YOU ADD?

AUTOWIRE="ON"

Default autowire in a bean what attribute do you add?

autowire="off"

7. NOT ADDRESSED BY SPRING JDBC?

if in question DataSourceAccess, then answer DataSourceAccess

otherwise

DAO

8. TO USE JAVA.UTIL.LIST, WHAT IS THE SPRING COLLECTION ELEMENT THAT U NEED?

<list>

9. TO USE BEAN DEFINATION TEMPLATE, YOU SHOULD ADD WHAT ATTRIBUTE IN THE BEAN?

A. ABSTRACT="TRUE"

B. INTERFACE="TRUE"

C. PUBLIC ="TRUE"

A

10. USER MANAGED TRANSACTION HAS TO BE COMMITTED EXPLICITLY?

SET AUTOCOMMIT(FALSE)

## 11. PROGRAMatic TRANSACTION

C

## 12. Spring INHERITANCE

to share common values, properties or configurations.

## 13. SPRING RESOLVE

AT RUNTIME

## 14. JDBC DAO

@REPOSITORY

## 15. JAVA BASED WHAT WE USE

@CONFIGURATION

## 16. HOLLYWOOD

DONT CALL US WE WILL CALL YOU

## 17. modes of AUTOWIRING

FIVE

types of autowire

4

18. MINIMIZE XML config file

AUTOWIRING

19. EACH REQUIREMENT

PROTOTYPE

20. CONSTRUCTOR IN PROGRAM

CONSTRUCTOR TYPE

21. SCOPE OF BEAN

SESSION, REQUEST, PROTOTYPE, SINGLETON, GLOBAL SESSION

22. HOW DO YOU DECLARE MVC ANNOTATION IN BEAN IN XML FILE?

<MVC: ANNOTATION-CONFIG/>

23. ALTERNATIVE OF @AUTOWIRED

USING AOP

24. USE OF DEPENDENCY INJECTION

## A & C (LOOSE COUPLING AND EASY TESTING)

### 25. NON INVASIVE

#### AUTOWIRED

26. Which of the following beans isn't significant for the WebApplicationContext?

1. Locale resolver.
2. Handler exception resolver.
3. Multipart file.
4. None of these.
5. Resolver.

4

27. Which of the following statements is true for @RequestMapping annotation?

1. Can be applied to the controller classes.
2. Can be applied to the controller methods.
3. None of these.
4. Used to map web requests onto specific handler methods.
5. Used to map web requests onto specific handler classes.

3

if multiple ans then 4,5

28. which annotation is used for setter method?

`@autowired`

29. SOME PROGRAM

`@AUTOWIRED SOMETHING OBJECT`

30. AUTOWIRING IS AN ALTERNATIVE TO

something with aop

31.ANNOTATION FOR CONFIGURATION

`@CONFIGURATION`

32. BASIC PACKAGES

`org.springframework.context.ApplicationContext;`

`org.springframework.context.support.ClassPathXMLApplicationContext;`

or

`//     ORG.SPRINGFRAMEWORK CONTEXT AND  
ORG.SPRINGFRAMEWORK.APPLICATION`

33. CODE IN QUESTION regarding HelloWorld



OPTION WITH PROTOTYPE

(MAYB OPTION B)

34. PROGRAMTIC TRANSACTION MANAGEment (with the help of normal programming)

Declarative TRANSACTION MANAGEMENT(AOP)

35. JAVA-BASED CONFIGURATION

a) <Context: annotation-config>

b) @Configuration public class HelloWorldConfig{...}

c) @Autowired

d) ApplicationContext context= new  
ClassPathXMLApplicationContext("Bean.XML")

Ans: b)@Configuration public class HelloWorldConfig{...}

36. TRADITIONAL TRANSACTION MANAGEMENT

USING TEMPLATE CLASS

37. GENERATOR GENERATOR....PROGRAM

LOOSE COUPLING

TIGHT COUPLING

DECOUPLING

if new opearator used--> tight coupling

if this. used--> loose coupling

38. DAO ACCESS

@REPOSITORY

39. MYBEAN= NULL

A. TRUE

B. FALSE

C. COMPILATION ERROR

D. NULL

D.NULL

40. program with spell checker

SETTER DI

Constructor DI

Setter DI

\*41.what does @ Transactional annotation mean

Transaction = true (option b)

42. current bean for request scope

43. I am object A

null

```
<bean id="helloWorld" class="com.tutorialspoint.HelloWorldId"
scope="prototype"></bean>
```

44. class String MyBeans

```
{
    if(MyBeans.equals(MyBean)
        sop(true);
    else
        sop(false)
}
```

- A. TRUE
- B. FALSE
- C. NULL POINTER
- D. CLASS CAST

c

45. types of autowire-4

diff modes of autowiring=5

46. set map id class

id class

47. difference between autowire and autowiring

48. @ autodiscovery

need to eliminate <property> or <constructor tag> (option a)

49. autowire->

50. autowire+auto discovery

-> reduce java code

51. Spellchecker

-> setterDI

52. INNER BEAN?

-> DEFINED INSIDE CONSTRUCTOR ARG OR PROPERTY TAG

```
53. public class TextEditor {  
    private SpellChecker spellChecker;  
  
    public TextEditor(SpellChecker spellChecker) {  
        this.spellChecker = spellChecker;  
    }  
}
```

Which DI?

a.CONSTRUCTOR DI

b.Getter DI

c. Setter DI

d. NOne

c

54. WHAT BASIC PACKAGES ARE IMPORTED IN JAVA PROG IN SPRING

`org.springframework.context.ApplicationContext;`

`org.springframework.context.support.ClassPathXMLApplicationContext;`

55. Mandatory attribute for bean

id & class

56.what mandatory attribute does aop include?

concern

float

advice

pointcut

concern mayb

57.....object myObject;

`if(myObject==null)`

`sop(true)`

`else`

`(false)`

`true`

false

null

compilation error

null(mayb)

58. order of inner bean

ans --><constructor-arg> or <property>

59. programmatic transaction management

a. pojo based model

d. aop

60. auto-discovery

1st option eliminate<property>

61. mvc question

<mvc:annotation-driven>

62.abstract layer, init layer, dependency check

long answer

63. default bean and its functionality

singleton and single ioc container

64. @autowired is used in

constructor destructor option

65.What is the purpose of Spring Inheritance?

Ans: to share common values, properties or configurations.

66.How to handle Named Query Parameters?

Ans: NamedParameterJdbcTemplate

67. In the bean configuration file for Hibernate (i.e., beans-hibernate.xml), you have to declare a HibernateTransactionManager instance for this application and enable declarative transaction via:-

- a) tx:annotation
- b) tx:annotationdriven
- c) tx:annotation-driven
- d) none of the mentioned

Ans: c) tx:annotation-driven

68. Which of the below option is not a valid collection type?

- a) list
- b)Set
- c)props

d)property

Ans: property

-----

Q 1 - Which of the following is correct about dependency injection?

A - It helps in decoupling application objects from each other.

B - It helps in deciding the dependencies of objects.

C - It stores objects states in database.

D - It stores object states in file system.

Answer : A

Q 2 - Which are the correct implementation classes of ApplicationContext?

A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext

B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext

C - AdvancedApplicationContext, FileApplicationContext



D - FileSystemApplicationContext, ClassPathApplicationContext

Answer : A

Q 3 - What is true about <set> collection configuration elements?

A - This helps in wiring a list of values, allowing duplicates.

B - This helps in wiring a list of values but without any duplicates.

C - This can be used to inject a collection of name-value pairs where name and value can be of any type.

D - This can be used to inject a collection of name-value pairs where the name and value are both Strings.

Answer : B

Q 4 - What is constructor mode of autowiring?

A - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

B - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

C - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in configuration file.

D - Similar to byType, but type applies to constructor arguments. If there is not exactly one bean of the constructor argument type in the container, a fatal error is raised.

Answer : D

Q 5 - What is true about @Autowired annotation?

A - The @Autowired annotation can be used to autowire bean on the setter method.

B - This annotation provides more fine-grained control over where and how autowiring should be accomplished.

C - The @Autowired annotation can be used to autowire bean on the methods with arbitrary names and/or multiple arguments.

D - All of above.

Answer : B

Q 6 - What are the different points where weaving can be applied?

A - Compile time, load time

B - Compile time, run time

C - Run time

D - Compile time, load Time, Run time

Answer : D

Q 7 - What is Spring MVC framework?

A - Spring MVC framework is Model-Value-Class architecture and used to bind model data with values.

B - The Spring web MVC framework provides model-view-controller architecture and ready components that can be used to develop flexible and loosely coupled web applications.

C - Spring MVC framework is used for Transaction management for Web Applications.

D - Spring MVC framework is used for AOP for Web Applications.

Answer : B

Q 8 - By default a bean is lazily loaded.

A - True

B - False

Answer : B

Q 9 - What is the scope of bean in portlet context?

A - session

B - global-session

C - prototype

D - request

Answer : B

Q 10 - Which of the following class can be used to call Stored Procedures in spring?

A - SPHelper

B - JdbcTemplateCall

C - JdbcTemplate

D - SimpleJdbcCall

Answer : D

=====

=====

Q 1 - What is ContextRefreshedEvent event?

A - This event is published when the Servlet Context is either initialized or refreshed.

B - This event is published when the HTTP Request is received.

C - This event is published when the HTTP Response is returned.

D - This event is published when the ApplicationContext is either initialized or refreshed.

Answer:D

Q 2 - What is Introduction?

A - An introduction represents a point in your application where you can plug-in AOP aspect.

B - This is used to inject values in objects.

C - This is not invoked during program execution by Spring AOP framework.

D - An introduction allows you to add new methods or attributes to existing classes.

Answer:D

Q 3 - What is @Controller annotation?

A - The @Controller annotation indicates that a particular class serves the role of a controller.

B - The @Controller annotation indicates how to control the transaction management.

C - The @Controller annotation indicates how to control the dependency injection.

D - The @Controller annotation indicates how to control the aspect programming.

Answer:A

Q 4 - What is prototype scope?

A - This scopes a single bean definition to have any number of object instances.

B - This scopes the bean definition to a single instance per HTTP Request.

C - This scopes the bean definition to a single instance per HTTP Session.

D - This scopes the bean definition to a single instance per HTTP Application/ Global session.

Answer:A

Q 5 - What is true about <props> collection configuration elements?

A - This helps in wiring a list of values, allowing duplicates.

B - This helps in wiring a list of values but without any duplicates.

C - This can be used to inject a collection of name-value pairs where name and value can be of any type.

D - This can be used to inject a collection of name-value pairs where the name and value are both Strings

Answer:A

Q 6 - How to handle shut down of IoC container?

A - Using shutdownHook()

B - Using shutdownHandler()

C - Using registerHook()

D - Using registerShutdownHook()

Answer:D

Q 7 - By default a bean is eagerly loaded.

A - False

B - True

Answer:B

Q 8 - What is true about BeanPostProcessor?

A - It is a concrete class.

B - It is an interface.

C - It is an abstract class.

D - None of the above.

Answer:B

Q 9 - How can you inject Java Collection in Spring?

A - Using list, set, map or props tag.

B - Using list, set, map or collection tag.

C - Using list, set, props or collection tag.

D - Using list, collection, map or props tag.

Answer:A

Q 10 - Which class is used to map a database row to a java object in spring?



A - ResultSet

B - RowMapper

C - RowSetMapper

D - ResultSetMapper

Answer:B

Q 11 - How bean life cycle can be controlled?

A - Using init() only

B - Using InitializingBean class only

C - Using DisposableBean class only

D - Using All of above

Answer:D

Q 12 - What is the scope of bean in portlet context?

A - session

B - global-session

C - prototype

D - request

Answer:B

Q 13 - What is ContextStoppedEvent event?

A - This event is published when the Servlet Context is either initialized or refreshed.

B - This event is published when the ApplicationContext is stopped using the stop() method on the ConfigurableApplicationContext interface.

C - This event is published when the HTTP Request is received.

D - This event is published when the HTTP Response is returned.

Answer:B

Q 14 - Which ORM Spring supports ?

A - Hibernate

B - iBatis

C - JPA

D - All of above.

E - None of above.

Answer:D

Q 15 - What is true about <map> collection configuration elements?

A - This helps in wiring a list of values, allowing duplicates.

B - This helps in wiring a list of values but without any duplicates.

C - This can be used to inject a collection of name-value pairs where name and value can be of any type.

D - This tag is not supported.

Answer:C

Q 16 - Core container has AOP as one of its module.

A - True

B - False

Answer:B

Q 17-Can you inject null and empty string values in Spring?

A - Yes

B - No

Answer:A

Q 18 - What stands true for spring framework?

A - Spring framework is a light weight framework.

B - Spring framework is a heavy weight framework.

Answer:A

Q 19 - Which class acts as IoC Container?

A - ServletContext

B - DispatcherServlet

C - ApplicationContext

D - None of the above

Answer:C

Q 20 - What is the scope of stateless bean?

A - global-session

B - singleton

C - prototype

D - request

Answer:B

=====

1. Beans defined in spring framework are by default?

---->singleton

2. which of the following is not a spring module

--->HTML/JSP

3. spring is a?

---->an open source network

4. which of the following is true

--->ApplicationContext extends BeanFactory

5. controller in spring is

---> an interface

6. what are different autowire types?

---->byName, byType,constructor and autodetect

7. types of bean injections?

-->consttuctor and setter

8. which are different points where weaving can be applied?

---> compile time,classload time,runtime

9. IOC or dependent injection is a

----> Design pattern

10. which exception class is related to all exceptions that are thrown in spring applications?

--->DataAccessException

=====

1. what is stand for spring MVC?

-->Model view controller

2.what is the type of proxy in spring framework?

-->2(sttaic,dynamic)

3.spring MVC is a request-based framework?

--->YES

4.what is the stand of AOP?

--->aspect oriented programming

5.how many types of dynamic proxy is available in spring?

--->2

6.how many types of injection method in spring?

--->2

=====

1.how to auto-inject into a field a bean by its name?

---> by using both the @autowired and the @qualifier spring annotations

2.given the following spring configuration file,what is correct answer:

```
<bean class="com.spring.service.MyServiceImpl">
```

```
<property name="repository" ref="jpaDao"/>
```

```
</bean>
```

```
<bean class="com.spring.repository.JpaDao"/>
```

Answer:-->the second declared bean JpaDao is missing an Id must be named as jpaDao

3.how could you externalize constants from a spring configuration file or aspring annotation into a .properties file?

--->By using the context:property-placeholder tag

4.what are the main advantages of using interfaces when describing bussiness services?

---->Mocking or stubbing the service

5.how is named the bean that is defined in the following configuration class?

---->client repository

6.what statement is not correct in live environment?

--->All the above

7.what one is not the reight affirmations about the @postconstruct, @resource and the @predestroy notations?

---->the spring framework embedded those annotation

8. what is/are typically case(s) where you usually need to manually instanciatedan applicaton context?

---->in a standalone application started with a main method

=====  
=====

1. Spring-WS provides various abstract endpoint classes for you to process the request.

a) org.springframework.ws.server.endpoint - ANS

b) org.springframework.ws.server

- c) org.springframework.\*
- d) none of the mentioned

2. Which method is used to gracefully shutdown all the bean processes after closing the spring container?

- Ans A. shutdownHook
- B. destroy method
- C. none of the mentioned
- D. all of the mentioned

3. Which one is the default scope of the beans?

- A. Prototype
- B. Session
- C. Request

Ans- D. Singleton

4. Declaring bean form object properties can be done using:-

- Ans - A. PropertyPathFactoryBean
- B. util:constant
- C. None of the mentioned
- D. All of the mentioned



5. What are the ways to create custom Property Editors?

Ans A. implement PropertyEditor interface

Ans B. extend PropertyEditorSupport Class

C. none of the mentioned

- D. all of the mentioned

6. Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.

A.true Ans

B.false

7. Which of the among is not a special bean in WebApplicationContext?

a.Locale resolver

b.Theme resolver

c.multipart file

d.resolver

e.Handler exception resolver

f.Handler mappings

g. Non of the above -Ans

8. How the Spring DispatcherServlet supports for returning the last-modification-date, as specified by the Servlet API ?

- a.By implementing the LastModified interface --Ans
- b.By implementing the LastSessionModified interface
- c.By implementing the LastDateModified interface
- d.By implementing the LastModifiedDate interface

9. @RequestMapping annotation is used to map a HTTP request method (GET or POST) to a specific class or method in the controller which will handle the respective request?

- 1. false
- 2. true --Ans

10. Which of the following exception classes is related to all the exceptions thrown in spring applications?

- 1. DataAccessException.---Ans
- 2. ArrayIndexOutOfBounds.
- 3. NullPointerException.
- 4. SpringException.
- 5. None of these.

11. Which of the following is true for Controllers in Spring?

- 1. An abstract class.
- 2. An interface.---Ans

3. Concrete class.
4. Final class.
5. None of these.

12. Annotations work as metadata for classes and Spring uses annotations as an alternative to XML for declarative configuration. Is it true or false?

1. false
2. true----Ans

13. Which of the following specify the points where weaving can be applied?

1. Compile Time , Classload Time , load time.
2. Compile Time , Classload Time , Runtime-----Ans
3. Compile Time and Runtime.
4. Compile Time and Classload Time.
5. None of these.

14. Which of the following layers the @Controller annotation is used in?

1. Business layer
2. Service layer.
3. Presentation layer..----Ans
4. Session layer.
5. None of these.

15. Which Of The Following Is Not A Valid Difference Between @Inject And @Autowired Annotations?

1. @Inject annotation is part of Java EE 7 s Context and Dependency Injection framework while @Autowired is the Spring Frameworks own implementation
2. @Autowired is a legacy annotation whereas @Inject is its latest version..-----Ans
3. There is no difference and both can be used interchangeably.
4. None of these.

16.Which of the following correctly mentions the Autowire types?

1. None of these.
2. byName , byMethod, constructor and autodetect.
3. byName , byType, destructor and autodetect.
4. byName , byType, constructor and autodetect.-----Ans
5. byName , byType, constructor and autocorrect.-----

17. Which of the following statements is true?

1. BeanFactory extends ApplicationContext.
2. ApplicationContext extends BeanFactory.-----Ans
3. ApplicationContext implements BeanFactory.
4. BeanFactory implements ApplicationContext
5. None of these.

18. Which of the following is advice supported by Aspect Annotation?

- A. @Before B. @After C. @AfterReturning
- D. All of the mentioned---Ans

19. Which advice is executed only when join point returns or throws an exception?

- A. @Before
- B. @After
- C. @AfterReturning -----Ans
- D. @AfterThrowing

20. Special compiler used during weaving

- A. jvm B. gcc
- C. ajc -----Ans
- D. none of the mentioned

21. How to inject Spring bean into domain objects

- A. AOP
- B. XML C. AspectJ
- D. Java Based-----Ans

22. Method used to process bean before initialization callback

- A. scope
- B. postProcessAfterInitialization() -----Ans
- C. postProcessBeforeInitialization()
- D. it s own constructor

23. Which attribute is used to set the scope of the bean?

- A. setScope B. scope-----Ans
- C. getScope D. none of the mentioned

24. Which attribute is used to specify classname of the bean?

- A. name B. id C. class -----Ans
- D. constructor-args

25. Which tag is also allowed by static field?

- A. util:constant -----Ans
- B. list C. set D. constructor-args

1. Which is correct for Spring Framework?

- ☐ a. Spring framework is light-weight solution.
- ☐ b. Spring framework is heavy-weight solution.

2. AOP is the part of Core Container in Spring Framework.

- ☐ a. True
- ☐ b. False

3. Expression Language is the part of Core Container in Spring Framework.

- ☐ a. True
- ☐ b. False

4. Which is the part of Data Access layer in Spring Framework?

- ☐ a. Aspects



b. JMS

5. Can we integrate spring with struts?



a. Yes

- ☐ b. No

6. How to get the object of DAO in spring framework?

- ☐ a. Using new keyword



b. Using Spring Dependency injection

7. In which spring version, java 5 features has been introduced?

☐ a. Spring 2.5

☒ b. Spring 3.0

8. In which spring version has Spring Expression Language been supported?

☐ a. Spring 2.5

☐ b. Spring 3.0

9. Which class represents the IoC container?

☐ a. ApplicationContext

☐ b. ServletContext

10. How to use idref in spring framework?

☐ a. With setter method only

☒ b. With setter method and constructor argument both

11. How to use <ref> in spring framework?

☒ a. <ref> is used with bean id.

☐ b. <ref> is used with string values.

12. How to define Inner bean in spring?

☐ a. Use <property/> only

☐ b. Use <property/> or <constructor-arg/>.

13. Which property is replaced by p-namespace in spring?

☐ a. <property/>



- ☐ b. <constructor-arg>

14. Which property is replaced by c-namespace in spring?

- ☐ a. <property/>
- ☐ b. <constructor-arg>

15. What is the scope of a bean by default in spring?

- ☐ a. Prototype scope
- ☐ b. Singleton scope

16. By default a bean is eagerly initialized in spring?

- ☐ a. True
- ☐ b. False

17. By default a bean is not autowired?

- ☐ a. True
- ☐ b. False

18. What is the role of ApplicationContextAware in spring?

- ☐ a. Dependency injection is performed.
- ☐ b. Makes a bean aware to the container.

19. Can the class which spring container makes subclass be final?

- ☐ a. No
- ☐ b. Yes

20. What is the scope of stateless bean in spring?

- ☐ a. Singleton scope
- ☐ b. Prototype scope

21. What is the scope of statefull bean in spring?

- ☐ a. Singleton scope
- ☐ b. Prototype scope

22. In which spring scope is a single bean instance created per IoC container?

- ☐ a. Singleton scope
- ☐ b. Request scope

23. In which spring scope can any number of instances of bean be created?

- ☐ a. Request scope
- ☐ b. Prototype scope

24. When a bean has scope limited to only HTTP request, that is called

- ☐ a. Request scope
- ☐ b. Session scope

25. When a bean has scope limited to HTTP session, that is called

- ☐ a. Request scope
- ☐ b. Session scope

26. Thread scope bean has been introduced in spring from the version

☐ a. Spring 2.5

☐ b. Spring 3.0

27. Which bean scope is used in portlet context in spring?

☐ a. Session scope

☐ b. Global session scope

28. InitializingBean and DisposableBean is used to customize the nature of bean in spring.

☐ a. Yes

☐ b. No

29. How to control a bean lifecycle in spring?

☐ a. Using init() method only.

☐ b. Using InitializingBean and DisposableBean or init() method.

30. How to shutdown the Spring IoC container gracefully in non-web applications?

☐ a. Using shutdownNow()

☐ b. registerShutdownHook()

31. can we use index and type attribute together ?

- ☐ Yes
- ☐ No

Which of the following is not a module in Spring Framework ?

- ☐ DAO
- ☐ ORM
- ☐ MVC
- ☐ AOP
- ☐ IOC
- ☐ JEE
- ☐ Non of these

32.Which of the following statement is false ?

- ☐ The Corepackage is the most fundamental part of the framework and provides the IoC and Dependency Injection features
- ☐ TheDAO package provides a JDBC-abstraction layer that removes the need to do tedious JDBC coding and parsing of database-vendor specific error code
- ☐ The ORM package provides integration layers for popular object-relational mapping APIs, including JPA,JDO,Hibernate, and iBatis
- ☐ AOP takes out the direct dependency of crosscutting tasks from classes which, we can achieve through normal object oriented programming model. or example, method-interceptors and pointcuts to cleanly decouple code implementing functionality that
- ☐ Spring's Web package provides basic web-oriented integration features, such as multipart file-upload functionality .

☒ Spring's MVC package provides a Model-View-Controller implementation for web-application MVC package provides a Model-View-Controller implementation for web-application

☐ Non of these

33. Which of the following is false?

☐ A BeanFactory pretty much just instantiates and configures beans.

☐ An ApplicationContext also instantiates bean, and it provides the supporting infrastructure to enable lots of enterprise-specific features such as transactions and AOP, message resource handling (for use in internationalization), event propagation.

☐ Application-layer specific contexts such as the WebApplicationContext use in web application

☐ none of the above

34. A bean is simply an object that is instantiated, assembled and otherwise managed by a Spring IoC container.

☐ true

☐ false

35. Which of the following is actual representation of the Spring IoC container ?

☐ ApplicationContext

☐ XmlBeanFactory

☐ BeanFactory

☐ WebApplicationContext

36. How container load configuration metadata from a variety of external resources such as the local file system from the Java CLASSPATH ?

☐ ApplicationContext constructor

☐ Init()

- ☐ Context Contructor
- ☐ XmlClassPath Contructor

37.Which of following is correct way to load bean definitions from another file ?

- ☐ <bean id="bean1" class=".." import="something.xml"/>
- ☐ <resources name="something.xml" import ="true"/>
- ☒ <import resource="something.xml"/>
- ☐ all the above

```
package com.bullraider;  
  
public class TestBean {  
  
    private int year;  
  
    private String happy;  
  
    public TestBean( String happy,int year) {  
  
        this.year = year;  
  
        this.happy = happy;  
  
    }  
  
}
```

38.What is the correct way to write constructor injection?

- ☐ <bean name="testClass" class="com.bullraider.TestBean">  
<constructor-arg type="java.lang.String" value="Happy new year"/>  
<constructor-arg type="int" value="2015"/>  
</bean>
- ☐ <bean name="testClass" class="com.bullraider.TestBean">  
  
<constructor-arg index="0" value="Happy new year"/>  
  
<constructor-arg index="1" value="2015"/>

</bean>

☐ <bean name="testClass" class="com.bullraider.TestBean">

<constructor-arg index="1" value="Happy new year"/>

<constructor-arg index="0" value="2015"/>

</bean>

☐ <bean name="testClass" class="com.bullraider.TestBean">  
<constructor-arg type="int" value="2015"/>  
<constructor-arg type="java.lang.String" value="Happy new year"/>  
</bean>

☐ All the above

39.Which of the following are correct statements ?

☐ Form Controllers handle form interaction

☒ Multipart resolver to handle file uploads

☒ Dynamic binding of data to domain model

☒ Interaction with velocity,jsp, thymeleaf,pdf, Excel

☒ Sending email remote accessing via Soap,Burlap

40.What do you mean by 'bean' ?

☐ component

☒ object

- ☐ class
- ☐ container
- ☐ xml file

41. What are responsibilities included for bean factory do ?

- ☐ Instantiating or sourcing application objects
- ☒ configuring such objects
- ☒ Assembling the dependencies between these objects
- ☐ Manage lifecycle of spring

42. What are the responsibilities of XmlBeanFactory included ?

- ☐ This implementation allows you to express the objects that compose your application
- ☒ Resolve inter-dependencies between such objects, in terms of XML
- ☒ The XmlBeanFactory takes this XML configuration metadata and uses it to create a fully configured system or application
- ☐ Non of these

43. Which of the following are the correct way to instantiating a Spring IoC container ?

- ☐ `ApplicationContext context = new ClassPathXmlApplicationContext(new String[] {"beans-definitions.xml"})`

`BeanFactory factory = context;`

- ☒ `XmlBeanFactory beanFactory = new XmlBeanFactory(new ClassPathResource("beans-definitions.xml"));`

- ☒ `BeanFactory fac = new ClassPathXmlApplicationContext("beans-definitions.xml");`



☒ Resource res=new Classpathresource("beans-definitions.xml");

BeanFactory fac=new XmlBeanFactory(res);

44.What are the correct ways to instantiate bean?

- ☐ Using default (empty) constructor
- ☐ Instantiation using a static factory method
- ☐ Instantiation using an instance factory method
- ☐ Using the container
- ☐ All these above are true

45.Which statements are true about IOC(Inversion of Control) ?

☒ Decoupling of the execution of a certain task from implementation.

☒ Each module can focus on what it is designed for.

☒ Replacing modules should not affect to other modules.

- ☐ Modules should observe about what other systems do .
- ☐ All the above

46.What do you mean by inner bean?

☐ A <bean/> element inside the <bean/> elements is used to define a so-called inner bean.

☐ A <bean/> element inside the <property/> elements is used to define a so-called inner bean.

☐ A <bean/> element inside the <constructor-arg/> elements is used to define a so-called inner bean.

☒ A <bean/> element inside the <property/> or <constructor-arg/> elements is used to define a so-called inner bean.

Q - How before advice works?

[A - Run advice before a class loads.](#)

B - Run advice before a method execution.

[C - Run advice before http response is to be returned.](#)

[D - Run advice before http request is to be processed.](#)

**Answer : B**

**Explanation**

before advice runs before a method execution.

Q - What is Introduction?

[A - An introduction represents a point in your application where you can plug-in AOP aspect.](#)

[B - This is used to inject values in objects.](#)

[C - This is not invoked during program execution by Spring AOP framework.](#)

[D - An introduction allows you to add new methods or attributes to existing classes.](#)

**Answer : D**

Q - How to use ref keyword in beans.

xml?

[A - Using setter method only.](#)

[B - Using constructor argument only.](#)

C - Using setter method and constructor argument both.

[D - None of the above.](#)

**Answer : C**

**Explanation**

ref is used to reference already defined bean and it can

Q - How can you inject Java Collection in Spring?

A - Using list, set, map or props tag.

[B - Using list, set, map or collection tag.](#)

[C - Using list, set, props or collection tag.](#)

[D - Using list, collection, map or props tag.](#)

**Answer : A**

**Explanation**

Spring supports list, set, map or props tags to inject java collections.

Q - What is ContextStartedEvent event?

[A - This event is published when the Servlet Context is either initialized or refreshed.](#)

[B - This event is published when the HTTP Request is received.](#)

C - This event is published when the ApplicationContext is started using the start() method on the ConfigurableApplicationContext interface.

[D - This event is published when the HTTP Response is returned.](#)

**Answer : C**

**Explanation**

ContextStartedEvent event is published when the ApplicationContext is started using the start() method on the ConfigurableApplicationContext interface.

Q - How do you turn on annotation wiring?

[A - Add <annotation-context:config /> to bean configuration.](#)

[B - Add <annotation-config /> to bean configuration.](#)

[C - Add <annotation-context-config /> to bean configuration.](#)

D - Add <context:annotation-config/> to bean configuration.

**Answer : D**

**Explanation**

Add <context:annotation-config/> to bean configuration to turn on annotation wiring.

Q - What is Join point?

[A - This represents a point in your application which joins two objects.](#)

[B - This represents a point in your object where you join values.](#)

[C - This represents a point in your object where you join injected values.](#)

D - This represents a point in your application where you can plug-in AOP aspect.

**Answer : D**

**Explanation**

Join point represents a point in your application where you can plug-in AOP aspect.

Q - What are the different points where weaving can be applied?

[A - Compile time, load time](#)

[B - Compile time, run time](#)

[C - Run time](#)

D - Compile time, load Time, Run time

**Answer : D**

**Explanation**

Weaving can be applied at Compile time, load Time and Run time.

Q - If a bean is scoped to HTTP session, scope is

[A - global-session](#)

B - session

[C - prototype](#)

[D - request](#)

**Answer : B**

**Explanation**

If a bean is scoped to HTTP session, scope is session.

Q - Which of the following database is not supported using jdbcTemplate?

[A - MySql](#)

[B - PostgresSql](#)

C - NoSql

[D - Oracle](#)

**Answer : C**

**Explanation**

NoSql is not supported using jdbcTemplate.

Q - What is bean scope?

A - Bean scope forces Spring to produce a new bean instance as per the scope defined.

B - Bean scope defines the accessibility of bean in a java class.

C - Bean scope defines the accessibility of bean in a java package.

D - Bean scope defines the accessibility of bean in a web application.

**Answer : A**

**Explanation**

Bean scope instructs Spring Container to produce a new bean instance as per the scope defined.

Q - What is bean autowiring?

A - Autowiring lets Spring resolve collaborators (other beans) for your bean by inspecting the contents of the BeanFactory without using <constructor-arg> and <property> elements.

B - Autowiring injects values in spring beans.

C - Autowiring injects one bean into another.

D - Autowiring helps in wiring a list of values, allowing duplicates.

Q - Which of the following aspect implementation spring supports?

A - XML Schema based aspect implementation

B - @AspectJ based aspect implementation

C - Both of above.

D - None of above.

**Answer : C**

**Explanation**

Spring supports both XML Schema based and @AspectJ based aspect implementation.

Q - What is byType mode of autowiring?

A - Default setting which means no autowiring and you should use explicit bean reference for wiring.

B - Autowiring by property name. Spring tries to match and wire its properties with the beans defined by the same names in the configuration file.

C - Spring first tries to wire using autowire by constructor, if it does not work, Spring tries to autowire by byType.

D - Autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in configuration file.

**Answer : D**

### **Explanation**

byType mode of autowiring is autowiring by property type. Spring tries to match and wire a property if its type matches with exactly one of the beans name in configuration file.

Q - What is true about cross-cutting concerns?

A - The functions that span multiple points of an application are called cross cutting concerns.

B - Cross-cutting concerns are conceptually separate from the application's business logic.

C - Logging is one of the examples of cross cutting concerns.

D - All of the above.

**Answer : A**

### **Explanation**

Cross-cutting concerns are the functions that span multiple points of an application.

Q - What is Pointcut?

A - This represents a point in your application where you can plug-in AOP aspect.

B - This is a set of one or more joinpoints where an advice should be executed.

C - This is used to inject values in objects.

D - This is invoked during program execution by Spring AOP framework.

**Answer : B**

### **Explanation**

Pointcut is a set of one or more joinpoints where an advice should be executed.

Q - Core container has AOP as one of its module.

A - True

B - False

**Answer : B**

**Explanation**

AOP is not the part of spring core container.

Q - Which are the IoC containers in Spring?

A - BeanFactory, ApplicationContext

[B - BeanFactory, ApplicationContext, locContextFactory](#)

[C - BeanFactory, BeanContext, locContextFactory](#)

[D - BeanFactory, ApplicationContext, BeanContext](#)

**Answer : A**

**Explanation**

BeanFactory, ApplicationContext classes acts as IoC containers in Spring.

Q - Which of the following stands true for spring beans?

[A - Spring beans are managed by the Spring IoC container.](#)

[B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.](#)

[C - Spring beans are simple POJOs.](#)

D - All of the above.

**Answer : D**

**Explanation**

All of the above statements are correct.

Q - How to get object of a service in spring framework?

[A - Using new keyword](#)

B - Using dependency injection

**Answer : B**

**Explanation**

In spring, dependency injection is used to get any object.

The string "api/route" in Express is an example of a...

- A. Middleware
- B .Route
- C. Error Handler

ANS : B

2. Which library allows the use of the "require" keyword in front-end code?

- A. guid
- B. Browserify
- C. React
- D. jQuery

ANS : B

3. The \_\_\_\_\_ attribute of a React form is called when the main action button inside the form is pressed.

- A. onSubmit
- B. onChange
- C. onComplete

ANS : A

4. Which of the following are automation tools?

- A. Gulp
- B. JSX
- C. Express

ANS : A

5. Which of the following is not a method of a dispatcher?



- A. register
- B. vector
- C. dispatch

ANS : B

6. What React function is used to define a new element tag?

- A. renderComponent
- B. createClass
- C. setInnerHTML

ANS : B

7. Isomorphic means to focus exclusively on an app's back-end.

- A. True
- B. False

ANS : B

8. React's abstraction of the document object model is often called...

- A. Virtual DOM
- B. Meta DOM
- C. Virtualized Model

ANS : A

9. True or false : The only thing that can change the data inside a store is the store itself.

- A. false
- B. true

ANS : B

10.

The Gulp task which takes a stream and outputs a file is...

- A. src
- B. task
- C. dest

ANS : C

11. The best way to detect when a user has completed a form is

- A. to handle the onClick event on the form's submit element
- B. to handle the onBlur event of the last form element
- C. to handle the form's onSubmit event
- D. to accumulate the onChange events for each form element

ANS : C

12. React is

- A A library that represents asynchronous data streams with observables
- B A comprehensive client-side application framework
- C An optimizing compiler for JavaScript
- D A client-side library providing rendering and event handling

ANS : D

13. Routing can be added to a React application by

- A. adding an HTML5 or hash fragment based routing library
- B. rendering the application on the server using react-page
- C. handling the window.onLocationChanged event
- D. using the React.route function

ANS : A

14. Behavior is shared between components using

A. propTypes

B. state

C. mixins

D. props

ANS : C

15. An advantage of React is

A. jQuery syntax

B. static type checking

C. speed

D. easy integration with other JavaScript libraries

ANS : C

16. The JSX pre-processor is packaged as a

A. dynamic link library

B. Nuget package

C. Gem

D. npm module

ANS : D

17. The just-in-time JSX transformer is good for

A. integrating with JavaScript tools

B. simple development workflow

C. reusing behavior

D. performance

ANS : B

18. The function that creates React components is

- A. React.component.extend
- B. React.createComponent
- C. React.renderComponent
- D. React.createClass

ANS : D

19. How does React Router make URL params accessible to components?

- A. parseUrlParams
- B. State
- C. getUrlParams
- D. Props

ANS : D

20. Why is the Link component useful?

- A. It allows you to hyperlink to routes by name
- B. It enhances page load performance
- C. It removes invalid characters from the URL
- D. It provides faster navigation than hyperlinks

ANS : A

21. How do Flux stores find out about Flux actions?

- A. They query the server via AJAX
- B. They poll HTML5 localStorage
- C. They register a callback with the dispatcher.
- D. They call action creators directly

ANS : C

22. Where should mutable data for React components be stored?

- A. props
- B. reactData
- C. state
- D. React.cookie

ANS : C

23. What mechanism is recommended for passing data down to child components in React?

- A. props
- B. sendChildData
- C. getDefaultProps
- D. state

ANS : A

24. What is JSX?

- A. A language that looks like JavaScript that compiles to HTML.
- B. A styling language that compiles down to CSS.
- C. A language for querying RESTful web services.
- D. A language that looks like HTML that compiles down to JavaScript.

ANS : D

25. What function should all your JSX be placed in?

- A. render
- B. compile
- C. jsx

D. emit

ANS : A

Q 1 Which features does TypeScript support natively?

Support for standard JavaScript code

Classes and modules

Static typing

Interfaces

Constructors

F. All of the above

ANS : F

Q 2. Which of the following is not a type keyword in TypeScript?

float

bool

all are valid types in TypeScript

number

string

ANS : float

Q 3. Interfaces defined in TypeScript are included in the JavaScript that is generated?

True

False

ANS : B

Q 4. How do you extend an interface in TypeScript?

interface ITruckOptions extends IAutoOptions

interface ITruckOptions => IAutoOptions

interface ITruckOptions : IAutoOptions

interface ITruckOptions implements IAutoOptions

ANS : A

Q 5. Benefits offered by TypeScript include:

Code encapsulation

All of these

More maintainable code

Strongly-typed code

ANS :B

Q 6. How do you automatically generate fields in a TypeScript class?

Add the public keyword to a constructor parameter

None of these

Use the auto keyword

TypeScript doesn't support automatically generated fields

ANS : A

Q 7.Which TypeScript function will accept 2 parameters that must be numbers and return their product?

`var func = (x: number, y: number) => x * y;`

`var func = (x, y) => x * y;`

None of these

`var func = function(x, y) => x * y;`

All of these

ANS : A



Q 8. Which statement will allow your code to reference an internal module in a file named shapes.ts using TypeScript?

/// <reference="shapes.ts" />

/// <reference path="shapes.ts" />

import "shapes"

/// shapes.ts

ANS : C

Q 9. Which TypeScript statement will import an external module in a file named viewmodels.ts that is in the same folder as your code?

import vm = module(viewmodels);

import vm = module('../viewmodels');

import module('viewmodels');

import vm = module('viewmodels');

ANS : D

Q 10. Which TypeScript compiler parameter will emit JavaScript that is AMD compliant?

--AMD

--module AMD

Neither of these

ANS : B

Q 11.What keyword is used to apply all of the properties on one interface to another interface?

implements

extends

class

furtherers

references

ANS B

Q 12.What value will be output at the end of the following code block?

```
enum Color { Red = 5, Green = 10, Blue = 15 };
```

```
console.log(Color.Green);
```

2

Color.Green

1

Green

10

ANS : E

Q 13. Which of the following best defines a "class"?

Objects with no implementation details

Object with no properties or methods

JSON objects with methods

Collection of private, reusable functions

Template for creating objects

ANS : E

Q 14. What term describes the concept of treating an object as if it were a particular type, even if it weren't declared as that type?

Extension Typing

Duck Typing

Interface Typing

Generic Typing

Ruby Typing

ANS : B

Q 15.What access modifier limits the accessibility of a class member to its own class and subclasses?

export

internal

protected

private

sensitive

ANS : C

Q 16.What is the primary difference between a generic class and a non-generic class?

Generic classes do not accept constructor parameters.

Generic classes are always declared abstract.

Generic class may not implement interfaces.

Generic classes accept a type parameter.

Generic classes must inherit from a base class.

ANS : D

Q17.Which TypeScript compiler options lets you specify the location of the tsconfig.json file to be used for the compilation?

--project

--json

--outDir

--input

--source

ANS : A

Q 18.Which of the following symbols is used to denote that a function parameter is a rest parameter?

<>

>>>

!!

...

||

ANS : D

Q 19.Which of the following will correctly import the items exported by this export statement?

```
export { GetUser as GetLibraryUser, GetBook as GetLibraryBook };
```

```
import TypeScript from './library';
```

```
import namespace from './library';
```

```
import { GetLibraryUser, GetLibraryBook } from './library';
```

```
import { GetUser, GetBook } from './library';
```

```
import all from './library';
```

ANS : C

Q 6. How do you automatically generate fields in a TypeScript class?

- A.Add the public keyword to a constructor parameter
- B.None of these
- C.Use the auto keyword
- D.TypeScript doesn't support automatically generated fields

Copilot

The correct answer is D. TypeScript doesn't support automatically generated fields. Unlike some other languages, TypeScript does not provide built-in automatic field generation based on constructor parameters or any other keywords. Developers need to explicitly define class properties and their initialization within the class body.

5.To change the datatype from one type to another,we require

- A.TypeOperator
- B.TypeAssertion
- C.InferredTyping

ANS : B

6.For a variable,Without Specifying any data type is

- A.InferredTyping
- B.TypeOperator
- C.TypeAssertion

ANS : A

typescript file?

A.ts abc.ts

B.t abc.ts

C.tsc abc.ts

D.tst abc.ts

ANS : C

8.What will be output of : var a:string=47;

console.log("Value of a="+a);

A.Value of a=47

B.Value of a=0

C.Value of a=

D.None of the above

C

9.Internal Modules are known as namespaces in Typescript.

A.True

B.False

Ans : A

11.function fun1(...params) {

    console.log(params.length);

}

fun1();

fun1(5);

fun1(5, 6, 7);

Which concept resemble the above program

A.String Interpolation

B.Tagged Templates

C.Spread Operator/Rest Parameters

D.Object Destructuring

ANS : C

24) What is async/await?

A.New syntax to write JavaScript promises

B.New syntax to consume JavaScript promises

C.New syntax to consume asynchronous callbacks

ANS: B

25)Which of the following is an attribute css selector?

A.[selector]

B. .selector

C. #selector

ANS: A

-----  
26) When is it recommended to pass this.setState a function instead of an object

A.When the new state should completely replace the old state .

B.When the new state does not depend on the old state.

C.When the new state depends on the old state.



When using `this.setState` in React, there are scenarios where passing a function instead of an object is recommended. Let's explore the options:

A. When the new state should completely replace the old state: In this case, you can directly use an object to update the state. For example:

JavaScript

```
this.setState({ quantity: 2 });
```

AI-generated code. Review and use carefully. More info on [FAQ](#).

This approach is straightforward and readable.

B. When the new state does not depend on the old state: If the updated state doesn't rely on the previous state or any other external factors, using an object is sufficient.

C. When the new state depends on the old state: Here's where passing a function becomes useful. When the updated state relies on the current state (or props), using a function ensures safety and prevents race conditions. For instance:

JavaScript

```
this.setState((prevState) => ({ counter: prevState.counter + 1 }));
```

AI-generated code. Review and use carefully. More info on [FAQ](#).

In this example, we access the previous state (`prevState`) to calculate the next state.

In summary:

Use an object when the state update is straightforward.

Use a function when the update depends on the previous state or props1234.

Therefore, the correct answer is C. When the new state depends on the old state. 😊

-----

30) Where should you sort and filter data?

- A. In a directive
- B. In a pipe
- C. In your component
- D. In a filter

ANS: C

33) When bootstrapping an Angular app, how do you specify the main, top-level app component to be used?

- A. You add the component as a package in your SystemJs config.
- B. You load the component directly in your index.html file.
- C. You add the component to the bootstrap array in your main app module.
- D. You load the component directly from your main SystemJs package (e.g. main.ts).

ANS : C

35) What syntax do you use to render the App component into an HTML element with id "root"?

- A. ReactDOM.render(App, document.getElementById("root"));
- B. React.render(App, document.getElementById("root"));
- C. ReactDOM.mountNode(<App />, document.getElementById("root"));
- D. ReactDOM.render(<App />, document.getElementById("root"));

ANS : D

38) What is the name of the file that manages TypeScript projects?

A.tsproj.config

B.tsmeta.json

C.tsconfig.json

D.package.json

ans C

-----  
43) How would you define a string type variable with the value bar in TypeScript?

A.let string: foo = 'bar';

B.let foo:string = 'bar';

C.let string foo = 'bar';

D.let foo (string) = 'bar';

Copilot

The correct way to define a string type variable with the value “bar” in TypeScript is:

B. let foo: string = 'bar';

In TypeScript, you declare a variable by specifying its name, followed by a colon and the data type. In this case, we use string as the data type for the variable foo, and then assign the value 'bar' to it. 😊

-----

5) What theorem or principal MongoDB follows?

A.APAC

B.Always-Sync

C.ASCII

D.CAP

ANS : D

8) Upon running the following command, what all the fields will be returned in the output?

```
db.collection.find( { qty: { $gt: 6 } }, { name: 1, _id: 0 } )
```

A.\_id

B.name

C.\_id, name

D.name1, \_id0

ANS : B

10) In the following command, what does the second parameter represent?

```
db.collection.find( { qty: { $gt: 6 } }, { name: 1, wins: 1 } )
```

Read Concern

Projection

Write Concern

Query

ANS : B

3. EventEmitter class's simple interface, which basically encompass two methods . . . . .  
. . . . can therefore be used to trigger custom events and listen to events as well, both  
synchronously or asynchronously.

- A) exit()
- B) superscript()
- C) subscribe()
- D) emit()

Ans: C

5. Three main components of Routing are . . . . .

- A) RouteObject
- B) RouterOutlet
- C) RouterState
- D) RouterLink

Ans: A,B,D

7. To use observables, Angular uses a third-party library called . . . . .

- A) ng2-bootstrap
- B) angular-cli
- C) numeral
- D) Reactive Extensions (RxJS)

Ans: D

8. Observables help you manage . . . . . data.

- A) synchronous
- B) asynchronous

C) Both asynchronous & synchronous

D) None of above

Ans: B

11. The promise becomes . . . . . after resolving or rejecting.

A) mutable

B) ignored

C) immutable

D) recurring

Ans: C

12. Which of the following is not built-in pipe in Angular

A) DatePipe

B) CurrencyPipe

C) DataPipe

D) PercentPipe

Ans: C

13. If you chain multiple pipes together, they are executed . . . . .

A) in parallel

B) LIFO order

C) in the order in which you specify them

D) none of above

Ans: C

1. arithmetic exception and io exception are of which type?? runtime error

2. what is the sequence of java servlet deployment descriptor??

ans: coding, compiling, deploying

3. which is used to submit query?? options: statement

4. abstract method doesn't follow which statement??

options: it instantiates its object

abstract class should have at least one abstract method

none of the above

5. what are the implicit objects in jsp??

ans: page, this

6. valid status codes ans: 200-299

7. super class of all classes ans: object

8. which thread method defines all threads? ans: daemon

9. jdbc belongs to which package? ans: java.sql\*

10. which is not a primitive datatype? ans: string

11. println belongs to class printstream

12. void method deosnot return value

13. which method reverse the string?? reverse()

14. session belongs to which class?? httpServletRequest

15. frame has default border type and plain has default flow type

16. which interface is used to connect database to application?

17. list of file returns wat? ans:an array of string is returned

18.which is a valid declaration??

ans: public void int A()-false

public int test2(int o);

public void test int(int i,intj);-false

19. public class B()

static int x;

psvm()

{

sop (B.x);

}

ans: prints 0

20.which is not a directive??



21. what jsp syntax are following scriptlet, expression, directive?

22. `<jsp:usebean id="name" class="java.lang.string"><%=name%>`

options: prints 0

prints null

compile time error

none of the above

23. which invoke method will print base class constructor?

ans: super(10)

24. q18 in previous paper

25. how session object can be destroyed?

26. public class test()

{static int i, static double d, static char c, static long l, static short s,

static string s;

psvm{sop(i+d+c+l+s+S);

}

}

27. `<%!int i=8%>`

`<%int i=10%>`

`<%=i>`

options: prints 8, prints 10

28. `class.forName()` throws `class not found` exception

29. `seek` method is used in random access file to specify the new position

30. which is not a valid declaration??

options: `char c=a;`

`char d =\u00432;`

`char c= \uae262;`

31. `public class test ()`

`static int i=012;`

`psvm(srting args[])`

`{sop (i);`

`}`

ans: prints 10

32. which is not valid?

ans: `int i[] = new int []`

`int i [][] = new int[] {1,2}`

33. `public class B()`

`boolean a,b,c;`

`boolean =a||(b=true)&&(c=true)`

`sop(a+","+b+","+c);`

options: `false,false,false`

`false,true,true`

34. `inputstreamreader i=new inputstreamreader (vds cv);`

bufferedReader br = new bufferedreader (i);

35. how ll u create the copy of the same object?? ans: clone()

36.all servlet object implements 1.servlet context 2.web context 3. both 1&2

37.submitting a query is done by ans:statement object

38.which math object is used for pointing 4 while x=4.8

options: math.ceil,math.floor

39.wat is feature of java?? OOPS

40.how does applet communicate with browser?? ans: init and database document

### TYPESCRIPT MCQS

1. Which are the different Data Types supported by Typescript?

A. Boolean var bValue: boolean = false

B. Number var age: number = 16

C. String var name: string = "jon"

D. All of the above mentioned

Ans : d

=====

2. What are the variable scopes available in TypeScript?

A. Global Scope

B. Class Scope

C. Local Scope

D. All of the above

Ans : d

=====

3. Which object oriented terms are supported by Typescript?

- A. Modules
- B. Classes
- C. Interfaces
- D. All of these

Ans : d

=====

4. In typescript, it calls an javascript internally and then performs validations

- A. True

=====

5. To change the datatype from one type to another, we require

- A. TypeOperator
- B. TypeAssertion
- C. InferredTyping

Ans: B

=====

6. For a variable, Without Specifying any datatype is

- A. InferredTyping
- B. TypeOperator
- C. TypeAssertion

Ans: a

=====

7. Which of the following is valid command to compile typescript file?

A.ts abc.ts

B.t abc.ts

C.tsc abc.ts

D.tst abc.ts

Ans : c

=====

8.What will be output of : var a:string=47; console.log("Value of a="+a);

A.Value of a=47

B.Value of a=0

C.Value of a=

D.None of the above

Ans : d

=====

9.Internal Modules are known as namespaces in Typescript.

A.True

=====

10.Which of the following Arrowfunctions are true

A.var reflect = value => value;

B.var sum = (num1, num2) => num1 + num2;

C.var getName = () => "Nicholas";

D.var sum = (num1, num2) => { return num1 + num2; };

E.All

=====

```
11.function fun1(...params) {  
  console.log(params.length);  
}
```

fun1();

fun1(5);

fun1(5, 6, 7);

Which concept resemble the above program

A.String Interpolation

B.Tagged Templates

C.Spread Operator/Rest Parameters

D.Object Destructuring

Ans: c

=====

12.Typescript is a

A.Strongly typed

B.Object oriented

C.Compiled Language

D.All the above

Ans: D

=====

13.Typescript is

A.Language

B.set of tools

C.Both

D.None

Ans: c

=====

14. Select user defined datatypes in typescript

- A. arrays
- B. String
- C. Boolean
- D. Void

Ans: A

=====

15. Which of the following are true

- A. we can use printf in ts(typescript)
- B. we can use console.log in ts(typescript)

Ans: B

=====

16. Which of the following are array methods

- A. Unshift()
- B. Sort()
- C. Pop()
- D. All the above

Ans: D

=====

17. Supertype of all datatypes in typescript

- A. number
- B. Enum
- C. Void
- D. Any

Ans: D

=====

18. Which of the following are true

A.var a=new String(hii)

B.var b=new array[1]

C.var a:String=hii;

D.All the above

Ans:B

=====

19. List some features of Typescript?

A. Typescript can be compiled to all major versions of Javascript(ES3,ES5,ES6,ES7)

B. Typescript can be used for cross –browser development and is an open source project

C. Typescript is a superset of JavaScript that provides typed nature to your code

D. All of the above

Ans: D

=====

20.What are the components of TypeScript?

A. TypeScript Language

B. TypeScript Compiler

C. TypeScript Language Service

D. All of these

Ans: D

=====

21.

static code analysis for angular

ng build

ng serve

ng lint-ans



Default scope of the bean

singleton-ans

log4j package are known

logger

parcel

none of the above

how to add annotation

3.Annotation added as an input parameter to the handler method.

a) @PathVariable

b) @Path

c) @PathLocale

d) None of the mentioned

ans: a

. Which library allows the use of the "require" keyword in front-end code?

A. guid

B. Browserify

C. React

D. jQuery

7.Which of the following API is a MUST for every React Component ?

A. getInitialState.

B. render

C.renderComponent.

ANS : C.

0. Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows

into a single object? Select a unique answer.

a. ResultSetExtractor

b. RowMapper

c. RowCallbackHandler

d. ResultSetMapper

ans: B

Q 26 - By default a bean is lazily loaded.

A - True

B - False

\*

1)2 object p ={name:"annu"} o={name:"annu"} what is the output of p.equalsOf(O)

2)given "text" how to match this with the pattern ans-pattern.test("text")

3)fulx is one way binding true/false

4)route can have multiple child true/false

5)microservice at the gateway what it will do login/fault tolerance/

6)in js we use addEventListener if we want to exit what what is the command  
exit()/abort()/default prevent /prevent listener

### Spring Questions

1. Shutdown of IOC containers - Using registerShutDownHook
2. <list> - wiring values and allows duplicates
3. Log4J parlance each package name - logger
4. Hibernate uses HibernateException and Jpa uses Persistence Exception - True
5. Unit test is used to validate single domain class - True
6. Lifecycle of JPA - all of the above
7. Bean pattern names is used only in XML based where as it is not used in AspectJ based - True
8. call stored procedures - simpleJDBCCall

9. You indicate a class selector with

A.the hash (#) symbol

B.the dot (.)

C.the tag

ANS : B

micro service -all

docktype html

props and states

49. In Angular routing , which of these tags is used to show the selected route component dynamically ?

- A. <router></router>
- B.<router-output></router-output>
- C.<router-outlet></router-outlet>
- D.<router-input></router-input>

ANS : C

row mapper

render component

40.Which of the following is the correct syntax for two-way binding?

- A.([ngModel])='listFilter'
- B.[(ngModel)]='listFilter'
- C.{{listFilter}}
- D.ngModel='listFilter'

ANS : B

42. When you apply an Angular 'pipe' , it changes the value of the underlying component's member variable as well.

- A . TRUE.
- B. FALSE.

ANS :- B.

44. In Angular, how can you set a header in every http request ?

- A. Use HttpClientInterceptor.
- B. Extend HttpClient class.
- C. Set as a request header parameter.

ANS : A.

47. In Angular , one can create a local HTML reference of an HTML tag using a variable , which starts with character \_\_\_\_ .

- A. @
- B. #
- C. \*
- D. &

49. In Angular routing , which of these tags is used to show the selected route component dynamically ?

- A. <router></router>
- B. <router-output></router-output>
- C. <router-outlet></router-outlet>
- D. <router-input></router-input>

ANS : C

Q 6 - How to handle shut down of IoC containers?

- A - Using shutdownHook()
- B - Using shutdownHandler()
- C - Using registerHook()
- D - Using registerShutdownHook()

Answer:D

10 - Which class is used to map a database row to a java object in spring?

A - ResultSet

B - RowMapper

C - RowSetMapper

D - ResultSetMapper

Answer:B

Q 18 - Which is the way to provide configuration metadata to spring?

A - XML Based configuration file.

B - Annotation based configuration.

C - Java based configuration.

D - All of the above.

D

75.) The four kinds of class members are\_\_\_\_\_

a) Instance methods, Instance fields, Static method, Dynamic method

b) Instance fields,Instance methods, Class fields, class methods

c) Instance fields, non-Instance fields, Dynamic methods, Global methods

d) Global methods, Local methods, Dynamic methods, Static methods

Ans: B

What is the premise of equality for IdentityHashMap?

Name equality

Hashcode equality

Length equality

Reference equality

ANS :- D

5) Which Annotation is used to refer to Pointcuts.

A. @PointCut.

B. @PointCutExecution.

C. @PointCutBefore.

ANS :- A

14) A logger location where logging information is sent is called

1. appender.

2. logger

3. component.

ANS : appender A

38) Which of the following statements is true about microservices architecture ?

i. Microservices applications are required to be deployed on the same physical host.

ii. Microservices architecture supports high availability of individual microservices.

iii. Microservices cannot be used if a company is embracing DevOps.

iv. Microservices are designed using a bounded context that can communicate with other bounded contexts.

A. Only i

B. Only ii

C. Only ii and iv

D. All of the above.

Ans -> D

53) To use the Angular HttpClient component, you must import the \_\_\_\_ module.

A. HttpClientModule.

B. HttpModule.

C. Http

Ans :- A

54) In Angular , one can create local HTML reference or HTML tag using variable which starts with character \_\_\_\_

A. @

B. #

C. “

D. &

ANS :- B

2. In Java, which of the following can be marked static?

a) Methods, Variables, Initialization Blocks, Outer Classes, and Nested Classes

b) Methods, Variables, Initialization Blocks, and Nested Classes

c) Methods, Variables, Initialization Blocks, and Outer Classes

d) Methods, Variables, and Initialization Block

Answer: b) Methods, Variables, Initialization Blocks, and Nested Classes

7. What is the max capacity of a Java Blocking Queue?



- a) Integer.INFINITY
- b) Integer.MAX\_VALUE
- c) BigDecimal.MAX\_VALUE

Answer: b) Integer.MAX\_VALUE

31. Which Java Set implementation is sorted and synchronized?

- a) TreeSet
- b) LinkedHashSet
- c) HashSet
- d) ConcurrentSkipListSet

Answer: d) ConcurrentSkipListSet

43. Which one of the following statements on Java String is correct?

- a) replace() method replaces only first occurrences of a character in invoking string with the given character.
- b) replace() method replaces last occurrence of a character in invoking string with the given character.
- c) replace() method replaces all occurrences of one character in invoking string with the given character.
- d) replace() method replaces all the characters in invoking string with the given character.

Answer: c) replace() method replaces all occurrences of one character in invoking string with the given character.

What annotation wiring element is added to scan packages to find and

register beans with in the Application Context?

a) <context:annotation-config/>

b) <context:annotation-scan/>

c) <context:component-config/>

3. Spring

7

d) <context:component-scan>

Answer

d. <context:component-scan

54.Which of the following API is a MUST for every React component? render

getInitialState

render

renderComponent

ANS : B

51.What are the limitations of React?

React is only for view layer of the app so we still need the help of other technologies to get a complete tooling set for development.

React uses inline templating and JSX. This can seem awkward to some developers.

The library of React is too large.

All of the above

ANS : D

Which class is introduced in Java 8 to get the current milliseconds?

- a. Clock
- b. Current Time
- c. Date Time
- d. Time

v Explanation:

The class introduced in Java 8 to get the current milliseconds is:

- 1. Clock

->Redux is a single dimension insertion

->Flux has only one dispatcher

->How can you access the state of a component from inside of a member function in React?

`this.getState()`

`this.prototype.stateValue`

`this.state`

`this.values`

->we can use ES6 classes for creating components (T/F)

->which of the following are proper Routing for homepage

`<Route exact path="/" render={()=>{}}>`

`<Route path="/" render={()=>{}}>`

`<Route path="/" component={<Homepage/>}>`

->states can be passed from outside components using - props

->Docker container is an instance of docker image - (T/F)

->Docker cannot have multiple containers of a single image - (T/F)

->What is the replacement for PermGen space in java8?

- a. FixedGen
- b. Metaspace
- c. MetaGenspace

-> In Java, which of the following data structure classes CANNOT be type parameterized?

- a. List
- b. Set
- c. Array
- d. Map

->What is the difference between peek() and dequeue()?

- a. Peek() returns the next item in line but dequeue() removes and returns the next item in line
- b. peek() and dequeue() return the next item in line
- c. peek() and dequeue remove and return the next item in line

->Dependency injection Router qn

->PointCut cannot be???

->how to get background zebra striped (HTML)

->how to mark checkbox in input (HTML)

->SpringBoot can use repository from (Spring)

Maven

Gradle

npm

->you can create object using new keyword and \_\_\_\_ method - (JS)

->Which PointCut annotation executes after a joinpoint is over?

1.Which of the following is not built-in pipe in Angular

A) DatePipe

B) CurrencyPipe

C) DataPipe (correct)

D) PercentPipe

in angular <router-outlet> atmost only one (t or f)

2.A logger location where logging information is sent is called

- 1. appender.
- 2. logger
- 3. component.

ANS : appender

3.Which element is used to restrict the access to a particular URL within Spring Security ?

- i. restrict-url
- ii.intercept-url (correct)
- iii.intercept-restrict.

4.What theorem or principal MongoDB follows? (like this but question diff)

- A.APAC
- B.Always-Sync
- C.ASCII
- D.CAP (correct)

5.Which of the following is the correct CSS syntax to set the background color of an element?

Ans: `element { background-color: red; }`

6.Which class indicates a dropdown menu?

- a) .dropdown-list
- b) .select
- c) .dropdown (correct)

7.which syntax is correct :

ans-limit()

8.How can we render JSX in the browser ?

- i. Using Babel
  - ii. Using Typescript.
  - iii. Using Flux.
  - iv. Using Redux.
- A. Only i and ii
  - B. Only iv
  - C. Only i,ii and iv
  - D. Only iii and iv.

ANS : A

9.What is the max capacity of a Java Blocking Queue?

- a) Integer.INFINITY
- b) Integer.MAX\_VALUE (correct)
- c) BigDecimal.MAX\_VALUE

10.Which one is NOT a spring data MongoDB annotation ?

- A. @Document
- B. @Field

C. @Entity

D. @Query.

ANS : @Entity

11.Which of the following is NOT a valid keyword in JavaScript?

Options:

A) use strict

B) debugger

C) with

D) if

Answer: A) use strict

12.Can a single Docker image be used to create only one container?

Options:

A) True

B) False

Correct Answer: B) False

13.A Docker container contains all the application files but does not include the instructions required to run the application.

Options:



- A) True
- B) False

Correct Answer: B) False

14.Which of the following statements about Join Point and Advice in Aspect-Oriented Programming (AOP) is correct?

ans: Join Point is a point in program execution where an aspect can be applied, and Advice is the action performed at that point.

15.Which of the below methods are NOT a part of React Component's lifecycle ?

- A. Initial Rendering.
- B. Render / Re-Render.
- C. unMounting.
- D. Remote URL calls. (correct)

8.Which button class is used to create a large button?

- a).btn-lg b).btn-xl c).btn-l d).btn-large

9.Which class indicates a dropdown menu?

- a).dropdown-list b).dropdown c).select

Where can the <script> tag be added in HTML?

Options:

- \* Head
- \* Body
- \* Both

Answer: Both head and body

How to change bold formate

font-size B)font-style C)text-style D)text-format

15.Syntax for large buttons in Bootstrap?

A).large

B).button-large

Bt

5.Which of the following CSS Property controls how an element is positioned?

A.position

B.set

C.static

ANS : A

12) Which of the following rules allows users to import style rules from other style sheets?

A.@media

B.@important

C.@import

.container question

mongodb remove() qn

which relationship is used in mongodb to connect document - 1:1 embedded document

1:many embedded document

1:many document reference

in sql table similiar in mongodb? collections

How to check if the pattern matches with the string "text" in JavaScript - Ans.  
`pattern.test("text")`

What is closure in JavaScript - Ans. Both Function Object and Scope ...

What JSONP stands for - JSON with padding

```
const obj = {  
    obj.foo;  
}  
obj.bar;  
error----ans
```

scope of var is block

scope of let is block

what defines docker - all of the above

48. Spring Boot: What is the default embedded server?

- a) Jetty
- b) Tomcat
- c) Undertow
- d) Netty

Answer: b) Tomcat

Where is the correct place to insert a Javascript ?

- A. The <head> section
- B. Both the <head> section and the <body> section are correct.
- C. The <body> section.
- D. The <footer> section.

You can also use an ES6 class to define a component in React.

True

False

What are the variable scopes available in TypeScript?

- A. Global Scope
- B. Class Scope
- C. Local Scope
- D. All of the above

Ques. Bootstrap's grid system allows up to

- a. 6 columns across the page
- b. 12 columns across the page
- c. columns across the page
- d. columns across the page

) What does it mean by publishing a REST service in Spring ?

- i. Publishing an application's data as Json over a REST service.
- ii. Publishing an application's data as XML over a REST service.
- lii. Sharing Database within two microservices using Spring Data Access Object (DAO).

A. Only 1

- B. Only 1 and 2
- C. Only 2
- D. All of the above.

A logger location where logging information is sent is called

- 1. appender.
- 2. logger
- 3. component.

In MongoDB, which of the following methods return one document ?

- A. findOne()
- B. findOne1()
- C. selectOne()

Identify the incorrect statement on MongoDB.

- A. Secondary indices are not available in MongoDB.
- B. MongoDB supports search by field, range queries, and regular expression searches.
- C. MongoDB can store the business subject in the minimal number of documents.

How can we render JSX in the browser ?

- i. Using Babel
- ii. Using Typescript.
- iii. Using Flux.
- iv. Using Redux.

- A. Only i and ii
- B. Only iv
- C. Only i,ii and iv
- D. Only iii and iv.

Is the following a valid Java code snippet ?

```
Optional< String > aValue = Optional.of("Not Empty");  
if(aValue.isPresent()) {  
    System.out.println("Got Not Empty");  
}
```

A. True

B. False

In Java `Optional.ofNullable()` method of the `Optional` Class, returns a Non-empty `Optional` if the given object has a value, otherwise it returns an empty `Optional`.

A. true

B.false

Load Balancing improves the distribution of workloads across multiple computing resources , such as computers or a computer cluster.

A. True.

B. False.

what are the components of redux - action reducer and store

Which is the correct CSS Syntax ?

A. body color-black

B. {body color-black}

C. body {color:black;}

D. {body;color:black}

Which Bootstrap class adds zebra-stripes to a table ?

- A. table-bordered.
- B. table-zebra.
- C. table-stripped
- D. even and odd

How can you detect the client's browser name in Javascript ?

- A. class.navName
- B. navigator.appName
- C. browser.name
- D. window.browser.

In Java, which of the following can be marked static?

- a) Methods, Variables, Initialization Blocks, Outer Classes, and Nested Classes
- b) Methods, Variables, Initialization Blocks, and Nested Classes
- c) Methods, Variables, Initialization Blocks, and Outer Classes
- d) Methods, Variables, and Initialization Block.

Java system properties can be set or modified at runtime.

- a) True
- b) False

Which one will allow traversing a Java collection in each direction?

- a) ListIterator
- b) Iterator
- c) MapIterator
- d) SetIterator

In Java, it is NOT possible to override default serialization.

- a⌘ True
- b⌘ False

Java Functional Interface can be defined as\_\_\_\_\_.

- a⌘ Simple Abstract Markup
- b⌘ Simple Active Markup
- c⌘ Single Ambivalue Method
- d⌘ Single Abstract Method

Which class is introduced in Java 8 to get the current milliseconds?

- a⌘ Clock
- b⌘ Current Time
- c⌘ Date Time
- d⌘ Time

In Java, Which of these class types supports sequential and parallel processing of data?

- a⌘ Queue
- b⌘ Lambda
- c⌘ ExecutorService
- d⌘ Streams

Java Stream does NOT store the elements. It simply performs the aggregate operations to get the desired stream of data.

- a⌘ True



b ⚡ False

What is the return type of Java Lambda expressions?

a ⚡ Void

b ⚡ Lambda

c ⚡ Function

d ⚡ Regex

Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.

a ⚡ true

b ⚡ false

A collection value of the type `java.util.Properties` is defined by what element?

a ⚡ `<props>`

b ⚡ `<prop>`

c ⚡ `<properties>`

d ⚡ `<property>`

Rakesh

Q 8 - By default a bean is lazily loaded.

A - True

B - False

Answer : B

6. What React function is used to define a new element tag?

A. renderComponent

B. createClass

C. setInnerHTML

ANS : B

12. React is

A A library that represents asynchronous data streams with observables

B A comprehensive client-side application framework

C An optimizing compiler for JavaScript

D A client-side library providing rendering and event handling

ANS : D

27.Which of the following API is a MUST for every React Component ?

A. getInitialState.

B. render

C.renderComponent.

ANS : C.

30. What is React in MVC ?

A. Controller

B. Middleware

C. Model

D. Router

ANS :- None of these. Updated ans is Controller. Option A

31) What is React?

Just a server-side framework

Just a user-interface framework

Both a server-side framework as well as a user-interface framework

ANS : C

32) What are the limitations of React?

React is only for the view layer of the app so we still need the help of other technologies to get a complete tooling set for

development.

React uses inline templating and JSX. This can seem awkward to some developers.

The library of React is too large.

All of the above

ANS :- D

33) How can you access the state of a component from inside of a member function in React?

`this.getState()`

`this.prototype.stateValue`

`this.state`

`this.values`

ANS : D

34) You can also use an ES6 class to define a component in React.

True

False

ANS : A

35) Which of the following API is a MUST for every React component?

getInitialState

render

renderComponent

ANS : C

16.Which of the following are array methods

A.Unshift()

B.Sort()

C.Pop()

D.All the above

12.Which of the following is not built-in pipe in Angular

A) DatePipe

B) CurrencyPipe

C) DataPipe

D) PercentPipe

Ans: C

41. In Angular / RxJs , when should you unsubscribe from a Subscription ?

A. There is no need to unsubscribe . Angular routing handles it.

B. Unsubscribe them in ngOnDestroy.

ANS :- B.

12. Publishing an application's data as a REST service requires.

a) @RequestMapping

b) @PathVariable

c) All of the mentioned

d) None of the mentioned

ans: a b

Q 8 - By default a bean is lazily loaded.

A - True

B - False

Answer : B

2. \_\_\_\_\_ contains the navigation menu, or other navigation functionality for the page

A.section

B.header

C.nav

D.aside

ANS : C

25. What is advice? Select a unique answer.

- a. An action taken by an aspect at a particular join point
- b. A point during the execution of a program
- c. An aspect and a pointcut
- d. A predicate that matches join points

ans: a

70.) ----- is an HTML specification used to add more information to HTML tags.

A) Macrodata

B) Microdata

c)Minidata

d) Modifydata

Ans: B

3) How would you limit the results of a find to 10 items?

`adb.collection.find({}, {limit: 10})`



`db.collection.find().limit(10)`

`db.collection.limit(query, 10)`

This is not possible in the node.js driver

ANS : B

28) Which of the following is not a callback method of React ?

A. `render()`

B. `componentDidMount()`

C. `componentFailedToMount()`

D. `componentWillunMount()`

ANS : C -> failedtomount

32) Which of the following statements are true about Java ?

i. A finally block is executed, only after the catch block is executed.

ii. A finally block is executed whether an exception is thrown or not.

iii. A finally block is NOT Mandatory.

iv. A finally block is executed, only if an exception occurs.

A. i

B. iv

C. i and iv

D. ii and iii

ANs :- D

33) Which Java Class has natural sorting implemented ?

A. LinkedHashSet.

B. TreeSet

C. HashSet

ANS :- B -> TreeSet

49) Which of the following methods CANNOT be used to add array elements in Javascript ?

A. push()

B. splice()

C. shift()

D. All can be used.

ANS : D

53) To use the Angular HttpClient component, you must import the \_\_\_\_ module.

A. HttpClientModule.

B. HttpModule.

C. Http

Ans :- A

• ☹️ 🤖 Identify the parts of lambda expression in Java.

a 🤖 Name

b 🤖 Parameterlist

c 🤖 Body

d 🤖 ReturnType

1.i

2.iv

3.i and iv

4.

ii and iii

• • ρ Java Functional Interface can be defined as\_\_\_\_\_.

a ρ Simple Abstract Markup

b ρ Simple Active Markup

c ρ Single Ambivalue Method

d ρ Single Abstract Method

• • ρ Which of the following types of Java methods CANNOT be Type

parameterized?

a ρ Final methods

b ρ Overriding methods

22

## 1. Core Java

c) Overloaded methods

- • ) What is the replacement for PermGen space in java8?

a. FixedGen

b. Metaspace

c. MetaGenspace

- ☹ ) What is the return type of Java Lambda expressions?

a) Void

b) Lambda

c) Function

d) Regex

Q 32 - What is the scope of stateless beans?

A - global-session

B - singleton

C - prototype

D - request

B

Js- Ananya

1.closure

2.code snippet:

```
const obj{
```

```
  obj.foo}
```

```
  obj.bar;
```

3.

57. DevOps: What is Infrastructure as Code? a) Manual server setup b) Defining infrastructure using scripts c) Database management d) Application deployment

React is server-side rendering?

Options:- True/False

ANS;- True

what is the feature of the canvas -to design for graphics (something)

which is not used to add the elements to an array

push

shift

splice

all---ans

which of the following is the consider to be end rest ?

|

,

{}

which one will allow traversing a java collection in each direction?

ListIterator----ans

Iterator

MapIterator

SetIterator

The \_\_\_ javascript method like the mathematical "for all" quantifier, which resumes true if the function returns true for all elements in the array.

every()---ans

some()

all()

forall()

which of these angular services can be injected in your component to dynamic mode navigation?

Routing

RouterService

RoutingService

Router-----ans

```
Optional<String>aValue=optional.of("Not Empty");
```

```
If(aValue.isPresent()){
```

```
System.out.println("Got Not Empty");
```

```
}
```

True

False-----ans

what is the difference between peak() and dequeue()?

peek() returns the next item in line but dequeue() removes and returns the next item in line-----ans

peek() and dequeue() return the next item in line

peek() and dequeue remove and return the next item in line



14. Pick the correct Java method reference usages.

- a) object::instanceMethod
- b) Class::staticMethod
- c) Class::new
- d) All of the above

Answer: d) All of the above

. Java Functional Interface can be defined as \_\_\_\_\_.

- a) Simple Abstract Markup
- b) Simple Active Markup
- c) Single Ambivalue Method
- d) Single Abstract Method

Answer: d) Single Abstract Method

```
class First {  
    public First() {  
        System.out.println("F");  
    }  
}  
  
class Second extends First {  
    public Second() {  
        System.out.println("S");  
    }  
}  
  
class Third extends Second {  
    public Third() {  
        System.out.println("T");  
    }  
}
```

```
    }  
}  
public class TestClass {  
    public static void main(String[] args) {  
        Third c = new Third();  
    }  
}
```

a) TSF

b) T

c) F

d) FST

Answer: d) FST

28. Pick the correct Java Stream API's?

a) Stream.allMatch

b) Stream.noneMatch

c) Stream.anyMatch

d) All the above

Answer: d) All the above

What is the return type of Java Lambda expressions?

a) Void

b) Lambda

c) Function

d) Regex

Answer: c) Function

HTML numbers = <ol>

Redux is an exclusive feature of the React store ?

A. TRUE

B. FALSE

ANS : B

14. Pick the correct Java method reference usages.

a) object::instanceMethod

b) Class::staticMethod

c) Class::new

d) All of the above

Answer: d) All of the above

2. \_\_\_\_\_ contains the navigation menu, or other navigation functionality for the page

A.section

B.header

C.nav

D.aside

ANS : C

4. Which of the following are automation tools?

A. Gulp

B. JSX

C. Express

ANS : A

27. Which of the following API is a MUST for every React Component ?

A. `getInitialState`.

B. `render`

C. `renderComponent`.

ANS : C.

29. How does React handle the Web Accessibility Initiative - Accessible Rich Internet Applications (WAI-ARIA) standard ?

A. `aria-*` HTML attributes are fully supported in JSX where most DOM properties and attributes in React are camelCased, these

attributes should be lowercase.

B. React processes `aria-*` attributes separately and updates the DOM for accessibility if the user requires it.

C. `aria-*` attributes should be converted to camelCase like other attributes; For example: `className`, `onChange` and so on.

D. React is yet to support the WAI-ARIA standard.

ANS : D.

30. What is React in MVC ?

A. Controller

B. Middleware

C. Model

D. Router

ANS :- None of these. Updated ans is Controller. Option A

31) What is React?

Just a server-side framework

Just a user-interface framework

Both a server-side framework as well as a user-interface framework

ANS : C

Q 11.What keyword is used to apply all of the properties on one interface to another interface?

implements

extends

class

furtheres

references

ANS B

Q.what does pop() do

ans: it deletes the frist element

Q 26 - By default a bean is lazily loaded.

A - True

B - False

4) What is continuous testing?

Testing continuously for 24 hours

Manually testing each build, end-to-end

Automated testing

Automated testing on each continuous deployment

ANS : D

14) GIT can be integrated with Jenkins.

True

False

ANS : A

8) Which element is used to restrict the access to a particular URL within Spring Security ?

i. restrict-url

ii.intercept-url

iii.intercept-restrict.

ANS : B

16) Sharding a MongoDB database across many server instances can be achieved with

\_\_\_\_\_

A.LAN

B.SAN.

C.MAN

ANS : LAN A

31) Which of the following is the immutable Java List static factory method ?

A. List.of

B. List.immutable.

C. List.create

ANS : - A

33) Which Java Class has natural sorting implemented ?

A. LinkedHashSet.

B. TreeSet

C. HashSet

ANS :- B -> TreeSet

30. What is React in MVC ?

A. Controller

B. Middleware

C. Model

D. Router

31) What is React?

Just a server-side framework

Just a user-interface framework

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ANS : C

12. Which of the following is not built-in pipe in Angular

A) DatePipe

B) CurrencyPipe

C) DataPipe

D) PercentPipe

Ans: C

5) Which Annotation is used to refer to Pointcuts

A. @PointCut.

B. @PointCutExecution.

C. @PointCutBefore.

ANS :- A

20.REST service endpoint comprises of which address? Starts with http:// and may end with?

Starts with http:// and may end with?

No certain URL is specified

Depends upon the platform used

ANS : A

23.Microservices is most closely related to which other established acronym?

API

AWS

GCP

PAAS

ANS : A

27.What is the return type of Java lambda expressions?

void

Lambda

Function

Regex

ANS : C

Which tag is used to define a clickable image map in HTML5?

a) <map>

b) <area>



c) <img>

d) <figure>

Answer: a) <map>

6. Spring Web MVC (and in Spring in general) is the "Open for extension, closed for modification" principle. The reason that this principle is being mentioned here is because a number of methods in the core classes in Spring Web MVC are marked final.

A.true

B.false

Answer: True

95. Which implementation of Set would you choose if you want the iterator of Set to give objects in the order they were inserted?

a) LinkedHashSet

b) TreeSet

c) HashSet

Answer: a) LinkedHashSet

19. REST service endpoint comprises of which address?

Ans : Starts with http:// and may end with ?

37. Q.)Overriding can be done in?

Poly

Abstr

Inher

Encaps

Ans. polymorphism

48. Which class indicates a dropdown menu?

a) .dropdown-list

b) .select

c) .dropdown

Ans: C

Q) answer Array.splice()

Q) @After

@Before

@After return

All of the above---- answer

Q) o= fun

console(o.constructure === fun)

true

false

0

1

19.Which of the following will correctly import the items exported by this export statement?

```
export { GetUser as GetLibraryUser, GetBook as GetLibraryBook };
```

```
import TypeScript from './library';
```

```
import namespace from './library';
```

```
import { GetLibraryUser, GetLibraryBook } from './library';
```

```
import { GetUser, GetBook } from './library';
```

```
import all from './library';
```

ANS : C

What type of management does “Capped Collections” provide for inserted documents in MongoDB ?

A. in order to specifying them

B. LIFO

C. LRU

D. MRU

ANS :- in order to specifying them

⌚ Java Functional Interface can be defined as\_\_\_\_\_.

a ⌚ Simple Abstract Markup

b ⌚ Simple Active Markup

c ⌚ Single Ambivalue Method

d ⌚ Single Abstract Method

